

PHYSICAL VARIABLE MODELING OF MULTIPHASE INDUCTION MACHINES

Liwei Wang¹, Sina Chini Foroosh¹, Juri Jatskevich¹ and Ali Davoudi²

¹University of British Columbia, Canada

²University of Illinois at Urbana-Champaign, USA

ABSTRACT

Models of non-standard induction machines with number of phases higher than 3 are not commonly available and require special effort to implement. This paper presents a model of induction machine with arbitrary number of phases on the stator and rotor, which has been implemented in Matlab-Simulink. It is shown that machines with number of phases higher than 3 have less severe torque pulsation and the stator current increase following a loss of phase(s). The relative location of the faulted phases has an impact on redistribution of currents and resulting torque. The proposed model can be used to represent induction motors for transient studies involving multiple faults and system-level reconfiguration.

Index Terms— coupled-circuit model, induction machine, multiphase, phase-domain model.

1. INTRODUCTION

Induction machines with large number of phases and magnetic poles may offer advantages for electrical subsystems of vehicles, ships, and aircraft [1]–[8]. Such advantages include lower nominal phase currents (which allows the use of inverters with lower ratings), reduced torque pulsation, increased reliability and possible operation with one or more faulted phases [8]. A simplified circuit diagram depicting a multiphase (typically more than three) machine-inverter system is shown in Fig. 1. A modular design [6] can be used wherein the inverters may be arranged to supply the stator winding from the same or different DC sources for increased redundancy.

The detailed computer models of electric drives are particularly useful for predicting the electro-mechanical and electrical transients. Depending on the purpose of study, the induction motors may be modeled with different levels of accuracy using various techniques. At a very approximate level, the steady-state calculations of the torque ripple and spatial harmonics may be sufficient [1]–[3]. At the design stage, the finite element analysis models are often used. However, for the transient and system-level studies, the qd models [9] are typically used, but the generalized

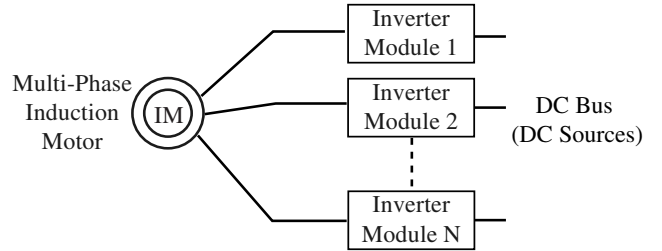


Fig. 1. Simplified diagram of multiphase induction motor drive system.

qd models for machines with high number of phases are generally not available in commonly used simulation packages (EMTP, PSCAD, VTB, Matlab/Simulink, etc.).

The model presented in this paper is based on representing the induction machine with arbitrary configuration of windings and phases as a collection of coupled circuits. Instead of using the transformed qd variables, the general model is expressed in the physical machine's variables and implemented using the state variable approach and the toolbox [10]. The resulting dynamic models are easier to generalize and interconnect to the power electronic networks for investigating various transients and reconfiguration scenarios.

2. MULTIPHASE INDUCTION MACHINE MODEL

The multiphase induction machine model can be derived by extending the 3-phase case [9] to an arbitrary number of phases. In this paper, the number of stator and rotor phases is denoted by N_{ph}^s and N_{ph}^r , respectively. For the purpose of generality, $N_{ph}^s \neq N_{ph}^r$. The corresponding voltage equations in machine physical variables can be written as

$$\begin{bmatrix} v_s \\ v_r \end{bmatrix} = \begin{bmatrix} \mathbf{r}_s & 0 \\ 0 & \mathbf{r}_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_s \\ \mathbf{i}_r \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} \boldsymbol{\lambda}_s \\ \boldsymbol{\lambda}_r \end{bmatrix} \quad (1)$$

where $\mathbf{v}_s$, $\mathbf{i}_s$, and $\boldsymbol{\lambda}_s$ are the vectors consisting of stator phase voltages, currents, and flux linkages, all of dimension N_{ph}^s ; $\mathbf{v}_r$, $\mathbf{i}_r$, and $\boldsymbol{\lambda}_r$ are the vectors consisting of rotor winding voltages, currents, and flux linkages, all of dimension N_{ph}^r ; and $\mathbf{r}_s$ and $\mathbf{r}_r$ are the matrices that

contain stator and rotor winding resistances, respectively. The flux linkages can be expressed as

$$\begin{bmatrix} \lambda_s \\ \lambda_r \end{bmatrix} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}_{sr}(\theta_r) \\ \mathbf{L}_{sr}^T(\theta_r) & \mathbf{L}_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_s \\ \mathbf{i}_r \end{bmatrix} \quad (2)$$

In (2), $\mathbf{L}_s$ and $\mathbf{L}_r$ are the stator and rotor inductance matrices. Since the machine is round and symmetrical, these matrices are constant. The coupling between the rotor and stator is represented by matrix $\mathbf{L}_{sr}(\theta_r)$, which has dimensions $N_{ph}^s \times N_{ph}^r$ and depends on the rotor position θ_r . The electromagnetic torque is calculated as

$$T_e = \frac{P}{2} \mathbf{i}_s^T \frac{\partial \mathbf{L}_{sr}(\theta_r)}{\partial \theta_r} \mathbf{i}_r \quad (3)$$

where P is the number of magnetic poles.

For machines with small number of phases, e.g. 3 and 5 phases, all windings may be viewed as single group. However, when 6 and more phases are considered, the phases may be grouped for better utilization of the phase zones. For example, instead of having a 6-phase stator winding with 60 electrical degrees between the adjacent phases, it is advantageous to represent this winding as two 3-phase groups that are shifted relative to each other by 30 electrical degrees. This reduces the overall phase zone from 60 down to 30 electrical degrees.

Multiple groups of windings are often used with 6-, 9-, 12-, 15-, and 18-phase machines. Multiple groups are depicted in Fig. 2 where it is assumed that the stator/rotor winding is composed of several 3-phase groups denoted by N_g^s and N_g^r , respectively. The number of phases in each of

the stator and rotor group is denoted by $N_{g,ph}^s$ and $N_{g,ph}^r$, respectively, which in general may be different (other than three).

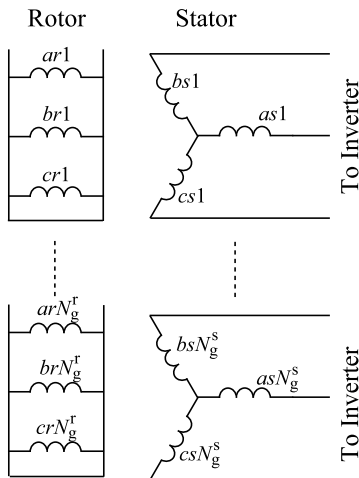


Fig. 2. Multiphase induction machine coupled circuits considering multiple Y-groups on the stator and rotor.

Such stator configuration has many advantages in terms of spatial harmonics as well as current ripple when the machine is fed by an inverter. Other advantages may include possible reconfiguration and reliability that also depends on the inverter(s) configuration [6], [8]. Using notations presented here, the overall number of phases in the stator and rotor can be expressed as

$$N_{ph}^s = N_{g,ph}^s N_g^s \text{ and } N_{ph}^r = N_{g,ph}^r N_g^r \quad (4)$$

3. SELF AND MUTUAL INDUCTANCES

To complete the model (1)–(3), the machine inductances must be specified. The inductances are derived based on the magnetic axes of the stator and rotor windings as shown in Fig. 3. Here, a spatial angle between the adjacent phases in the same group is defined as

$$\alpha_s = \frac{2\pi}{N_{g,ph}^s} \text{ and } \alpha_r = \frac{2\pi}{N_{g,ph}^r} \quad (5)$$

for the stator and rotor phases, respectively, (which for the 3-phase groups becomes $\alpha_s = \alpha_r = 2\pi/3$). The angular displacement between the stator groups is denoted by β_s , and between the rotor groups is β_r , respectively. All rotor magnetic axes are shifted by the angle θ_r . If $N_{g,ph}^s$ and $N_{g,ph}^r$ are odd numbers, then in order to minimize the phase zones, a good choice for these angle would be

$$\beta_s = \frac{\alpha_s}{2N_{g,ph}^s} \text{ and } \beta_r = \frac{\alpha_r}{2N_{g,ph}^r} \quad (6)$$

Based on (5)–(6), we express the spatial angle $\gamma_{i,j}$ between any i^{th} and j^{th} magnetic axes. In particular, for the stator magnetic axes this angle is

$$\gamma_{i,j}^s = \alpha_s(n-l) + \beta_s(m-k); \quad (7)$$

where
$$\begin{cases} i = N_{g,ph}^s(k-1) + l \\ j = N_{g,ph}^s(m-1) + n \end{cases}$$

$$k, m = 1 \dots N_g^s, \text{ and } l, n = 1 \dots N_{g,ph}^s$$

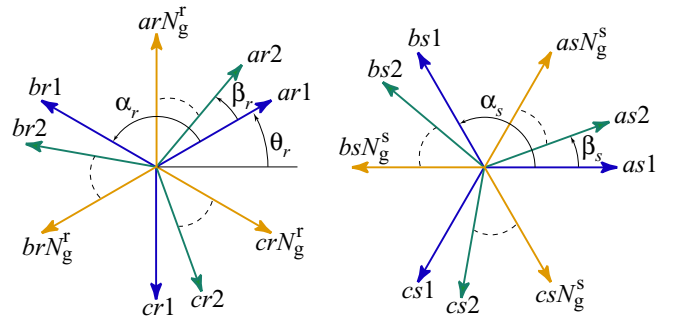


Fig. 3. Rotor and stator magnetic axes considering multiple 3-phase groups.

Similarly, for the rotor axes we have

$$\gamma_{i,j}^r = \alpha_r(n-l) + \beta_r(m-k);$$

$$\text{where } \begin{cases} i = N_{g,ph}^r(k-1) + l \\ j = N_{g,ph}^r(m-1) + n \end{cases}; \quad (8)$$

$$k, m = 1 \dots N_{g,ph}^r, \text{ and } l, n = 1 \dots N_{g,ph}^r$$

Taking into account the machine's symmetry and (7)–(8), the stator and rotor inductance matrices are calculated as

$$\mathbf{L}_s(i, j) = \begin{cases} L_{ms} + L_{ls}, & \text{if } i = j \\ L_{ms} \cos(\gamma_{i,j}^s), & \end{cases} \quad (9)$$

$$\mathbf{L}_r(i, j) = \begin{cases} L_{ms} + L_{lr}, & \text{if } i = j \\ L_{ms} \cos(\gamma_{i,j}^r), & \end{cases} \quad (10)$$

where $L_{ms} = L'_{mr}$, and L_{ls} , L_{lr} are the magnetizing and leakage inductances, respectively.

The angle between rotor and stator axes is found similarly. In particular, at $\theta_r = 0$, we have

$$\gamma_{i,j}^{sr} = -\alpha_s(l-1) - \beta_s(k-1) + \alpha_r(n-1) + \beta_r(m-1);$$

$$\text{where } \begin{cases} i = N_{g,ph}^s(k-1) + l \\ j = N_{g,ph}^r(m-1) + n \end{cases}; \quad (11)$$

$$k = 1 \dots N_{g,ph}^s, \text{ and } l = 1 \dots N_{g,ph}^s$$

$$m = 1 \dots N_{g,ph}^r, \text{ and } n = 1 \dots N_{g,ph}^r$$

Finally, the rotor-to-stator inductance matrix is calculated as

$$\mathbf{L}_{sr}(i, j) = L_{ms} \cos(\theta_r + \gamma_{i,j}^{sr}) \quad (12)$$

4. SIMULATION STUDIES

Equations (1)–(3), (7)–(12) were programmed in MATLAB-Simulink C-MEX S-function together with the automated state model generator toolbox module [10] and built into a general purpose single block-model. This block is then masked to provide a GUI interface for specifying the machine parameters, number of phases, phase groups, etc. The per-unit parameters of the base case 3-phase 100-HP motor used here are summarized in Appendix, from which the parameters for the 6- and 9-phase motors are calculated assuming the same rotating mmf. The stator windings are assumed to have a common neutral point that is left floating.

Several scenarios are considered with loss of one and multiple phases, which may happen due to failure of the winding(s) and/or the inverter leg(s). The results are summarized in Tables 1 and 2. Fig. 4 shows the transient observed in the stator currents (instantaneous and rms values) of the 3-phase base case motor. As can be seen, after one phase is opened, the current in the remaining phases increases above 2pu (above 90%). The results of studies

comparing the 3, 6, and 9-phase motors response to a loss of one phase are summarized in Table 1. As expected, the machines with higher number of phases have less severe torque pulsation and the stator current increase following a loss of one phase. In general, the re-distribution of currents is uneven among the remaining phases and only the maximum increase is given in Table 1. These results confirm the expected advantage of the 9-phase motor.

A simultaneous loss of two adjacent phases $a1$ and $a2$ from the first and second groups in the 9-phase induction motor is included here. The resulting transients are shown in Figs. 5 and 6. Because the faulted phases are close to each other (separated by 20 degrees), the effect on the current increase (111.07%) and the torque pulsation (48.5%) is significant. The decrease in the average torque and speed is also pronounced, which shows that the loss of adjacent phases is very undesirable. However, when phases $a1$ and $b1$ from the same group $abc1$ are faulted, the effect on the current increase (46.35%) and the torque pulsation (14.85%) are much less severe. This is because the faulted phases $a1$ and $b1$ are significantly apart (separated by 120 degrees).

Table 1: Response to a single winding open circuit.

Num. of phases	Torque pulsation, %	Change in average torque, %	Change in speed, %	Max. increase in phase current, %
3	204.27	2.51	1.27	91.19
6	29.73	0.239	0.12	64.65
9	16.0	0.125	0.063	36.96

Table 2: Nine-phase motor response to a loss of multiple phase windings.

Open phases	Torque pulsation, %	Change in average torque, %	Change in speed, %	Max. increase in phase current, %
1, (a1)	16.05	0.125	0.063	36.96
2, (a1, b1)	14.85	0.264	0.133	46.35
2, (a1, a2)	48.5	0.420	0.211	111.07
3, (abc1)	0	0.416	0.209	52.18

5. CONCLUSION

This paper presented a coupled-circuit physical-variable model for multiphase induction machines. The generalized equations for the self and mutual inductance matrices have been derived. The developed interface in Simulink makes it possible to readily implement an induction machine with arbitrary number of phases and/or phase groups on the stator and rotor in a very straightforward way. The simulation studies demonstrate that the motors with higher number of phases have less severe torque pulsation and the stator current increase following the loss of one or more phases making such motors especially suitable for applications with high reliability requirements.

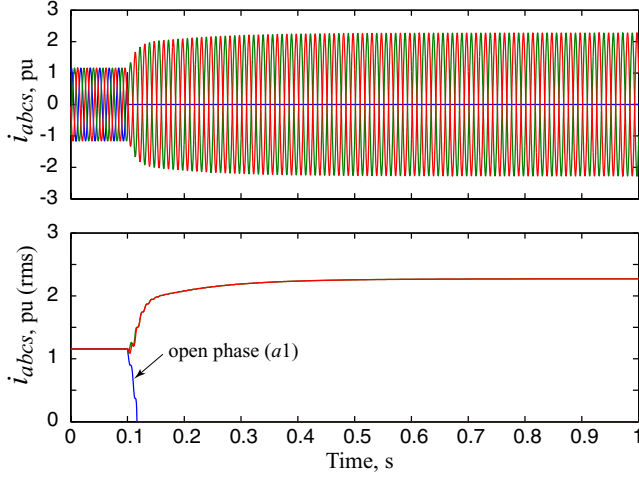


Fig. 4. Three-phase motor stator currents response to the loss of $a1$ winding.

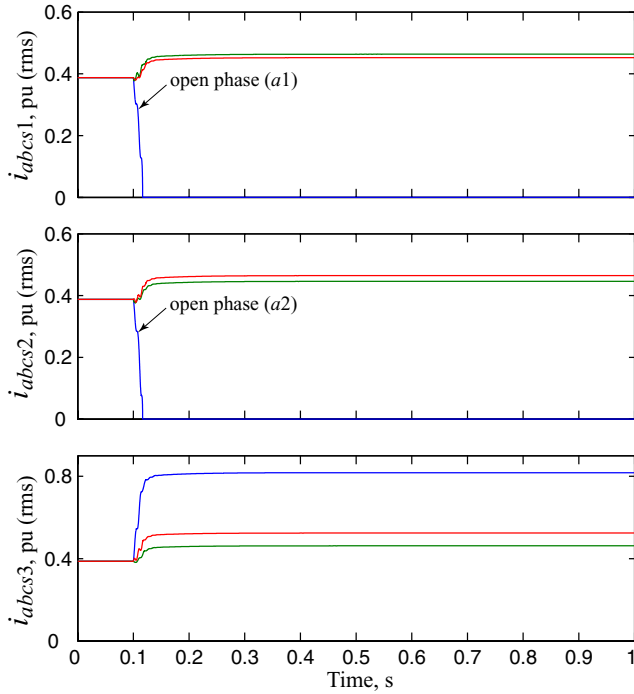


Fig. 5. Nine-phase motor stator currents response to the simultaneous loss of $a1$ and $b1$ windings.

APPENDIX

Induction machine parameters:

$$P_b = 74.6 \text{ kW}, T_b = 415.4 \text{ N} \cdot \text{m}, \omega_b = 2 \cdot \pi \cdot 60 \text{ rad/s}, \\ V_b = 265.6 \text{ V}, I_b = 93.6 \text{ A}, r_s = 0.0153, x_{ls} = 0.0532 \\ r_r = 0.0402, x_{lr} = 0.0532, x_m = 2.306, H = 0.8 \text{ s}.$$

Mechanical load torque-speed characteristic:

$$T_m = 0.0092 \cdot \omega_r + 1.1016 \cdot \omega_r^2$$

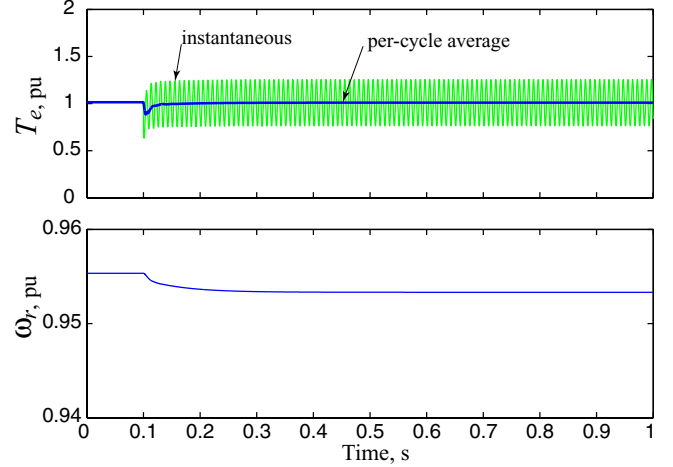


Fig. 6. Nine-phase motor torque and speed response to the simultaneous loss of $a1$ and $b1$ windings.

REFERENCES

- [1] S. Williamson and S. Smith, "Pulsating torque and losses in multiphase induction machines," *IEEE Trans. Industry Applications*, vol. 39, pp. 986–993, July-Aug. 2003.
- [2] J. Apsley and S. Williamson, "Analysis of multiphase induction machines with winding faults," *IEEE Trans. Industry Applications*, vol. 42, pp. 465–472, Mar-Apr 2006.
- [3] A. C. Smith, S. Williamson, and C. G. Hodge, "High torque dense naval propulsion motors," In *proc. IEEE Electric Machines and Drives Conf. (IEMDC'03)*, vol. 3, 2003, pp. 1421–1427.
- [4] C. Lewis, "The Advanced Induction Motor," In *Proc. IEEE Power Engineering Society Summer Meeting*, vol. 1, 2002, pp. 250–253.
- [5] M. D. Maksimcev, "Electric ship system integration – Designing reliable, low maintenance electric ships," In *proc. IEEE Electric Ship Technologies Symp. (ESTS'05)*, 25-27 July, 2005.
- [6] E. Schroeder, M. Pichot, A. Ouroua, and J. Beno, "Development of Electric Propulsion Motors with Integrated Power Electronics," Presented at Electric Machine Technology Symposium (EMTS), Philadelphia, PA, January 27-29, 2004.
- [7] Xu Huangsheng H. A. Toliyat, and L. J. Petersen, "Five-phase induction motor drives with DSP-based control system," *IEEE Trans. Power Electronics*, vol. 17, pp. 524–533, Jul. 2002.
- [8] J. R. Fu and T. A. Lipo, "Disturbance-free operation of a multiphase current-regulated motor drive with an opened phase," *IEEE Trans. Industry Applications*, vol. 30, pp. 1267–1274, Sept.-Oct. 1994.
- [9] P. C. Krause, O. Wasynczuk, and S. D. Sudhoff, *Analysis of Electric Machinery and Drive Systems*, IEEE Press, Wiley, 2002.
- [10] Automated State Model Generator (ASMG), Reference Manual, Version 2, PC Krause and Associates Inc., 2002. Available at: www.pcka.com

A SIMPLE INDUCTION MACHINE MODEL FOR PREDICTING LOW FREQUENCY DYNAMICS

Sina Chini Foroosh, Liwei Wang, and Juri Jatskevich

University of British Columbia, Vancouver, Canada

ABSTRACT

Conventional models of induction machine for steady-state and dynamic analysis use fixed value of the rotor resistance; whereas for representing the deep-rotor-bar effect a second rotor branch is typically required therefore increasing the model complexity and order. This paper proposes a simple but very effective modification of traditional model, wherein a single-branch rotor resistance is changed as a function of speed. The new model parameters are readily identified using standard experiments and tests. Experimental results and simulation studies demonstrate the improved accuracy and advantage of the proposed model in predicting the low-frequency transients. More importantly, the improvement is achieved without increasing the model order.

Index Terms— induction machine, transient model, deep-rotor-bar effect, low frequency dynamics.

1. INTRODUCTION

There have been a large number of models proposed in the literature for representing various phenomena in induction machines. The modeling approaches range from highly accurate finite-element based models, magnetic-equivalent circuit, and coupled electrical circuit models which require the least amount of computational resources and therefore are widely used [1].

This paper is focused on modeling squirrel cage induction machines using coupled electrical circuit approach [1]. To accurately model a deep-rotor-bar effect, the rotor can be represented as a high-order transfer function. A corresponding steady-state equivalent circuit is depicted in Fig. 1(a). Based on this approach, an advanced dynamic qd model accounting for saturation and arbitrary rotor network has been proposed in [2]. This model is capable of predicting both high- and low-frequency phenomena with excellent accuracy as observed in currents and developed torque. Such models may therefore be used to study machine-inverter interactions, switching transients, and torque harmonics.

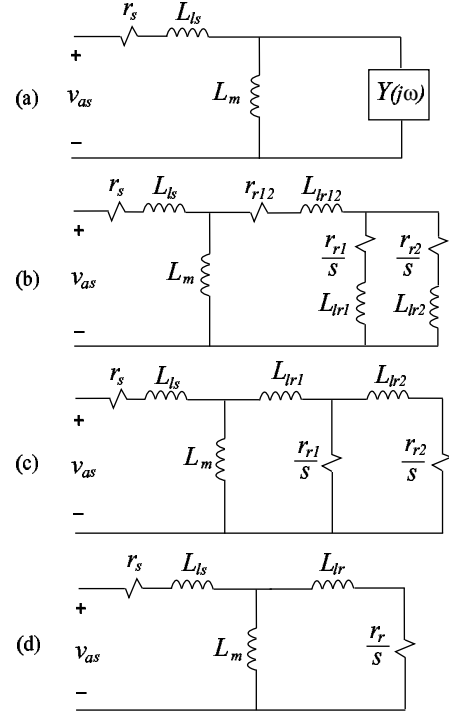


Fig. 1. Induction machine models with different configurations to represent the rotor circuit: (a) Generalized high-order transfer function; (b) and (c) Variations of two branches with resistive and inductive elements to represent the double-cage effect; and (d) Traditional single-rotor-branch configuration.

It is also possible to include the saturation and deep-rotor-bar effect directly into the three phase coupled circuit model as was demonstrated in [3]. However, the increased structural complexity of such models, number of required parameters and test procedures for their determination, etc., may be quite challenging [4]. Although these models provide great accuracy for many cases, they are generally harder to implement as compared to the standard qd models that are even available as built-in library components in many simulation packages.

Several simpler models have been proposed to represent the so-called double-cage rotor circuit in earlier publications, e.g. [5]–[8]. The approach is based on representing the rotor circuit using several resistive and inductive elements that are arranged in two rotor branches. This is schematically depicted in equivalent circuits in Fig.

1(b) and (c). In [7], several dynamic models with similar configurations have been compared. These and similar variations proposed in the literature increase the order of the circuit by one additional branch with the goal of capturing the changes in equivalent rotor resistance, but are still not sufficient for predicting the high-frequency dynamics.

The conventional *qd* model and its steady state equivalent circuit shown in Fig. 1(d) have the advantage of low-order and simplicity, which explains its wide acceptance and use. Since in general it is not possible to have a model that is accurate in wide range of frequencies without increasing its dynamic order as in [2], we propose to modify the conventional low-order model with a single rotor branch [1] to better predict at least the low-frequency deep-rotor-bar effect during electromechanical transients. The motivation of this approach is to improve the accuracy of traditional model, without adding structural complexity as in [5] – [8], such that it can be readily adopted by many engineers and possibly implemented in commonly-used simulation packages.

2. PROPOSED MODEL

In the following discussion, all rotor parameters are referred to the stator side. As well known, the equivalent rotor resistance, denoted here as r_r , changes with the frequency of rotor currents and slip due to the deep-rotor-bar effect, which depends on the rotor design [8]. During electromechanical transients in wide range of speeds (from stall to nominal), the effective changes in r_r may be quite significant. In general, at very low slip frequency, the equivalent resistance will be smaller and it will increase with the frequency due to the inductance and skin effect in the rotor bars.

For the purpose of discussion in this paper, the values of the rotor resistance at low slip ($s \approx 0$) and at stall ($s \approx 1$) are denoted by r_{r1} and r_{r2} , respectively, as depicted in Fig. 2. The high value of the rotor resistance r_{r2} may be readily estimated by performing the standard blocked-rotor test at which the rotor frequency is 60 Hz.

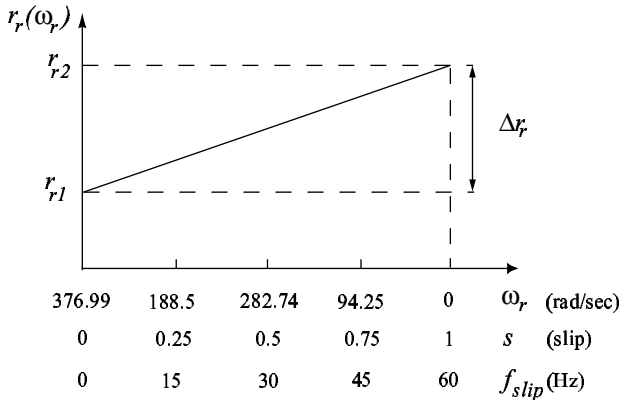


Fig. 2. Approximated speed-dependent equivalent rotor resistance.

At low-slip (close to nominal speed), the equivalent rotor resistance r_{r1} may be calculated using the equivalent circuit Fig. 1(d) and the following approach. The corresponding steady-state torque-speed characteristic is

$$T_e = 3 \frac{P}{2} \frac{V_{th}^2}{\omega_e} \cdot \frac{r_r/s}{(R_{th} + r_r/s)^2 + (X_{th} + X_r)^2}. \quad (1)$$

Here, ω_e is the electrical frequency of the source, and P is the number of poles. The Thevenin equivalent voltage V_{th} , resistance R_{th} , and reactance X_{th} are derived from the equivalent circuit of Fig. 1(d) [8]. At very low slips, the torque may be approximated as

$$T_e \approx 3 \frac{P}{2} \frac{V_{th}^2}{\omega_e} \cdot \frac{r_r/s}{(r_r/s)^2} = 3 \frac{P}{2} \frac{V_{th}^2}{\omega_e} \cdot \frac{s}{r_r}. \quad (2)$$

which is linearly proportional with the slip. Based on this approximation, rotor resistance r_{r1} can be estimated by measuring the torque and speed of a slightly loaded motor. Although it is also possible to find the values of the equivalent rotor resistance anywhere in between $s \approx 0$ and $s \approx 1$ by applying different frequency, for the purpose of this paper all intermediate values are obtained using linear interpolation between r_{r1} and r_{r2} as depicted in Fig. 2.

This paper suggests that as long as objectives of the modeling are focused on predicting the low frequency electromechanical transients, the frequency-dependent equivalent rotor resistance r_r depicted in Fig. 2 may be represented as a function of rotor speed ω_r . Based on linear approximation of Fig. 2, this dependency can be compactly expressed in the following form

$$r_r(\omega_r) = r_{r1} + \Delta r_r \cdot s \quad (3)$$

where $\Delta r_r = r_{r2} - r_{r1}$. This result is then substituted into the standard equivalent circuit of Fig. 1(d) to obtain a modified steady-state equivalent circuit shown in Fig. 3.

To illustrate the effect of such speed-dependent rotor resistance $r_r(\omega_r)$ on the steady-state torque-speed characteristic, we calculated the torque for three models and the resulting curves are shown in Fig. 4. Here, Model 1 and Model 2 are based on standard equivalent circuit of Fig. 1(d) and use fixed rotor resistances. In particular, in Model 1 the rotor resistance has been set to its low-slip value r_{r1} , and in Model 2 the rotor resistance corresponds to the standstill value r_{r2} , respectively.

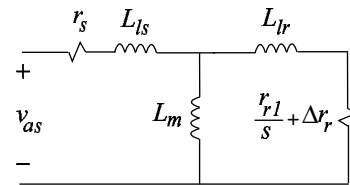


Fig. 3. Modified steady-state equivalent circuit including variation of rotor resistance.

The third Proposed Model corresponds to the modified equivalent circuit depicted in Fig. 3. The characteristics shown in Fig. 4 correspond to the induction machine considered in this paper. An important observation is that the proposed equivalent circuit of Fig. 3 captures both high starting torque (due to increased rotor resistance at stall) and steep torque at nominal speed (due to reduced rotor resistance at low slip), as desired.

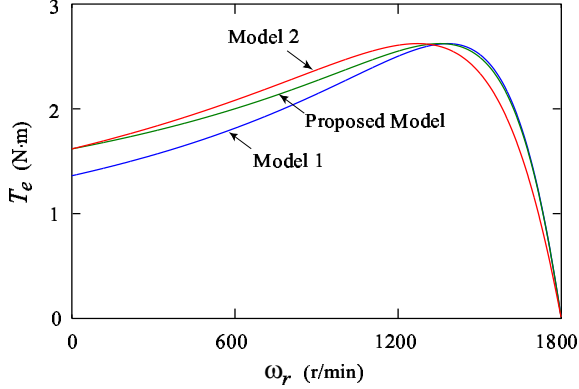


Fig. 4. Steady-state torque-speed characteristic for three models with different rotor resistances.

The results of speed-dependent equivalent rotor resistance demonstrated in steady state in Figs. 2 – 4 can be further applied to a transient model. In this paper we show a very straightforward method of including such speed-dependent rotor resistance in a classical qd model [1]. For consistency, the model equations are included here. In particular, the voltage equations in arbitrary reference frame are

$$v_{qs} = r_s i_{qs} + \omega \lambda_{ds} + p \lambda_{qs} \quad (4)$$

$$v_{ds} = r_s i_{ds} - \omega \lambda_{qs} + p \lambda_{ds} \quad (5)$$

$$v_{qr} = r_r (\omega_r) i_{qr} + (\omega - \omega_r) \lambda_{dr} + p \lambda_{qr} \quad (6)$$

$$v_{dr} = r_r (\omega_r) i_{dr} - (\omega - \omega_r) \lambda_{qr} + p \lambda_{dr} \quad (7)$$

where ω is the reference frame speed. The corresponding flux linkages are given as

$$\lambda_{qs} = L_{ls} i_{qs} + \lambda_{mq} \quad (8)$$

$$\lambda_{ds} = L_{ls} i_{ds} + \lambda_{md} \quad (9)$$

$$\lambda_{qr} = L_{lr} i_{qr} + \lambda_{mq} \quad (10)$$

$$\lambda_{dr} = L_{lr} i_{dr} + \lambda_{md} \quad (11)$$

$$\lambda_{mq} = L_m (i_{qs} + i_{qr}) \quad (12)$$

$$\lambda_{md} = L_m (i_{ds} + i_{dr}) \quad (13)$$

The developed electromagnetic torque is calculated as

$$T_e = \frac{3P}{4} (\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}). \quad (14)$$

It should be noted that, by making the rotor resistance speed-dependent, the modified qd model now includes the deep-rotor-bar effect at least in the range of low frequency applicable to electromechanical transients.

3. EXPERIMENTAL VERIFICATION

The induction motor considered in this paper has been characterized using standard no-load and blocked-rotor tests. Several additional measurements in the low-slip region were taken to estimate the equivalent rotor resistance at low frequency based on (2) by loading the motor. The machine parameters are summarized in Appendix.

To compare several models with the actual hardware, we have implemented a startup transient of the motor-dynamometer system available in our lab. The motor was fed from a low-voltage three phase Variac and step-down transformer. A comprehensive custom-built data acquisition system based on NI card interfaced with PC was used to capture and record all three phase currents, voltages, and speed during the acceleration of the motor. For simulation studies, the conventional qd model and the proposed model were implemented in MATLAB-Simulink [9]. To include the effects of non-idealities in the voltage source, i.e. source impedance and harmonics, the actual measured and recorded stator voltages have been imported in the Simulink models to reproduce the same excitation as applied to the hardware and the models. The applied voltage was a bit below the nominal level as to reduce the effect of saturation. Two conventional qd models have been considered. In Model 1 the rotor resistance has been set to its low-slip value r_{r1} , and in Model 2 the rotor resistance has been set to the standstill value r_{r2} , respectively.

The predicted transient responses and the hardware measurements are superimposed in Figs. 5–7. Fig. 5 shows the stator current i_{as} during the initial transient going into a steady state. Fig. 6 shows the same stator current magnified in the middle of transient for clarity. During this time, the motor speed is significantly below its nominal value and the slip is high too. This explains why the Model 1 which uses constant rotor resistance r_{r1} (and Model 2 which uses r_{r2}) overestimates (underestimates) the stator current. As was observed, the proposed model has also a better match with the measurement during both the initial transient and the steady state period. Fig. 7 illustrates the corresponding measured and predicted mechanical speed of the motor. The electromagnetic torque predicted by the three models is shown in Fig. 8. Model 1 with the smallest rotor resistance r_{r1} initially develops slightly less torque, which explains its lower speed in Fig. 7. Model 2 predicts the initial torque better, but results in a lower steady state speed due to higher rotor resistance r_{r2} . However, the proposed model demonstrates the stator currents as well as the speed that are more consistent with the measurements.

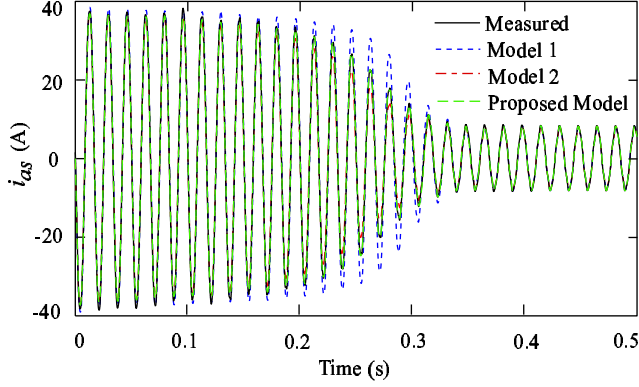


Fig. 5. Stator currents as measured and as predicted by the three models.

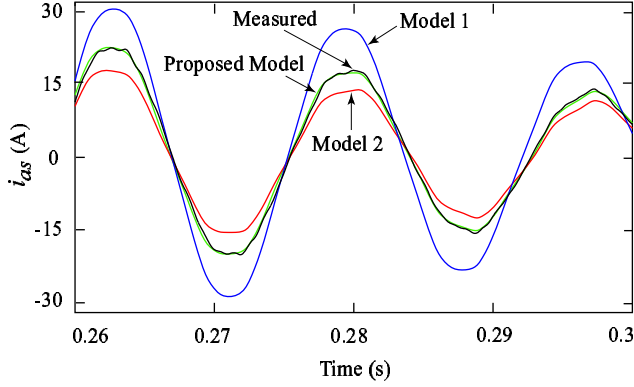


Fig. 6. Magnified stator currents as measured and as predicted by the three models.

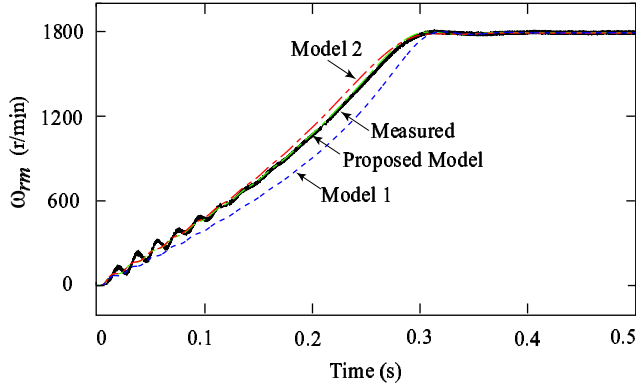


Fig. 7. Rotor speeds as measured and as predicted by the three models.

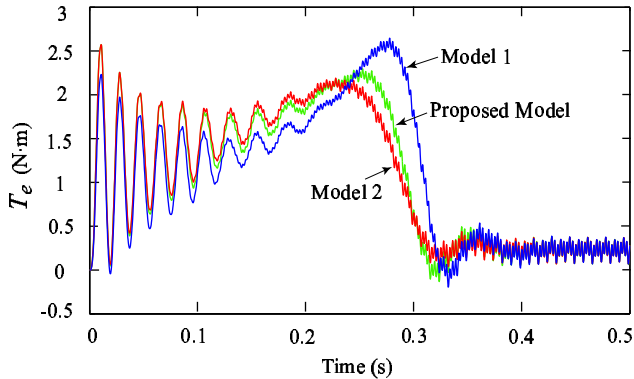


Fig. 8. Electromagnetic torques as predicted by the three models.

4. CONCLUSION

The proposed induction machine model includes the speed-dependent equivalent rotor resistance, which is shown to be adequate for modeling the deep-rotor-bar effect during the low-frequency electromechanical transients. The proposed methodology does not require increasing the order of the model and/or adding branches in the rotor equivalent circuit. The advantageous features include simplicity, ease of implementation, and readily measurable parameters.

5. APPENDIX

Machine parameters: 0.25 hp, 34V, 60 Hz, $P=4$, 1750 rpm, $r_s = 0.17\Omega$, $X_{ls} = 0.19\Omega$, $X_m = 3.1\Omega$, $r_{r1} = 0.07\Omega$, $r_{r2} = 0.12\Omega$, $X_{lr} = 0.19\Omega$, $J = 0.0023 \text{ kg} \cdot \text{m}^2$.

Speed (rpm)	170	450	680	1150	1680
Friction Torque ($N \cdot m$)	0.16	0.19	0.20	0.21	0.22

6. REFERENCES

- [1] P. C. Krause, O. Wasynczuk, and S. D. Sudhoff, *Analysis of Electric Machinery and Drive Systems*, 2nd Edition, IEEE Press, Piscataway, NJ, 2002.
- [2] S. D. Sudhoff, D. C. Aliprantis, B. T. Kuhn, and P. L. Chapman, "An induction machine model for predicting inverter-machine interaction," *IEEE Trans. on Energy Conversion*, vol. 17, no. 2, pp. 203–210, Jun. 2002.
- [3] M. Ikeda, and T. Hiyama, "Simulation studies of the transients of squirrel-cage induction motors," *IEEE Trans. on Energy Conversion*, vol. 22, no. 2, pp. 233–239, Jun. 2007.
- [4] S. D. Sudhoff, D. C. Aliprantis, B. T. Kuhn, and P. L. Chapman, "Experimental characterization procedure for use with an advanced induction machine model," *IEEE Trans. on Energy Conversion*, vol. 18, no. 1, pp. 48–56, Mar. 2003.
- [5] A. C. Smith, R. C. Healey, and S. Williamson, "A transient induction motor model including saturation and deep-rotor-bar effect," *IEEE Trans. on Energy Conversion*, vol. 11, no. 1, pp. 8–15, Mar. 1996.
- [6] J. R. Willis, and B. K. Johnson, "Tailoring induction motor analytical models to fit known motor characteristics and satisfy particular study needs," *IEEE Trans. on Power Systems*, vol. 6, no. 3, pp. 959–965, Aug. 1991.
- [7] W. Levy, C. F. Landy and M. D. McCulloch, "Improved models for the simulation of deep bar induction motors," *IEEE Trans. on Energy Conversion*, vol. 5, no. 2, pp. 393–400, Jun. 1990.
- [8] P. C. Sen, *Principles of electric machines and power electronics*, 2nd Edition, John Wiley & Sons, 1996.
- [9] *Simulink Dynamic System Simulation Software – Users Manual*, MathWorks, Inc., Natick, Massachusetts, 2005.

REDUCED-ORDER REPRESENTATION OF THE DEEP-ROTOR-BAR PHENOMENA IN INDUCTION MACHINES

Sina Chiniforoosh, Hamid Atighechi, and Juri Jatskevich

Electrical and Computer Engineering, University of British Columbia, Vancouver, Canada

ABSTRACT

To accurately model the deep-rotor-bar phenomena in squirrel-cage induction machines for a wide range of frequencies, the distributed rotor may be generally represented as a high-order transfer function or an equivalent lumped-parameter ladder network. However, when lower-frequency electromechanical transients are of interest, an effective reduced-order model, in which a single-branch rotor resistance is changed as a function of rotor speed, can take these effects into account without increasing the model complexity and order. In this paper, the reduced- and high-order formulations are compared with the classic models and experimental results. It is found that the proposed reduced-order model provides very good accuracy for predicting low-frequency dynamics. This model can be particularly useful for studying larger systems composed of a number of electromechanical subsystems.

Index Terms— induction machine, transient model, deep-rotor-bar effect, low-frequency dynamics.

1. INTRODUCTION

There have been a large number of models proposed in the literature for representing various phenomena in induction machines. The modeling approaches range from highly accurate finite-element-based models, magnetic-equivalent circuit, and coupled electrical circuit models. The last type of models requires the least amount of computational resources and therefore is widely used [1].

This paper is focused on modeling the squirrel-cage induction machines using the coupled electrical circuit approach [1]. To accurately model the deep-rotor-bar effect in a wide range of frequencies, the actual distributed circuit of the rotor may be generally represented as a high-order transfer function as shown in Fig. 1(a). Based on this approach, an advanced dynamic qd model accounting for saturation and arbitrary rotor network has been proposed in [2]. This model is capable of predicting both high- and low-frequency phenomena with excellent accuracy as observed in currents and developed torque. Such models may therefore be used to study machine-inverter interactions, switching transients, and torque harmonics.

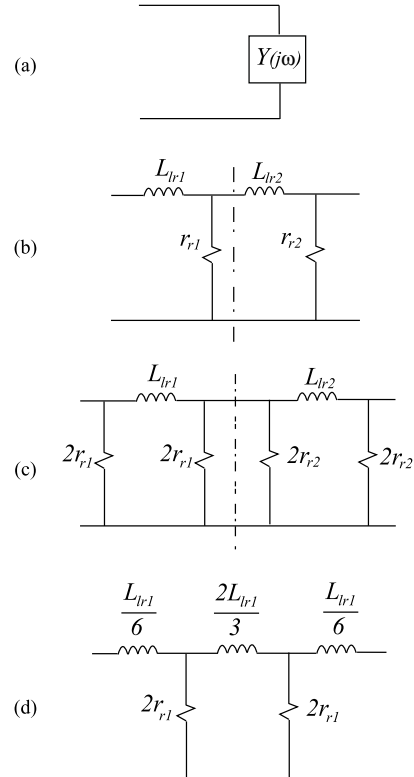


Fig. 1. Modeling the distributed rotor circuit in squirrel-cage induction machines: (a) generalized high-order transfer function; (b) equivalent L-circuit; (c) equivalent Pi-circuit; and (d) improved equivalent Pi-circuit.

However, the increased structural complexity of the models, number of required parameters and test procedures for their determination, etc., may be quite challenging [3].

As an alternative to the general high-order transfer function, several simpler structures may be used to represent the distributed rotor circuit in a desired range of frequencies [4]. Namely, the L-, Pi-, and improved Pi-circuit configurations are depicted in Figs. 1(b)-(d). Generally, the details of the machine cage design have to be considered; the rotor bars are then divided into several segments and each segment is represented by its lumped-parameter equivalent configuration. In Figs. 1(b) and (c), for simplicity of the figure, only two adjacent segments have been illustrated which are characterized by their parameters: r_{r1} , L_{lr1} , and r_{r2} , L_{lr2} , respectively. In more complicated structure of Fig. 1(d), only one segment has been shown.

The final structure may be composed of any number of segments to resemble the distributed rotor in a limited frequency range. The number of required segments and hence the order of the model is determined such that the desired accuracy is achieved in this range of frequencies. However, since the details of the cage design are typically required to identify the parameters of such high-order networks, it poses practical constraints for such approaches.

In order to eliminate the above limitation, alternative approaches have been proposed in which an appropriate structure is defined for the rotor network and is then parameterized numerically using an iterative simulation scheme. The final goal is to adjust the parameters of the network iteratively such that the resulting characteristics match those of the actual machine which are in turn either supplied by the manufacturer or extracted directly from hardware measurements and/or standard tests [5], [6]. The structures used are typically similar to the ones depicted in Fig. 1 although the parameterization procedures are different. It is also possible to include the saturation and deep-rotor-bar effect directly into the three phase coupled circuit model as demonstrated in [7]. Although these models may provide enough accuracy for many cases, they are generally harder to implement as compared to the standard qd models that are commonly available as built-in library components in many simulation packages.

The conventional qd model and its steady state equivalent circuit shown in Fig. 2(a) have the advantages of low-order and simplicity which explains their wide acceptance and use. However, these models may not provide sufficient accuracy in many cases since they use fixed values of rotor parameters and hence neglect the deep-rotor-bar phenomena associated with the frequency of the electrical signals in the rotor circuit. In order to include these effects for predicting the lower-frequency dynamics such as start-up transients, the classic model may be modified to have a second order rotor network similar to the so-called double-cage rotor circuit of Fig. 2(b). Such a structure is typically sufficient for predicting the low-frequency dynamics and there is no need to increase the order of the system any further. However, even this structure and its variations increase the order of the circuit by one additional branch with the goal of capturing the changes in equivalent rotor resistance, and are still not sufficient for predicting the high-frequency dynamics.

Since in general it is not possible to have a model that is accurate in a wide range of frequencies without increasing its dynamic order as in [2], the authors have proposed to modify the conventional low-order model of Fig. 2(a) with a single rotor branch to better predict at least the low-frequency deep-rotor-bar effect during electromechanical transients [8]. The motivation of this approach is to improve the accuracy of the traditional model without increasing the order and structural complexity as in [4]-[6].

In this paper, the proposed reduced-order representation of the deep-rotor-bar phenomena is compared with the

simplest higher-order counterpart, two classic fixed-parameter models, and hardware measurements. It is demonstrated that the reduced-order representation is very effective as it provides superior accuracy for the lower-frequency electromechanical transient studies while its parameterization and implementation are much simpler than the higher-order models and quite similar to the classic models.

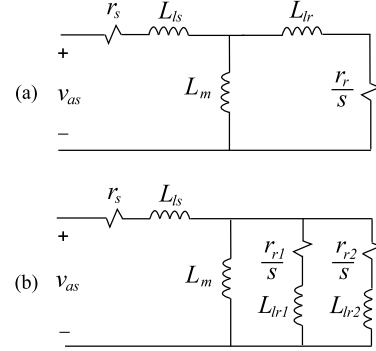


Fig. 2. Two common equivalent circuits for induction machines: (a) The traditional circuit with a first-order rotor circuit and (b) A typical double-cage configuration with a second-order rotor circuit.

2. HIGH-ORDER AND REDUECD-ORDER CONFIGURATIONS

Herein, it is assumed that all rotor parameters have been referred to the stator side. As well known, the equivalent rotor resistance, denoted here as r_r , changes with the frequency of rotor currents and slip due to the deep-rotor-bar effect, which depends on the rotor design [9]. During electromechanical transients in wide range of speeds (from stall to nominal), the effective changes in r_r may be quite significant. In general, at very low slip frequency, the equivalent resistance will be smaller and it will increase with the frequency due to the inductance and skin effect in the rotor bars. The variation of rotor resistance and leakage inductance for the machine may be provided by the manufacturer or extracted directly from hardware measurements.

For the purpose of discussion in this paper, the values of the rotor resistance at low slip ($s \approx 0$) and at stall ($s = 1$) are denoted by r_{rs0} and r_{rs1} , respectively. The high value of the rotor resistance r_{rs1} may be readily estimated by performing the standard blocked-rotor test at which the rotor frequency is 60 Hz. At low-slip (close to nominal speed), the equivalent rotor resistance r_{rs0} may be calculated using the equivalent circuit Fig. 2(a) and the following approach. The corresponding steady-state torque-speed characteristic is

$$T_e = 3 \frac{P V_{th}^2}{\omega_e} \cdot \frac{r_r/s}{(R_{th} + r_r/s)^2 + (X_{th} + X_r)^2} \quad (1)$$

Here, ω_e is the electrical frequency of the source, and P is the number of poles. The Thevenin equivalent voltage V_{th} , resistance R_{th} , and reactance X_{th} are derived from the equivalent circuit of Fig. 2(a) [9]. At very low slips, the torque may be approximated as

$$T_e \approx 3 \frac{P V_{th}^2}{2 \omega_e} \cdot \frac{r_r/s}{(r_r/s)^2} = 3 \frac{P V_{th}^2}{2 \omega_e} \cdot \frac{s}{r_r}, \quad (2)$$

which is linearly proportional with the slip. Based on this approximation, the rotor resistance r_{rs0} can be estimated by measuring the torque and speed of a slightly loaded motor.

In the classic configuration of Fig. 2(a), only one value may be defined as the rotor resistance. For the sake of comparison, in this paper, two models are then considered based on the standard equivalent circuit of Fig. 2(a). These models are referred to as the Classic Models 1 and 2, and use the fixed rotor resistance values of r_{rs0} and r_{rs1} , respectively.

For the high-order configuration of Fig. 2(b), the parameters of the rotor network should be obtained using an iterative fitting technique such that the overall resistance of the rotor network is equal to the above values at 60 Hz and very small frequencies, respectively. The variation of the overall rotor network resistance with respect to slip for the high-order model is depicted in Fig. 3 (top plot). As seen there, the rotor resistance value changes between the above-mentioned fixed values at low-slip and standstill operation which have been used in the Classic Models 1 and 2.

As long as the objectives of modeling are focused on predicting the low-frequency electromechanical transients, the frequency-dependent equivalent rotor resistance r_r may be represented as a function of rotor electrical speed ω_r .

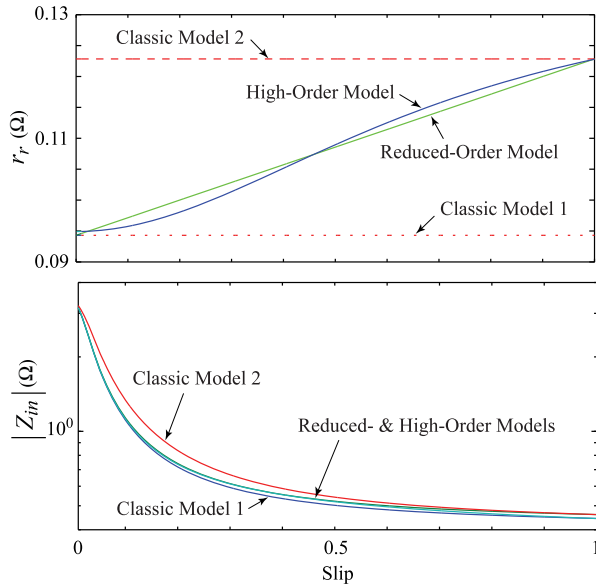


Fig. 3. Variation of rotor resistance (top plot) and input impedance (bottom plot) with slip as predicted by different models.

Using a linear approximation as shown in Fig. 3 (top plot), this dependency can be compactly expressed in the following form

$$r_r(\omega_r) = r_{rs0} + \Delta r_r \cdot s, \quad (3)$$

where $\Delta r_r = r_{rs1} - r_{rs0}$. This result is then substituted into the standard equivalent circuit of Fig. 2(a) to obtain a modified steady-state equivalent circuit shown in Fig. 4.

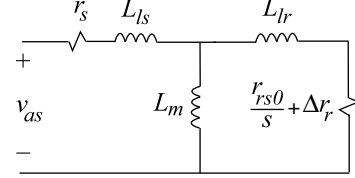


Fig. 4. Modified steady-state equivalent circuit including variation of rotor resistance with rotor speed.

The overall input impedance of the machine looking into stator terminals have been depicted for different models in Fig. 3 (bottom plot). Due to the significant difference between the values of this impedance at low and high slips, the logarithmic scale has been used for the y -axis to pronounce the deviation among the characteristics predicted by different models. As seen there, the characteristics predicted by both reduced- and high-order models approach those of the Classic Models 1 and 2 at low and high slip, respectively.

A transient model corresponding to each of the above-mentioned steady-state equivalent circuit models may be established based on the classic qd model in arbitrary reference frame [1]. The high-order model is represented by one set of stator and two sets of rotor state equations, as well as the output equations as

$$p\lambda_{qd0s} = \mathbf{v}_{qd0s} - r_s \mathbf{i}_{qd0s} - \omega \lambda_{dqs}, \quad (4)$$

$$p\lambda_{qd0r1} = \mathbf{v}_{qd0r1} - r_{r1} \mathbf{i}_{qd0r1} - (\omega - \omega_r) \lambda_{dqr1}, \quad (5)$$

$$p\lambda_{qd0r2} = \mathbf{v}_{qd0r2} - r_{r2} \mathbf{i}_{qd0r2} - (\omega - \omega_r) \lambda_{dqr2}, \quad (6)$$

$$\lambda_{qd0s} = L_{ls} \mathbf{i}_{qd0s} + \lambda_{mqd}, \quad (7)$$

$$\lambda_{qd0r1} = L_{lr1} \mathbf{i}_{qd0r1} + \lambda_{mqd}, \quad (8)$$

$$\lambda_{qd0r2} = L_{lr2} \mathbf{i}_{qd0r2} + \lambda_{mqd}, \quad (9)$$

$$\lambda_{mqd} = L_m (\mathbf{i}_{qds} + \mathbf{i}_{qdr1} + \mathbf{i}_{qdr2}). \quad (10)$$

In the above equations, $\mathbf{f}_{qd0x} = [f_{qx} \ f_{dx} \ f_{0x}]^T$, where f can be voltage, v , current, i , or flux linkage, λ , and x can be any of the stator, s , or rotor, r_1 , r_2 branches. Also, ω denotes the reference frame speed. In the vector $\mathbf{f}_{dqx}$, the third element is set to zero, and $\mathbf{f}_{dqx} = [f_{dx} \ -f_{qx} \ 0]^T$.

The results of the speed-dependent equivalent rotor resistance demonstrated in steady state in Fig. 4 can also be further applied to a transient model. In this model, the stator equations are the same as the high-order model, but only

one set of rotor branches is included. The final model is hence expressed by (4), (7), and the following equations:

$$p\lambda_{qd0r} = \mathbf{v}_{qd0r} - r_r(\omega_r) \mathbf{i}_{qd0r} - (\omega - \omega_r)\lambda_{dqr}, \quad (11)$$

$$\lambda_{qd0r} = L_{lr} \mathbf{i}_{qd0r} + \lambda_{mqd}, \quad (12)$$

$$\lambda_{mqd} = L_m (\mathbf{i}_{qds} + \mathbf{i}_{qdr}). \quad (13)$$

It should be noted that, by making the rotor resistance speed-dependent, the modified qd model now includes the deep-rotor-bar effect at least in the range of low frequencies applicable to electromechanical transients.

The transient models corresponding to the Classic Models 1 and 2 are described by the exact same equations as the reduced-order model except that the value $r_r(\omega_r)$ is fixed and equal to r_{rs0} and r_{rs1} , respectively. This demonstrates that the reduced-order representation is essentially of the same order and complexity as the classic models and no further modifications or complex parameterization is required unlike the high-order model. For convenience, the developed electromagnetic torque may be represented in terms of stator quantities only so that in all models the following equation can be used for modeling the mechanical subsystem:

$$T_e = \frac{3P}{4} (\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}). \quad (14)$$

Note that including the zero sequence and mechanical equation, the overall order of the system for high-order and reduced-order representations are 10 and 7, respectively.

3. MODEL COMPARISON AND DISCUSSION

The induction motor considered in this paper has been characterized using standard no-load and blocked-rotor tests. Several additional measurements in the low-slip region were taken to estimate the equivalent rotor resistance at low frequency based on (2) by loading the motor. The machine parameters are summarized in Appendix.

To compare several models with the actual hardware, we have implemented a startup transient of the motor-dynamometer system. The motor was fed from a low-voltage three phase Variac and step-down transformer. A comprehensive custom-built data acquisition system based on NI card interfaced with PC was used to capture and record all three phase currents, voltages, and speed during the acceleration of the motor. For simulation studies, the Classic Models 1 and 2, the high-order model, and the reduced-order model with speed-dependent equivalent rotor resistance were all implemented in MATLAB-Simulink [10]. To include the effects of non-idealities in the voltage source, i.e. source impedance and harmonics, the actual measured and recorded stator voltages have been imported in the Simulink models to reproduce the same excitation as applied to the hardware and the models. The applied voltage was a bit below the nominal level as to reduce the effect of saturation.

The predicted transient responses and the hardware measurements for the start-up transient are superimposed in Fig. 5. The measured and predicted mechanical speed of the motor are shown in the top plot, and the middle plot illustrates the corresponding stator current i_{as} . For better clarity and comparison, the magnified views of these plots in the middle of the transient are provided in Fig. 6. During this time, the motor speed is significantly below its nominal value and the slip is high. This explains why the Classic Model 1 which uses constant rotor resistance r_{rs0} (and Classic Model 2 which uses r_{rs1}) overestimates (underestimates) the stator current. However, the high-order and reduced-order models both demonstrate the stator current waveforms that are more consistent with the measurement. The electromagnetic torque predicted by all models during this study is shown in the bottom plot of Fig. 5. In this plot, the predicted waveforms are shown only since the electromagnetic torque cannot be measured directly.

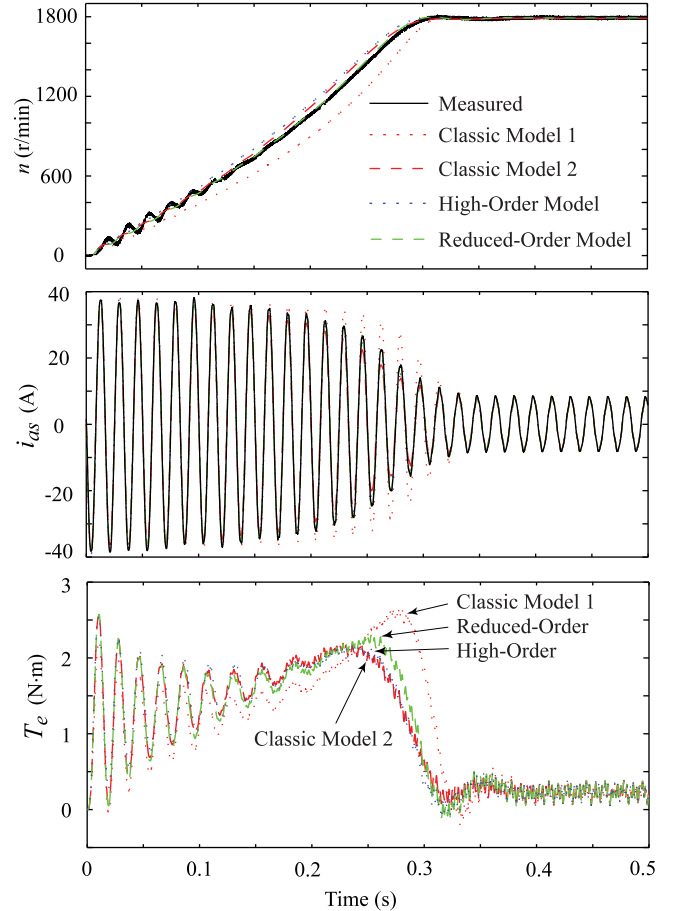


Fig. 5. Measured and predicted start-up transient waveforms.

The Classic Model 1, with the smallest rotor resistance r_{rs0} , initially develops slightly less torque, which explains its lower speed in Figs. 5 and 6. The Classic Model 2

predicts the initial torque better, but results in a lower steady state speed due to the higher rotor resistance value r_{rs1} . The torque characteristics predicted by the reduced- and high-order models lie between these two extreme cases.

As observed in the results of the start-up transient study, both high-order and reduced-order models provide very good match with the measurement during the initial transient as well as the steady state period. Therefore, for this and similar low-frequency transient studies, increasing the order and complexity of the system as well as the additional efforts for parameterization and implementation of the high-order model seem unnecessary. This suggests that, especially in the case of a larger system with a large number of machines, employing the higher-order models would impose additional burden on the simulation without providing any critical improvement in the results. This demonstrates the effectiveness of the reduced-order representation.

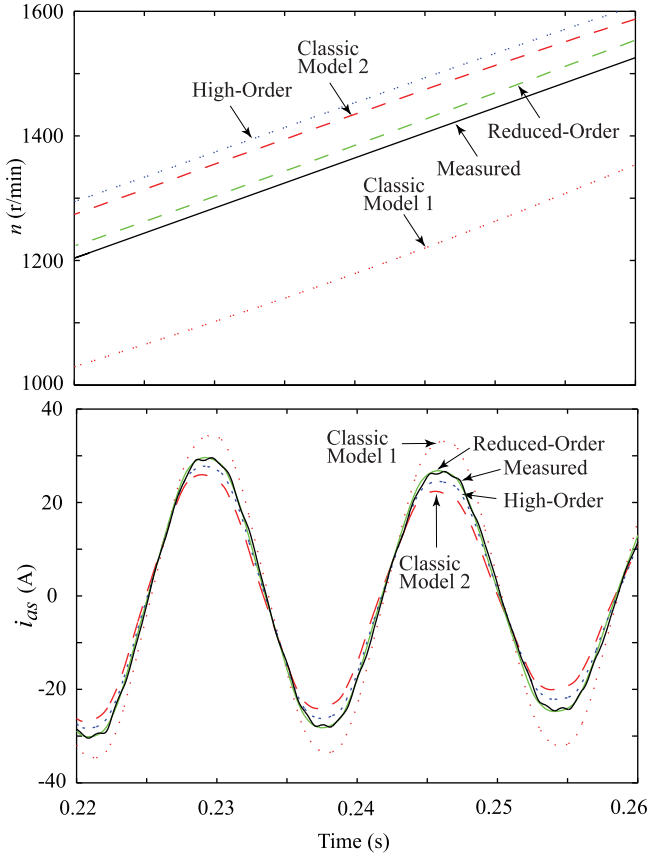


Fig. 6. Rotor speed (top plot) and stator current (bottom plot) as measured and as predicted by all models.

4. CONCLUSION

Although accurate modeling of the deep-rotor-bar effect, especially at high frequencies, would require a high-order formulation for the rotor network, the reduced-order representation, employing the speed-dependent equivalent rotor resistance, is shown to be quite adequate during the electromechanical transients. The advantages of such representation include simplicity, ease of implementation, and readily measurable parameters. It was demonstrated that increasing the model order does not provide any further improvement in predicting the lower-frequency dynamics despite imposing additional burden on the simulation.

5. APPENDIX

Machine parameters: 0.25 hp, 34V, 60 Hz, P=4, 1750 rpm, $J = 0.0023 \text{ kg} \cdot \text{m}^2$, $r_s = 0.17\Omega$, $X_{ls} = X_{lr} = 0.186\Omega$, $X_m = 3.1\Omega$, $r_{rs0} = 0.094\Omega$, $r_{rs1} = 0.123\Omega$, $r_{r1} = 0.289\Omega$, $r_{r2} = 0.141\Omega$, $X_{lr1} = 0.222\Omega$, $X_{lr2} = 0.393\Omega$.

Speed (rpm)	170	450	680	1150	1680
Friction Torque ($N \cdot m$)	0.16	0.19	0.20	0.21	0.22

6. REFERENCES

- [1] P. C. Krause, O. Wasynczuk, and S. D. Sudhoff, *Analysis of Electric Machinery and Drive Systems*, 2nd Edition, IEEE Press, Piscataway, NJ, 2002.
- [2] S. D. Sudhoff, D. C. Aliprantis, B. T. Kuhn, and P. L. Chapman, "An induction machine model for predicting inverter-machine interaction," *IEEE Trans. on Energy Conversion*, vol. 17, no. 2, pp. 203–210, Jun. 2002.
- [3] S. D. Sudhoff, D. C. Aliprantis, B. T. Kuhn, and P. L. Chapman, "Experimental characterization procedure for use with an advanced induction machine model," *IEEE Trans. on Energy Conversion*, vol. 18, no. 1, pp. 48–56, Mar. 2003.
- [4] W. Levy, C. F. Landy and M. D. McCulloch, "Improved models for the simulation of deep bar induction motors," *IEEE Trans. on Energy Conversion*, vol. 5, no. 2, pp. 393–400, Jun. 1990.
- [5] J. R. Willis, and B. K. Johnson, "Tailoring induction motor analytical models to fit known motor characteristics and satisfy particular study needs," *IEEE Trans. on Power Systems*, vol. 6, no. 3, pp. 959–965, Aug. 1991.
- [6] A. C. Smith, R. C. Healey, and S. Williamson, "A transient induction motor model including saturation and deep-rotor-bar effect," *IEEE Trans. on Energy Conversion*, vol. 11, no. 1, pp. 8–15, Mar. 1996.
- [7] M. Ikeda, and T. Hiyama, "Simulation studies of the transients of squirrel-cage induction motors," *IEEE Trans. on Energy Conversion*, vol. 22, no. 2, pp. 233–239, Jun. 2007.
- [8] S. Chini Foroosh, L. Wang and J. Jatskevich, "A simple induction machine model for predicting low-frequency dynamics," in *Proc. Canadian Conf. Elec. and Comp. Eng., CCECE'08*, Niagara Falls, ON, Canada, 2008.
- [9] P. C. Sen, *Principles of electric machines and power electronics*, 2nd Edition, John Wiley & Sons, 1996.
- [10] *Simulink Dynamic System Simulation Software – Users Manual*, MathWorks, Inc., Natick, Massachusetts, 2005.

EXPLICIT AND IMPLICIT REPRESENTATION OF SATURATION FOR INDUCTION MACHINES MODELLING IN STATE-VARIABLE TRANSIENT SIMULATORS

Francis Therrien¹, Liwei Wang² and Juri Jatskevich¹

¹Electrical and Computer Engineering, The University of British Columbia, Vancouver, Canada

²ABB Sweden Corporate Research, Vasteras, Sweden

Emails: {francist, juri.j}@ece.ubc.ca; liwei.wang@se.abb.com

ABSTRACT

Low-order qd saturable induction machine models are available as built-in component models in commonly used state-variable-based transient simulation packages/toolboxes. The paper investigates the implementation of the main flux saturation in symmetrical squirrel-cage induction machines using the flux correction (FC) method, wherein implicit (with algebraic loop) and explicit (without algebraic loop) implementations are possible. The presented computer studies demonstrate accuracy and efficiency of several commonly used low-order models and approaches. The explicit FC method is proposed as a very effective and accurate way of representing the effects of saturation even at large time steps, which is a very desirable property.

Index Terms— Induction machines, modeling, saturation, state-variable approach, transient simulation.

1. INTRODUCTION

Taking into account the magnetic saturation in the models of induction machines can improve the accuracy of simulations in both transient and steady state. Fairly accurate machine models that include magnetic saturation and structural/material properties can be constructed for instance using finite element (FE) [1] or magnetic equivalent circuits (MEC) [2] methods. While modeling the overall machine with a high degree of precision, these approaches have very significant computational cost making them prohibitive for general purpose system-level studies. They also usually require more input data than what is generally available to utilities. For these cases, the low-order coupled electrical circuit representation of the induction machine is typically preferred. Many general purpose machine models have been developed in the literature [3]–[7]. Most are based on the qd reference frame [3]–[5] whereas some are developed in the (abc) phase coordinates [6], [7] in order to achieve direct interfacing.

The simpler models such as [3] are based on a number of assumptions, including machine symmetry, harmonic-free MMF wave, linear magnetic characteristic, constant rotor impedance, etc. Numerous coupled electrical circuit models have been further developed with fewer limitations. For instance, the iron core saturation has been modeled in [3]–

[5], [7]. In these models, only the fundamental component of the magnetizing main flux is represented. This assumption has been shown to give some errors [8], but remains a good compromise between simplicity and accuracy for system studies.

The paper investigates the implementation of saturation of the fundamental component of the main flux in squirrel-cage induction machines using the flux correction (FC) method [3], wherein implicit (with algebraic loop) and explicit (without algebraic loop) implementations are possible. Whereas the FC method corrects the unsaturated magnetizing flux according to the machine's saturated vs. unsaturated characteristics, other methods such as [5] instead opt to directly modify the magnetizing inductance, in this case with a time step relaxation. The saturable induction machine models available as built-in component models in commonly used state-variable-based transient simulation packages/toolboxes in Matlab/Simulink, namely the SimPowerSystems (SPS) [9] and Piece-wise Linear Electrical Circuit Simulation (PLECS) [10], are also compared. As a reference for accuracy of representing the dynamic and steady state effects of saturation, the Levi's generalized flux space vector approach [4] is considered. The explicit FC method is proposed as a very effective and accurate way of representing the effects of saturation even at large time steps, which is a very desirable property.

2. MAIN FLUX SATURATION REPRESENTATION

Due to space limitations, the differential and algebraic equations of the low-order qd linear induction machine model (with flux linkages as state variables) are not presented here. They can be found in [3]. Herein, all rotor variables are referred to the stator and motor sign convention is used. It is usually assumed that an induction machine has an isotropic rotor. Therefore, the main magnetizing current i_m is aligned with the main flux linkage λ_m as shown in Fig. 1, and these variables can easily be expressed in term of their q and d axis projections, as

$$\lambda_m = \sqrt{\lambda_{mq}^2 + \lambda_{md}^2} \quad (1)$$

$$i_m = \sqrt{i_{mq}^2 + i_{md}^2} \quad (2)$$

where

$$i_{mq} = i_{qs} + i_{qr} \quad (3)$$

$$i_{md} = i_{ds} + i_{dr}. \quad (4)$$

The above equations inherently take into account the well understood qd cross-saturation effect.

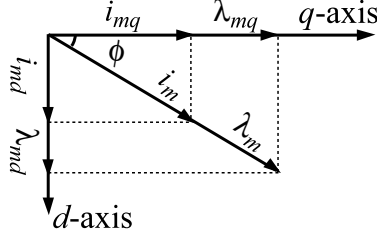


Fig. 1. Vector representation of the main flux linkage.

The simplest way to obtain the main flux saturation characteristic of an induction machine is with a no-load test. Pairs of line-to-neutral stator RMS voltage $V_{as,NL}$ and line stator RMS current $I_{as,NL}$ are measured experimentally. Their corresponding i_m and λ_m are then given by

$$i_m = \sqrt{2} I_{as,NL} \quad (5)$$

$$\lambda_m = \sqrt{2} \frac{(V_{as,NL} - \omega_b I_{as,NL} L_{ls})}{\omega_b}. \quad (6)$$

A typical saturation characteristic $\lambda_m(i_m)$ is presented in Fig. 2, which shows that for smaller fluxes, the iron core is not saturated and the saturated and unsaturated characteristics are thus identical. By increasing the flux, the saturated region is reached and more current is required. The slope of the curves is analogous to the equivalent incremental magnetizing inductance L_m .

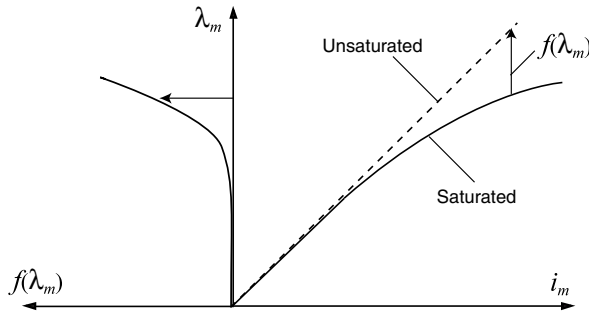


Fig. 2. Main flux saturation characteristic.

The main flux saturation function $\lambda_m(i_m)$ depicted in Fig. 2 can be represented using different methods, e.g. piece-wise linear interpolation (widely used in EMTP-type software), higher order polynomial interpolation/approximation, arctangent functions [11], etc.

The FC method is presented in [3]. First, the difference between the saturated and unsaturated fluxes at a given i_m

is denoted by the function $f(\lambda_m)$, which is displayed in Fig. 2. Applying a few algebraic manipulations to the basic induction machine equations and using the properties of Fig. 1 [3]:

$$\lambda_{mq} = \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{qr}}{L_{lr}} \right) L_{aq} - \frac{\lambda_{mq}}{\lambda_m} \frac{L_{aq}}{L_m} f\left(\sqrt{\lambda_{mq}^2 + \lambda_{md}^2}\right) \quad (7)$$

$$\lambda_{md} = \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{dr}}{L_{lr}} \right) L_{ad} - \frac{\lambda_{md}}{\lambda_m} \frac{L_{ad}}{L_m} f\left(\sqrt{\lambda_{mq}^2 + \lambda_{md}^2}\right) \quad (8)$$

with $L_{aq} = L_{ad}$ as defined in [3].

This method is very simple as it requires only a slight change from the linear model, i.e. correcting the unsaturated fluxes, and can be efficiently implemented using a 1-D lookup table as $f(\lambda_m)$ is a monotonically increasing function. It keeps the machine parameters constant. However, its main drawback is that it forms an algebraic loop as (7) and (8) are implicit algebraic equations. Simulink is able to solve these equations iteratively, which leads to increasing the amount of flops per time step. Moreover, other simulation tools may not have this option.

An explicit method can be obtained by extending the approach set forth in [13] to induction machines. Another quick fix is to break the algebraic loop by adding a one time step delay to (7), (8), i.e. computing λ_{mq}^{k+1} and λ_{md}^{k+1} using $f(\lambda_m^k)$. With the synchronous reference frame, the steady state solution will be exact as λ_m remains constant. On the other hand, an error will appear during transients, which is expected to grow with the size of the time step.

The Levi's generalized flux space vector approach for induction machines [4] is algebraically identical to the implicit FC method. This approach allows the user to select any mix of currents and fluxes as the state variables of a single machine. It is explicit. Due to the apparition of currents as state variables (as opposed to FC), the derivative of the magnetizing inductance is required at every time step. This allows to exactly model dynamic cross-saturation for any choice of state variables [12]. Additional algebraic manipulations are required to obtain matrices in standard state-variable form.

The saturated induction machine model in PLECS [10] uses a simplified version of the generalized flux space vector approach. Based on the conclusions of [12], the model, which uses i_{qs} , i_{ds} , λ_{md} and λ_{mq} as state variables, neglects the dynamic cross-saturation terms. Under this assumption, no derivative of the magnetizing inductance is required. The saturation curve is conveniently represented by an arctangent function [11]. The saturable induction machine model in SPS uses the flux linkages as state variables, as in [3]. Therein, the saturation is represented using an explicit formulation with a 5th order polynomial approximation, built very similarly to FC.

3. COMPUTER STUDIES

The system comprises a single squirrel-cage induction motor connected to an infinite-bus. The machine parameters are summarized in Appendix a). At $t = 0$ s, the motor is assumed to operate in steady state under no-load condition. At $t = 0.02$ s, the voltage of the infinite-bus changes from $0.8pu$ to $1pu$, bringing the machine into the saturated region. The saturation characteristics for Levi's and PLECS' models are entered using the arctangent function [11] with parameters summarized in Appendix b). For the FC method, the saturation function is represented by 13 pairs of (i_m, λ_m) , also shown in Appendix c). Finally, for SPS, the aforementioned pairs of (i_m, λ_m) are converted to $I_{as,NL}$ and $\sqrt{3}V_{as,NL}$ using (5), (6).

To evaluate the accuracy of various models considered in this paper, a reference solution is obtained using the Levi's Type 2 ($1/\Lambda$) model [4], which has been solved using a very small time step of $1\mu s$ and the ode5 (Dormand-Prince) solver. This solver is used with every model.

To verify the consistency of all models, the study is first executed using $\Delta t = 50\mu s$, a small time step. The corresponding transient solutions obtained by all considered models are superimposed in Figs. 3-5. As it can be observed, all models but PLECS' exhibit nearly similar and convergent solutions in both steady state and transient. Fig. 4 shows in a clearer way PLECS' discrepancy during the first peak of the transient. The same observation can be made for λ_m in Fig. 5. This error is mainly attributed to neglecting the dynamic cross-saturation. The amplitude of this error is nonetheless small, as previously proven in [12].

The computational costs of the models for the $\Delta t = 50\mu s$ study ran in Simulink on a personal computer are summarized in Table I. As expected, the implicit FC formulation has higher CPU time due to the algebraic loop. The explicit Levi's model is somewhat almost 3 times faster. The most efficient model is the explicit FC which shows the smallest CPU time among all the models. SPS is the slowest despite its explicit formulation. PLECS is also slower than Levi's even though it uses a simplified version of Levi's. Simulink commercial toolboxes models have long been known to be slower than homemade models.

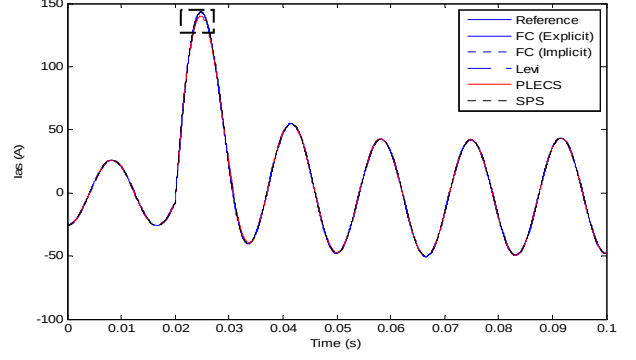


Fig. 3. Stator current transient solution using $\Delta t = 50\mu s$.

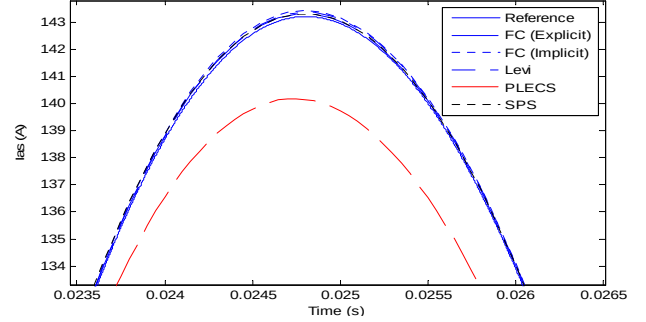


Fig. 4. First peak stator current solution using $\Delta t = 50\mu s$.

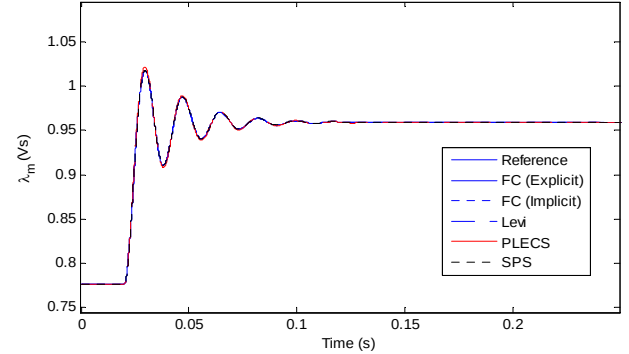


Fig. 5. Main flux linkage transient solution using $\Delta t = 50\mu s$.

Table I Computational cost for transient study using $\Delta t = 50\mu s$.

Models	Levi	FC (imp.)	FC (exp.)	SPS	PLECS
CPU times	0.50s	1.52s	0.38s	2.25s	1.01s

To further compare the accuracy offered by these models, the same study is repeated using a much larger time step, $\Delta t = 1.67ms$. The results obtained by various models are shown in Figs. 6-8. As expected, there is now a visible difference among some of the solutions predicted by the various models. The explicit FC method produces very good results for the current during the first transient peak, but its main flux linkage is less accurate than the implicit FC method. The SPS model gives results nearly identical to implicit FC and Levi's (for the same Δt), which all happen

to be close from the reference case. The results obtained with PLECS are different from all others, but are not necessarily that far from the reference case.

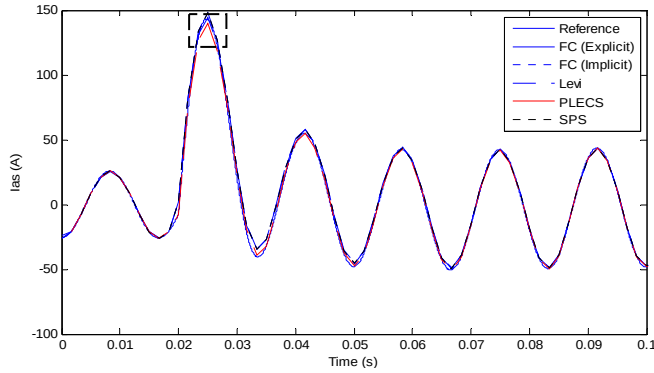


Fig. 6. Stator current transient solution using $\Delta t = 1.67\text{ms}$.

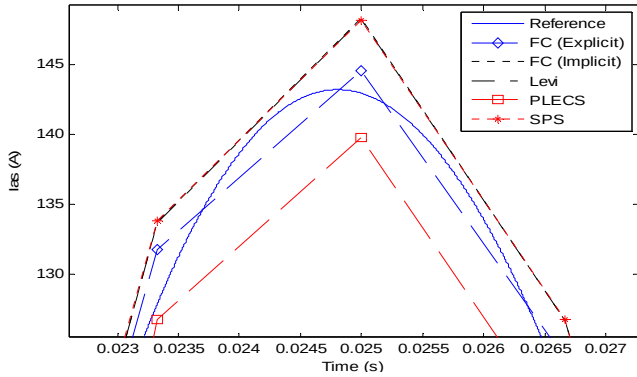


Fig. 7. First peak stator current solution, using $\Delta t = 1.67\text{ms}$.

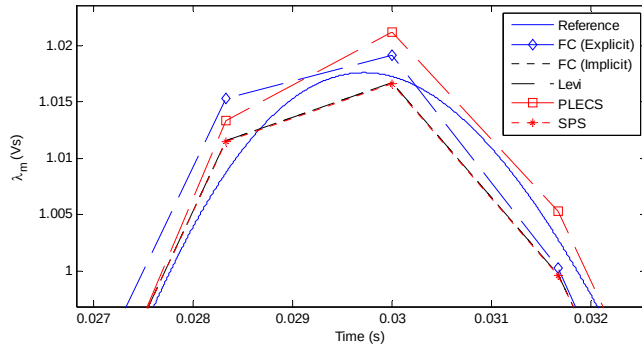


Fig. 8 First peak main flux linkage, using $\Delta t = 1.67\text{ms}$.

4. CONCLUSIONS

This paper discussed several different methods for representing magnetic saturation of the main flux in state variable induction machines models. The saturable induction machine models available as built-in component models in SPS and PLECS are also compared. The methods that are shown to give consistent and accurate results include the flux correction method and the generalized flux space vector approach, as does SPS' saturable induction machine. It is also shown that neglecting dynamic cross-

saturation (as in PLECS) gives a small error with an appropriate choice of state variables. The proposed explicit flux correction method is shown to be most efficient computationally among the considered models, and yet also very accurate for a large time step Δt .

APPENDIX

a) Machine parameters [3]: 50HP, 460V, 4 poles,

$$J = 1.662\text{kg} \cdot \text{m}^2, r_s = 0.087\Omega, r_r = 0.228\Omega,$$

$$X_{ls} = 0.302\Omega, X_{lr} = 0.302\Omega, X_m = 13.08\Omega \text{ (unsat.)}.$$

b) Saturation representation for the arctangent-function:

$$\lambda_T = 0.82\text{V} \cdot \text{s}, \quad \tau_T = 20(1/\text{V} \cdot \text{s}), \quad f_T = 3.261,$$

$$M_a = 88.95(1/H), \quad M_d = 62.75(1/H).$$

c) Equivalent saturation representation points (i_m, λ_m):

λ_m (V·s)	0.38	0.48	0.60	0.68	0.75	0.79	0.82
i_m (A)	11.20	14.33	18.33	21.31	24.42	26.74	29.07
λ_m (V·s)	0.86	0.92	0.96	1.01	1.06	1.10	
i_m (A)	33.21	40.78	46.21	53.21	60.33	66.10	

REFERENCES

- [1] S. J. Salon, *Finite Element Analysis of Electrical Machines*, Kluwer Academic Publishers, 1995.
- [2] G. R. Slemon, "An equivalent circuit approach to analysis of synchronous machines with saliency and saturation," *IEEE Trans. Energy Convers.*, vol. 5, no. 3, pp. 538–545, Sept. 1990.
- [3] P. C. Krause, O. Wasynczuk, and S. D. Sudhoff, *Analysis of Electric Machine*, 2nd Edition, IEEE Press, Piscataway, NJ, 2002.
- [4] E. Levi, "A unified approach to main flux saturation modeling in D-Q axis models of induction machines," *IEEE Trans. Energy Convers.*, vol. 10, no. 3, pp. 455–461, Sept. 1995.
- [5] R. Wamkeue, D. Aguglia, M. Lakehal and P. Viarouge, "Two-step method for identification of nonlinear model of induction machine," *IEEE Trans. Energy Convers.*, vol. 22, no. 4, pp. 801–809, Dec. 2007.
- [6] J.R. Marti and T. O. Myers, "Phase-domain induction motor model for power system simulators," *IEEE Conference Communications, Power, and Computing*, vol. 2, pp. 276–282, May 1995.
- [7] M. Ikeda and T. Hiyama, "Simulation studies of the transients of squirrel-cage induction motors," *IEEE Trans. Energy Conversion*, vol. 22, no. 2, pp. 233–239, Jun. 2007.
- [8] X. Tu, L.-A. Dessaint, R. Champagne and K. Al-Haddad, "Transient modeling of squirrel-cage induction machine considering air-gap flux saturation harmonics," *IEEE Trans. Ind. Electron.*, vol. 55, no. 7, pp. 2798–2809, Jul. 2008.
- [9] SimPowerSystems 4-User Guide, The MathWorks, Inc., Natick, Massachusetts [Online]. Available at: www.mathworks.com, 2007.
- [10] Piece-wise Linear Electrical Circuit Simulation (PLECS) User Manual Version 1.5, Plexim GmbH [Online]. Available at: www.plexim.com, 2006.
- [11] K. A. Corzine, B. T. Kuhn, S. D. Sudhoff, H. J. Hegner, "An improved method for incorporating magnetic saturation in the q-d synchronous machine model," *IEEE Trans. Energy Conversion*, vol. 13, no. 3, pp. 270–275, Sept. 1998.
- [12] E. Levi, "Impact of cross-saturation on accuracy of saturated induction machine models", *IEEE Transactions on Energy Conversion*, Vol. 12, No. 3, Sept. 1997.

- [13] L. Wang, J. Jatskevich, N. Ozog, A. Davoudi, "A Simple Explicit Method of Representing Magnetic Saturation of Salient-Pole Synchronous Machines in Both Rotor Axes using Matlab-Simulink," *IEEE Canadian Conference on Electrical and Computer Engineering*, Vancouver, BC, April 22 – 26, 2007.

Efficient Explicit Representation of AC Machines Main Flux Saturation in State-Variable-Based Transient Simulation Packages

Francis Therrien, *Student Member, IEEE*, Liwei Wang, *Member, IEEE*, Juri Jatskevich, *Senior Member, IEEE*, and Oleg Wasynczuk, *Fellow, IEEE*

Abstract—Inclusion of magnetic saturation in low-order qd models of induction and synchronous machines improves the accuracy of system studies. Over the years, numerous explicit and implicit methods of representing magnetic saturation have been proposed in the literature, some of which find their use in commonly used transient simulation packages. This paper presents a straightforward and improved flux correction method of modeling saturation explicitly in classical qd state variable ac machine models. The approach requires limited additional information and avoids the need for a dynamic inductance, which is present in many other approaches. The method is first developed for induction machines and then extended to salient-pole synchronous machines as a more general case. Computer studies demonstrate the accuracy of the proposed method for predicting the dynamic and steady-state cross-saturation phenomena, as well as its computational advantages.

Index Terms—AC machines, cross-saturation, dynamic simulation, explicit methods, induction machine, magnetic saturation, synchronous machine.

I. INTRODUCTION

MODELING of ac electrical machines has been an active area of research for a long time. Depending on degree of detail and required accuracy for machine models, one may find rich literature on finite-element analysis [1], [2], magnetic equivalent circuit models [3], and lumped-parameter coupled equivalent circuit-based models [4], [12]–[31]. This paper is focused on the so-called general-purpose qd machine models [4]. Although such models are derived based on many assumptions, their accuracy is generally considered sufficient for system-level transient studies. Due to their simplicity, small number of required parameters, and relative computational efficiency, similar models are found in many commercial transient simulation packages and toolboxes including the nodal-analysis-based

electromagnetic transient program (EMTP-type [5]–[8]) and state-variable-based simulators (SimPowerSystems (SPS) [9], PLECS [10], ASMG [11]) as built-in component models.

The modeling of magnetic saturation is a frequent optional addition to the basic ac machine models in commercial packages [5]–[10]. For induction machines, the saturation is typically applied to the main flux, therefore taking into account the so-called qd axes cross-saturation [4]–[10]. For synchronous machines, depending on the rotor construction, the saturation is sometimes applied to the d -axis only, [8], [9], [12], [13], to the q - and d -axes independently [5], [7], or to the main flux [5]–[7], [9], [10], [13]. As suggested by experiments [14] and finite-element analysis [2], the saturation of the main flux results in better accuracy [15]. For the qd salient-pole synchronous machine model, single-saturation factor [17], [18], [21], or two-saturation factor approaches [14], [15] can be utilized depending on the availability of the q -axis and/or intermediate directions of the saturation characteristic [16], [20]. However, in practice, only the d -axis saturation characteristic can be easily obtained experimentally [16]. Therefore, a single saliency factor approach is often considered sufficient for the general-purpose models [17]–[19].

This paper is focused on the general-purpose state-variable-based models of ac machines wherein a number of approaches have previously been developed to incorporate main flux saturation using a single-saturation factor methodology. A simple approach is to modify the equivalent steady-state magnetizing inductance according to the level of saturation at the previous time step (a time step relaxation) [21], [22]. The generalized flux space-vector approach [17], [18], [23], [24] incorporates the main flux saturation and accounts for the steady state and dynamic cross-saturation. Therein, for any set of independent state variables (with the exception of all winding flux linkages), a dynamic inductance is required (adding time-varying nonzeros in matrix formulation). An approximate model can be achieved by using certain state variables and ignoring these additional terms [25], [26], as is done in PLECS [10].

The original flux correction (FC) approach [4], [27]–[29] is based on first expressing the unsaturated magnetizing fluxes and then correcting them according to the saturation characteristic. The FC approach is very straightforward, easy to append to the unsaturated model (as an option that can be turned on/off), and is therefore very convenient to use in state-variable transient simulation programs. However, in its exact formulation, this approach results in implicit equations that require iterative

Manuscript received August 13, 2012; revised December 22, 2012; accepted January 24, 2013. Date of publication March 7, 2013; date of current version May 15, 2013. This work was supported by the Natural Science and Engineering Research Council of Canada under the Discovery Grant. Paper no. TEC-00422-2012.

F. Therrien and J. Jatskevich are with the Department of Electrical and Computer Engineering, University of British Columbia, Vancouver, BC V6T 1Z4, Canada (e-mail: francist@ece.ubc.ca; juri.j@ece.ubc.ca).

L. Wang is with ABB Sweden Corporate Research, SE-721 78 Västerås, Sweden (e-mail: liwei.wang@se.abb.com).

O. Wasynczuk is with Purdue University, West Lafayette, IN 47907 USA (e-mail: wasynczu@ecn.purdue.edu).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TEC.2013.2245332

solution at every time step. An approximate FC approach was proposed in [30], which is explicit but has reduced accuracy.

The goal of this paper is to present an FC approach that is both explicit and accurate, and due to its simplicity and computational efficiency, can be easily incorporated into state-variable transient simulation programs that are widely used by engineers and researchers in industry and academia. More specifically, the properties of the proposed FC approach and the contributions of this paper can be summarized as follows.

- 1) First, the FC approach [4], [27]–[29] is rederived for a symmetrical induction machine in explicit form without any approximations.
- 2) The FC approach is then extended to a salient rotor synchronous machine using the single saliency factor approach. Several formulations are presented including implicit, approximate, and explicit (with 2-D and 1-D lookup tables).
- 3) The accuracy of the proposed explicit FC method is verified against several established models in commonly used simulation packages. The proposed model is shown to accurately incorporate steady state and dynamic cross-saturation, as well as to offer numerical/computational advantages.

II. INDUCTION MACHINES

A. Magnetically Linear Model

To set the stage for the derivation of the proposed approach and for completeness, the magnetically linear model is briefly described first. The electrical and mechanical dynamics of a general-purpose magnetically linear squirrel-cage induction machine can be represented by a set of differential equations [4]. For the purpose of this paper, all rotor variables are referred to the stator side and the variables are expressed in the arbitrary reference frame. The motor sign convention is used. Throughout this paper, the operator p denotes the derivative with respect to time. Neglecting zero sequence, the voltage equations are as follows [4]:

$$v_{qs} = r_s i_{qs} + \omega \lambda_{ds} + p \lambda_{qs} \quad (1)$$

$$v_{ds} = r_s i_{ds} - \omega \lambda_{qs} + p \lambda_{ds} \quad (2)$$

$$0 = r_r i_{qr} + (\omega - \omega_r) \lambda_{dr} + p \lambda_{qr} \quad (3)$$

$$0 = r_r i_{dr} - (\omega - \omega_r) \lambda_{qr} + p \lambda_{dr}. \quad (4)$$

The corresponding flux linkage equations are

$$\lambda_{qs} = L_{ls} i_{qs} + \lambda_{mq} \quad (5)$$

$$\lambda_{ds} = L_{ls} i_{ds} + \lambda_{md} \quad (6)$$

$$\lambda_{qr} = L_{lr} i_{qr} + \lambda_{mq} \quad (7)$$

$$\lambda_{dr} = L_{lr} i_{dr} + \lambda_{md}. \quad (8)$$

For the unsaturated machine, the magnetizing fluxes are

$$\lambda_{mq} = L_{mu} i_{mq} = L_{mu} (i_{qs} + i_{qr}) \quad (9)$$

$$\lambda_{md} = L_{mu} i_{md} = L_{mu} (i_{ds} + i_{dr}), \quad (10)$$

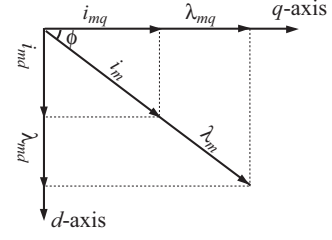


Fig. 1. Projections of the main flux linkage and magnetizing current onto qd -axes for symmetrical induction machines.

where L_{mu} is the unsaturated magnetizing inductance. The developed electromagnetic torque T_e , mechanical load torque T_m , and rotor speed ω_r are related by

$$T_e = \left(\frac{3}{2}\right) \left(\frac{P}{2}\right) (\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}) \quad (11)$$

$$T_e - T_m = J \left(\frac{2}{P}\right) p \omega_r \quad (12)$$

where P and J denote the number of poles and combined rotor-load inertia, respectively.

B. Representation of Main Flux Saturation

For smooth air-gap symmetrical induction machines, the main flux λ_m and magnetizing current i_m are aligned and are related to their qd -axes projections as depicted in Fig. 1 and expressed by the following:

$$\lambda_m = \sqrt{\lambda_{mq}^2 + \lambda_{md}^2} \quad (13)$$

$$i_m = \sqrt{i_{mq}^2 + i_{md}^2}. \quad (14)$$

The main flux saturation characteristic defines the nonlinear relationship between λ_m and i_m . It is also important to point out that, due to the choice of scaling in the reference frame transformation [4], and throughout this paper, the variables λ_m and i_m refer to peak values. The saturation characteristic can be generated experimentally from a standard no-load test. Specifically, neglecting stator and core losses, the measured rms stator line currents $I_{as,NL}$ and rms line-to-neutral voltages $V_{as,NL}$ are related to λ_m and i_m by

$$\lambda_m \approx \sqrt{2} \frac{(V_{as,NL} - \omega_e I_{as,NL} L_{ls})}{\omega_b} \quad (15)$$

$$i_m \approx \sqrt{2} I_{as,NL} \quad (16)$$

where ω_e and ω_b are the operating and base frequencies, respectively. Thereafter, curve-fitting techniques such as piecewise linear interpolates, polynomials, arctangent functions [31], etc., can be used to represent the saturation characteristic. Regardless of the method, for the purpose of this paper, this characteristic is represented by the function (lookup table)

$$i_m = F(\lambda_m) \quad (17)$$

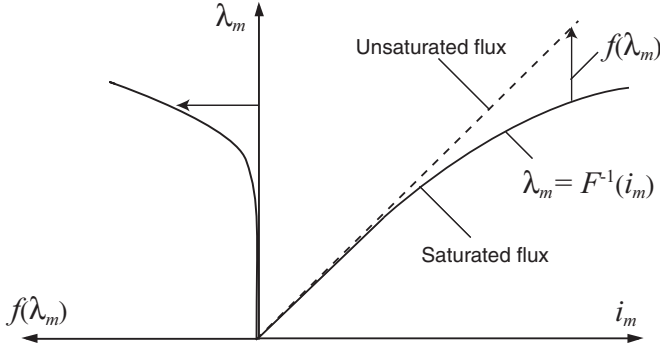


Fig. 2. Main flux saturation characteristic and FC function.

or conversely,

$$\lambda_m = F^{-1}(i_m). \quad (18)$$

The relationship between the saturated main flux λ_m and magnetizing current i_m is qualitatively depicted in Fig. 2 (right-hand side), wherein an unsaturated flux is also shown as a straight line.

C. Implicit FC Method

The basic idea of the FC approach is to define the correction function $f(\lambda_m)$ as the difference between the saturated and unsaturated curves as shown in Fig. 2 [4]. Extending this approach to qd -axes, the corrected (saturated) flux linkages in each axis become

$$\lambda_{mq} = L_{mu}(i_{qs} + i_{qr}) - f_q(\lambda_m) \quad (19)$$

$$\lambda_{md} = L_{mu}(i_{ds} + i_{dr}) - f_d(\lambda_m) \quad (20)$$

where for a symmetrical induction machine

$$f_q(\lambda_m) = \frac{\lambda_{mq}}{\lambda_m} f(\lambda_m) \quad (21)$$

$$f_d(\lambda_m) = \frac{\lambda_{md}}{\lambda_m} f(\lambda_m). \quad (22)$$

Substituting (5), (7), (21) in (19) and (6), (8), (22) in (20), and solving for λ_{mq} and λ_{md} yields

$$\lambda_{mq} = L_{aqu} \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{qr}}{L_{lr}} \right) - \frac{L_{aq}}{L_{mu}} \frac{\lambda_{mq}}{\lambda_m} f(\lambda_m) \quad (23)$$

$$\lambda_{md} = L_{adu} \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{dr}}{L_{lr}} \right) - \frac{L_{ad}}{L_{mu}} \frac{\lambda_{md}}{\lambda_m} f(\lambda_m) \quad (24)$$

where

$$L_{aqu} = L_{adu} = \left(\frac{1}{L_{mu}} + \frac{1}{L_{ls}} + \frac{1}{L_{lr}} \right)^{-1}. \quad (25)$$

Replacing (9) and (10) by (23) and (24), respectively, the classical qd induction machine model now takes into account main flux saturation in steady-state and dynamic sense exactly as defined by (17) and (18). However, λ_{mq} and λ_{md} , which are not state variables, are dependent on $f(\lambda_m)$. Therefore, (23) and (24) are implicit nonlinear equations requiring iterative solution (e.g., Newton–Raphson) and additional computing time. Although some software packages (e.g., Simulink [32]) may

have the capability to solve implicit equations, it is still very desirable to reformulate the saturation model such that it requires only explicit and straightforward to calculate equations, improving the computational efficiency.

D. Explicit FC Method

It is possible to reformulate the FC method by introducing intermediate variables. In particular, considering the q -axis first, substituting (5) and (7) in (9), i_{mq} can be rewritten as a function of state variables (λ_{qs} and λ_{qr}) and the magnetizing flux linkage λ_{mq} as

$$i_{mq} = \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{qr}}{L_{lr}} \right) - \left(\frac{\lambda_{mq}}{L_{ls}} + \frac{\lambda_{mq}}{L_{lr}} \right). \quad (26)$$

Similarly, for the d -axis, substituting (6) and (8) in (10), i_{md} is rewritten as

$$i_{md} = \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{dr}}{L_{lr}} \right) - \left(\frac{\lambda_{md}}{L_{ls}} + \frac{\lambda_{md}}{L_{lr}} \right). \quad (27)$$

Defining the intermediate variables i_{zq} and i_{zd} , (26) and (27) can be simplified to

$$i_{zq} \equiv \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{qr}}{L_{lr}} \right) = i_{mq} + \frac{\lambda_{mq}}{L_{\Sigma l}} \quad (28)$$

$$i_{zd} \equiv \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{dr}}{L_{lr}} \right) = i_{md} + \frac{\lambda_{md}}{L_{\Sigma l}} \quad (29)$$

where

$$L_{\Sigma l} = \left(\frac{1}{L_{ls}} + \frac{1}{L_{lr}} \right)^{-1}. \quad (30)$$

Based on the diagram in Fig. 1 and (28), (29), we have the following relationships:

$$\cos(\phi) = \frac{i_{mq} + \lambda_{mq}/L_{\Sigma l}}{i_m + \lambda_m/L_{\Sigma l}} = \frac{i_{zq}}{i_m + \lambda_m/L_{\Sigma l}} \quad (31)$$

$$\sin(\phi) = \frac{i_{md} + \lambda_{md}/L_{\Sigma l}}{i_m + \lambda_m/L_{\Sigma l}} = \frac{i_{zd}}{i_m + \lambda_m/L_{\Sigma l}}. \quad (32)$$

Applying a basic trigonometric identity to (31) and (32), after some manipulations, we obtain

$$\left(\frac{i_{zq}}{i_m + \lambda_m/L_{\Sigma l}} \right)^2 + \left(\frac{i_{zd}}{i_m + \lambda_m/L_{\Sigma l}} \right)^2 = 1 \quad (33)$$

and

$$i_z \equiv \sqrt{i_{zq}^2 + i_{zd}^2} = i_m + \lambda_m/L_{\Sigma l}. \quad (34)$$

It is now clear that there is a unique mapping between the state-variable-dependent i_z [see (28) and (29)] and the pair of (i_m, λ_m) [see (34)]. Moreover, based on (34) and (17), it is possible to define the explicit relationship between i_z and saturated flux λ_m . This new function is denoted as

$$\lambda_m = G(i_z) \quad (35)$$

which can be readily calculated from the measured data points (17) using (34), and subsequently implemented using a 1-D lookup table. While it is not necessary for the simulation, the

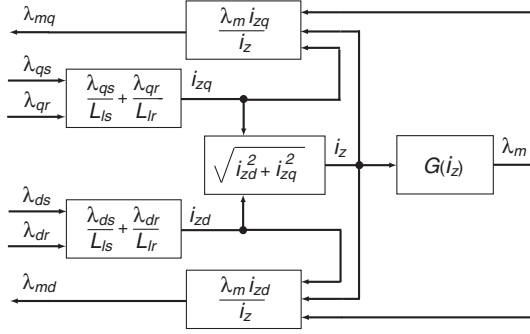


Fig. 3. Implementation of induction machine main flux saturation using the explicit FC approach.

magnetizing current i_m can also be evaluated using (34), if desired. Finally, using (34) and Fig. 1, the resulting saturated magnetizing fluxes in the q - and d -axes can be computed using

$$\lambda_{mq} = \lambda_m \frac{i_{zq}}{i_z} \text{ and } \lambda_{md} = \lambda_m \frac{i_{zd}}{i_z}. \quad (36)$$

Replacing (9) and (10) with (36) yields another method of accounting for main flux saturation in the classical qd model. A block diagram depicting all algebraic equations for this implementation of the FC method is shown in Fig. 3. As it can be seen, as opposed to the method described in Section II-C, here λ_{mq} and λ_{md} are defined in terms of intermediate variables i_{zq} and i_{zd} , which are computed from the state variables according to (28) and (29), yielding an explicit formulation.

To better understand the proposed approach, the function (35) and the saturation characteristic (18) for a 50-HP induction machine [4], [34] are plotted in Fig. 4(a), where we observe that $G(i_z)$ appears very linear, with only a small change in the slope in the saturated region. For clarity, this segment is magnified and plotted in Fig. 4(b). To explain this phenomenon, (9) and (10) are first rewritten for a saturated machine

$$\lambda_{mq} = L_{ms} i_{mq} = L_{ms} (i_{qs} + i_{qr}) \quad (37)$$

$$\lambda_{md} = L_{ms} i_{md} = L_{ms} (i_{ds} + i_{dr}) \quad (38)$$

where L_{ms} is the saturated magnetizing inductance, which depends on λ_m . Solving for i_{qs} , i_{ds} , i_{qr} , and i_{dr} in (5)–(8) and substituting the results into (37), (38), we obtain

$$\lambda_{mq} = L_{aqs} \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{qr}}{L_{lr}} \right) = L_{aqs} i_{zq} \quad (39)$$

$$\lambda_{md} = L_{ads} \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{dr}}{L_{lr}} \right) = L_{ads} i_{zd} \quad (40)$$

where

$$L_{aqs} = L_{ads} = \left(\frac{1}{L_{ms}} + \frac{1}{L_{ls}} + \frac{1}{L_{lr}} \right)^{-1}. \quad (41)$$

Therefore, $G(i_z)$ is equivalent to the product of L_{aqs} (or L_{ads}) and i_z . According to (41), L_{aqs} and L_{ads} appear as the parallel combination of the leakage inductances and the saturation-dependent magnetizing inductance. Since the magnetizing inductance of a rotating machine is typically much larger than its leakage inductances, L_{aqs} and L_{ads} will change

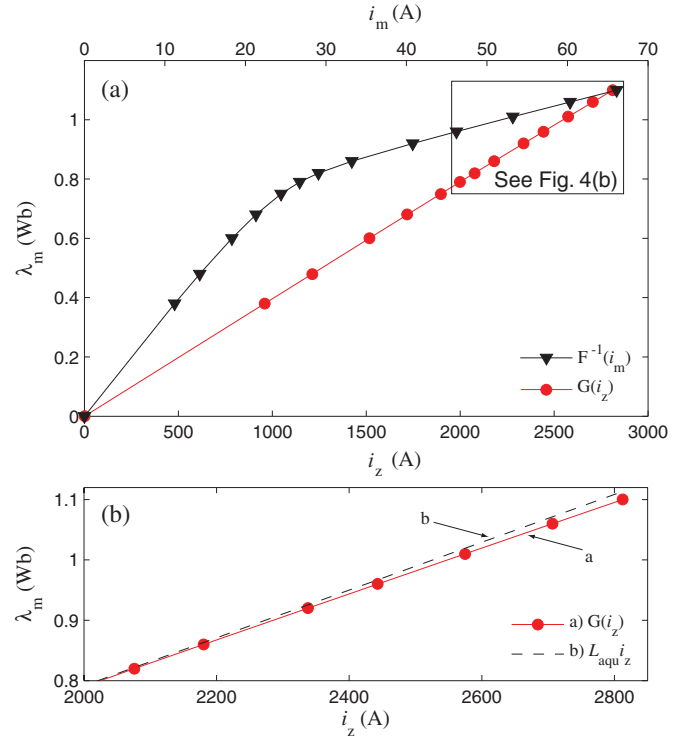


Fig. 4. (a) Saturation characteristic $F^{-1}(i_m)$ and equivalent explicit function $G(i_z)$ for a 50-HP induction machine [4], [34]; and (b) magnified view of $G(i_z)$ in the saturation region.

very little from no saturation to full saturation. As a result, it may visually appear that $G(i_z)$ remains almost constant through the operating region. However, the detailed plot in Fig. 4(b) shows a small slope with respect to a straight line produced by using the constant L_{aqu} or L_{adu} .

III. SYNCHRONOUS MACHINES

A. Magnetically Linear Model

To extend the concept developed in Section II-D, a general-purpose qd synchronous machine model [4] is now considered. Without loss of generality, we consider one d -axis field winding fd , one d -axis damper winding kd , and two q -axis damper windings $kq1$ and $kq2$, respectively. Either round-rotor or salient-pole machines can be considered (depending on the magnetizing inductances L_{mq} and L_{md}), and if needed, damper windings can easily be added or eliminated. All rotor variables are referred to the stator side, and motor sign convention is used herein. Throughout this section, the rotor reference frame is assumed, whereas the superscript “ r ” is omitted for notational compactness. The corresponding voltage equations are

$$v_{qs} = r_s i_{qs} + \omega_r \lambda_{ds} + p \lambda_{qs} \quad (42)$$

$$v_{ds} = r_s i_{ds} - \omega_r \lambda_{qs} + p \lambda_{ds} \quad (43)$$

$$v_{0s} = r_s i_{0s} + p \lambda_{0s} \quad (44)$$

$$v_{kq1} = r_{kq1} i_{kq1} + p \lambda_{kq1} \text{ and } v_{kq2} = r_{kq2} i_{kq2} + p \lambda_{kq2} \quad (45)$$

$$v_{fd} = r_{fd} i_{fd} + p \lambda_{fd} \text{ and } v_{kd} = r_{kd} i_{kd} + p \lambda_{kd}. \quad (46)$$

The flux linkage equations are

$$\lambda_{qs} = L_{ls} i_{qs} + \lambda_{mq} \quad (47)$$

$$\lambda_{ds} = L_{ls} i_{ds} + \lambda_{md} \quad (48)$$

$$\lambda_{0s} = L_{ls} i_{0s} \quad (49)$$

$$\lambda_{kq1} = L_{lkq1} i_{kq1} + \lambda_{mq} \text{ and } \lambda_{kq2} = L_{lkq2} i_{kq2} + \lambda_{mq} \quad (50)$$

$$\lambda_{fd} = L_{lfd} i_{fd} + \lambda_{md} \text{ and } \lambda_{kd} = L_{lk d} i_{kd} + \lambda_{md} \quad (51)$$

where the unsaturated magnetizing fluxes are given by

$$\lambda_{mq} = L_{mqu} i_{mq} = L_{mqu} (i_{qs} + i_{kq1} + i_{kq2}) \quad (52)$$

$$\lambda_{md} = L_{mdu} i_{md} = L_{mdu} (i_{ds} + i_{fd} + i_{kd}). \quad (53)$$

The electromagnetic torque and mechanical equations are given by (11) and (12). The rotor angle is defined as

$$\delta = \theta_r - \theta_e = \int_0^t (\omega_r - \omega_e) dt \quad (54)$$

where θ_r and θ_e are the rotor and bus voltage angles, respectively, and ω_e is the bus voltage (operating) frequency.

B. Saliency Factor Approach for Main Flux Saturation

Herein, it is assumed that the d -axis saturation characteristic of synchronous machines can be obtained experimentally under steady-state open-circuit conditions by recording the measured pairs of the field current I_{fd} (properly referred/converted to the stator side) and rms stator line-to-neutral voltages $V_{as,NL}$. Based on these measurements, assuming that the core losses can be neglected, the values for the magnetizing flux λ_{md} and current i_{md} can be calculated as follows:

$$\lambda_{md} = \sqrt{2} \frac{V_{as,NL}}{\omega_b} \quad (55)$$

$$i_{md} \approx I_{fd}. \quad (56)$$

As before, the variable λ_{md} refers to the peak values of the flux. Thereafter, the d -axis saturation characteristic can be defined by the function

$$i_{md} = F_d(\lambda_{md}). \quad (57)$$

Due to the shape of the salient-pole synchronous machine air gap, the q -axis saturation characteristic is also required to adequately model the main flux saturation. However, since this information is not typically provided by the manufacturers and the procedures for obtaining it experimentally are not simple [15], [16], [20], for the purpose of system-level studies, a common methodology is to use a saliency factor approach where scaling is applied to the q -axis in order to obtain an equivalent isotropic machine [17], [18], [21]. Assuming the q - and d -axes saturate at the same level, the saliency factor m is defined as follows:

$$m = \sqrt{\frac{L_{mqu}}{L_{mdu}}} = \sqrt{\frac{L_{mqs}}{L_{mds}}} \quad (58)$$

where L_{mqu} , L_{mdu} , L_{mqs} , and L_{mds} are the unsaturated and saturated q - and d -axis magnetizing inductances, respectively.

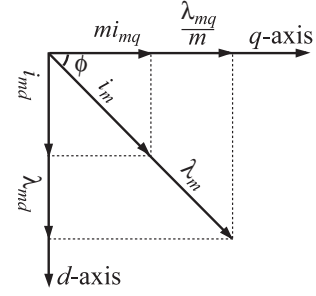


Fig. 5. Projections of the main flux linkage and magnetizing current onto qd -axes for synchronous machines considering the saliency factor approach.

The main flux and magnetizing current are then calculated according to the diagram depicted in Fig. 5 as

$$\lambda_m = \sqrt{\left(\frac{\lambda_{mq}}{m}\right)^2 + \lambda_{md}^2} \quad (59)$$

$$i_m = \sqrt{(i_{mq}m)^2 + i_{md}^2}. \quad (60)$$

After this rescaling, the d -axis saturation function (57) is applied to the main flux saturation as

$$i_m = F_d(\lambda_m) \quad (61)$$

or conversely

$$\lambda_m = F_d^{-1}(i_m). \quad (62)$$

As it can be seen from (58)–(60), this formulation automatically covers both salient-pole and round-rotor machines. For the latter case, $m = 1$, and the equations become similar to those presented in Section II-B.

C. Implicit FC Method

For the implementation of saturation in the model, the FC function $f(\lambda_m)$ depicted in Fig. 2 is redefined using the synchronous machine saturation characteristic (61). It is then possible to write the corrected (saturated) magnetizing flux for each axis as

$$\lambda_{mq} = L_{mqu}(i_{qs} + i_{kq1} + i_{kq2}) - m \cdot f_q(\lambda_m) \quad (63)$$

$$\lambda_{md} = L_{mdu}(i_{ds} + i_{fd} + i_{kd}) - f_d(\lambda_m) \quad (64)$$

where

$$f_q(\lambda_m) = \frac{\lambda_{mq}}{m \cdot \lambda_m} f(\lambda_m) \quad (65)$$

$$f_d(\lambda_m) = \frac{\lambda_{md}}{\lambda_m} f(\lambda_m). \quad (66)$$

Substituting (47), (50), and (65) into (63) and (48), (51), and (66) into (64), and solving for λ_{mq} and λ_{md} yields

$$\lambda_{mq} = L_{aqu} \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}} \right) - \frac{L_{aq}}{L_{mqu}} \frac{\lambda_{mq}}{\lambda_m} f(\lambda_m) \quad (67)$$

$$\lambda_{md} = L_{adu} \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lk d}} \right) - \frac{L_{ad}}{L_{mdu}} \frac{\lambda_{md}}{\lambda_m} f(\lambda_m) \quad (68)$$

where

$$L_{aqu} = \left(\frac{1}{L_{mq u}} + \frac{1}{L_{ls}} + \frac{1}{L_{lk q1}} + \frac{1}{L_{lk q2}} \right)^{-1} \quad (69)$$

$$L_{adu} = \left(\frac{1}{L_{md u}} + \frac{1}{L_{ls}} + \frac{1}{L_{lfd}} + \frac{1}{L_{lk d}} \right)^{-1}. \quad (70)$$

Replacing (52) and (53) by (67) and (68), respectively, the general-purpose qd synchronous machine model now takes into account the main flux saturation for both salient- and round-rotor construction by correcting the unsaturated flux using the implicit algebraic equations (67) and (68).

D. Explicit FC Method

The methodology to achieve explicit FC formulation involves similar steps to those described in Section II-D. In particular, considering the q -axis first, substituting (47) and (50) into (52), i_{mq} is expressed as a function of state variables (λ_{qs} , λ_{kq1} , and λ_{kq2}) and λ_{mq} . For the d -axis, substituting (48) and (51) into (53), i_{md} is rewritten as a function of state variables (λ_{ds} , λ_{fd} , and λ_{kd}) and λ_{md} . The result is

$$i_{mq} = \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{kq1}}{L_{lk q1}} + \frac{\lambda_{kq2}}{L_{lk q2}} \right) - \left(\frac{\lambda_{mq}}{L_{ls}} + \frac{\lambda_{mq}}{L_{lk q1}} + \frac{\lambda_{mq}}{L_{lk q2}} \right) \quad (71)$$

$$i_{md} = \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lk d}} \right) - \left(\frac{\lambda_{md}}{L_{ls}} + \frac{\lambda_{md}}{L_{lfd}} + \frac{\lambda_{md}}{L_{lk d}} \right). \quad (72)$$

Introducing the intermediate variables i_{zq} and i_{zd} , (71) and (72) are simplified to

$$i_{zq} \equiv \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{kq1}}{L_{lk q1}} + \frac{\lambda_{kq2}}{L_{lk q2}} \right) = i_{mq} + \frac{\lambda_{mq}}{L_{\Sigma lq}} \quad (73)$$

$$i_{zd} \equiv \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lk d}} \right) = i_{md} + \frac{\lambda_{md}}{L_{\Sigma ld}} \quad (74)$$

where

$$L_{\Sigma lq} = \left(\frac{1}{L_{ls}} + \frac{1}{L_{lk q1}} + \frac{1}{L_{lk q2}} \right)^{-1} \quad (75)$$

$$L_{\Sigma ld} = \left(\frac{1}{L_{ls}} + \frac{1}{L_{lfd}} + \frac{1}{L_{lk d}} \right)^{-1}. \quad (76)$$

It can be observed that generally, $L_{\Sigma lq} \neq L_{\Sigma ld}$. From the vector diagram in Fig. 5 and (73), (74), we obtain

$$\cos(\phi) = \frac{i_{mq} + \lambda_{mq}/L_{\Sigma lq}}{i_m/m + m\lambda_m/L_{\Sigma lq}} = \frac{i_{zq}}{i_m/m + m\lambda_m/L_{\Sigma lq}} \quad (77)$$

$$\sin(\phi) = \frac{i_{md} + \lambda_{md}/L_{\Sigma ld}}{i_m + \lambda_m/L_{\Sigma ld}} = \frac{i_{zd}}{i_m + \lambda_m/L_{\Sigma ld}}. \quad (78)$$

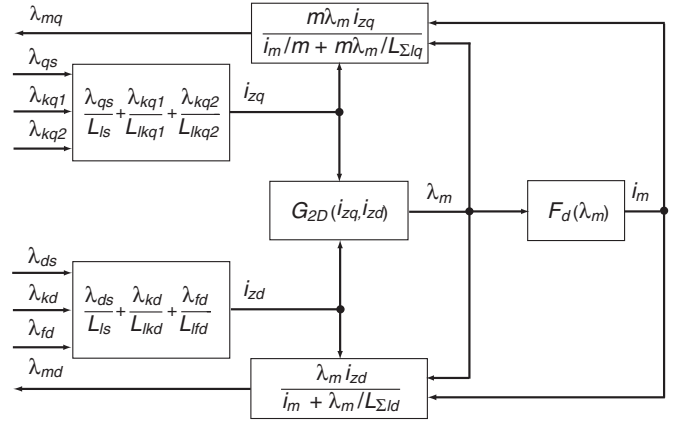


Fig. 6. Explicit implementation of synchronous machine saturation using the FC approach with a 2-D lookup table.

Applying a trigonometric identity to (77) and (78), we have

$$\left(\frac{i_{zq}}{i_m/m + m\lambda_m/L_{\Sigma lq}} \right)^2 + \left(\frac{i_{zd}}{i_m + \lambda_m/L_{\Sigma ld}} \right)^2 = 1. \quad (79)$$

Unlike in (33), the denominators of the two left-hand side terms in (79) are different due to the saliency factor m and the inequality between $L_{\Sigma lq}$ and $L_{\Sigma ld}$. Therefore, unlike it was for an induction machine in Section II-D, (79) cannot be solved easily for i_m and λ_m . Consequently, several methods to obtain the desirable explicit formulation are described here.

1) *Two-Dimensional Explicit FC*: Based on (73), (74), and (79), it can be observed that for any pair of intermediate variables (i_{zq} , i_{zd}), there exists a unique pair of (i_m , λ_m), as they are related by the saturation function. This unique mapping can be implemented using a 2-D lookup table with i_{zq} and i_{zd} as the inputs and λ_m as the output. Symbolically,

$$\lambda_m = G_{2D}(i_{zq}, i_{zd}). \quad (80)$$

Assuming that this function (lookup table) is obtained, the corresponding explicit formulation can be realized according to Fig. 6. Therein, the intermediate variables i_{zq} and i_{zd} are calculated from the state variables (fluxes). Then, the magnetizing flux λ_m and current i_m are evaluated using the lookup tables (80) and (61), respectively. The q -axis and d -axes magnetizing fluxes are computed using a combination of (77), (79), and the projections depicted in Fig. 5 as follows:

$$\lambda_{mq} = \frac{m\lambda_m i_{zq}}{i_m/m + m\lambda_m/L_{\Sigma lq}} \text{ and } \lambda_{md} = \frac{\lambda_m i_{zd}}{i_m + \lambda_m/L_{\Sigma ld}}. \quad (81)$$

The disadvantages of this method are twofold. First, not all simulation packages provide a convenient way of implementing multidimensional lookup tables. The second disadvantage relates to the difficulty of generating $G_{2D}(i_{zq}, i_{zd})$, which is done during the initialization stage. Assuming that the vectors i_{zq} and i_{zd} have a and b components, respectively, $a \times b$ values of λ_m must be calculated by solving the nonlinear equations that include (80) using an iterative approach, which may be time-consuming.

2) *Approximate FC*: An explicit but approximate FC method using only 1-D lookup tables was presented in [30]. For completeness, its main steps are briefly summarized here. In particular, algebraically manipulating (79) yields

$$i_z \equiv \sqrt{i_{zd}^2 + k^2 i_{zq}^2} = i_m + \lambda_m / L_{\Sigma ld} \quad (82)$$

where

$$k = \left(\frac{i_m + \lambda_m / L_{\Sigma ld}}{i_m / m + m \lambda_m / L_{\Sigma lq}} \right). \quad (83)$$

It is observed that due to the saliency factor and the asymmetry of the equivalent leakage inductances, the coefficient k is function of λ_m , making the underlying equations implicit. However, assuming $1/L_{\Sigma lq} \gg 1/L_{mq}$ and $1/L_{\Sigma ld} \gg 1/L_{md}$, (83) can be simplified to obtain the following approximate constant coefficient:

$$\tilde{k} = \left(\frac{L_{\Sigma lq}}{m L_{\Sigma ld}} \right). \quad (84)$$

Using (84), the intermediate variable i_z is reformulated as

$$i_z \equiv \sqrt{i_{zd}^2 + \tilde{k}^2 i_{zq}^2} \approx i_m + \lambda_m / L_{\Sigma ld} \quad (85)$$

from which a unique mapping between i_z and λ_m becomes apparent. Symbolically, this new mapping defines a function

$$\lambda_m = G(i_z). \quad (86)$$

This new lookup table (86) along with (85) is used to replace the 2-D lookup table in Fig. 6, which constitutes the approximate explicit implementation.

3) *One-Dimensional Explicit FC*: To avoid the errors introduced by (84) and (85) [30], a corrected methodology is proposed here. First, it is noted that (82) and (61) capture the unique relation between λ_m and i_z . Using (83), this can first be extended to obtain a mapping between k and i_z , herein defined as

$$k = H(i_z) \quad (87)$$

which can easily be realized as a 1-D lookup table. However, (87) cannot be used directly since the definition of i_z in (82) is also a function of k , which would again result in an implicit formulation. Instead, the approach presented here uses the unsaturated constant coefficient defined as

$$k_u = \left(\frac{i_{mu} + \lambda_{mu} / L_{\Sigma ld}}{i_{mu} / m + m \lambda_{mu} / L_{\Sigma lq}} \right) \quad (88)$$

where (λ_{mu}, i_{mu}) is usually the first pair of the data (λ_m, i_m) given by (61). Then, another intermediate variable i_z^* is calculated using

$$i_z^* = \sqrt{i_{zd}^2 + k_u^2 i_{zq}^2}. \quad (89)$$

If the flux λ_m is in the unsaturated region, the variable i_z^* will be close to the exact i_z as defined by (82). However, in the saturated region and more generally, it will not be the case. Thus, another step is needed, which is achieved by evaluating k as follows:

$$k = H(i_z^*). \quad (90)$$

It is then possible to compute i_z using (82) with this new k . After that, the saturated magnetizing flux λ_m is computed with (86), and the projections λ_{mq} and λ_{md} are calculated using (61) and (81).

A block diagram that shows the implementation of the proposed explicit FC approach for modeling saturation in synchronous machines is depicted in Fig. 7. As it can be seen, this approach requires three 1-D lookup tables (or two lookup tables if (61) is a fitted function), but otherwise, the formulation is very straightforward and similar to that depicted in Fig. 6.

Finally, as it was done for the induction machine, the magnetizing fluxes can be expressed in terms of state variables. Replacing L_{mq} and L_{md} in (52) and (53) by L_{mqs} and $L_{m ds}$, respectively, and substituting (50) and (51), we obtain

$$\lambda_{mq} = L_{aqs} \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}} \right) \equiv L_{aqs} i_{zq} \quad (91)$$

$$\lambda_{md} = L_{ads} \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lk d}} \right) \equiv L_{ads} i_{zd} \quad (92)$$

where

$$L_{aqs} = \left(\frac{1}{L_{mqs}} + \frac{1}{L_{ls}} + \frac{1}{L_{lkq1}} + \frac{1}{L_{lkq2}} \right)^{-1} \quad (93)$$

$$L_{ads} = \left(\frac{1}{L_{m ds}} + \frac{1}{L_{ls}} + \frac{1}{L_{lfd}} + \frac{1}{L_{lk d}} \right)^{-1}. \quad (94)$$

Therefore, the relation between λ_{mq} and i_{zq} , and also the one between λ_{md} and i_{zd} , will again be almost linear—similar to that depicted in Fig. 4. Consequently, the function $G(i_z)$ defined for synchronous machines in (86) will also visually appear close to being linear due to the fact that L_{mqs} and $L_{m ds}$, as defined by (93) and (94), will change very little from no to full saturation.

IV. CASE STUDIES

Three case studies are presented in this section to validate the proposed explicit FC approach and compare the induction and synchronous machine models with the reference models. For consistency, all studies were executed on a personal computer with 2.8-GHz Intel CPU and 4 GB of RAM.

A. Induction Machine

For the purpose of validation, the proposed explicit FC induction machine model (see Section II-D) has been implemented in MATLAB/Simulink environment along with the implicit FC approach (see Section II-C) and the built-in qd induction machine models in commercial toolboxes SPS [9] and PLECS [10]. All these models except PLECS' use winding flux linkages as the state variables. It is also important to note that PLECS' model does not include dynamic cross-saturation based on the results of [26]. A reference solution is obtained by implementing the generalized flux space-vector approach (with $\mathbf{i}_s$ and λ_s as the state variables [24]) in the same environment, as this approach has been previously verified experimentally and is considered to possess sufficient accuracy for the general-purpose models considered in this paper. In the reference model [24], the dynamic

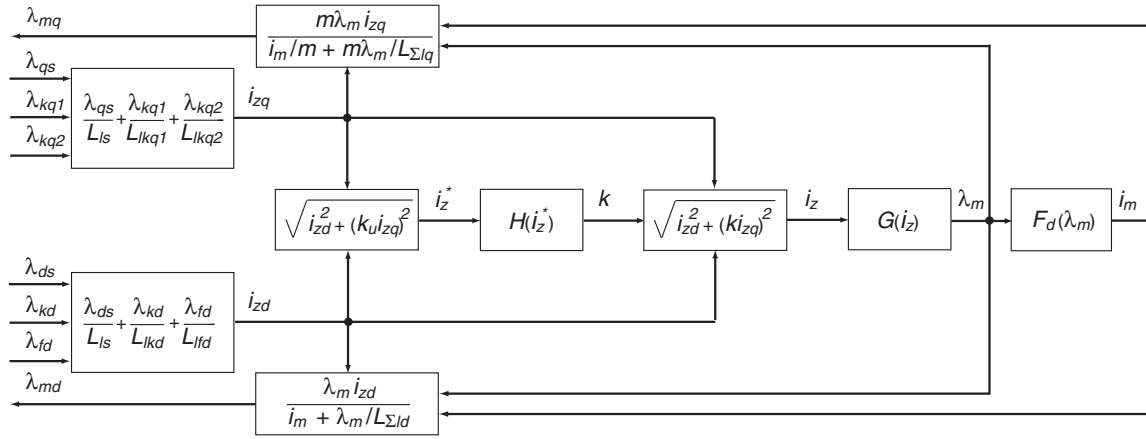
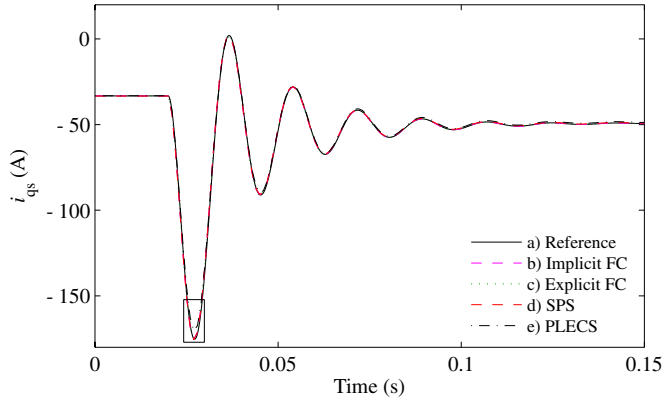
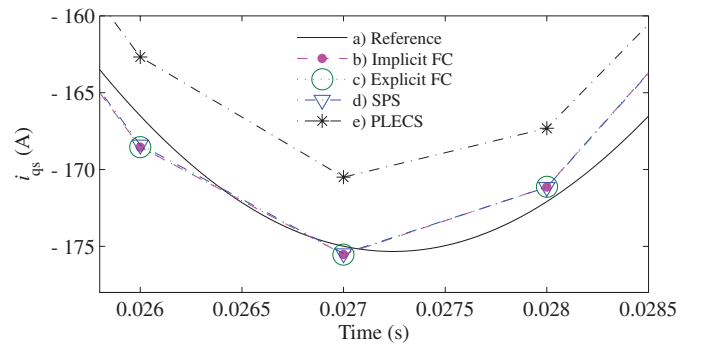


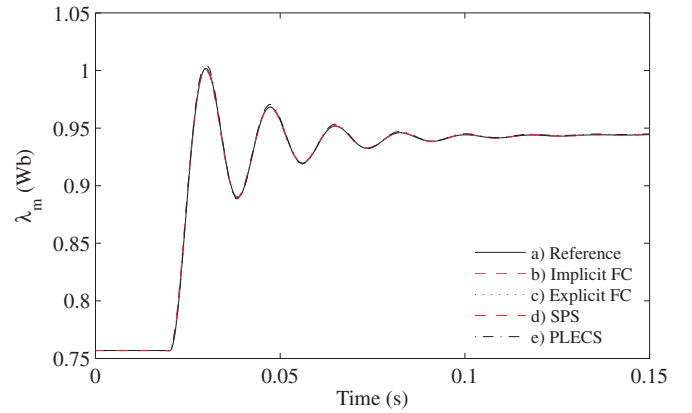
Fig. 7. Implementation of the proposed explicit FC approach for modeling saturation in synchronous machines using a saliency factor approach.


 Fig. 8. Induction machine transient in stator current i_{qs} due to change in stator voltage.

 Fig. 9. Magnified view of the first peak of stator current i_{qs} transient due to change in stator voltage (induction machine).

inductance is included, and is computed using Lagrange polynomial interpolation for nonuniformly spaced points [33]. To obtain a very accurate reference solution, the reference model was solved using the ode5 solver (Dormand–Prince) with a very small fixed time step of 1 μ s. For comparison, all other models were solved using the same solver but with a larger time step of 1 ms. For consistency, all models were implemented in the synchronous reference frame.

1) Change of Stator Voltage: In this study, a 50-HP induction motor [4] (see Appendix A) is connected to an infinite bus. Initially, the machine operates in a steady state with 150-N·m mechanical load. At $t = 0.02$ s, the bus voltage is increased from 0.8 to 1 p.u. This fast electrical transient should allow the observation of the static and dynamic cross-saturation effects.

The predicted transient responses of i_{qs} , λ_m , and ω_r are plotted in Figs. 8–11. As it can be seen in Fig. 8, the models that include the dynamic cross-saturation (implicit FC; explicit FC; and SPS) predict very similar responses in the q -axis stator current. A magnified view of the first peak in i_{qs} is shown in Fig. 9. Here, the small error between the reference model, the implicit FC, explicit FC, and SPS models is mostly due to the larger step. However, the PLECS model, where the dynamic cross-saturation is neglected, shows a small but noticeable error. This error vanishes in steady state. Similar observations can be made


 Fig. 10. Transient in main flux λ_m due to change in stator voltage (induction machine).

for the transient response in main flux λ_m shown in Fig. 10. Moreover, based on Figs. 4 and 10 (and the saturation data in Appendix A), it can also be observed that the machine operates significantly in the saturated region. Finally, Fig. 11 shows the transient rotor speed ω_r , which is accurately predicted by all models.

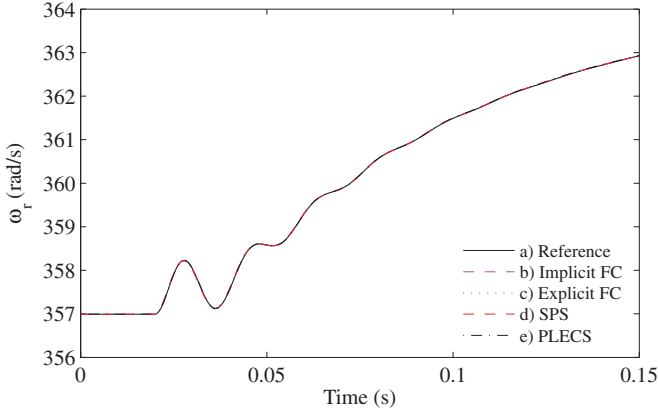


Fig. 11. Transient in electrical rotor speed ω_r due to change in stator voltage (induction machine).

B. Synchronous Machine

The proposed explicit FC synchronous machine model using 1-D lookup tables (see Section III-D3) has been implemented using MATLAB/Simulink along with the implicit FC (see Section III-C) and the approximate FC (see Section III-D2) models. For benchmarking with other existing models, the saturable synchronous machine model from the commercial toolbox SPS [9] has been implemented as well. As with the induction machine study, all these models use winding flux linkages as the state variables. The generalized flux space-vector method [18] (with i_s , i_{fd} , and λ_m as the state variables) serves again as the reference model. The dynamic inductance of the reference model [18] is implemented as described in Section IV-A. The reference model is solved using the ode5 solver with a small time step of $10 \mu\text{s}$. All other models were simulated with the same solver but using a step size of 1 ms.

To obtain a more general case, a salient-pole synchronous machine [4] has been considered. The machine parameters and saturation curve are summarized in Appendix B. To keep the focus on the accuracy of machine models, the subject machine is assumed to be connected to an infinite bus.

1) *Change of Load Torque*: In the following study, the machine is assumed to initially operate in a steady state under nominal excitation and terminal voltages in idle mode (without mechanical torque). At $t = 0.2$ s, the mechanical torque of 1.2 p.u. is applied. Following a relatively slow transient, the machine becomes heavily loaded and the projection of the main flux onto the q -axis and the effect of cross-saturation become significant. The corresponding transient responses in d -axis stator current i_{ds} , q -axis magnetizing flux λ_{mq} , rotor angle δ , and main flux λ_m predicted by various models are plotted in Figs. 12–16. As it can be seen in these figures, the SPS model predicts responses that are visibly different from the reference and all other methods because the SPS salient-pole synchronous machine model considers saturation along the d -axis only. Since this model neglects the q -axis saturation and cross-saturation, it therefore predicts higher rotor angle δ even in steady state, which results in higher flux projection in the q -axis λ_{mq} and higher d -axis stator current i_{ds} , respectively. All other models that utilize

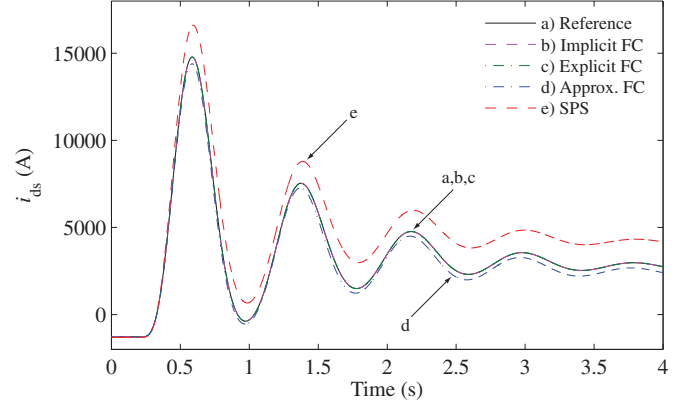


Fig. 12. Transient in stator current i_{ds} due to change of load torque (synchronous machine).

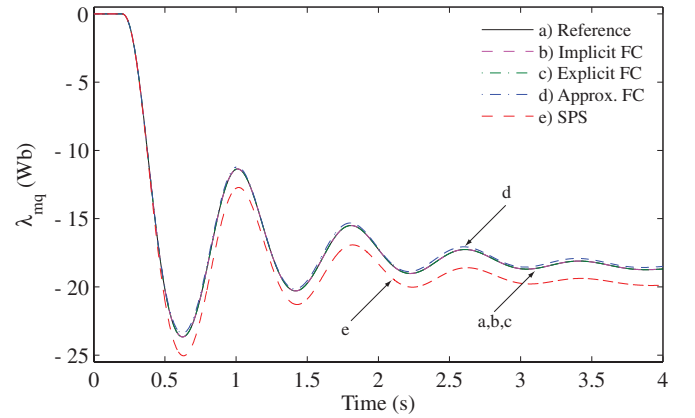


Fig. 13. Transient in magnetizing flux λ_{mq} due to change of load torque (synchronous machine).

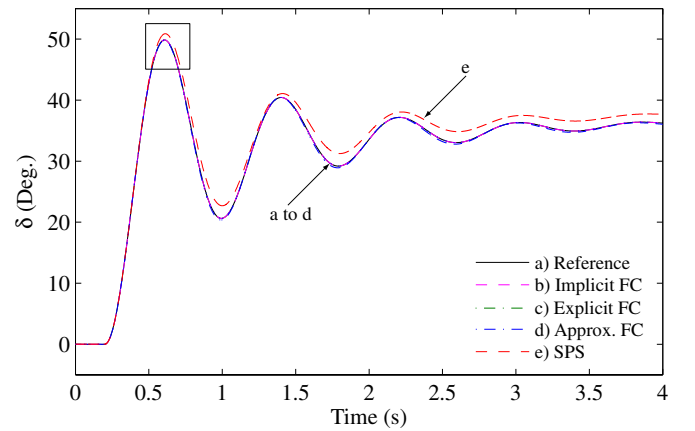


Fig. 14. Transient in rotor angle δ due to change of load torque (synchronous machine).

TABLE I
2-NORM RELATIVE ERRORS OF i_{ds} , λ_{mq} , AND δ IN CHANGE OF LOAD TORQUE TRANSIENT STUDY (SYNCHRONOUS MACHINE)

Variable	Imp. FC	Exp. FC	App. FC	SPS
i_{ds}	0.0789 %	0.0715 %	5.752 %	27.43 %
λ_{mq}	0.0173 %	0.0166 %	0.954 %	6.514 %
δ	0.0203 %	0.0198 %	0.551 %	3.964 %

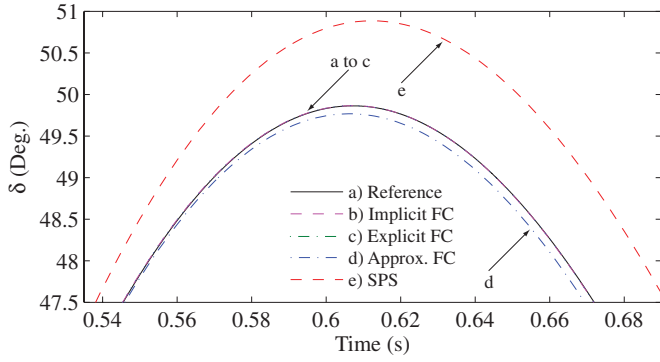


Fig. 15. Magnified view of the first peak of rotor angle δ transient due to change of load torque (synchronous machine).

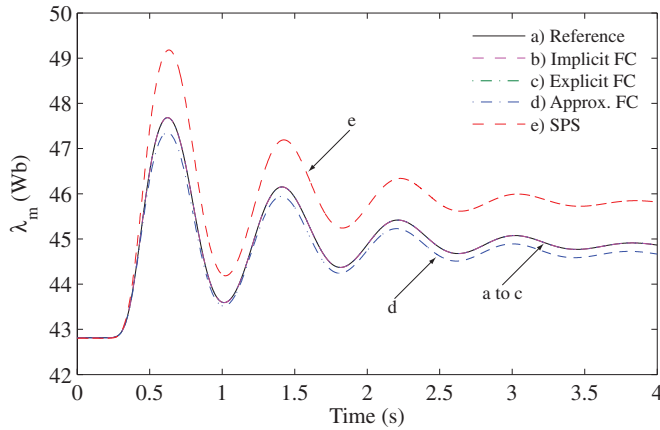


Fig. 16. Transient in main flux λ_m due to change of load torque (synchronous machine).

the FC approach are much closer to the reference solution, with the approximate FC method giving slightly different results. It can also be observed from Fig. 16 and the saturation data given in Appendix B that the machine is saturated in both steady state and transient. The 2-norm (cumulative) relative error [33] between the solution trajectories and the reference trajectory is used here to compare the accuracy achieved by the various models. Specifically, the 2-norm relative errors of i_{ds} , λ_{mq} , and δ with respect to the reference solution are summarized in Table I. As it can be seen in Table I, the SPS model produces by far the largest errors due to neglecting the q -axis saturation and cross-saturation. Next, the approximate FC method also gives noticeable errors (5.752%, 0.954%, and 0.551%, respectively) due to approximations in the calculation of i_z . The implicit FC method gives very small errors for all three variables (0.789%, 0.0173%, and 0.0203%, respectively). These errors are almost identical to the errors obtained with the proposed explicit FC approach, wherein the small differences are mostly caused by the large integration time step.

2) *Change of Stator Voltage*: In the following study, the machine is assumed to initially operate in a steady state under nominal excitation and terminal voltages, and rated mechanical torque. At $t = 0.2$ s, the stator terminal voltages are increased from 1 to 1.05 p.u., thus pushing the machine deeper into the saturated region. Similar fast electrical transients have been used

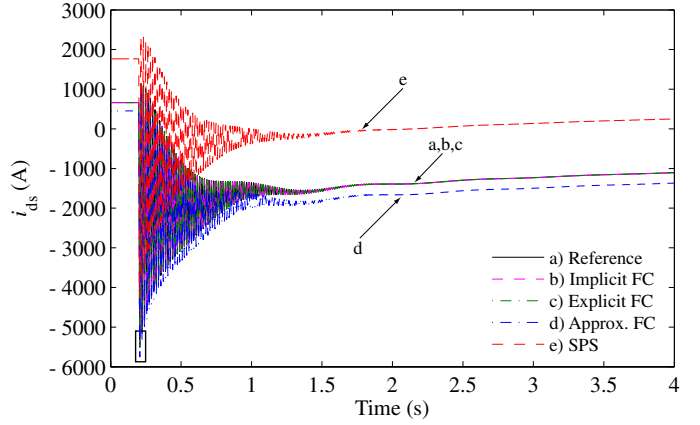


Fig. 17. Transient in stator current i_{ds} due to change of stator voltage (synchronous machine).

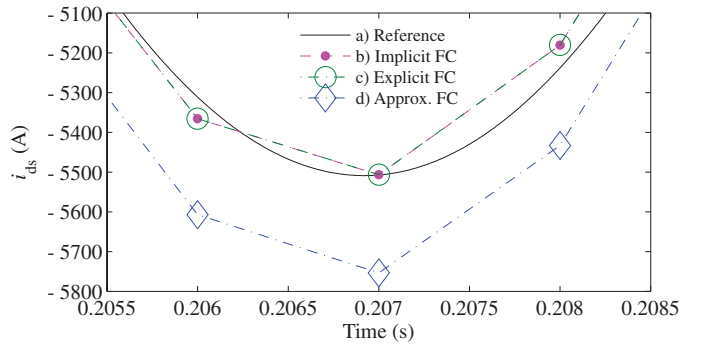


Fig. 18. Magnified view of the first peak of stator current i_{ds} transient due to change of stator voltage (synchronous machine).

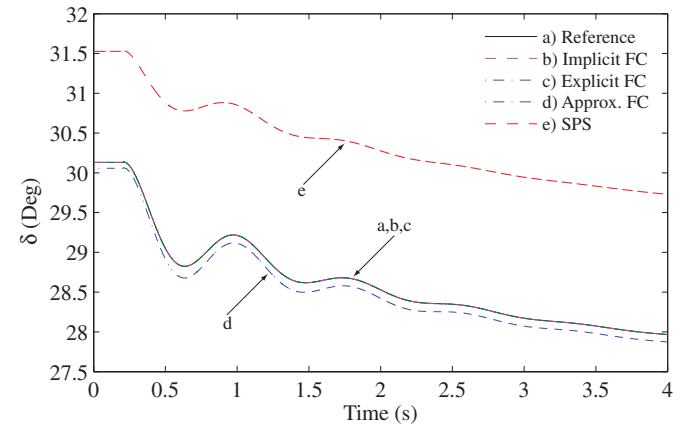


Fig. 19. Transient in rotor angle δ due to change of stator voltage (synchronous machine).

in [17], [18], and [25] for investigating synchronous machine saturation and are considered very appropriate and informative. The corresponding transient responses produced by the considered models are shown in Figs. 17–19, wherein the stator current i_{ds} and the rotor angle δ are shown. For clarity, a magnified view of the first peak of i_{ds} from Fig. 17 is shown in Fig. 18, whereas the rotor angle δ is plotted in Fig. 19. As it can be seen in Figs. 17 and 19, the SPS model predicts different steady states before and after the transient as a consequence of neglecting

TABLE II
2-NORM RELATIVE ERRORS OF i_{ds} AND δ IN CHANGE OF STATOR VOLTAGE
TRANSIENT STUDY (SYNCHRONOUS MACHINE)

Variable	Imp. FC	Exp. FC	App. FC	SPS
i_{ds}	1.588 %	1.584 %	19.47 %	96.86 %
δ	0.004 %	0.004 %	0.387 %	5.924 %

TABLE III
NUMERICAL EFFICIENCY OF THE MODELS FOR DIFFERENT SOLVERS

	Solver	Imp. FC	Exp. FC	App. FC	SPS
CPU time (s)	ode5	1.100	0.217	0.213	0.766
	ode15s	0.475	0.113	0.113	0.483
	ode45	0.387	0.084	0.083	0.403
No. of time steps	ode5	4001	4001	4001	4001
	ode15s	2624	2621	2621	2601
	ode45	915	906	905	881
No. of calls to sat. fun.	ode5	303 964	24 001	24 001	24 001
	ode15s	112 313	7969	7971	7899
	ode45	94 555	6338	6341	6167

the q -axis saturation and cross-saturation. The approximate FC model also shows noticeable differences in i_{ds} and δ with respect to the reference. At the same time, both explicit FC and implicit FC models predict results that are almost indistinguishable from the reference. The 2-norm relative errors with respect to the reference solution for this transient study are summarized in Table II, which shows that explicit FC and implicit FC models have almost identical small errors. Again, such small errors are mostly introduced by the larger time step.

V. COMPUTATIONAL PERFORMANCE

Presented studies demonstrate that both explicit FC and implicit FC models have the best accuracy in predicting the saturation phenomena during slow and fast transients even when using a fairly large time step of 1 ms. To investigate the computational (numerical) difference between the implicit and explicit FC formulations and the available models, the transient study of Section IV-B2 (change of synchronous machine stator voltage) is repeated using fixed-step (ode5) and variable-step (ode15s and ode45) solvers. For all variable-step solvers, the maximum and minimum time steps are 10^{-2} and 10^{-8} s, respectively, and the absolute and relative error tolerances are set to 10^{-4} . The CPU time, the number of time steps, and the number of times the saturation function was called for this study are summarized in Table III. When the fixed-step solver (ode5) is used, the number of calls to the saturation function for all explicit models (i.e., Exp. FC, App. FC, and SPS) is the same (24 001). However, the implicit FC model contains an algebraic loop that requires iterations and thus many more calls to the saturation function (303 964), resulting in the longest computing time of 1.1 s. The proposed explicit FC model runs as fast as the approximate FC model, and is about five times faster than the implicit FC model while achieving the same accuracy (see Table II). The same trend is observed when the variable-step solvers (ode15s and ode45) are used. Since the same error tolerances are used for all models, the number of time steps taken by each model when using ode15s (about 2600) and when using ode45 (about 900)

is consistent and similar for each model. However, the implicit FC model requires about 12 to 15 times more calls to the saturation function, which results in slower simulation time. The SPS model is also explicit and takes approximately the same number of time steps and calls to the saturation function as the other explicit models (i.e., App. FC and Exp. FC). However, this model is slower (perhaps due to its interface/overhead) as well as less accurate (due to neglecting the q -axis saturation and cross-saturation). At the same time, the proposed explicit FC model preserves the numerical efficiency of the approximate FC and the accuracy of the implicit FC model while correctly including cross-saturation.

VI. DISCUSSION

A. Comparison With Generalized Flux Space Vector

To better understand the difference between the proposed FC and the generalized flux space-vector approaches, it is instructive to give a brief summary of the latter. In the generalized flux space-vector approach [17], [18], [23], [24], a flux space vector $\underline{\lambda}$, which may or may not physically exist, is defined as the linear combination of almost any set of independent state variables $\underline{x}_i$. Here, the underlined variables represent space vectors as in the original source references [17], [18], [23], [24]. This flux space vector is aligned with the magnetizing current $\underline{i}_m$ and both are related by

$$\underline{\lambda} = \Lambda \underline{i}_m \quad (95)$$

where Λ is a saturation-dependent equivalent inductance, which also may or may not have direct physical meaning.

To implement this approach in a state-variable-based environment, it is necessary to first write the state equations, i.e., (1)–(4) and (42)–(46) for induction and synchronous machines, respectively, in terms of state variables and inputs

$$\mathbf{A}(\mathbf{p}\mathbf{x}) + \mathbf{B}\mathbf{x} = \mathbf{u} \quad (96)$$

where $\mathbf{u}$ is a vector of input, i.e., stator and rotor voltages. It is necessary to solve (96) for the state derivatives $\mathbf{p}\mathbf{x}$ [15], [17], yielding the following standard state-space formulation:

$$\mathbf{p}\mathbf{x} = -\mathbf{A}^{-1}\mathbf{B}\mathbf{x} + \mathbf{A}^{-1}\mathbf{u}. \quad (97)$$

On the one hand, for the specific case where all state variables are winding flux linkages (as in the proposed FC model), $\mathbf{A}$ becomes a diagonal matrix, which is easily inverted analytically. Moreover, the product term $\mathbf{A}^{-1}\mathbf{B}$ (which can also be defined analytically) will have the same number of nonzeros as $\mathbf{B}$. On the other hand, for any other choice of state variables, $\mathbf{A}$ will not be diagonal. More importantly, $\mathbf{A}$ will also be time varying due to the terms associated with the dynamic inductance. In a brute-force implementation, at every integration time step, it will be necessary to invert $\mathbf{A}$ and execute the matrix multiplication $\mathbf{A}^{-1}\mathbf{B}$ (which will result in a full matrix). Therefore, a significant overhead may result in the implementation and computation of (97) if state variables other than winding flux linkages are used.

The generalized flux space-vector approach [17], [18], [23], [24] is a systematic, theoretically sound, and experimentally

verified method to obtain explicit saturated machine models using almost any set of independent state variables. By properly selecting the state variables, it is thus possible to minimize (or even eliminate) the number of algebraic equations.

It has also been shown in [17] that the number of times the time-varying dynamic inductance appears in $\mathbf{A}$ for a mixed current-flux state-space model may be lower than that for the traditional all winding current model. To simplify the model formulation, it is also possible to completely ignore the dynamic cross-saturation in the generalized flux space-vector approach [25], [26], rendering a constant (but not diagonal) matrix $\mathbf{A}$. The error due to such approximation may be large for most sets of state variables. However, in a few specific cases, the predicted transients were close to those obtained with the full algebraically exact model.

Based on these considerations, it appears that the model that uses only winding flux linkages as state variables is the most efficient and simplest to implement in commonly used simulation packages such as SPS, PLECS, etc. It was further shown in Sections IV and V that the proposed explicit FC method provides the best combination of efficiency and accuracy amongst the saturable models of the same type. The required saturation functions (see Figs. 3, 6, and 7) are also easy to append to the standard magnetically linear machine models, as these models use the same dynamic equations. This may be another practical advantage for commercial simulation programs where, depending on the required accuracy, the saturation may be turned ON/OFF by the users.

B. Generalized Current Space-Vector Approach

The proposed algebraically exact explicit FC induction machine model (see Section II-D) creates a fictitious current space vector $\underline{i}_z = i_{zd} + j i_{zq}$, which is a linear combination of the state variables, and is aligned with the main flux as

$$\Lambda \underline{i}_z = \underline{\lambda}_m. \quad (98)$$

The proposed explicit FC model may be considered a specific case of a generalized current space vector ($\underline{i}_z$) concept, where $\underline{i}_z$ may or may not have direct physical connotation. Hence, this idea could be extended to different sets of independent state variables. However, for any choice of state variables other than winding flux linkages, a dynamic inductance would appear. In [24], it was shown that in the generalized flux space-vector approach, for some sets of state variables, $\underline{\lambda} \equiv \underline{\lambda}_m$, and consequently $\Lambda \equiv L_{ms}$. In these cases, according to (98), the generalized current and generalized flux space-vector approaches would yield the same exact set of equations. However, as shown in (39) and (40), for the pure winding flux linkage model, $\Lambda \neq L_{ms}$; therefore, the two approaches yield different (yet algebraically equivalent) equations.

VII. CONCLUSION

This paper presents a straightforward and improved flux correction method for modeling saturation explicitly in classical general-purpose qd state-variable models of induction and synchronous machines with flux linkages as state variables. The sat-

uration characteristic is assumed to be fully defined by data obtained from simple experimental procedures, namely the no-load and open-circuit tests for induction and synchronous machines, respectively. The proposed explicit FC approach is compared to the implicit and, for synchronous machines, approximate formulations. Computer studies demonstrate that the proposed explicit FC method predicts the dynamic and steady-state cross-saturation phenomena with the same accuracy as the implicit FC approach, which represents an appreciable improvement over the approximate FC or the models that neglect the q -axis saturation and cross-saturation. At the same time, since the proposed formulation is explicit, it avoids the algebraic loop and iterative solution required by the implicit FC formulation, which represents a computational advantage. The proposed explicit FC method can therefore be used for efficient and accurate implementation of the general-purpose induction and synchronous machine component models in state-variable-based transient simulation programs.

APPENDIX A

- 1) Induction machine parameters [4]:
50 HP, 460 V, 4 poles, 60 Hz, $J = 1.662 \text{ J}\cdot\text{s}^2$, $r_s = 0.087 \Omega$, $X_{ls} = 0.302 \Omega$, $r_r = 0.228 \Omega$, $X_{lr} = 0.302 \Omega$, $X_m = 13.08 \Omega$.
- 2) Saturation data (peak values) [34]

λ_m (Wb)	0.38	0.48	0.60	0.68	0.75	0.79	0.82
i_m (A)	11.20	14.33	18.33	21.31	24.42	26.74	29.07
λ_m (Wb)	0.86	0.92	0.96	1.01	1.06	1.10	
i_m (A)	33.21	40.78	46.21	53.21	60.33	66.10	

APPENDIX B

- 1) Salient-pole synchronous machine parameters [4]:
325 MVA, 20 kV, 64 poles, 60 Hz, $J = 35.1 \cdot 10^6 \text{ J}\cdot\text{s}^2$, $r_s = 0.00234 \Omega$, $X_{ls} = 0.1478 \Omega$, $r_{fd} = 0.0005 \Omega$, $X_{lfd} = 0.2523 \Omega$, $r_{kq2} = 0.01675 \Omega$, $X_{lkq2} = 0.1267 \Omega$, $r_{kd} = 0.01736 \Omega$, $X_{lkd} = 0.1970 \Omega$, $X_{md} = 0.8989 \Omega$, $X_{mq} = 0.4433 \Omega$.
- 2) Saturation data (peak values) [35]

λ_m (Wb)	10.8	22.5	31.0	35.3	38.2
i_m (kA)	4.5143	9.4981	13.255	15.261	16.707
λ_m (Wb)	40.9	42.5	45.3	47.9	50.3
i_m (kA)	18.198	19.210	21.340	23.652	25.930

REFERENCES

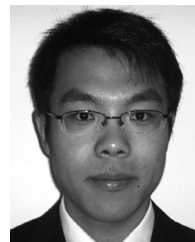
- [1] S. J. Salon, *Finite Element Analysis of Electrical Machines*. Norwell, MA, USA: Kluwer, 1995.
- [2] D. C. MacDonald, A. B. J. Reece, and P. J. Turner, "Turbine-generator steady-state reactances," *IEE Proc. C., Gen., Trans. Distrib.*, vol. 132, no. 3, pp. 101–108, May 1985.
- [3] G. R. Slemon, "Equivalent circuits for transformers and machines including non-linear effects," *Proc. IEE*, vol. 100, part iv, pp. 129–143, Jul. 1953.

- [4] P. C. Krause, O. Wasynczuk, and S. D. Sudhoff, *Analysis of Electric Machinery and Drive Systems*, 2nd ed. Piscataway, NJ, USA: IEEE Press, 2002.
- [5] CEA Technologies, Inc. (2007). *Electromagnetic Transient Program, EMT-P-RV*. [Online]. Available: <http://www.emtp.com>
- [6] *MicroTran Reference Manual*, MicroTran Power System Analysis Corp., Vancouver, BC, Canada 1997 [Online]. Available: <http://www.microtran.com>
- [7] ATP-EMTP, ATP User Group. (2007). *Alternative Transients Programs* [Online]. Available: <http://www.emtp.org>
- [8] Manitoba HVDC Research Centre and RTDS Technologies, Inc., Winnipeg, MB, Canada. (2009). *PSCAD/EMTDC V4.2 On-Line Help* [Online]. Available: <http://www.pscad.com>
- [9] *SimPowerSystems R2009a—User's Guide*, The MathWorks, Inc., Natick, MA, USA, 2009 [Online]. Available: www.mathworks.com
- [10] *Piece-wise Linear Electrical Circuit Simulation (PLECS) User Manual Version 3.2*, Plexim GmbH, Zurich, Switzerland, 2012 [Online]. Available: www.plexim.com
- [11] *Automated State Model Generator (ASMG) Reference Manual*, P. C. Krause and Associates, Inc., West Lafayette, IN, USA, 2002.
- [12] P. M. Anderson and A. A. Fouad, *Power System Control and Stability*. Piscataway, NJ, USA: IEEE Press, 1994.
- [13] P. Kundur, *Power System Stability and Control*. New York, NY, USA: McGraw-Hill, 1994.
- [14] A. M. El-Serafi, A. S. Abdallah, M. K. El-Sherbiny, and E. H. Badawy, "Experimental study of the saturation and the cross-magnetizing phenomenon in saturated synchronous machines," *IEEE Trans. Energy Convers.*, vol. 3, no. 4, pp. 815–823, Dec. 1988.
- [15] S.-A. Tahan and I. Kamwa, "A two-factor saturation model for synchronous machines with saliency and saturation," *IEEE Trans. Energy Convers.*, vol. 10, no. 4, pp. 609–616, Dec. 1995.
- [16] N. C. Kar and A. M. El-Serafi, "Measurement of the saturation characteristics in the quadrature axis of synchronous machines," *IEEE Trans. Energy Convers.*, vol. 21, no. 3, pp. 690–698, Sep. 2006.
- [17] E. Levi, "State-space d-q axis models of saturated salient pole synchronous machines," *Proc. Inst. Elect. Eng., Elect. Power Appl.*, vol. 145, no. 3, pp. 206–216, May 1998.
- [18] E. Levi, "Saturation modeling in d-q axis models of salient pole synchronous machines," *IEEE Trans. Energy Convers.*, vol. 14, no. 1, pp. 44–50, Mar. 1999.
- [19] V. Brandwajn, "Representation of magnetic saturation in the synchronous machine model in an electro-magnetic transients program," *IEEE Trans. Power App. Syst.*, vol. PAS-99, no. 5, pp. 1996–2002, Sep./Oct. 1980.
- [20] A. M. El-Serafi and N. C. Kar, "Methods for determining the intermediate-axis saturation characteristics of salient-pole synchronous machines from the measured d-axis characteristics," *IEEE Trans. Energy Convers.*, vol. 20, no. 1, pp. 88–97, Mar. 2005.
- [21] R. Wamkeue, I. Kamwa, F. Baetscher, and J. L. Hayek, "A new and efficient approach for analysis of a saturated synchronous generator under the load rejection test," *Elect. Power Compon. Syst.*, vol. 34, no. 5, pp. 539–563, 2006.
- [22] R. Wamkeue, D. Aguglia, M. Lakehal, and P. Viarouge, "Two-step method for identification of nonlinear model of induction machine," *IEEE Trans. Energy Convers.*, vol. 22, no. 4, pp. 801–809, Dec. 2007.
- [23] E. Levi, "Modelling of magnetic saturation in smooth air-gap synchronous machines," *IEEE Trans. Energy Convers.*, vol. 12, no. 2, pp. 151–156, Jun. 1995.
- [24] E. Levi, "A unified approach to main flux saturation modeling in d-q axis models of induction machines," *IEEE Trans. Energy Convers.*, vol. 10, no. 3, pp. 455–461, Sep. 1995.
- [25] E. Levi and V. A. Levi, "Impact of dynamic cross-saturation on accuracy of saturated synchronous machine models," *IEEE Trans. Energy Convers.*, vol. 15, no. 2, pp. 224–230, Jun. 2000.
- [26] E. Levi, "Impact of cross-saturation on accuracy of saturated induction machine models," *IEEE Trans. Energy Convers.*, vol. 12, no. 3, pp. 211–216, Sep. 1997.
- [27] C. H. Thomas, "Discussion of analogue computer representations of synchronous generators in voltage-regulation studies," *AIEE Trans. (Power App. Syst.)*, vol. 75, pp. 1182–1184, Dec. 1956.
- [28] T. A. Lipo and A. Consoli, "Modeling and simulation of induction motors with saturable leakage reactances," *IEEE Trans. Ind. Appl.*, vol. IA-20, no. 1, pp. 180–189, Jan./Feb. 1984.
- [29] J. O. Ojo, A. Consoli, and T. A. Lipo, "An improved model of saturated induction machines," *IEEE Trans. Ind. Appl.*, vol. 26, no. 2, pp. 212–221, Mar./Apr. 1990.
- [30] L. Wang, J. Jatskevich, N. Ozog, and A. Davoudi, "A simple explicit method of representing magnetic saturation of salient-pole synchronous machines in both rotor axes using MATLAB-Simulink," in *Proc. Can. Conf. Electr. Comput. Eng.*, Apr. 22–26, 2007, pp. 256–259.
- [31] K. A. Corzine, B. T. Kuhn, S. D. Sudhoff, and H. J. Hegner, "An improved method for incorporating magnetic saturation in the q-d synchronous machine model," *IEEE Trans. Energy Convers.*, vol. 13, no. 3, pp. 270–275, Sep. 1998.
- [32] *Simulink Dynamic System Simulation Software—Users Manual*, The MathWorks, Inc., Natick, MA, USA, 2009 [Online]. Available: <http://www.mathworks.com>
- [33] U. M. Ascher and C. Greif, *A First Course in Numerical Methods*. Philadelphia, PA, USA: SIAM, 2011.
- [34] L. Wang and J. Jatskevich, "Including magnetic saturation in voltage-behind-reactance induction machine model for EMT-type solution," *IEEE Trans. Power Syst.*, vol. 25, no. 2, pp. 975–987, May 2010.
- [35] L. Wang and J. Jatskevich, "Magnetically saturable voltage-behind-reactance synchronous machine model for EMT-type solution," *IEEE Trans. Power Syst.*, vol. 26, no. 4, pp. 2355–2363, Nov. 2011.



Francis Therrien (S'12) received the B.Eng. degree in electrical engineering from the Université de Sherbrooke, Sherbrooke, QC, Canada, in 2010. He is currently working toward the M.A.Sc. degree in electrical and computer engineering at the University of British Columbia, Vancouver, BC, Canada.

He was a Power Systems Researcher with CYME International T&D until 2011. His research interests include modeling of electrical distribution systems, electrical machines, and simulation of electromagnetic transients.



Liwei Wang (S'04–M'10) received the M.S. degree in electrical engineering from Tianjin University, Tianjin, China, and the Ph.D. degree in electrical and computer engineering from the University of British Columbia, Vancouver, BC, Canada, in 2004 and 2010, respectively.

From February 2009 to July 2009, he was an Internship Researcher at ABB Corporate Research Center, Baden-Dättwil, Switzerland. He was a Postdoctoral Research Fellow in the Department of Electrical and Computer Engineering, University of British Columbia, from February 2010 to July 2010. In August 2010, he joined ABB Corporate Research Center, Västerås, Sweden, as a Research Scientist. He is a holder of nine patent filings in the area of high power electronic converter topologies and high power converter designs. His research interests include power system analysis, operation and simulation, electrical machine and drives, power electronic converter design, control and topology, power semiconductors modeling and characterization, utility power electronics applications, HVdc and FACTS, renewable energy sources, and distributed generation.



Juri Jatskevich (M'99–SM'07) received the M.S.E.E. and the Ph.D. degrees in electrical engineering from Purdue University, West Lafayette, IN, USA, in 1997 and 1999, respectively.

Since 2002, he has been a Faculty Member at the University of British Columbia, Vancouver, BC, Canada, where he is currently a Professor of electrical and computer engineering. His research interests include power electronic systems, electrical machines and drives, modeling and simulation of electromagnetic transients.

Dr. Jatskevich is currently the Editor of the IEEE TRANSACTIONS ON ENERGY CONVERSION and the IEEE POWER ENGINEERING LETTERS, and an Associate Editor of the IEEE TRANSACTIONS ON POWER ELECTRONICS. He is the Chair of the IEEE Task Force on Dynamic Average Modeling, under Working Group on Modeling and Analysis of System Transients Using Digital Programs. He Chaired the IEEE CAS Power Systems and Power Electronic Circuits Technical Committee in 2009–2010.



Oleg Wasynczuk (M'76–SM'88–F'10) received the B.S.E.E. degree in electrical engineering from Bradley University, Peoria, IL, USA, in 1976, and the M.S.E.E. and Ph.D. degrees in electrical engineering from Purdue University, West Lafayette, IN, USA, in 1977 and 1979, respectively.

He is currently a Professor at the School of Electrical and Computer Engineering, Purdue University. He is the author or coauthor of more than 100 technical papers, and the coauthor of two books entitled *Analysis of Electric Machinery* and *Electromechanical Motion Devices* (IEEE press).

His current research interests include power system dynamics and control, and the analysis and design of electromechanical devices.

Dr. Wasynczuk was the Past Chair of the Generator Subcommittee of the Electric Machinery Committee of the IEEE.

Modeling of Induction Machines Using a Voltage-Behind-Reactance Formulation

Liwei Wang, *Student Member, IEEE*, Juri Jatskevich, *Senior Member, IEEE*, and Steven D. Pekarek, *Member, IEEE*

Abstract—Over the past several years, there has been renewed interest in modeling electrical machines using phase (*abc*) variables. This paper considers modeling induction machines using phase variables in a voltage-behind-reactance (VBR) formulation. Specifically, three VBR models are proposed wherein the rotor electrical subsystem is modeled using flux linkages as state variables expressed in the *qd* reference frame. The stator electrical dynamics are represented in *abc* phase coordinates that enable direct interface of the machine model to an external network. Such a direct interface is advantageous when the machine is fed from a power electronic converter and/or when the modeling is carried out using circuit-based simulators. Computer studies of an induction machine demonstrate that the proposed VBR models achieve a 740% improvement in computational efficiency as compared with the traditional coupled-circuit phase-domain model.

Index Terms—Coupled-circuit (CC) model, dynamic simulation, induction machine, *qd* model, voltage-behind-reactance (VBR) model.

I. INTRODUCTION

THE MODELING of induction machines dates back to the early years of the 20th century [1]–[3]. Depending on the objective of studies and the required level of fidelity, the modeling approaches may be roughly divided into three categories: finite element (or difference) method [4]; equivalent magnetic circuit approach [5], [6]; and coupled electric circuit approach. This paper mainly considers the last approach, which leads to a relatively small number of equations. In particular, this paper focuses on the so-called general purpose models that are available in numerous simulation languages as built-in library components and by far the most commonly used by engineers and researchers in industry or academia (or at least attempted first, prior to constructing custom models). From this point of view, improving numerical accuracy and efficiency of such models even by a fraction may result in very significant savings of the engineering time worldwide.

Among the many methods developed, a key concept has been to transform physical (*abc*) variables of the machine into ficti-

tious (*qd*) variables using rotating reference frames, e.g., stationary, rotor, or synchronous. The arbitrary reference frame (ARF) theory proposed in [7] generalized the *qd* modeling approach by showing that one can assign an appropriate reference speed ω to transform machine physical variables to a particular reference frame. The ARF provides a direct means of obtaining the machine equations in the aforementioned frames of reference.

The advantages of the *qd* induction machine models include the following: 1) the time-varying inductances between stator and rotor windings are eliminated; 2) the flux linkage equations are decoupled; 3) zero sequence quantities disappear for balanced operation and may be removed to reduce the number of state equations [8]; 4) the design of controls for machine drive systems can be conveniently developed in a reference frame in which variables become constant in the steady state [9]; and 5) the average-value modeling of machine-converter systems is simplified when expressing the machine in terms of *qd* variables [10]–[14].

Despite the many important advantages of using reference frame transformations when modeling systems that contain more than just a single machine, the *qd* model requires an interface with the external components or power electronics circuits. In many instances, the external components are modeled in *abc* physical variables (coordinates). Therefore, an interface is formed by the transformation (and inverse transformation), which poses an algebraic constraint.

For circuit- and/or nodal-analysis-based simulation languages, it is not easy to interface the *qd* equivalent circuits of the machine with the external network. Therefore, in simulation programs such as Electromagnetic Transients Program (EMTP) [15], prediction methods are often used [16], which reduce the simulation accuracy and efficiency by requiring small time steps and/or iterations. Interested reader may find more detailed discussions of the challenges associated with interfacing the *qd* model in [17]–[20], wherein the loss of stability and numerical accuracy have been documented.

In the state-variable languages, such as the widely used packages SimPowerSystem (SPS) [21] and Piece-wise Linear Electrical Circuit Simulation (PLECS) [22], the *qd* models are typically represented as voltage-controlled current sources [22], [23]. Therefore, such built-in *qd* models may not be directly interfaced with external circuits terminating with inductive branches, in which case, fictitious snubbers are often used to create the necessary voltages and enable the interface [21]–[24]. Discretizing the machine models separately from the main system-circuit is possible to avoid algebraic loops. For example, [21] permits discretizing the machine models using the Forward Euler method, while the external circuit may be

Manuscript received April 17, 2007; revised August 24, 2007. This work was supported in part by the Natural Sciences and Engineering Research Council (NSERC) of Canada under the Discovery Grant and in part by the Power Engineering Education and Research Grant from British Columbia Transmission Corporation and BC Hydro Inc. Paper no. TEC-00124-2007.

L. Wang and J. Jatskevich are with the Department of Electrical and Computer Engineering, University of British Columbia, Vancouver, BC V6T 1Z4, Canada (e-mail: liweiw@ece.ubc.ca; juri@ece.ubc.ca).

S. D. Pekarek is with the Department of Electrical and Computer Engineering, Purdue University, West Lafayette, IN 47907 USA (e-mail: spekarek@purdue.edu).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TEC.2008.918601

discretized using a fixed-step trapezoidal (Tustin) integration rule.

To simplify the machine model–network interface, several authors have suggested using the coupled-circuit (CC) phase-domain machine models [25]–[27], wherein the stator and rotor circuits are represented in abc phase coordinates. Using such an approach enables a direct connection between a machine and an external circuit. However, this approach comes at the cost of using a time-varying inductance matrix, which complicates the model and reduces its simulation efficiency. A hybrid approach that uses both abc phase variables for the stator circuit and qd stationary reference frame for the rotor circuit was proposed in [28]–[30]. Although the resulting machine inductance matrix does not depend on rotor position, these models use structural coupling between the stator and rotor and do not have a convenient equivalent-circuit representation. Therefore, it is a challenge to integrate these models with the external network systems using commonly available simulation languages.

A voltage-behind-reactance (VBR) model formulation was proposed for synchronous machines in [31] and [32] that also achieves the direct interface sought by the CC phase-domain model. As documented in [19], the VBR-based models are also very convenient for implementation in various nodal analysis- and circuit-based simulation packages and offer advantages in terms of numerical accuracy and efficiency.

This paper makes the following contributions in the area of modeling and numerical analysis of electrical machines:

- 1) We show that it is possible to extend the VBR formulation to a full-order induction machine model. In this formulation, the stator is expressed in abc direct-phase coordinates as three-phase dependent subtransient emf sources behind an RL circuit, and the rotor is expressed in transformed qd coordinates using fluxes as the independent variables.
- 2) It is also shown that the resulting VBR model can be expressed in several different forms, wherein, depending on the need, the RL circuit may be coupled or decoupled with an option of including the zero sequence wherever appropriate.
- 3) The proposed VBR model formulation enables a very straightforward and efficient interconnection of the stator RL circuit with the external circuit network, which is an advantage over all previously developed induction machine models.
- 4) The computer studies and eigenvalue analysis demonstrate that the proposed VBR machine model is computationally more efficient and accurate than the traditional CC model (that is typically used to achieve the direct interface) and/or the classical qd model when it is interfaced using fictitious snubbers.

II. COUPLED-CIRCUIT MACHINE MODEL

To better understand the proposed advanced models, and for consistency purposes, CC model is reviewed here. Without the loss of generality, this paper assumes a three-phase, wye-connected, induction machine [9], whose cross-sectional view is shown in Fig. 1.

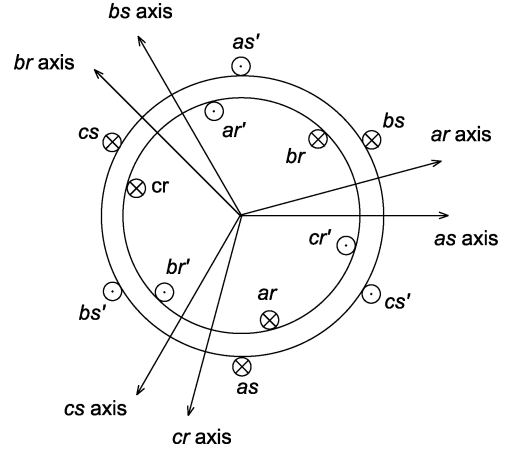


Fig. 1. Basic structure of an induction machine model.

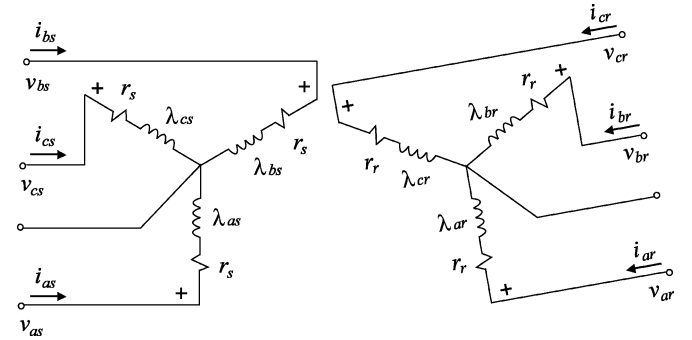


Fig. 2. CC model of induction machine.

The stator and rotor windings are assumed to be symmetric, sinusoidally distributed with resistance denoted by r_s and r_r , respectively. The positive direction of the magnetic axes corresponds to the direction of flux linkages induced by the positive phase currents. The induction machine may be represented in terms of coupled electric circuits, shown in Fig. 2. In this paper, motor convention is used, which assumes that positive stator and rotor currents flow into the machine when positive phase voltages are applied, as depicted in Fig. 2. For convenience, all rotor variables and parameters are referred to the stator side using an appropriate turns ratio.

The corresponding voltage equation may be expressed in matrix form as

$$\begin{bmatrix} \mathbf{v}_{abcs} \\ \mathbf{v}_{abcr} \end{bmatrix} = \begin{bmatrix} \mathbf{r}_s & \\ & \mathbf{r}_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{abcr} \end{bmatrix} + p \begin{bmatrix} \lambda_{abcs} \\ \lambda_{abcr} \end{bmatrix} \quad (1)$$

where the stator and rotor diagonal resistance matrices are 3×3 and denoted by $\mathbf{r}_s$, $\mathbf{r}_r$, respectively. The operator p denotes d/dt . The corresponding flux linkage equation is

$$\begin{bmatrix} \lambda_{abcs} \\ \lambda_{abcr} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}_{sr} \\ \mathbf{L}_{sr}^T & \mathbf{L}_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{abcr} \end{bmatrix} \quad (2)$$

where the stator and rotor self-inductance matrices are $\mathbf{L}_s$ and $\mathbf{L}_r$, respectively, and are constant due to machine symmetry. The expressions for $\mathbf{L}_s$ and $\mathbf{L}_r$ can be found in [9] and are not included here due to space limitations. The mutual inductance

matrix has the following form:

$$\mathbf{L}_{sr} = L_m \begin{bmatrix} \cos \theta_r & \cos\left(\theta_r + \frac{2\pi}{3}\right) & \cos\left(\theta_r - \frac{2\pi}{3}\right) \\ \cos\left(\theta_r - \frac{2\pi}{3}\right) & \cos \theta_r & \cos\left(\theta_r + \frac{2\pi}{3}\right) \\ \cos\left(\theta_r + \frac{2\pi}{3}\right) & \cos\left(\theta_r - \frac{2\pi}{3}\right) & \cos \theta_r \end{bmatrix}. \quad (3)$$

The developed electromagnetic torque may be expressed in direct physical machine variables as

$$T_e = \left(\frac{P}{2}\right) (\mathbf{i}_{abcs})^T \frac{\partial}{\partial \theta_r} [\mathbf{L}_{sr}] \mathbf{i}_{abcr}. \quad (4)$$

III. QD MACHINE MODEL IN ARF

The CC induction machine model is often transformed into the qd arbitrary reference frame [9], where the flux linkages become decoupled. For convenient derivation of the VBR models, the qd model is included here in decoupled form. In particular, the voltage equations in the ARF are given as

$$v_{qs} = r_s i_{qs} + \omega \lambda_{ds} + p \lambda_{qs} \quad (5)$$

$$v_{ds} = r_s i_{ds} - \omega \lambda_{qs} + p \lambda_{ds} \quad (6)$$

$$v_{0s} = r_s i_{0s} + p \lambda_{0s} \quad (7)$$

$$v_{qr} = r_r i_{qr} + (\omega - \omega_r) \lambda_{dr} + p \lambda_{qr} \quad (8)$$

$$v_{dr} = r_r i_{dr} - (\omega - \omega_r) \lambda_{qr} + p \lambda_{dr} \quad (9)$$

$$v_{0r} = r_r i_{0r} + p \lambda_{0r}. \quad (10)$$

The flux linkage equations are expressed as

$$\lambda_{qs} = L_{ls} i_{qs} + \lambda_{mq} \quad (11)$$

$$\lambda_{ds} = L_{ls} i_{ds} + \lambda_{md} \quad (12)$$

$$\lambda_{0s} = L_{ls} i_{0s} \quad (13)$$

$$\lambda_{qr} = L_{lr} i_{qr} + \lambda_{mq} \quad (14)$$

$$\lambda_{dr} = L_{lr} i_{dr} + \lambda_{md} \quad (15)$$

$$\lambda_{0r} = L_{lr} i_{0r} \quad (16)$$

where magnetizing fluxes are defined as

$$\lambda_{mq} = L_m (i_{qs} + i_{qr}) \quad (17)$$

$$\lambda_{md} = L_m (i_{ds} + i_{dr}) \quad (18)$$

and

$$L_m = \frac{3}{2} L_{ms}. \quad (19)$$

The developed electromagnetic torque in terms of transformed qd variables is given as

$$T_e = \frac{3P}{4} (\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}). \quad (20)$$

IV. VOLTAGE-BEHIND-REACTANCE MODELS IN ARF

To derive the VBR model, it is necessary to formulate the stator voltage equation so that it appears similar to that of RL -branches, whereas the contribution due to the rotor subsystem is expressed in terms of dependent voltage sources. Since for an induction machine such model formulation is not unique, several models have been derived that may be of practical use, depending on the application.

A. Voltage-Behind-Reactance Model I

Derivation of the first model is performed by first solving (14) and (15) for currents and substituting the result into (17) and (18). The magnetizing fluxes are then expressed as

$$\lambda_{mq} = L_m'' \left(i_{qs} + \frac{\lambda_{qr}}{L_{lr}} \right) \quad (21)$$

$$\lambda_{md} = L_m'' \left(i_{ds} + \frac{\lambda_{dr}}{L_{lr}} \right) \quad (22)$$

where

$$L_m'' = \left(\frac{1}{L_m} + \frac{1}{L_{lr}} \right)^{-1}. \quad (23)$$

Substituting (21) and (22) into (11) and (12), respectively, the stator flux linkage equations may be rewritten as

$$\lambda_{qs} = L'' i_{qs} + \lambda_q'' \quad (24)$$

$$\lambda_{ds} = L'' i_{ds} + \lambda_d'' \quad (25)$$

where the subtransient inductance is defined by

$$L'' = L_{ls} + L_m''. \quad (26)$$

The subtransient flux linkages are defined as

$$\lambda_q'' = L_m'' \frac{\lambda_{qr}}{L_{lr}} \quad (27)$$

$$\lambda_d'' = L_m'' \frac{\lambda_{dr}}{L_{lr}}. \quad (28)$$

Substituting (24) and (25) into (5) and (6), respectively, the stator voltage equations may be represented as

$$v_{qs} = r_s i_{qs} + \omega L'' i_{ds} + p L'' i_{qs} + \omega \lambda_q'' + p \lambda_q'' \quad (29)$$

$$v_{ds} = r_s i_{ds} - \omega L'' i_{qs} + p L'' i_{ds} - \omega \lambda_d'' + p \lambda_d''. \quad (30)$$

The rotor currents are derived from (14) and (15) and are given by

$$i_{qr} = \frac{1}{L_{lr}} (\lambda_{qr} - \lambda_{mq}) \quad (31)$$

$$i_{dr} = \frac{1}{L_{lr}} (\lambda_{dr} - \lambda_{md}). \quad (32)$$

From (8), (9) and (31), (32), the rotor voltage equations may be rewritten as the following state equations

$$p \lambda_{qr} = -\frac{r_r}{L_{lr}} (\lambda_{qr} - \lambda_{mq}) - (\omega - \omega_r) \lambda_{dr} + v_{qr} \quad (33)$$

$$p \lambda_{dr} = -\frac{r_r}{L_{lr}} (\lambda_{dr} - \lambda_{md}) + (\omega - \omega_r) \lambda_{qr} + v_{dr}. \quad (34)$$

The terms $p\lambda_q''$ and $p\lambda_d''$ in the respective stator voltage equations (29) and (30) may be eliminated by taking the derivatives of (27) and (28) and substituting (33) and (34) into the resulting equations. After some algebraic manipulations, the stator voltage equations (29) and (30) may be rewritten in the following form

$$v_{qs} = r''i_{qs} + \omega L''i_{ds} + pL''i_{qs} + e_q'' \quad (35)$$

$$v_{ds} = r''i_{ds} - \omega L''i_{qs} + pL''i_{ds} + e_d'' \quad (36)$$

where

$$r'' = r_s + \frac{L_m''^2}{L_{lr}^2} r_r \quad (37)$$

and

$$e_q'' = \omega_r \lambda_d'' + \frac{L_m'' r_r}{L_{lr}^2} (\lambda_q'' - \lambda_{qr}) + \frac{L_m''}{L_{lr}} v_{qr} \quad (38)$$

$$e_d'' = -\omega_r \lambda_q'' + \frac{L_m'' r_r}{L_{lr}^2} (\lambda_d'' - \lambda_{dr}) + \frac{L_m''}{L_{lr}} v_{dr}. \quad (39)$$

The stator voltage equations (7), (35), and (36) may now be transformed back into the abc phase coordinates by applying inverse arbitrary reference transformation $(\mathbf{K}_s)^{-1}$. This final step gives the voltage equation in VBR form as

$$\mathbf{v}_{abc s} = \mathbf{r}_{abc s}'' \mathbf{i}_{abc s} + \mathbf{L}_{abc s}'' p \mathbf{i}_{abc s} + \mathbf{e}_{abc s}'' \quad (40)$$

where

$$\mathbf{e}_{abc s}'' = [\mathbf{K}_s]^{-1} [e_q'' \quad e_d'' \quad 0]^T. \quad (41)$$

Here, the resistance matrix is given by

$$\mathbf{r}_{abc s}'' = \begin{bmatrix} r_S & r_M & r_M \\ r_M & r_S & r_M \\ r_M & r_M & r_S \end{bmatrix} \quad (42)$$

where

$$r_S = r_s + r_a \quad (43)$$

$$r_M = -\frac{r_a}{2} \quad (44)$$

$$r_a = \frac{2}{3} \frac{L_m''^2}{L_{lr}^2} r_r. \quad (45)$$

The new inductance matrix is

$$\mathbf{L}_{abc s}'' = \begin{bmatrix} L_S & L_M & L_M \\ L_M & L_S & L_M \\ L_M & L_M & L_S \end{bmatrix} \quad (46)$$

where

$$L_S = L_{ls} + L_a \quad (47)$$

$$L_M = -\frac{L_a}{2} \quad (48)$$

$$L_a = \frac{2}{3} L_m''. \quad (49)$$

Note that, in (40), subtransient resistance matrix (42) and inductance matrix (46) are constant due to machine symmetry, and are independent of any reference frame. These are

very desirable properties that make the VBR model more efficient than the CC model. Thus, equations (10), (33), (34), (38), (39), and (40) define the so-called VBR model formulation I (VBR-I).

Since in the VBR model formulation, there is no direct need to calculate the stator flux linkages λ_{qs} and λ_{ds} , the developed electromagnetic torque may be expressed using the magnetizing fluxes as [9]

$$T_e = \frac{3P}{4} (\lambda_{md} i_{qs} - \lambda_{mq} i_{ds}) \quad (50)$$

B. Voltage-Behind-Reactance Model II

The VBR-I contains a full resistance matrix (42), which may be difficult to implement in some simulation languages. As shown in [32], the model may be further simplified by using the diagonal stator resistance matrix $\mathbf{r}_s$ and moving the rotor resistance terms multiplied by stator currents i_{qs} and i_{ds} in (35) and (36), respectively, back into e_q'' and e_d'' . This formulation constitutes the second VBR model II (VBR-II), which is expressed as

$$\mathbf{v}_{abc s} = \mathbf{r}_s \mathbf{i}_{abc s} + \mathbf{L}_{abc s}'' p \mathbf{i}_{abc s} + \mathbf{v}_{abc s}''. \quad (51)$$

Here, other equations are the same as in VBR-I, except that the back emf $\mathbf{v}_{abc s}''$ is defined as

$$\mathbf{v}_{abc s}'' = [\mathbf{K}_s]^{-1} [v_q'' \quad v_d'' \quad 0]^T \quad (52)$$

where

$$v_q'' = \omega_r \lambda_d'' + \frac{L_m'' r_r}{L_{lr}^2} (\lambda_q'' - \lambda_{qr}) + \frac{L_m''}{L_{lr}} v_{qr} + \frac{L_m''^2 r_r}{L_{lr}^2} i_{qs} \quad (53)$$

$$v_d'' = -\omega_r \lambda_q'' + \frac{L_m'' r_r}{L_{lr}^2} (\lambda_d'' - \lambda_{dr}) + \frac{L_m''}{L_{lr}} v_{dr} + \frac{L_m''^2 r_r}{L_{lr}^2} i_{ds}. \quad (54)$$

Thus, the need for a mutual resistance matrix has been avoided [32].

C. Voltage-Behind-Reactance Model III

In both previous formulations, the stator inductive branches are coupled. A further simplification may be achieved if the stator inductance matrix (46) and resistance matrix (42) (in VBR-I) are made diagonal. The diagonalization may be carried out by first relating the stator current and zero-sequence current as [33]

$$i_{as} + i_{bs} + i_{cs} = 3i_{0s} \quad (55)$$

$$p i_{as} + p i_{bs} + p i_{cs} = 3p i_{0s}. \quad (56)$$

Here, we use (55) and (56) to reconsider the VBR-I formulation. The off-diagonal terms in the subtransient resistance and inductance matrices in (40) may be eliminated by expressing the stator currents associated with the off-diagonal entries in terms of the zero sequence current and the remaining phase (diagonal) current. After algebraic manipulations, (40) can be rewritten

as

$$\mathbf{v}_{abcs} = \begin{bmatrix} r_D & 0 & 0 \\ 0 & r_D & 0 \\ 0 & 0 & r_D \end{bmatrix} \mathbf{i}_{abcs} + \begin{bmatrix} L_D & 0 & 0 \\ 0 & L_D & 0 \\ 0 & 0 & L_D \end{bmatrix} p\mathbf{i}_{abcs} + \mathbf{e}_{abcs}'' + \left(3r_M \begin{bmatrix} i_{0s} \\ i_{0s} \\ i_{0s} \end{bmatrix} + 3L_M \begin{bmatrix} pi_{0s} \\ pi_{0s} \\ pi_{0s} \end{bmatrix} \right) \quad (57)$$

where

$$r_D = r_s - r_M \quad (58)$$

and

$$L_D = L_s - L_M. \quad (59)$$

The voltage equation for the stator zero sequence is

$$v_{0s} = r_s i_{0s} + L_{ls} p i_{0s} \quad (60)$$

which gives the following state equation

$$p i_{0s} = \frac{1}{L_{ls}} (v_{0s} - r_s i_{0s}). \quad (61)$$

Substituting (61) into (57), the VBR model formulation III (VBR-III) has the following form

$$\mathbf{v}_{abcs} = \begin{bmatrix} r_D & 0 & 0 \\ 0 & r_D & 0 \\ 0 & 0 & r_D \end{bmatrix} \mathbf{i}_{abcs} + \begin{bmatrix} L_D & 0 & 0 \\ 0 & L_D & 0 \\ 0 & 0 & L_D \end{bmatrix} p\mathbf{i}_{abcs} + \mathbf{e}_{abcs}'' + \left(3r_M - \frac{3L_M r_s}{L_{ls}} \right) \begin{bmatrix} i_{0s} \\ i_{0s} \\ i_{0s} \end{bmatrix} + \frac{3L_M}{L_{ls}} \begin{bmatrix} v_{0s} \\ v_{0s} \\ v_{0s} \end{bmatrix} \quad (62)$$

while other equations are the same as in VBR-I.

The same diagonalization procedure may be carried out with VBR-II to get its diagonal version, herein referred to as VBR-IV. However, it is not included here because the model structure of VBR-IV is very similar to VBR-III and no new features are added. As can be seen in (62), VBR-III has diagonal resistance and inductance matrices. A circuit-block diagram of the final VBR-III model is shown in Fig. 3. It should be noted that in this formulation, the RL circuit branches are decoupled from each other and have constant parameters. This feature provides a direct interface of the stator circuit with the external network, which makes it simpler to implement the induction machine in circuit-based simulation packages.

The VBR-III formulation depicted in Fig. 3 can be readily simplified if one assumes the wye-connected stator windings with a floating neutral, which is very common. In this case, the zero sequence variables simply disappear from (62) and Fig. 3, but the back emf $\mathbf{e}_{abcs}''$ remains the same as in VBR-I, (38), (39), and (41).

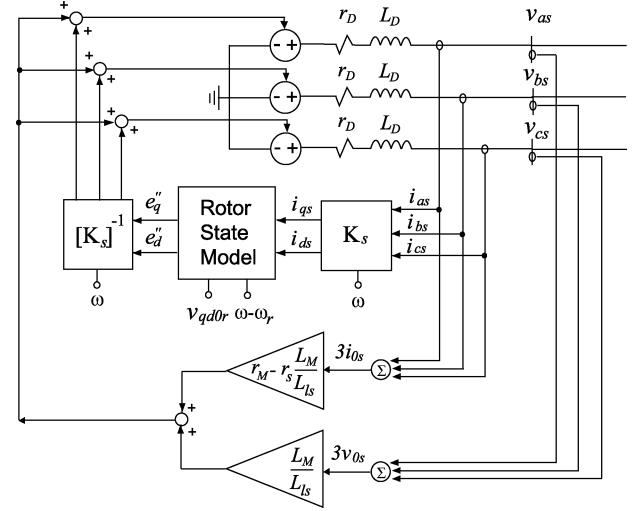


Fig. 3. Proposed VBR model formulation.

V. COMPUTER STUDIES

All previously described models were implemented using MATLAB/Simulink [34]. Two types of qd model were used. The first one was a proper state-variable model implemented using standard/basic Simulink blocks. The resulting model/subsystem requires voltages as inputs and currents as outputs and has no circuit interface. The second qd machine model was the built-in SPS model [24]. SPS is a circuits-based simulation toolbox widely used for power- and power-electronic-systems transient simulations using Matlab/Simulink environment. The VBR models and the CC model were implemented using the circuit-based approach described in [35] and the toolbox [36]. In general, similar to a synchronous machine ([31], see Section V), the CC model could be implemented using either currents or the flux linkages as the state variables. However, if the stator circuit is to be interfaced with the external network where the currents are used as the independent variables, then machine currents must be considered as the state variables.

An induction machine with parameters summarized in Appendix B is considered here for simulating a no-load startup transient. Since the focus of this section is on component level, in the studies presented here, the machine is directly fed from an ideal balance three-phase voltage source. To evaluate the accuracy and efficiency of all models considered in this paper, the simulation studies were carried out using variable- as well as fixed-time-step integration methods.

A. Variable-Step Method

To make a fair comparison among the models, the same solver, ODE15s, was used for each model. To ensure smooth and accurate solution, the relative and absolute error tolerances were set to 10^{-4} and 10^{-6} , respectively. Also, the maximum and minimum time-step limits were set to 10^{-3} and 10^{-10} s, respectively, while the initial step was set to 10^{-5} s.

The transient responses produced by the qd models, three types of VBR models, and the CC model using the variable-step

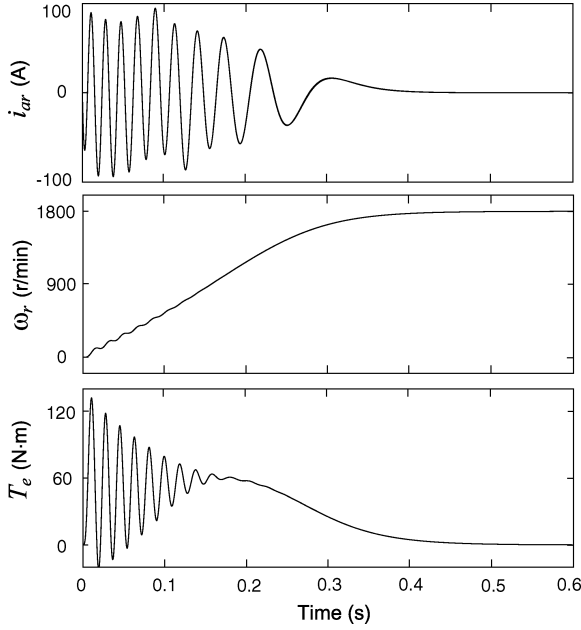


Fig. 4. Simulation results for six models using variable-step method.

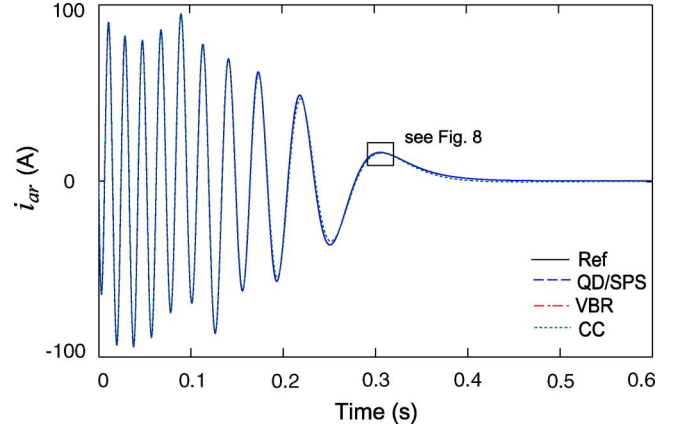
TABLE I
SIMULATION EFFICIENCY COMPARISONS

Model	CPU Time, s	Num. of Steps	CPU Time per Step, μ s
QD	0.062	744	83.3
CC	1.156	5283	218.8
SPS	0.313	1105	283.3
VBR-I	0.156	1036	150.6
VBR-II	0.157	1082	145.1
VBR-III	0.157	1075	146.0

integration method are depicted in Fig. 4. Due to space limitations, only the rotor current i_{ar} , the speed ω_r , and the electromagnetic torque T_e are shown. As can be seen in Fig. 4, all responses are visibly indistinguishable from each other. This result confirms that all models are consistent and equivalent, provided sufficiently small error tolerances and/or integration step size.

Assuming the same simulation accuracy for each model, the numerical efficiency of the models relative to each other may be evaluated by considering two factors: 1) the total number of integration steps taken to complete the study and 2) the computational load/cost per time step. The studies were run on a personal computer with an AMD 2200 processor using standard (not compiled, non-real-time) Simulink. The total CPU times, number of integration steps, and CPU times per step required to complete the study of Fig. 4 are summarized in Table I.

As can be seen in Table I, the bare-bones qd model, implemented in Simulink directly without circuit interface, has the fastest simulation speed (0.062 s) and the lowest computational cost requiring only 744 steps. The time-varying inductance matrix $\mathbf{L}(t)$ and the $p\mathbf{L}_{abc}$ term in the CC model (see Appendix A) leads to a significant increase in the CPU time per step (218.8 μ s) as compared with the qd model (83.3 μ s). In addition, the CC model requires significantly more time steps (5283) to achieve the same accuracy. The SPS machine model, when

Fig. 5. Rotor current i_{ar} with time step of 1 ms.

fed by ideal voltage sources, preserves the advantage of the qd model and takes small number of time steps (1105), although some computational overhead exist in SPS that cause its model run slower (0.313 s) than the bare-bones Simulink qd model (0.062 s).

At the same time, all three VBR models performed very similarly to each other, which is expected since these models are algebraically equivalent, with only structural differences. The VBR models required more steps (1036–1082) and CPU time (0.157 s) than the bare-bones qd model, but generally performed by a factor of 2 faster than the SPS model (0.313 s). The proposed VBR models also demonstrated about 7.4 times improvement as compared with the CC model in terms of the overall CPU time required for the given study.

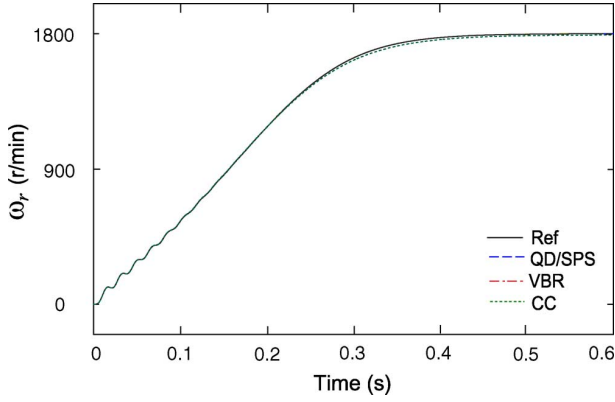
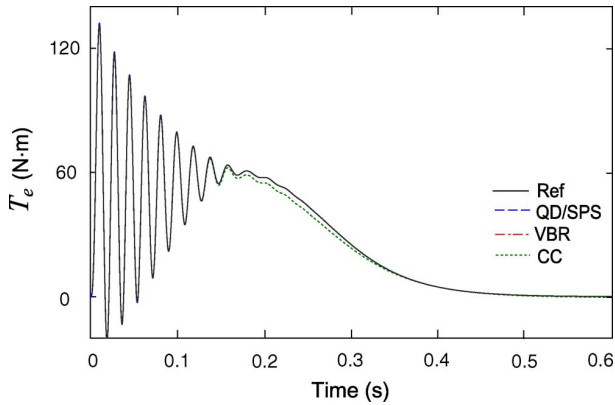
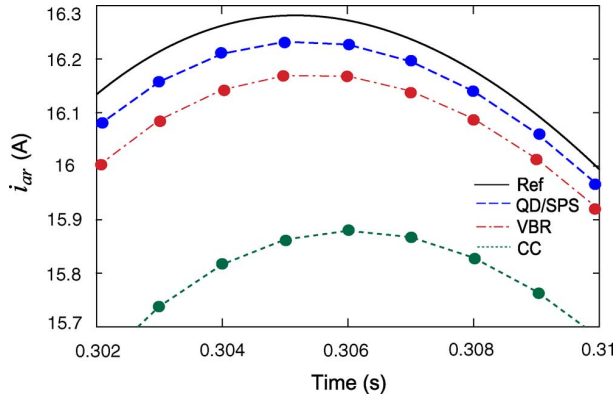
B. Fixed-Step Method

To further compare the numerical properties of the different models, the same case study is simulated with the fixed time-step integration method. To have a benchmark for comparing the simulation accuracy, a reference solution was obtained using the qd model with the fixed-step fourth-order Runge–Kutta integration method and a very small time step of 1 μ s. The study was repeated for all considered models using the same fourth-order Runge–Kutta method, but with a much larger time step of 1 ms. The simulation results are shown in Figs. 5–7, where it can be seen that all models predict a similar response. However, it now becomes noticeable that the CC model (dotted line) is the farthest from the reference solution (solid line). A detailed view of the rotor current plot is shown in Fig. 8, wherein the plot is magnified to clearly show this difference.

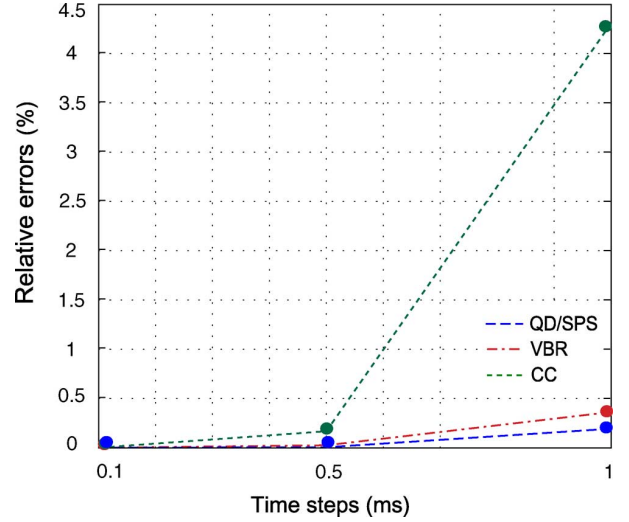
Without the loss of generality, the rotor current trajectory is considered here to further quantify the numerical accuracy, since the other variables show a similar trend. The cumulative relative error between the reference solution $\tilde{i}_{ar}$ and a given numerical solution i_{ar} , as defined in terms of 2-norm [37], is evaluated as

$$\varepsilon(h) = \frac{\|\tilde{i}_{ar} - i_{ar}\|_2}{\|\tilde{i}_{ar}\|_2} \times 100\% \quad (63)$$

for different time steps h .

Fig. 6. Rotor rotating speed ω_r with time step of 1 ms.Fig. 7. Electromagnetic torque T_e with time step of 1 ms.Fig. 8. Detailed view of rotor current i_{ar} with time step of 1 ms.

The simulation was run using time steps of 0.1, 0.5, and 1 ms. The calculated errors for the different models are shown in Fig. 9. As shown in Figs. 8 and 9, the bare-bones *qd* model and the SPS *qd* model produce exactly the same simulation results (shown in dashed line), and gave the smallest errors as compared with the other models. This is expected since the SPS uses the *qd* model internally, and the external circuit was just the ideal voltage sources. All of the VBR models (dash/dotted line) demonstrated identical results and accuracy, as all these models use similar state-space equations and differ only in the

Fig. 9. Numerical error propagation of i_{ar} .

form of stator branches. The least accurate performance was demonstrated by the CC model (dotted line).

VI. EIGENSYSTEM ANALYSIS

The *qd* models, the CC model, and the VBR models are all equivalent in the continuous time domain since these models are derived from each other without approximations. However, as demonstrated in Section V, the numerical properties of these models are quite different when the corresponding differential equations are discretized using an integration rule. In general, for nonlinear time-varying systems, the choice of state variables and/or the model structure may result in different eigenvalues of the system. The eigenvalues may influence the propagation of local errors, and therefore, affect the simulation accuracy when the differential equations are integrated numerically [38].

To gain some qualitative information and understanding of the numerical properties of the machine models described here, one may consider the state equations formed by the electrical circuit (see Fig. 2) of the machine [31]. In particular, if the rotor speed is assumed to be constant, the machine's nonlinear equations become linear, which allows one to express the corresponding system matrices. The resulting system matrices of various machine models are summarized in Appendix A, based on which the eigenvalues can be readily calculated. Alternatively, the respective models can be numerically linearized using a standard Matlab subroutine. Regardless of the approach, one should bear in mind that the eigenvalues of the induction machine model change quite significantly with speed [9, Ch. 8]. For comparison, the eigenvalues of all models were computed at an operating point close to steady state at no load (operating point corresponding to $t = 0.6$ s in the study shown in Fig. 4). The resulting eigenvalues are summarized in Table II.

In general, the negative real part of the eigenvalues is desirable for better damping, whereas positive eigenvalues tend to increase the local errors [39]. From this viewpoint, the *qd* model is well structured, since all of its eigenvalues are relatively far to the left side of the complex plane, as can be seen in Table II.

TABLE II
EIGENVALUES OF MACHINE MODELS

QD/SPS	CC	VBR-I, -II, -III
-19.52	1407	-0.016
$-89.3 + j315.9$	1406	$-90.9 + j54.2$
$-89.3 - j315.9$	0	$-90.9 - j54.2$
$-218.1 + j60.4$	-1723	$-226.3 + j322.9$
$-218.1 - j60.4$	-1724	$-226.3 - j322.9$
-217.5 / (N/A)	-217.5	-217.5
-408 / (N/A)	-408	-408

However, the CC model uses currents as state variables and has positive eigenvalues, which increases propagation of local errors, and therefore, reduces the simulation accuracy. A similar observation regarding the synchronous machine models has been made in [31].

An important observation can be made regarding the barebones *qd* model and the SPS *qd* model. In particular, the eigenvalues of these two models are identical, which also explains their identical simulation results in the previous studies. However, it should be noted that the SPS *qd* model ignores the stator and rotor zero sequence modes and that the two associated eigenvalues -217.5 and -408 are not present here.

Similar observation may be made for VBR models. Structurally, all three VBR models result in stator circuits (with constant stator inductances and resistances) that use currents as the independent variables, whereas the rotor subsystem uses flux linkages as the independent variables. Since these models are using the same set of state variables, their eigenvalues are identical, as shown in Table II. More importantly, the VBR models have negative eigenvalues, and therefore, their numerical properties are similar to those of the *qd* model, as was confirmed in Section V.

VII. DISCUSSION

The induction-machine models discussed in this paper may find their use depending on the modeling environment and application. For example, in differential-equation- or state-variable-based simulators (wherever there is no need to have a circuit interface), the machines may be considered as proper subsystems described by their corresponding equations. For these simulators, the *qd* model is perhaps the most accurate and simplest to use. However, for circuit-based simulation language, the machine stator windings must be made available for the interface with the external network. The CC phase-domain machine models are often used to fulfill this requirement; however, they are computationally more expensive and less accurate. The VBR models derived in this paper provide a good compromise among the models and offer very good structural as well numerical properties. Moreover, the final formulation VBR-III results in a decoupled (diagonal inductance and resistance matrices) time-invariant stator circuit, which allows a direct and simple interface of the stator branches with an external network.

To demonstrate the advantages of the VBR model over the *qd* model in terms of the model interface with the external network, it is sufficient to include small inductive impedance in the source, as shown in Fig. 10. Without the loss of generality, we

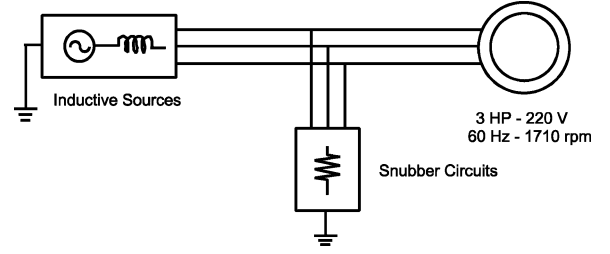
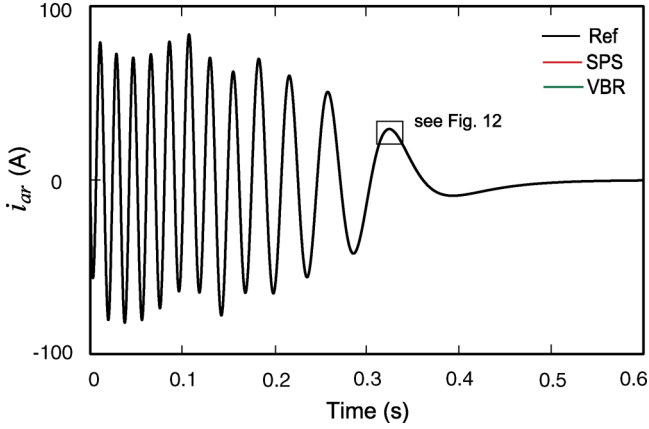
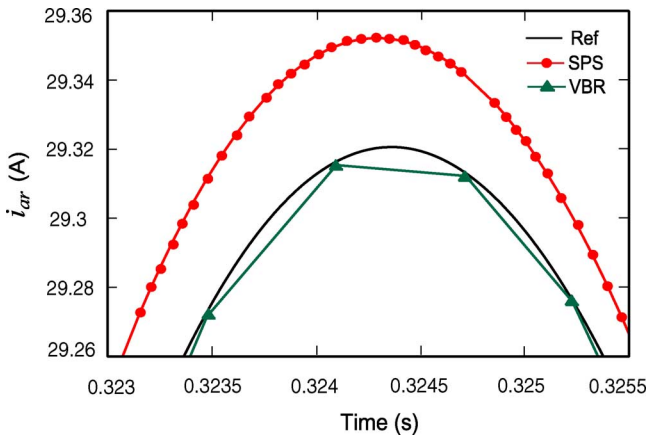


Fig. 10. Induction machine is fed from a voltage source with inductive impedance: artificial shunt resistors are used to interface the model.

conduct studies using the SPS and the proposed VBR model. The same machine as in Section V is used here, except that the voltage source has series inductance of 10^{-3} H in each phase. In the state-variable languages, the typical *qd* model is viewed and interfaced as a voltage-controlled current source (voltage input and current output). Therefore, the direct connection of machine terminals with inductive branches (or current sources) is not possible as it creates difficulties for forming the system's state space equations [21]–[24]. An approach that is often considered and suggested to the users for interfacing the machine models [22], [23] is to use an artificial shunt snubber circuit (very large resistors and/or small capacitors), which makes it possible to calculate the voltages at the interconnection point of the machine model and the external circuit, and therefore, completes the interface. In the studies presented here, to enable the SPS model interface, a shunt resistive branch was connected, as shown in Fig. 10. In general, the values of such artificial snubbers are chosen large enough as to minimize their effect on the simulation accuracy. However, when the VBR model is used, the stator *RL* branches are directly included into the external circuit network (see Fig. 3), and no artificial components are required.

The same startup transient study is performed for this case using the SPS *qd* model with snubber resistors of $10^3 \Omega$ and the proposed VBR-III model. The same variable-step ODE solver and integration parameters (error tolerances and time-step limits) as in Section V-A are used here. The predicted responses are overlapped with the reference solution and shown in Figs. 11 and 12. Due to space limitation, only the rotor current i_{ar} is shown here. The reference solution are obtained using the barebone *qd* model with the fixed-step fourth-order Runge–Kutta integration method and a very small time step of $1 \mu\text{s}$. In general, the transient trend is very similar to that shown in Section V (see Figs. 4 and 5), and the difference among the models is not noticed. This has been achieved by using relatively large values for the snubber resistors in the SPS model. However, the magnified plot of i_{ar} shown in Fig. 12 demonstrates that the SPS model actually converges to a different solution, which is shifted from the reference.

Another important observation that can be made based on Fig. 12 is that the SPS *qd* model uses significantly more time steps than the VBR model. To give the reader a better idea, the CPU times and the number of steps taken by the two models are summarized in Table III, where a very significant difference (an order of magnitude) can be seen.

Fig. 11. Rotor current i_{ar} with variable-step method.Fig. 12. Magnified portion of rotor current i_{ar} in Fig. 11.TABLE III
SIMULATION EFFICIENCY COMPARISONS

Model	CPU Time, s	Num. of Steps	CPU Time per Step, μ s
SPS	1.828	11504	158.9
VBR-III	0.156	1058	147.7

The presence of fictitious resistors also changes the network topology, as well as increases the total number of state variables due to the source inductance. Moreover, the resulting equations become very stiff, which can be noted by evaluating the eigenvalues summarized in Table IV, where it is shown that the SPS model now has three additional eigenvalues with very large magnitude on the order of 10^6 . At the same time, for the VBR model, the source inductances do not increase the numbers of the eigenvalues and/or affect the stiffness. It may be noted that the VBR model in Table III does not include the eigenvalues corresponding to stator and rotor zero sequences, as it does in Table II. For discussion in this section and consistency with the SPS model, here the zero sequence was not included in the VBR model.

The affects of the snubber circuits on the simulation accuracy and efficiency may be further investigated by varying the values of resistors from 10 to $10^5 \Omega$. Without the loss of generality, here we again consider the relative errors for rotor current i_{ar} , which

TABLE IV
EIGENVALUES OF MACHINE MODELS

SPS	VBR
-18.93	-0.016
$-33.44 + j275.7$	$-79.5 + j33.6$
$-33.44 - j275.7$	$-79.5 - j33.6$
$-212.45 + j23.47$	$-175.9 + j343.5$
$-212.45 - j23.47$	$-175.9 - j343.5$
-1.00e6	N/A
$-1.25e6 + j76.27$	N/A
$-1.25e6 - j76.27$	N/A

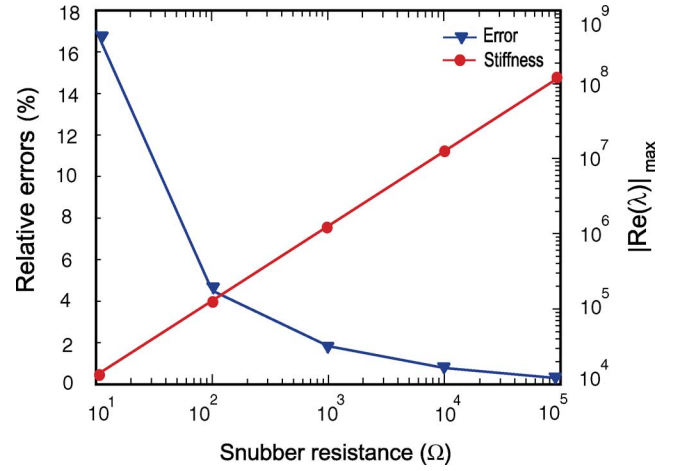


Fig. 13. Relative errors and stiffness of the SPS model for different snubber resistances.

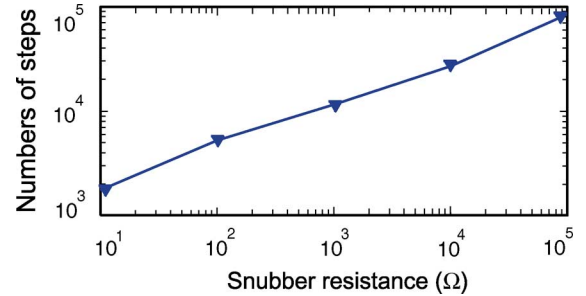


Fig. 14. Numbers of time steps required by the SPS model for different snubber resistances.

are calculated for the SPS model according to (63) and are plotted in Fig. 13. The magnitude of the largest negative eigenvalue is also plotted in Fig. 13 for the same range of snubber resistance. As can be seen here that it is possible to use very large resistors and achieve a high accuracy of such interface bring the errors below few percent. However, such improvement in accuracy comes at a price of making the system very stiff numerically.

The result of increasing numerical stiffness is that the ODE solver requires smaller and smaller time steps to satisfy the absolute and relative tolerances, which slows down the overall simulation. To give the reader a better idea, the number of time steps taken by the SPS model to complete the same transient study for the same range of snubber resistors is summarized in Fig. 14. As can be seen in Figs. 13 and 14, to achieve the solution within 1% of the reference solution, the SPS model requires the subbers of $10^5 \Omega$, which makes the system very stiff and bring

the number of time steps to the range of 10^5 . At the same time, as evident from Fig. 9, the VBR model can achieve an even better accuracy with time steps as large as 1 ms requiring 600 steps total.

VIII. CONCLUSION

This paper presents a VBR induction machine model that can be of great benefit in applications where a direct interface of the stator phase branches with an external network circuit is required. Traditionally, in such applications, the CC phase-domain models are used, which results in significant increase in computational cost due to time-varying inductances. A method of interfacing the *qd* machine models with external inductive circuits in state-variable languages often requires artificial snubbers that also increase computational burden. However, the proposed VBR model, in addition to achieving the required direct interface of the stator circuit, also provides an improved numerical accuracy with greatly reduced computational overhead.

APPENDIX A

The system state matrices of the *qd* model with flux linkages as state variables are represented as

$$\mathbf{A}_{qd} = -(\mathbf{R}\mathbf{L}_{qd}^{-1} + \mathbf{W}) \quad (\text{A1})$$

where

$$\mathbf{R} = \text{diag}[r_s, r_s, r_s, r_r, r_r, r_r] \quad (\text{A2})$$

$$\mathbf{L}_{qd0} = \begin{bmatrix} L_{ls} + L_m & 0 & 0 & L_m & 0 & 0 \\ 0 & L_{ls} + L_m & 0 & 0 & L_m & 0 \\ 0 & 0 & L_{ls} & 0 & 0 & 0 \\ L_m & 0 & 0 & L_{lr} + L_m & 0 & 0 \\ 0 & L_m & 0 & 0 & L_{lr} + L_m & 0 \\ 0 & 0 & 0 & 0 & 0 & L_{lr} \end{bmatrix} \quad (\text{A3})$$

$$\mathbf{W} = \begin{bmatrix} 0 & \omega & 0 & 0 & 0 & 0 \\ -\omega & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \omega - \omega_r & 0 \\ 0 & 0 & 0 & -(\omega - \omega_r) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (\text{A4})$$

The system state matrix of the CC model with currents as state variables may be expressed as

$$\mathbf{A}_{cc} = -\mathbf{L}_{abc}^{-1}(\mathbf{R} + p\mathbf{L}_{abc}) \quad (\text{A5})$$

where

$$\mathbf{L}_{abc} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}_{sr} \\ \mathbf{L}_{sr}^T & \mathbf{L}_r \end{bmatrix}. \quad (\text{A6})$$

For the VBR models, the state matrix is

$$\mathbf{A}_{vbr} = \begin{bmatrix} -\mathbf{L}_{abc}^{-1}\mathbf{r}_{abc}'' & -\mathbf{L}_{abc}^{-1}\mathbf{M}_1 \\ \mathbf{B}_2 & \mathbf{B}_1 \end{bmatrix} \quad (\text{A7})$$

where

$$\mathbf{M}_1 = \mathbf{K}_s^{-1} \begin{bmatrix} \frac{L_m'' r_r}{L_{lr}^2} \left(\frac{L_m''}{L_{lr}} - 1 \right) & \frac{\omega_r L_m''}{L_{lr}} & 0 \\ -\frac{\omega_r L_m''}{L_{lr}} & \frac{L_m'' r_r}{L_{lr}^2} \left(\frac{L_m''}{L_{lr}} - 1 \right) & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (\text{A8})$$

$$\mathbf{M}_2 = \begin{bmatrix} \frac{L_m''}{L_{lr}} & 0 & 0 \\ 0 & \frac{L_m''}{L_{lr}} & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (\text{A9})$$

$$\mathbf{B}_1 = \begin{bmatrix} \frac{r_r}{L_{lr}} \left(\frac{L_m''}{L_{lr}} - 1 \right) & -(\omega - \omega_r) & 0 \\ \omega - \omega_r & \frac{r_r}{L_{lr}} \left(\frac{L_m''}{L_{lr}} - 1 \right) & 0 \\ 0 & 0 & -\frac{r_r}{L_{lr}} \end{bmatrix} \quad (\text{A10})$$

$$\mathbf{B}_2 = \begin{bmatrix} \frac{r_r L_m''}{L_{lr}} & 0 & 0 \\ 0 & \frac{r_r L_m''}{L_{lr}} & 0 \\ 0 & 0 & 0 \end{bmatrix} \mathbf{K}_s. \quad (\text{A11})$$

APPENDIX B

Induction machine parameters [9]: 3 hp, 220 V, 1710 r/min, 4 poles, 60 Hz, $r_s = 0.435 \Omega$, $X_{ls} = 0.754 \Omega$, $X_m = 26.13 \Omega$, $r_r = 0.816 \Omega$, $X_{lr} = 0.754 \Omega$, $J = 0.089 \text{ kgm}^2$.

ACKNOWLEDGMENT

The authors would like to thank their UBC and Purdue colleagues, as well as the reviewers and editor of this paper, for insightful and inspiring discussions contributing to this manuscript.

REFERENCES

- [1] H. C. Stanley, "An analysis of the induction motor," *AIEE Trans.*, vol. 57, pp. 751–755, 1938.
- [2] D. S. Brereton, D. G. Lewis, and C. G. Young, "Representation of induction motor loads during power system stability studies," *AIEE Trans.*, vol. 76, pp. 451–461, Aug. 1957.
- [3] G. Kron, *Equivalent Circuit of Electric Machinery*. New York: Wiley, 1951.
- [4] S. J. Salon, *Finite Element Analysis of Electrical Machines*. Norwell, MA: Kluwer, 1995.
- [5] G. R. Slemon, "An equivalent circuit approach to analysis of synchronous machines with saliency and saturation," *IEEE Trans. Energy Convers.*, vol. 5, no. 3, pp. 538–545, Sep. 1990.
- [6] Y. Xiao, G. R. Slemon, and M. R. Iravani, "Implementation of an equivalent circuit approach to the analysis of synchronous machines," *IEEE Trans. Energy Convers.*, vol. 9, no. 4, pp. 717–723, Dec. 1994.
- [7] P. C. Krause and C. H. Thomas, "Simulation of symmetrical induction machinery," *IEEE Trans. Power App. Syst.*, vol. 84, pp. 1038–1053, Nov. 1965.

- [8] P. Kundur, *Power System Stability and Control*. New York: McGraw-Hill, 1994, Ch. 3.
- [9] P. C. Krause, O. Wasynczuk, and S. D. Sudhoff, *Analysis of Electric Machinery and Drive Systems*, 2nd ed. Piscataway, NJ: IEEE Press, 2002.
- [10] S. D. Sudhoff and O. Wasynczuk, "Analysis and average-value modeling of line-commutated converter-synchronous machine systems," *IEEE Trans. Energy Convers.*, vol. 8, no. 1, pp. 92–99, Mar. 1993.
- [11] S. D. Sudhoff, K. A. Corzine, H. J. Hegner, and D. E. Delisle, "Transient and dynamic average-value modeling of synchronous machine fed load-commutated converters," *IEEE Trans. Energy Convers.*, vol. 11, no. 3, pp. 508–514, Sep. 1996.
- [12] B. T. Kuhn, S. D. Sudhoff, and C. A. Whitcomb, "Performance characteristics and average-value modeling of auxiliary resonant commutated pole converter based induction motor drives," *IEEE Trans. Energy Convers.*, vol. 14, no. 3, pp. 493–499, Sep. 1999.
- [13] J. Jatskevich, S. D. Pekarek, and A. Davoudi, "Parametric average-value model of synchronous machine-rectifier systems," *IEEE Trans. Energy Convers.*, vol. 21, no. 1, pp. 9–18, Mar. 2006.
- [14] J. Jatskevich, S. D. Pekarek, and A. Davoudi, "Fast procedure for constructing an accurate dynamic average-value model of synchronous machine-rectifier systems," *IEEE Trans. Energy Convers.*, vol. 21, no. 2, pp. 435–441, Jun. 2006.
- [15] H. W. Dommel, *EMTP Theory Book*. Vancouver, BC, Canada: Micro-Tran Power System Analysis Corporation, May 1992, Ch. 8.
- [16] V. Brandwajn, "Synchronous generator models for the analysis of electromagnetic transients," Ph.D. thesis, Univ. British Columbia, Vancouver, BC, Canada, 1977.
- [17] X. Cao, A. Kurita, H. Mitsuwa, Y. Tada, and H. Okamoto, "Improvements of numerical stability of electromagnetic transient simulation by use of phase-domain synchronous machine models," *Electr. Eng. Jpn.*, vol. 128, no. 3, pp. 53–62, Apr. 1999.
- [18] A. B. Dehkordi, A. M. Gole, and T. L. Maguire, "Permanent magnet synchronous machine model for real-time simulation," presented at the Interracial Conf. Power Syst. Transients (IPST'05), Montreal, Canada.
- [19] L. Wang and J. Jatskevich, "A voltage-behind-reactance synchronous machine model for the EMTP-type solution," *IEEE Trans. Power Syst.*, vol. 21, no. 4, pp. 1539–1549, Nov. 2006.
- [20] L. Wang, J. Jatskevich, and H. W. Dommel, "Re-examination of synchronous machine modeling techniques for electromagnetic transient simulations," *IEEE Trans. Power Syst.*, vol. 22, no. 3, pp. 516–527, Aug. 2007.
- [21] SimPowerSystems 4–User Guide, The MathWorks, Inc., Natick, Massachusetts [Online]. Available at: www.mathworks.com, 2007.
- [22] Piecewise Linear Electrical Circuit Simulation (PLECS) User Manual, Version. 1.5, Plexim GmbH [Online]. Available at: www.plexim.com, 2006.
- [23] R. Champagne, L.-A. Dessaint, H. Fortin-Blanchette, and G. Sybille, "Analysis and validation of a real-time AC drive simulator," *IEEE Trans. Power Electron.*, vol. 19, no. 2, pp. 336–345, Mar. 2004.
- [24] SimPowerSystems 4–Reference Manual, The MathWorks, Inc., Natick, Massachusetts [Online]. Available at: www.mathworks.com, 2007.
- [25] J. R. Marti and T. O. Myers, "Phase-domain induction motor model for power system simulators," *IEEE Conf. Commun., Power, Comput.*, vol. 2, May 1995, pp. 276–282.
- [26] J. R. Marti and K. W. Louie, "A phase-domain synchronous generator model including saturation effects," *IEEE Trans. Power Syst.*, vol. 12, no. 1, pp. 222–229, Feb. 1997.
- [27] W. Gao, E. V. Solodovnik, and R. A. Dougal, "Symbolically aided model development for an induction machine in virtual test bed," *IEEE Trans. Energy Convers.*, vol. 19, no. 1, pp. 125–135, Mar. 2004.
- [28] R. W. Y. Cheung, H. Jin, B. Wu, and J. D. Lavers, "A generalized computer-aided formulation for the dynamic and steady state analysis of induction machine inverter drive systems," *IEEE Trans. Energy Convers.*, vol. 5, no. 2, pp. 337–343, Jun. 1990.
- [29] C. T. Liu and W. L. Chang, "A generalized technique for modeling switch-controlled induction machine circuits," *IEEE Trans. Energy Convers.*, vol. 7, no. 1, pp. 168–176, Mar. 1992.
- [30] P. Pillay and V. Levin, "Mathematical models for induction machines," in *Proc. Rec. 1995 IEEE Ind. Appl. Conf., 30th IAS Annu. Meet.*, vol. 1, no. 8–12, Oct. 1995, pp. 606–616.
- [31] S. D. Pekarek, O. Wasynczuk, and H. J. Hegner, "An efficient and accurate model for the simulation and analysis of synchronous machine/converter systems," *IEEE Trans. Energy Conv.*, vol. 13, no. 1, pp. 42–48, Mar. 1998.
- [32] W. Zhu, S. D. Pekarek, J. Jatskevich, O. Wasynczuk, and D. Delisle, "A model-in-the-loop interface to emulate source dynamics in a zonal DC distribution system," *IEEE Trans. Power Electron.*, vol. 20, no. 2, pp. 438–445, Mar. 2005.
- [33] R. R. Nucera, S. D. Sudhoff, and P. C. Krause, "Computation of steady-state performance of an electronically commutated motor," *IEEE Trans. Ind. Appl.*, vol. 25, no. 6, pp. 1110–1117, Nov./Dec. 1989.
- [34] Simulink Dynamic System Simulation Software–Users Manual, MathWorks, Inc., Natick, Massachusetts, 2005.
- [35] O. Wasynczuk and S. D. Sudhoff, "Automated state model generation algorithm for power circuits and systems," *IEEE Trans. Power Syst.*, vol. 11, no. 4, pp. 1951–1956, Nov. 1996.
- [36] Automated State Model Generator (ASMG), Reference Manual, Version 2, PC Krause and Associates Inc. [Online]. Available at: www.pcka.com, 2002.
- [37] W. Gautchi, *Numerical Analysis: An Introduction*. Boston, MA: Birkhauser, 1997.
- [38] L. O. Chua and P.-M. Lin, *Computer Aided Analysis of Electronic Circuit: Algorithms and Computational Techniques*. Englewood Cliffs, NJ: Prentice-Hall, pp. 43–45, 1975.
- [39] Uri Ascher and L. R. Petzold, *Computer Methods for Ordinary Differential Equations and Differential-Algebraic Equations*. Philadelphia, PA: SIAM, 1998, Ch. 2.



Liwei Wang (S'04) received the M.S. degree in electrical engineering from Tianjin University, Tianjin, China, in 2004. He is currently working toward the Ph.D. degree in electrical and computer engineering from the University of British Columbia, Vancouver, BC, Canada.

His current research interests include electrical machines, power, and power electronic systems simulation.



Juri Jatskevich (M'99–SM'07) received the M.S.E.E. and Ph.D. degrees in electrical engineering from Purdue University, West Lafayette, IN, in 1997 and 1999, respectively.

He was Post-Doctoral Research Associate and Research Scientist at Purdue University, as well as consulted for P C Krause and Associates, Inc. Since 2002, he has been a faculty member at the University of British Columbia, Vancouver, Canada, where he is now an Associate Professor of Electrical and Computer Engineering. His research interests include

electrical machines, power electronic systems, average-value modeling and simulation.

Dr. Jatskevich is presently a Secretary of IEEE CAS Power Systems & Power Electronic Circuits Technical Committee, Editor of IEEE TRANSACTIONS ON ENERGY CONVERSION, and Editor of IEEE POWER ENGINEERING LETTERS. He is also chairing the IEEE Task Force on Dynamic Average Modeling, under the Working Group on Modelling and Analysis of System Transients Using Digital Programs.



Steven D. Pekarek (M'96) received the Ph.D. degree in electrical engineering from Purdue University, West Lafayette, IN, in 1996.

During 1997–2004, he was an Assistant (Associate) Professor of Electrical and Computer Engineering at the University of Missouri-Rolla (UMR). He is currently a Professor of Electrical and Computer Engineering at Purdue University and is a Co-Director of the Energy Systems Analysis Consortium. As a faculty member, he has been the Principal Investigator on several research programs, including projects for the Navy, Airforce, Ford Motor Corporation, Motorola, and Delphi Automotive Systems. His current research interests include power electronic-based architectures for finite inertia power and propulsion systems.

Prof. Pekarek is an active member of the IEEE Power Engineering Society, the Society of Automotive Engineers, the IEEE Power Electronics Society, and he was Chairman of the 2005 International Future Energy Challenge.

Induction Machine Modeling Considering Magnetizing Flux Saturation with Air-Gap Harmonics

Navid Amiri, *Member, IEEE*, Seyyedmilad Ebrahimi, *Member, IEEE*, and Juri Jatskevich, *Fellow, IEEE*

Abstract—Accurate and computationally-efficient models of induction machines (IMs) are essential for many simulation programs and analysis of power systems transients. Conventionally, when modeling magnetic saturation in the classical qd models, only the fundamental component of magnetizing flux is considered. To represent the saturation air-gap flux harmonics, fairly complicated phase-domain and/or implicit models have been proposed in the prior literature. In this paper, a new explicit two-axis IM model is developed to include the air-gap harmonics caused by magnetic saturation. The new model is validated by experimental measurements and simulations in various steady state and transient scenarios and is shown to provide accurate results while being simple to implement.

Index Terms—Air-gap flux harmonics, induction machine, magnetic saturation, simulation, two-axis qd model.

I. INTRODUCTION

INDUCTION machines (IMs) are widely used in various power systems and industrial applications, and their basic models are often available in many simulation programs. Magnetic saturation in induction machines is a common phenomenon. Since in many applications IMs are operated in some level of saturation [1], accurate modeling of saturation is highly desirable. Modeling of magnetic saturation in IMs has been conventionally done by considering the nonlinear relationship between the magneto-motive force (MMF) and the resulting magnetizing flux [2]. For modeling simplicity, a large number of approaches consider only the fundamental component of the air-gap flux [2]–[7], which is often represented by variable (i.e., flux-dependent) inductances. While the assumption of sinusoidal flux distribution may be sufficient for many studies, it has been shown that magnetic saturation can cause distortions in air-gap flux distribution, resulting in spatial harmonics [8]–[12]. Such distortions may be attributed due to uneven saturation of stator/rotor teeth [8] and can introduce harmonics in currents and voltages [11]–[12]. Modeling harmonics in magnetizing flux may also be beneficial for studying various machine drives that use this phenomena for flux position estimation [12]–[16]. It is also worth mentioning that for modeling the air-gap flux distribution, typically only up to third harmonic component is considered (due to being the dominant), which is shown to be adequate [11]–[12].

The classical $qd0$ IM models [2] offer many advantages such as constant circuit parameters, simplicity, and straightforward

implementation in many electromagnetic transient (EMT) simulation programs, including the nodal-analysis-based programs (e.g., PSCAD/EMTDC, EMTP-RV, etc.), and the state-variable-based programs (e.g., MATLAB/Simulink, Simscape Electrical, PLECS, etc.), as well as real-time simulators (e.g., RTDS's RSCAD, Opal-RT's RT-LAB, Typhoon HIL, etc.).

Magnetic saturation in $qd0$ IM models may be considered using multitude of methods [2]–[7], wherein, typically only the fundamental component of the magnetizing flux is considered. A $qd0$ model that includes flux harmonic distortions due to saturation has been presented in [8]. In [8], the effect of third harmonic is represented using variable length air-gap function, which may not be true for a more general model of IMs or deep levels of saturation. The methodology [8] also requires solving implicit equations for transforming variables to synchronous reference frame, which adds complexity and computational cost. Furthermore, a steady-state single line equivalent circuit considering the third harmonic component of air gap flux is presented in [9], which might not be valid for transient studies, or drive applications.

Alternatively, a coupled-circuit phase-domain (CCPD) IM model has been presented in [10]–[12] to include the effect of flux harmonics caused by magnetic saturation. The model in reference [11] solves stator and rotor voltage equations in phase domain with stator and rotor fluxes as state-variables. However, the relation between stator and rotor currents and magnetizing flux components for each phase are defined in an explicit way. The explicit equation is then solved using an iterative procedure which adds to the computational cost of the model. Moreover, the application of saturable CCPD models is limited to simulation packages that permit variable inductances (e.g. PLECS [20] permits variable inductors, but Simscape Electrical [22] does not), which also comes at a cost of simulation speed.

To the best of authors' knowledge, explicit and constant-parameter $qd0$ models of IMs that also include the main flux saturation harmonics have not been proposed in the prior literature. In this paper, based on the CCPD model presented in [12] (whose accuracy has been verified experimentally), a new $qd0$ model is proposed for including the dominant third harmonic of the air-gap flux. The proposed $qd0$ model uses the explicit flux correction method [7] which is extended to include saturation harmonics. The new model is extensively validated in steady state and transient scenarios and compared with experimental measurements.

It is envisioned that the proposed model can become a valuable addition in various EMT simulation programs to provide higher fidelity modeling capabilities for studies with induction machines under saturation.

Manuscript received: 21 November 2020, revised on 14 April 2021, accepted 2 May 2021. This work was supported by the Natural Science and Engineering Research Council (NSERC) of Canada under the CRD Grant. Paper no. TEC-01165-2020. (Corresponding author: Juri Jatskevich.)

The authors are with the Electrical and Computer Engineering Department, University of British Columbia, Vancouver, BC V6T 1Z4, Canada (e-mail: navida@ece.ubc.ca; ebrahimi@ece.ubc.ca; juri@ece.ubc.ca).

II. MODELING SATURATION WITH AIR-GAP FLUX HARMONICS

In this paper, a three-phase squirrel cage induction machine is considered with air-gap flux harmonics up to third component. The modeling assumptions in this paper are based on the CCPD induction machine model defined in [12]. All machine parameters are referred to the stator side and motor sign convention is used. For notational consistency with [2], matrices, vectors and instantaneous quantities/variables are denoted using bold upper-case, bold lower-case, and lower-case italic letters, respectively.

A. Spatial Flux Harmonics Due to Saturation

Conventionally, for modeling saturation in electrical machines, it is often assumed that the magnetic flux density in the air-gap maintains a sinusoidal distribution along the stator circumference [2]. However, it has been shown in [8]–[12] that flux distribution can become considerably distorted as a result of different saturation levels in different parts/zones of the machine, which creates spatial harmonics in the distribution of magnetizing flux. A common cause is a deeper saturation in stator/rotor teeth/slots along the magnetization flux path compared to the sides [8], as depicted in Fig. 1. As a result, under saturation, the magnetic flux density will have a flattop waveform [8] containing odd harmonics, as depicted in Fig. 2.

In this paper, spatial harmonics are considered up to third component only, since it has the dominant effect. Higher-order harmonics may also be considered using the same approach, but their contribution is far less significant, as has been documented in [8]. As a result, two components (i.e., fundamental and third harmonic) are defined based on the magnetizing current vector angle and magnitude as [12]

$$\begin{aligned} B_{m1}(i_m, \theta_m, \phi) &= |B_1|(i_m) \cos(\phi - \theta_m) \\ B_{m3}(i_m, \theta_m, \phi) &= |B_3|(i_m) \cos(3\phi - 3\theta_m) \\ B_m(i_m, \theta_m, \phi) &= B_{m1}(i_m, \theta_m, \phi) + B_{m3}(i_m, \theta_m, \phi) \end{aligned} \quad (1)$$

where ϕ is the physical angle around the rotor circumference, and i_m and θ_m are the magnitude and angle of the magnetizing current vector, respectively. It is noted that the peaks of fundamental and third harmonic components of magnetic flux density are aligned with the magnetizing current vector as a result of non-salient structure of the IM.

The fundamental and third harmonic components of magnetizing flux magnitudes, λ_{m1} and λ_{m3} are defined as functions of the magnetizing flux density (1) (or magnetizing current) as

$$\begin{aligned} \lambda_{m1} &= U_1(|B_1|(i_m)) = F_1(i_m) \\ \lambda_{m3} &= U_3(|B_3|(i_m)) = F_3(i_m) \end{aligned} \quad (2)$$

where functions $U_1(\cdot)$ and $U_3(\cdot)$ are dependent on the machine's geometric properties. Functions $F_1(\cdot)$ and $F_3(\cdot)$ relate each of the magnetizing flux magnitude components to the magnetizing current amplitude. These functions can be directly extracted from standard characterization tests [12] and stored as look up tables for future use. For more details, see Appendix B.

Using (2), stator and rotor flux linkage components can be defined separately. Based on the stator winding functions, the following magnetizing flux linkages can be defined for each of the three phases [12] as

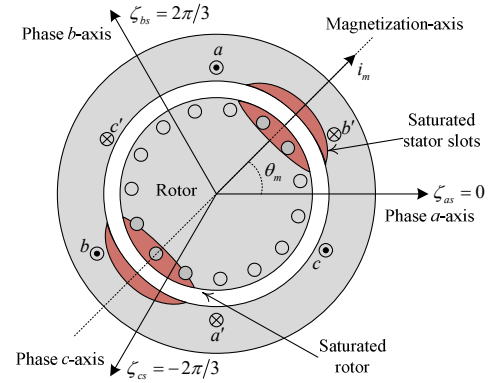


Fig. 1. Cross-section of a squirrel cage induction machine including magnetizing current vector and saturated stator and rotor slots/path [8].

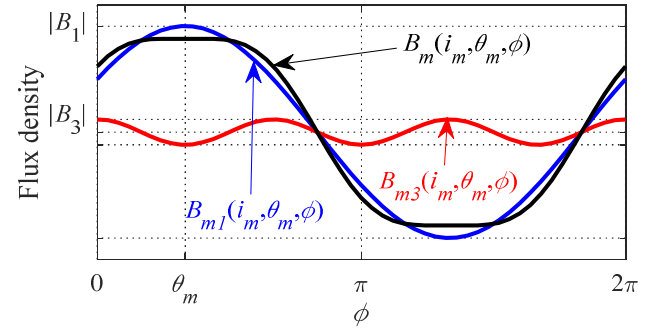


Fig. 2. Distorted air-gap flux density decomposed into fundamental and third harmonic components [8].

$$\lambda_{msx} = \lambda_{m1} \cos(\zeta_{xs} - \theta_m) + \lambda_{ms3} \cos 3\theta_m, \quad (3)$$

where $x \in \{a, b, c\}$, and ζ_{xs} corresponds each phase winding physical angle on the stator defined as

$$\zeta_{as} = 0, \quad \zeta_{bs} = 2\pi/3, \quad \zeta_{cs} = -2\pi/3. \quad (4)$$

In (3), λ_{ms3} corresponds to the magnitude of a fraction of the air-gap flux third harmonic component which is linked to the stator windings and is defined as

$$\lambda_{ms3} = \frac{k_{w3}}{3k_{w1}} \lambda_{m3} = \frac{k_{w3}}{3k_{w1}} F_3(i_m), \quad (5)$$

where k_{w1} and k_{w3} are the fundamental and third harmonic components of stator winding factors [12, Section II–B]. Since magnetizing current magnitude i_m is related to λ_{m1} according to (2), λ_{ms3} can be expressed in terms of λ_{m1} as

$$\lambda_{ms3} = \frac{k_{w3}}{3k_{w1}} F_3(F_1^{-1}(\lambda_{m1})) = G_{s3}(\lambda_{m1}), \quad (6)$$

In the subject squirrel-cage induction machine, the rotor consists of an arbitrary number of conductors. Therefore, as a result of only considering up to third flux harmonic, the rotor is represented as two separate three-phase winding sets, corresponding to the fundamental and third harmonic flux components, rotating at the electrical frequencies ω_r and $3\omega_r$, respectively [12]. As a result, the rotor flux linkages of rotor windings are defined as

$$\lambda_{mrx} = \lambda_{m1} \cos(\zeta_{xr1} - \theta_m) + \lambda_{mr3} \cos(\zeta_{xr3} - 3\theta_m), \quad (7)$$

where angles ζ_{xr1} and ζ_{xr3} correspond to each of the rotor winding component phases with respect to the stationary frame and are defined as [12]

$$\begin{aligned}\zeta_{ar1} &= \theta_r, & \zeta_{br1} &= \theta_r + 2\pi/3, & \zeta_{cr1} &= \theta_r - 2\pi/3 \\ \zeta_{ar3} &= 3\theta_r, & \zeta_{br3} &= 3\theta_r + 2\pi/3, & \zeta_{cr3} &= 3\theta_r - 2\pi/3\end{aligned}\quad (8)$$

where θ_r is the rotor electrical angle.

In (7), λ_{mr3} corresponds to the magnitude of a fraction of the air-gap flux third harmonic component which is linked to the rotor windings and is defined as

$$\lambda_{mr3} = \frac{1}{3}\lambda_{m3} = \frac{1}{3}F_3(i_m), \quad (9)$$

where the scaling factor $1/3$ is the ratio between the third and fundamental components of the rotor equivalent winding function [12, Section II-C]. Since magnetizing current i_m is related to λ_{m1} according to (2), λ_{mr3} can be expressed in terms of λ_{m1} as

$$\lambda_{mr3} = \frac{1}{3}F_3(F_1^{-1}(\lambda_{m1})) = G_{r3}(\lambda_{m1}). \quad (10)$$

Based on the defined flux linkage components, a complete coupled-circuit phase-domain model for the squirrel-cage induction machine is derived in [12] and validated using experimental tests.

B. Proposed Saturable $qd0$ Model

The proposed model with spatial flux harmonics is derived by transforming the CCPD model [12] to (in this case) synchronously rotating reference frame. The variables corresponding to the three-phase stator winding and the rotor winding fundamental component are transformed by the same Park's transformation [2], as

$$\mathbf{f}_{qd0k} = \mathbf{K}_s(\theta_s)\mathbf{f}_{abck}, \quad k = s, r \quad (11)$$

$$\mathbf{f}_{abck} = [f_{ak}, f_{bk}, f_{ck}]^T, \quad \mathbf{f}_{qd0} = [f_{qk}, f_{dk}, f_{0k}]^T, \quad (12)$$

where $\mathbf{K}_s(\theta_s)$ is Park's three-phase transformation matrix [2], and θ_s is the $qd0$ reference frame angle. Reference frame (and its corresponding angle θ_s) can be arbitrary, yet in this paper for the sake of simplicity and numerical accuracy, synchronous reference frame is considered. The variables corresponding to the rotor winding third harmonic component are transformed to a $qd0$ plane, rotating three times faster than the synchronous reference frame with the angle $3\theta_s$ as

$$\mathbf{f}_{qd03} = \mathbf{K}_s(3\theta_s)\mathbf{f}_{abcr3}, \quad (13)$$

where

$$\mathbf{f}_{abcr3} = [f_{ar3}, f_{br3}, f_{cr3}]^T, \quad \mathbf{f}_{qd03} = [f_{q3}, f_{d3}, f_{03}]^T. \quad (14)$$

The transformations applied to the rotor circuit are according to the generalized Park's transformation theory in [23] for arbitrary number of windings (which is the case in a squirrel cage rotor), where each harmonic component would transform into its own qd reference frame rotating with a speed corresponding to the harmonic order. It is also worth mentioning that since the stator consists of balanced three windings, the third harmonic component is transformed into the zero-sequence component. By applying the transformations in (11) and (13), to the CCPD machine model presented in [12], and considering only up to the third harmonic component of magnetizing flux as well as stator zero sequence components, stator and rotor voltage equations are obtained as

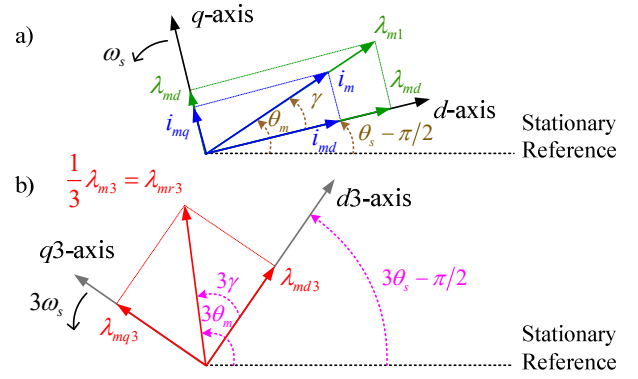


Fig. 3. Phasor diagram of magnetizing current and flux (fundamental and third harmonic components) vectors in the arbitrary qd reference frame.

$$v_{qs} = r_s i_{qs} + p\lambda_{qs} + \omega_s \lambda_{ds}, \quad (15)$$

$$v_{ds} = r_s i_{ds} + p\lambda_{ds} - \omega_s \lambda_{qs}, \quad (16)$$

$$v_{0s} = r_s i_{0s} + p\lambda_{0s}, \quad (17)$$

$$0 = r_r i_{qr} + p\lambda_{qr} + (\omega_s - \omega_r) \lambda_{dr}, \quad (18)$$

$$0 = r_r i_{dr} + p\lambda_{dr} - (\omega_s - \omega_r) \lambda_{qr}, \quad (19)$$

$$0 = r_r i_{qr3} + p\lambda_{qr3} + 3(\omega_s - \omega_r) \lambda_{dr3}, \quad (20)$$

$$0 = r_r i_{dr3} + p\lambda_{dr3} - 3(\omega_s - \omega_r) \lambda_{qr3}, \quad (21)$$

where p is the Heaviside's operator denoting time derivative ($p = d/dt$), and ω_s and ω_r are the reference frame angular electrical speed and rotor angular electrical frequency, respectively. It is worth mentioning that the third harmonic components of variables corresponding to the stator would be transformed into zero sequence component. Also, the rotor voltage equations for zero sequence are omitted since zero sequence current does not flow in a squirrel-cage rotor.

In (15)–(17), stator currents are defined as

$$i_{qs} = L_{ls}^{-1}(\lambda_{qs} - \lambda_{mq}), \quad i_{ds} = L_{ls}^{-1}(\lambda_{ds} - \lambda_{md}), \quad (22)$$

$$i_{0s} = L_{ls}^{-1}(\lambda_{0s} - \lambda_{m0}). \quad (23)$$

The rotor currents in (18)–(21) corresponding to fundamental and third harmonic components are expressed as

$$i_{qr} = L_{lr}^{-1}(\lambda_{qr} - \lambda_{mq}), \quad i_{dr} = L_{lr}^{-1}(\lambda_{dr} - \lambda_{md}), \quad (24)$$

$$i_{qr3} = L_{lr}^{-1}(\lambda_{qr3} - \lambda_{mq3}), \quad i_{dr3} = L_{lr}^{-1}(\lambda_{dr3} - \lambda_{md3}). \quad (25)$$

The main magnetizing current components in q - and d -axes are defined as

$$i_{mq} = i_{qs} + i_{qr}, \quad i_{md} = i_{ds} + i_{dr}, \quad (26)$$

$$i_m = \sqrt{i_{mq}^2 + i_{md}^2}. \quad (27)$$

In (22)–(24), the variables λ_{mq} and λ_{md} are the fundamental components of the magnetizing flux in q - and d -axes as shown in Fig. 3(a). Similarly, the variables λ_{mq3} and λ_{md3} are the third harmonic components of the magnetizing flux in $q3$ - and $d3$ -axes plane corresponding to rotor, respectively, as shown in Fig. 3(b). Since the magnetizing current and flux vectors are aligned [2], based on Fig. 3, the stator and rotor fundamental and rotor third harmonic magnetizing flux components $\lambda_{mq}, \lambda_{md}, \lambda_{mq3}, \lambda_{md3}$ can be expressed as

$$\lambda_{mq} = \lambda_{m1} \sin(\gamma), \quad \lambda_{md} = \lambda_{m1} \cos(\gamma), \quad (28)$$

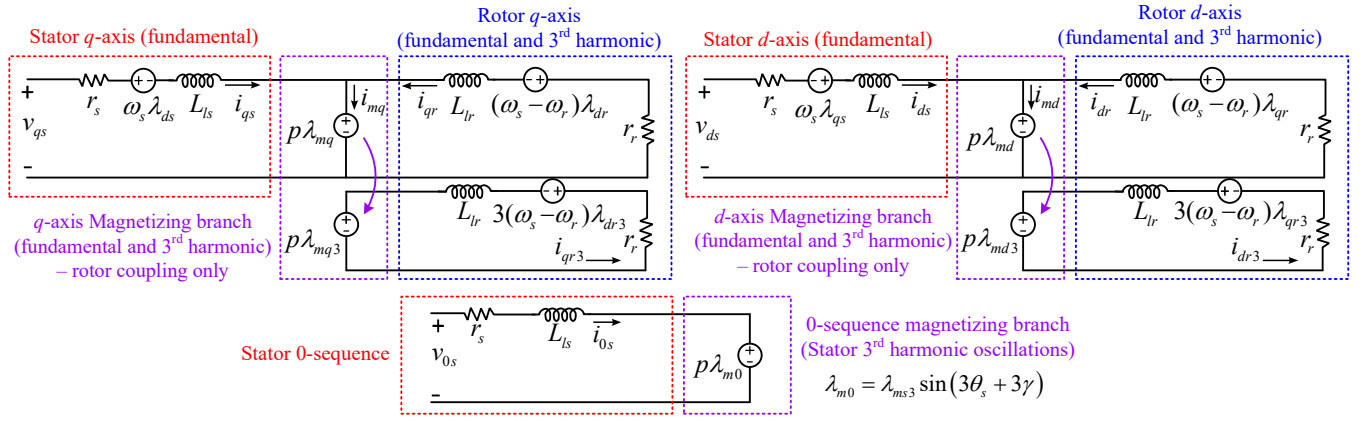


Fig. 4. The $qd0$ equivalent circuit of induction machine including third harmonic component of air-gap flux in synchronous reference frame. The third harmonic oscillation in the stator zero-sequence component is a function of the third harmonic component of the magnetizing flux.

$$\lambda_{mq3} = \lambda_{mr3} \sin(3\gamma), \quad \lambda_{md3} = \lambda_{mr3} \cos(3\gamma). \quad (29)$$

In (28)–(29), the angle γ is the magnetizing current vector angle in qd reference frame (with respect to d -axis). Based on Fig. 3, this angle is defined as

$$\gamma = \theta_m - \theta_s + \frac{\pi}{2} = \tan^{-1} \left(\frac{i_{mq}}{i_{md}} \right). \quad (30)$$

In addition, the third harmonic component of the magnetizing flux linkage on the stator becomes zero-sequence after being transformed to $qd0$ coordinates using (11) and is expressed as

$$\lambda_{m0} = \lambda_{ms3} \sin(3\theta_m) = \lambda_{ms3} \sin(3\theta_s + 3\gamma - 3\pi/2). \quad (31)$$

Since the third harmonic component of the magnetizing flux is rotating with $3\theta_m$ in the stationary reference frame, the transformed component in zero-sequence contains oscillatory terms with three-times the stator electrical frequency as shown in (31).

Choosing the stator and rotor fluxes as the state variables, and stator and rotor voltages as the inputs, the state-space model can be derived using (15)–(25). This model can be written in the following form:

$$p\lambda_{qs} = v_{qs} - r_s L_{ls}^{-1} (\lambda_{qs} - \lambda_{mq}) - \omega_s \lambda_{ds}, \quad (32)$$

$$p\lambda_{ds} = v_{ds} - r_s L_{ls}^{-1} (\lambda_{ds} - \lambda_{md}) + \omega_s \lambda_{qs}, \quad (33)$$

$$p\lambda_{0s} = v_{0s} - r_s L_{ls}^{-1} (\lambda_{0s} - \lambda_{m0}), \quad (34)$$

$$p\lambda_{qr} = -r_r L_{lr}^{-1} (\lambda_{qr} - \lambda_{mq}) - (\omega_s - \omega_r) \lambda_{dr}, \quad (35)$$

$$p\lambda_{dr} = -r_r L_{lr}^{-1} (\lambda_{dr} - \lambda_{md}) + (\omega_s - \omega_r) \lambda_{qr}, \quad (36)$$

$$p\lambda_{qr3} = -r_r L_{lr}^{-1} (\lambda_{qr3} - \lambda_{mq3}) - 3(\omega_s - \omega_r) \lambda_{dr3}, \quad (37)$$

$$p\lambda_{dr3} = -r_r L_{lr}^{-1} (\lambda_{dr3} - \lambda_{md3}) + 3(\omega_s - \omega_r) \lambda_{qr3}. \quad (38)$$

In order to form an explicit state space model, the magnetizing flux components in (32)–(38) (i.e., λ_{mq} , λ_{md} , λ_{m0} , λ_{mq3} , λ_{md3}) are required to be expressed in terms of state variables, i.e., stator and rotor flux components. Substituting stator and rotor currents in (26) with the ones defined in (22) and (24), respectively, the magnetizing current and fundamental flux components can be grouped and expressed as

$$i_{mq} + (L_{ls}^{-1} + L_{lr}^{-1}) \lambda_{mq} = L_{ls}^{-1} \lambda_{qs} + L_{lr}^{-1} \lambda_{qr}, \quad (39)$$

$$i_{md} + (L_{ls}^{-1} + L_{lr}^{-1}) \lambda_{md} = L_{ls}^{-1} \lambda_{ds} + L_{lr}^{-1} \lambda_{dr}. \quad (40)$$

To implement the non-linear relation (due to saturation) between the machine magnetizing flux and current magnitudes in a numerically efficient way, the explicit flux correction approach in [7] is utilized. By defining intermediate variables i_{zq} and i_{zd} in terms of stator and rotor fluxes as

$$i_{zq} = L_{ls}^{-1} \lambda_{qs} + L_{lr}^{-1} \lambda_{qr} = i_{mq} + (L_{ls}^{-1} + L_{lr}^{-1}) \lambda_{mq}, \quad (41)$$

$$i_{zd} = L_{ls}^{-1} \lambda_{ds} + L_{lr}^{-1} \lambda_{dr} = i_{md} + (L_{ls}^{-1} + L_{lr}^{-1}) \lambda_{md}, \quad (42)$$

current magnitude i_z can be defined using i_{zq} and i_{zd} as

$$i_z = \sqrt{i_{zq}^2 + i_{zd}^2} = \sqrt{(i_{mq} + (L_{ls}^{-1} + L_{lr}^{-1}) \lambda_{mq})^2 + (i_{md} + (L_{ls}^{-1} + L_{lr}^{-1}) \lambda_{md})^2}. \quad (43)$$

Since in an induction machine the magnetizing current and flux vectors are aligned [2], the intermediate current magnitude i_z can be expressed in terms of magnetizing flux and current magnitude as magnitude [7] as

$$i_z = i_m + (L_{ls}^{-1} + L_{lr}^{-1}) \lambda_{m1}. \quad (44)$$

By writing the magnetizing current in (44) in terms of magnetizing flux using (2), the intermediate variable i_z can be expressed as a non-linear function [denoted here as $F_z^{-1}(\cdot)$] of the machine magnetizing flux as

$$i_z = F_z^{-1}(\lambda_{m1}) + L_{lr}^{-1} \lambda_{m1} = F_z^{-1}(\lambda_{m1}). \quad (45)$$

The calculated fundamental component of the magnetizing flux magnitude in (45) can then be used to obtain the third harmonic components of both rotor and stator magnetizing flux linkage magnitudes according to (10) and (6), respectively. Moreover, by noting the fact the intermediate current vector defined using i_{zq} and i_{zd} in (41)–(42) are also aligned with the magnetizing current and flux vectors, the magnetizing flux components in (32)–(38) can ultimately be expressed in terms of i_z and magnetizing current angle γ using (28)–(31) and (45) as

$$\lambda_{mq} = F_z(i_z) \sin(\gamma), \quad \lambda_{md} = F_z(i_z) \cos(\gamma), \quad (46)$$

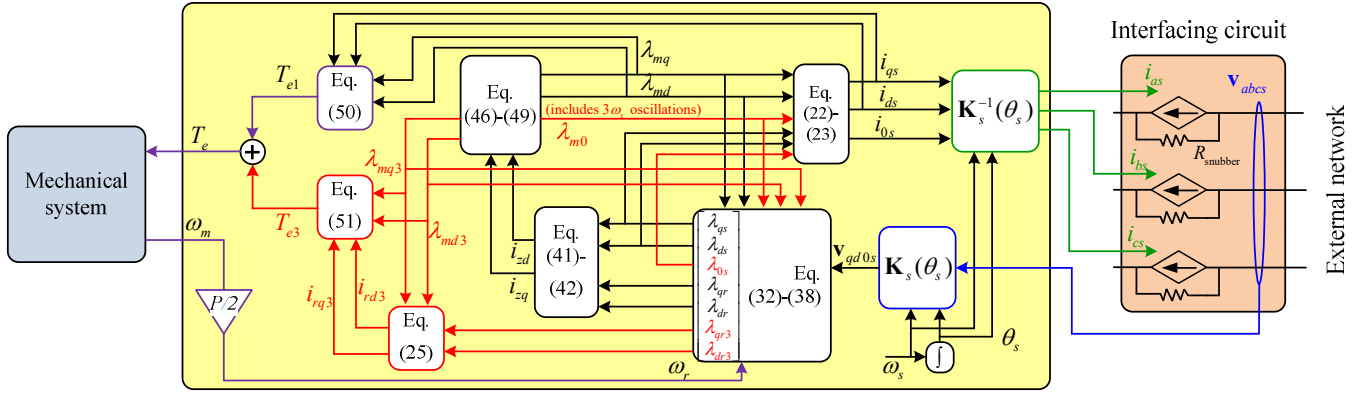


Fig. 5. Implementation and interfacing of the proposed saturable $qd0$ model in state-variable-based EMT simulation program. The additional calculations required for the third harmonic saturation modeling are highlighted in red, while black lines represent the conventional IM model implementation.

$$\lambda_{mq3} = G_{r3}(\lambda_{m1})\sin(3\gamma), \quad (47)$$

$$\lambda_{md3} = G_{r3}(\lambda_{m1})\cos(3\gamma), \quad (48)$$

$$\lambda_{m0} = G_{s3}(\lambda_{m1})\sin(3\theta_s + 3\gamma), \quad (49)$$

Since the intermediate variables i_{zq} and i_{zd} are functions of the state variables (as expressed in (41)–(42)), the magnetizing flux components in (46)–(49) can be used in (32)–(38) directly to form an explicit state-space model. Therein, stator and rotor current components are the outputs, which are defined in terms of machine fluxes in (22)–(25), and are used to interface the machine model with external network.

Based on (15)–(21), the $qd0$ equivalent circuit in synchronous reference frame is depicted in Fig. 4. In Fig. 4, for the sake of simplicity, the magnetizing branches have been represented with controlled voltage sources corresponding to respective mutual flux derivatives. This is also done since due to saturation, there are not fixed values of inductors to represent the corresponding magnetizing branches in the circuit. Instead, in the flux correction approach, the effect of variable/saturable inductances is implemented using fluxes (which also avoids the derivatives of inductances that would be required otherwise [24]).

As explained in Section II-A, the coupling between the fundamental and third harmonic components of magnetizing flux is also depicted in Fig. 4. It is shown that third harmonic components of the stator back-emf determines the dependent voltage source in the zero-sequence circuit (see Fig. 4).

Finally, the electromagnetic torque produced by the IM is comprised of the fundamental and third harmonic components of the air-gap flux, respectively, which are defined [12] as

$$T_{e1} = \left(\frac{P}{2}\right)\left(\frac{3}{2}\right)(\lambda_{md}i_{qs} - \lambda_{mq}i_{ds}), \quad (50)$$

$$T_{e3} = \left(\frac{P}{2}\right)\left(\frac{9}{2}\right)(\lambda_{mq3}i_{dr3} - \lambda_{md3}i_{qr3}), \quad (51)$$

$$T_e = T_{e1} + T_{e3}. \quad (52)$$

The torque T_{e1} in (50) is resulting from the fundamental component of the magnetizing flux and stator currents in q - and d - axes. Also, the third harmonic component of the magnetizing flux links with the rotor third harmonic component, which produces an additional torque T_{e3} in (51) in the same direction as of the fundamental component, contributing to the total

produced torque (52).

The implementation and interfacing of the proposed model is depicted in Fig. 5. To interface the proposed $qd0$ model with external network, the stator current $qd0$ components (which are the outputs) are transformed back to stator abc phase coordinates using invers Park's transformation $\mathbf{K}_s^{-1}(\theta_s)$ [2], and are then used as controlled current sources at the machine's interface. It is noted that snubber resistors are connected in parallel to the controlled current sources, as shown in Fig. 5, in order to obtain a compatible interface to external inductive or

power-electronic networks [18]. The value of snubbers is typically chosen high to keep the current through the snubber branch below a certain percentage (e.g. 1%) of the nominal machine's current.

III. MODEL VALIDATION

To verify the proposed model, a 230V 0.25hp squirrel-cage induction machine is considered for experimental measurements and validation. The machine parameters [25] along with the saturation characteristics for the fundamental and third harmonic of the magnetizing flux have been extracted according to the method presented Appendix and are summarized in Tables I–II therein. The extracted magnetizing flux characteristics are also shown in Fig. 6. The proposed model is implemented in MATLAB Simulink [21] using PLECS blockset [19], wherein the value of interfacing snubber resistors is chosen to be 28k Ω (corresponding to 100pu for the considered IM). It is assumed that IM is supplied from a three-phase grid (represented by voltage sources $\mathbf{v}_{abcg}$) behind Thévenin equivalent impedance (represented by r_{line} and L_{line}). The simulations are run on a PC with Intel® Core™ i7-8700K @ 3.7GHz processor using MATLAB ode15s variable step solver with relative and absolute error tolerances set to 10^{-4} .

A. Steady-State Operation

To see the effect of saturation including air-gap flux harmonics, three steady state operating points are considered in which the subject model is compared to hardware measurements under different stator winding configurations, in both no-load and full-load conditions. For more accurate comparison, the actual measured grid voltages have been recorded and fed into the simulation (instead of ideal sinusoidal sources). It is also worth mentioning that friction and other losses have been considered as equivalent load torque in

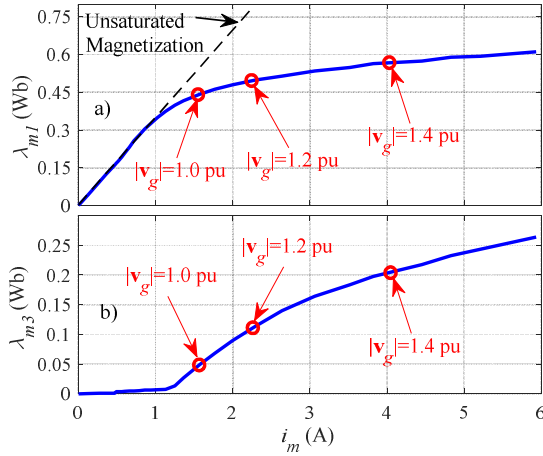


Fig. 6. Measured magnetization curves of the considered IM demonstrating the three considered operating points for: (a) fundamental component of flux, (b) third harmonic magnetizing flux component.

simulations. The voltage waveforms for each case-study were recorded under the same loading condition as in the simulation.

The studies are done under three different voltage levels, corresponding to nominal operating point [phase rms voltage $|v_g| = 133\text{V}$ (1pu)], slight saturation [phase rms voltage $|v_g| = 160\text{V}$ (1.2pu)], and deep saturation [phase rms voltage $|v_g| = 186\text{V}$ (1.4pu)], respectively. The variable voltages in hardware experiments have been produced using a three-phase auto-transformer. The considered operating points are shown in Fig. 6, depicting the fundamental and third harmonic magnetizing flux curves.

A-I) Wye-Connected Stator with Floating Neutral

In this study, the stator windings are wye-connected with floating neutral point as shown in Fig. 7. The simulation results obtained by the proposed model as well as the experimental measurements for the stator phase *a* voltage and current under no-load and 1Nm mechanical load conditions are depicted in Figs. 8–9, respectively.

As it can be seen in Figs. 8–9, the proposed model is able to accurately predict the measured variables. It is also seen that increasing the stator voltage leads to excessive stator current as a result of magnetic saturation. Moreover, it is also observed that higher stator voltages result in deeper saturation and create more pronounced distortions in phase voltages (which is a result of flux spatial harmonics).

Since the neutral point is floating, the stator currents do not contain 3rd harmonic (which correspond to zero sequence), and higher order harmonic distortions are mainly due to the non-ideal grid voltage containing such components.

A-II) Wye-Connected Stator with Neutral Connected

The same steady state scenario as in Section III-A-I is repeated with the neutral wire being connected to the source neutral point (also through impedance r_{line} and L_{line}), as shown in Fig. 10. The simulated and measured stator phase *a* voltage and current under both no-load and loaded operation are presented in Figs. 11–12, respectively.

As seen in Figs. 11–12, the increased stator voltages lead to considerable increase in stator current due to saturation, which is consistent between the model and measured results.

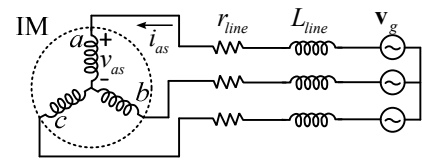


Fig. 7. Wye-connected IM with floating neutral supplied by the ac source.

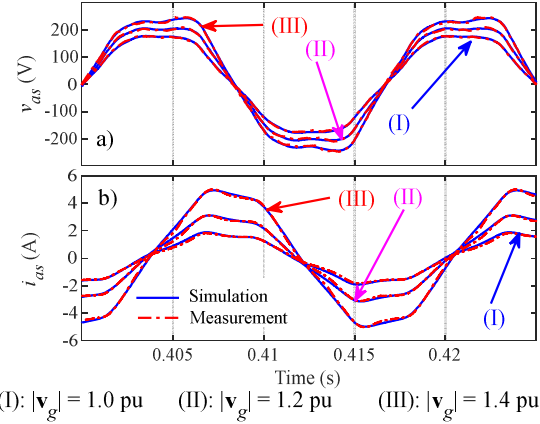


Fig. 8. Steady state waveforms of wye-connected IM with floating neutral at no-load condition under different stator voltages, as obtained by the proposed model and experiments for: (a) stator phase *a* voltage v_{as} , and (b) stator phase *a* current i_{as} .

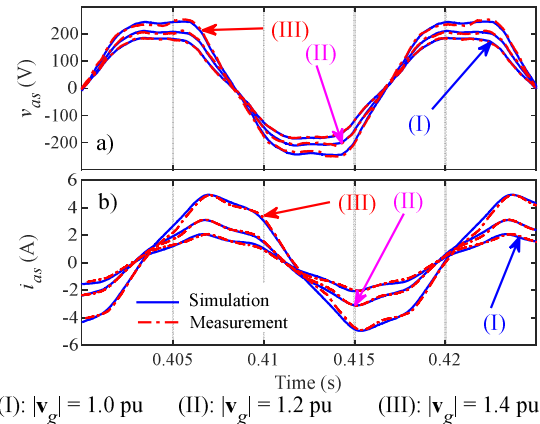


Fig. 9. Steady state waveforms of wye-connected IM with floating neutral at full-load condition under different stator voltages, as obtained by the proposed model and experiments for: (a) stator phase *a* voltage v_{as} , and (b) stator phase *a* current i_{as} .

However, as seen in Figs. 11(a) and 12(a), the addition of the neutral wire largely eliminates the third harmonic distortions from the stator phase voltages compared to the case with no neutral wire (due to small line impedances), and now provides the path for the third harmonic current to flow. Moreover, the third harmonic currents increase very rapidly with the increased phase voltage (due to third harmonic back-emfs in the stator windings).

A-III) Delta-Connected Stator Windings

As an extra simulation study, the stator windings are re-configured into Δ , as illustrated in Fig. 13. To match the same operating points shown in Fig. 6, the source phase voltages are reduced to the following: nominal operating point [phase rms voltage $|v_g| = 77\text{V}$ (1pu)], slight saturated [phase rms voltage $|v_g| = 92\text{V}$ (1.2pu)], and deep saturation [phase rms voltage $|v_g| = 108\text{V}$ (1.4pu)], respectively. Also, the same 1Nm mechanical load is considered.

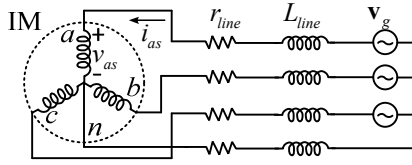
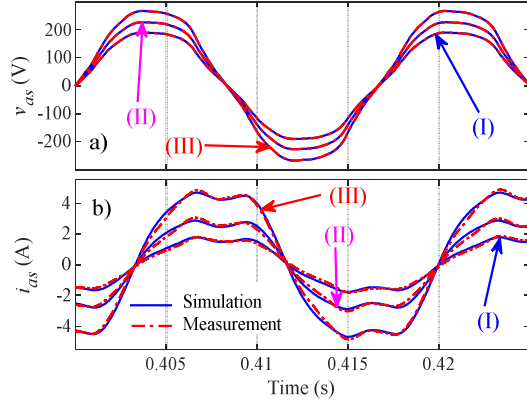
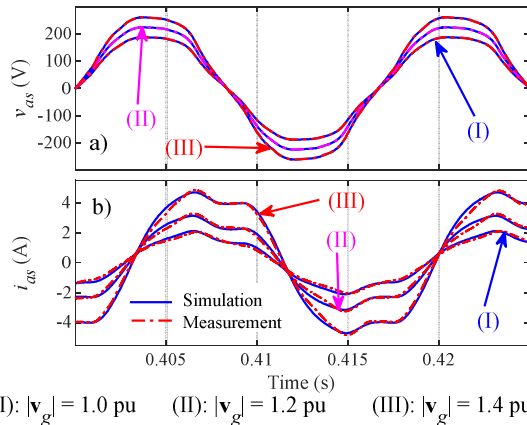


Fig. 10. Wye-connected IM with neutral wire connected to the ac source.



(I): $|v_g| = 1.0$ pu (II): $|v_g| = 1.2$ pu (III): $|v_g| = 1.4$ pu

Fig. 11. Steady state waveforms of wye-connected IM with neutral connected at no-load under different stator voltages, as obtained by the proposed model and experiments for: (a) stator phase a voltage v_{as} , and (b) stator phase a current i_{as} .



(I): $|v_g| = 1.0$ pu (II): $|v_g| = 1.2$ pu (III): $|v_g| = 1.4$ pu

Fig. 12. Steady state waveforms of wye-connected IM with neutral connected at loaded condition under different stator voltages, as obtained by the proposed model and experiments for: (a) stator phase a voltage v_{as} , and (b) stator phase a current i_{as} .

The simulated results for the stator phase a voltage and current are presented in Fig. 14–15 for no-load and loaded operating conditions, respectively. As observed in Figs. 14 and 15, as the stator voltage increases, the phase current becomes larger in amplitude and more distorted due to 3rd harmonic currents circulating through the Δ -connection. Since the laboratory IM does not provide the option for a Δ -connection, only simulation results are provided for this case.

A. Transient Study

For validating the proposed model under transients, two studies have been conducted including a startup transient and a voltage increase transient. In the startup transient, the machine stator is connected in a wye configuration including the neutral wire as depicted in Fig. 10.

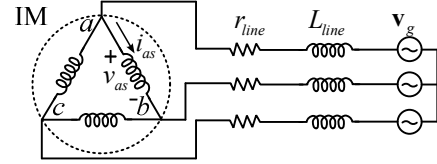
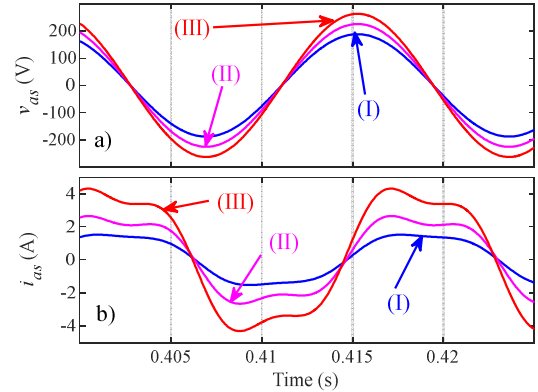
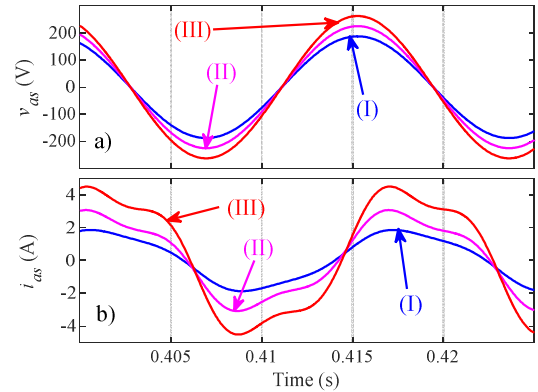


Fig. 13. Delta-connected IM supplied by the ac source.



(I): $|v_g| = 1.0$ pu (II): $|v_g| = 1.2$ pu (III): $|v_g| = 1.4$ pu

Fig. 14. Steady state waveforms of Δ -connected IM at no-load condition under different stator voltages, as obtained by simulations using the proposed model for: (a) stator phase a voltage v_{as} , and (b) stator phase a current i_{as} .



(I): $|v_g| = 1.0$ pu (II): $|v_g| = 1.2$ pu (III): $|v_g| = 1.4$ pu

Fig. 15. Steady state waveforms of Δ -connected IM at full-load condition under different stator voltages, as obtained by simulations using the proposed model for: (a) stator phase a voltage v_{as} , and (b) stator phase a current i_{as} .

The machine is initially standing still and is mechanically coupled to an unloaded DC dynamometer/generator. At $t = 0.99$ s, the stator is energized by a three-phase voltage source with a phase voltage of 146V (1.1pu) and the rotor begins to accelerate until it reaches a steady state, and stays in this state until the end of the simulation. The transient simulation results for the machine phase a current, phase a voltage, electromagnetic torque and mechanical speed as obtained by both the proposed model and hardware measurements are presented in Fig. 16. For consistency, the recorded grid phase voltage (see Fig. 16(a)) was also applied as the input to the simulated model.

As it can be seen in Fig. 16, the proposed model produces similar and consistent results with respect to the experimental measurements. In Fig. 16(b), it is observed that the machine draws high currents during the start-up with minimal harmonic distortions. This is due to fact that the magnetizing flux inside

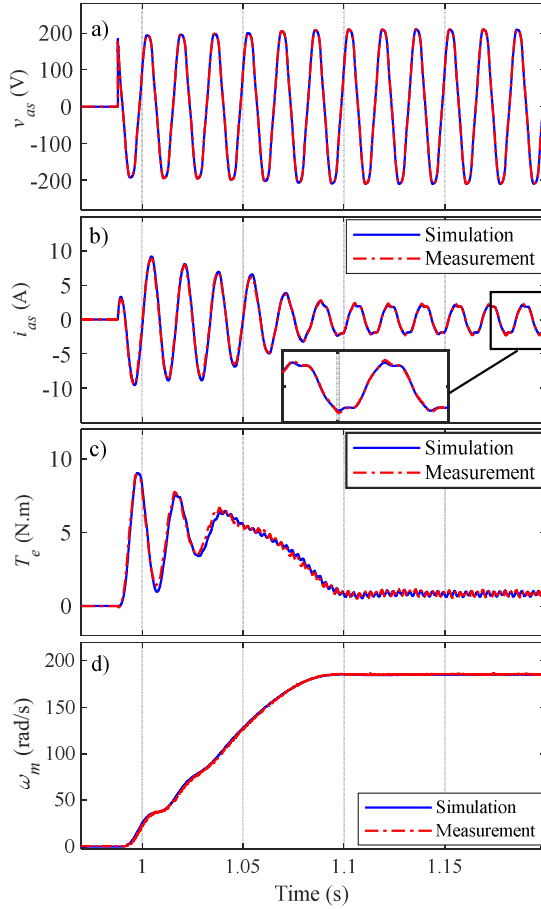


Fig. 16. Transient response of several variables of the IM with wye-connected stator with neutral wire during startup transient as obtained by the proposed saturable $qd0$ model and hardware measurements for: (a) stator phase a voltage, (b) stator phase a current, (c) electromagnetic torque, and (d) rotor speed.

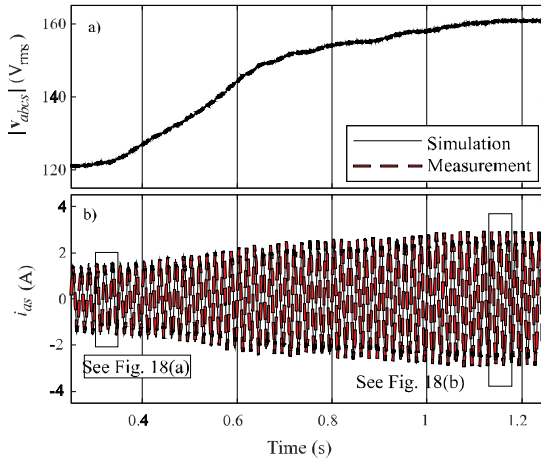


Fig. 17. Transient response of the IM with wye-connected stator with neutral wire during voltage increase as obtained by the proposed saturable $qd0$ model and hardware measurements for: (a) stator rms phase voltage, and (b) stator phase a current.

the machine does not change instantly and reaches the saturated levels (corresponding to 1.1pu stator phase voltage) only after some time. When the machine reaches steady-state, the third harmonic distortions begin to appear on the stator currents due to the neutral wire. Furthermore, Figs. 16(c)–(d) show the dynamic behavior of the machine electromagnetic torque and speed during this transient study.

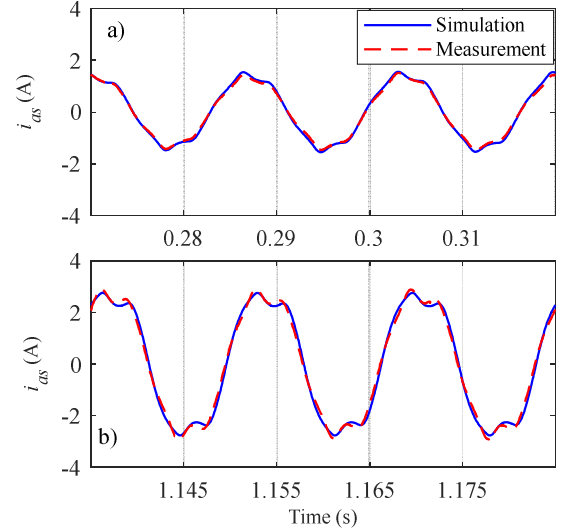


Fig. 18. Magnified view of stator phase a current corresponding to before and after voltage increase transient for when (a) stator rms phase voltage is 0.9pu, and (b) stator rms phase voltage is 1.2pu.

In the next transient, the machine is wye-connected with neutral wire, as depicted in Fig. 10, and the same mechanical load is present in the form of DC generator supplying resistive load. Initially, the stator phase voltage is 120V rms (0.9pu). At $t = 0.35$ s, the voltage is ramped up to 1.2pu which is achieved at $t = 1.1$ s, and the simulation continues. The resulting phase a current and rms voltage are shown in Fig. 17. It is noted that the applied voltage is supplied by an adjustable variac transformer, and the increase ramp (in Fig. 17(a)) is due to a manual operation. As can be seen in Fig. 17(b), the proposed model accurately predicts the experimental measurements.

Also, as a result of increased stator voltage, the stator phase current undergoes a significant increase in amplitude (due to main flux saturation) and distortions (due to the 3rd harmonic flux component). For a clearer comparison of stator current waveforms under 0.9pu and 1.2pu voltage, the magnified views from Fig. 17(b) are also provided in Fig. 18. This also to further clarifies the agreement between the proposed model and the actual machine.

IV. CONCLUSIONS

It has been shown in literature that air-gap flux distortions caused by magnetic saturation may be considerable in practical operation of induction machines. The saturation-caused air-gap flux harmonics have been previously considered in coupled circuit phase domain models and implicit $qd0$ models, which are hard to implement in commonly used EMT simulation programs. In this paper, a new and explicit $qd0$ model is proposed for including the magnetizing flux saturation in terms of its fundamental and 3rd harmonic components. The proposed model has been validated using extensive simulations and experimental studies and shown to provide accurate results. Due to its straightforward and explicit formulation, it is envisioned that the new model may become a useful addition to many EMT simulation programs (e.g., possibly as an extension to traditional/classical qd models) for providing higher fidelity when needed.

APPENDIX A

TABLE I. MEASURED INDUCTION MACHINE PARAMETERS [25]

Parameters	Value
Rated Voltage / Power / Poles	230 V / 0.25 hp / 4
L_m (unsaturated), L_{ls} , L_{lr}	350.1 mH, 26 mH, 11.1 mH
r_s , r_r	6.5 Ω , 6.6 Ω
Winding factors k_{w1} , k_{w3}	0.90, 0.33

TABLE II. MEASURED MAGNETIC SATURATION DATA

i_m (A)	0	0.12	0.27	0.35	0.47	0.49	0.51	0.56	0.6	0.66
λ_{m1} (mWb)	0	41.3	92.9	120.4	160.9	167.6	175	192	207	232.2
λ_{m3} (mWb)	0	0.6	0.9	0.9	0.9	3.3	3.6	3.6	3.9	4.5
i_m (A)	0.72	0.79	0.86	0.94	1.02	1.13	1.25	1.36	1.57	1.77
λ_{m1} (mWb)	256.7	280	304.3	326.1	349.3	374.6	398.3	416.2	442	461.6
λ_{m3} (mWb)	4.5	5.4	6	6.3	6.9	7.5	13.2	26.7	48.9	68.4
i_m (A)	2	2.3	2.64	3.06	3.52	3.82	4.45	4.83	5.21	5.93
λ_{m1} (mWb)	479.5	499.4	514.3	534.8	548.3	563.5	574.3	589.8	593.2	610.7
λ_{m3} (mWb)	90	114.3	139.8	164.4	183.6	197.7	217.8	232.5	243.3	263.7

In Table II, i_m , λ_{m1} , λ_{m3} correspond to the magnitudes of the magnetizing current, and fundamental and third harmonic components of magnetizing flux.

Line Impedance: $r_{line} = 0.04 \Omega$, $L_{line} = 300 \mu H$.

APPENDIX B

Obtaining Magnetizing Flux Saturation Characteristics:

The experimental setup used to obtain the saturation characteristic (summarized in Table II) is shown in Fig. 19. For this purpose, the induction machine is operated under no-load and is rotating at almost synchronous speed (i.e., slip $s \approx 0$). Based on this assumption, the rotor currents are also considered to be zero. The machine is supplied from a three-phase variable voltage source that can provide voltages in a wide range. It is important to note that access to the stator neutral point is required to establish the voltage measurement. Moreover, since the neutral wire is not connected to the source, the stator currents will not contain zero sequence and third harmonic components ($I_{s3} = 0$).

The instantaneous phase voltages $\mathbf{v}_{abcs}(t)$ and currents $\mathbf{i}_{abcs}(t)$ are recorded using the measurement probes and logging devices that are connected at the machine's terminals. Using the recorded stator phase voltages and currents, the fundamental and third harmonic components may be extracted using Fourier transform (i.e., using FFT function in MATLAB with appropriate calculations, which would be more accurate), or using simple rms measurements and calculations (less accurate).

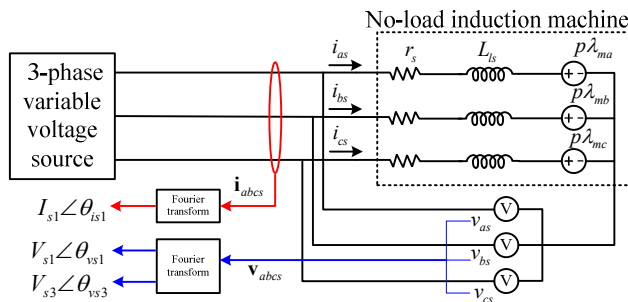


Fig. 19. Measurement setup for obtaining magnetizing flux fundamental and third harmonic saturation data.

Specifically, assuming phase a as a reference, based on the recorded $v_{as}(t)$ and $i_{as}(t)$ for a considered input voltage of the source, their corresponding fundamental component phasors (but with peak values as magnitude instead of traditional rms values, and corresponding angles) are extracted as $\tilde{V}_{s1}$ and $\tilde{I}_{s1}$, respectively. Since the stator resistance r_s and leakage inductance L_{ls} are known from previous tests [25], it becomes possible to calculate the fundamental component phasor of the voltage across the magnetizing branch, as

$$\tilde{E}_{s1} = \tilde{V}_{s1} - (r_s + j\omega_s L_{ls}) \tilde{I}_{s1}. \quad (A1)$$

The fundamental component of the flux and magnetizing current are then obtained as

$$\lambda_{m1} = \frac{|\tilde{E}_{s1}|}{\omega_s}, \quad (A2)$$

$$i_m = |\tilde{I}_{s1}|. \quad (A3)$$

Similarly, the 3rd harmonic of the phase voltage is extracted from $v_{as}(t)$ using Fourier transform, and its phasor, denoted by $\tilde{V}_{s3}$ (with peak as its magnitude). Since $I_{s3} = 0$, according to the zero-sequence voltage and flux equations in (17), (31) and (34), the zero-sequence component of the stator magnetizing flux linkage can be calculated as

$$\lambda_{ms3} = \frac{|\tilde{V}_{s3}|}{3\omega_s}. \quad (A4)$$

It is noted that since the third harmonic voltages are being measured from the stator terminals, according to (6) the winding factor ratio $k_{w3}/3k_{w1}$ is required to obtain the total third harmonic component of the magnetizing flux (i.e., λ_{m3}) as

$$\lambda_{m3} = \frac{3k_{w1}}{k_{w3}} \lambda_{ms3} = \frac{k_{w1}}{k_{w3}} \frac{|\tilde{V}_{s3}|}{\omega_s}. \quad (A5)$$

Alternatively, the 3rd harmonic flux magnitude can also be calculated using zero sequence voltage $v_{0s}(t) = (v_{as}(t) + v_{bs}(t) + v_{cs}(t))/3$. Then, the corresponding total third harmonic flux magnitude is obtained similarly as (A4)–(A5) as

$$\lambda_{m3} = \frac{k_{w1}}{k_{w3}} \frac{V_{0s}}{\omega_s}, \quad (A6)$$

where V_{0s} is the amplitude of $v_{0s}(t)$. The measurements are repeated for several points spanning a range of input voltages from approximately 0.5pu to 1.5pu, from which the data in Appendix A, Table II has been obtained. Note that the values of magnetizing current here are the peak (not rms), which is needed for $qd0$ model.

REFERENCES

- [1] D. de Almeida Souza, W. C. P. de Aragão Filho, and G. Costa Drumond Sousa, "Adaptive fuzzy controller for efficiency optimization of induction motors," *IEEE Trans. Ind. Electron.*, vol. 54, no. 4, pp. 2157–2164, Aug. 2007.
- [2] P. C. Krause, O. Wasynczuk, S. D. Sudhoff, and S. Pekarek, *Analysis of electric machinery and drive systems*, 3rd Edition, IEEE Press, Piscataway, NJ, 2013.

- [3] S. D. Sudhoff, D. C. Aliprantis, B. T. Kuhn, and P. L. Chapman, "An induction machine model for predicting inverter-machine interaction," *IEEE Trans. Energy Convers.*, vol. 17, no. 2, pp. 203–210, Jun. 2002.
- [4] E. Levi, "A unified approach to main flux saturation modeling in d-q axis models of induction machines," *IEEE Trans. Energy Convers.*, vol. 10, no. 3, pp. 455–461, Sep. 1995.
- [5] T. A. Lipo and A. Consoli, "Modeling and simulation of induction motors with saturable leakage reactances," *IEEE Trans. Ind. Appl.*, vol. IA-20, no. 1, pp. 180–189, Jan./Feb. 1984.
- [6] E. Levi, "Main flux saturation modelling in double-cage and deepbar induction machines," *IEEE Trans. Energy Convers.*, vol. 11, no. 2, pp. 305–311, Jun. 1996.
- [7] F. Therrien, L. Wang, J. Jatskevich, and O. Wasynczuk, "Efficient explicit representation of AC machines main flux saturation in state-variable-based transient simulation packages," *IEEE Trans. Energy Convers.*, vol. 28, no. 2, pp. 380–393, Jun. 2013.
- [8] J. C. Moreira and T. A. Lipo, "Modeling of saturated AC machines including air-gap flux harmonic components," *IEEE Trans. Ind. Appl.*, vol. 28, no. 2, pp. 343–349, Mar./Apr. 1992.
- [9] Y. Liao and T. Lipo, "Effect of saturation third harmonic on the performance of squirrel-cage induction machines," *Electr. Mach. Power Syst.*, vol. 22, no. 2, pp. 155–171, 1994.
- [10] V. Donescu, A. Charette, Z. Yao, and V. Rajagopalan, "Modeling and simulation of saturated induction motors in phase quantities," *IEEE Trans. Energy Convers.*, vol. 14, no. 3, pp. 386–393, Sep. 1999.
- [11] D. Bispo, L. M. Neto, J. Tarcisio de Resende, and D. Alves de Andrade, "A new strategy for induction machine modeling taking into account the magnetic saturation" *IEEE Trans. Ind. Appl.*, vol. 37, no. 6, pp. 1710–1719, Nov/Dec. 2001.
- [12] X. Tu, L. Dessaint, R. Champagne, and K. Al-Haddad, "Transient modeling of squirrel-cage induction machine considering air-gap flux saturation harmonics," *IEEE Trans. Ind. Electron.*, vol. 55, no. 7, pp. 2798–2809, Jul. 2008.
- [13] L. Kreindler, J. C. Moreira, A. Testa and T. A. Lipo, "Direct field orientation controller using the stator phase voltage third harmonic" *IEEE Trans. Ind. Appl.*, vol. 30, no. 2, pp. 441–447, Mar/Apr. 1994.
- [14] F. Profumo, G. Griva, M. Pastorelli, J. Moreira and R. De Doncker, "Universal field oriented controller based on air gap flux sensing via third harmonic stator voltage," *IEEE Trans. Ind. Appl.*, vol. 30, no. 2, pp. 448–455, Mar/Apr. 1994.
- [15] J. Holtz, "Sensorless control of induction machines—With or without signal injection?" *IEEE Trans. Ind. Electron.*, vol. 53, no. 1, pp. 7–30, Feb. 2006.
- [16] C. Caruana, G. M. Asher, and M. Sumner, "Performance of HF signal injection techniques for zero-low-frequency vector control of induction machines under sensorless conditions," *IEEE Trans. Ind. Electron.*, vol. 53, no. 1, pp. 225–238, Feb. 2006.
- [17] J. R. Marti and K. W. Louie, "A phase-domain synchronous generator model including saturation effects," *IEEE Trans. Power Syst.*, vol. 12, no. 1, pp. 222–229, Feb. 1997.
- [18] L. Wang *et al.*, "Methods of interfacing rotating machine models in transient simulation programs," *IEEE Trans. Power Delivery*, vol. 25, no. 2, pp. 891–903, Apr. 2010.
- [19] N. Amiri, S. M. Ebrahimi, M. Chapariha, J. Jatskevich and H. W. Dommel, "Constant parameter voltage-behind-reactance model of six-phase synchronous machines," *IEEE Trans. on Energy Convers.*, vol.32, no.2, pp. 548–559, Jun 2017.
- [20] (2018). *Piecewise Linear Electrical Circuit Simulation (PLECS) User Manual, Version 3.7* (Plexim GmbH). [Online]. Available: www.plexim.com
- [21] (2018). *Simulink Dynamic System Simulation Software Users Manual* (MathWorks Inc.). [Online]. Available: <http://www.mathworks.com>
- [22] (2018). *Simscape Electrical User's Guide (Electronics, Mechatronics, and Power Systems)* (The MathWorks Inc.). [Online]. Available: <http://www.mathworks.com>
- [23] A. A. Rockhill and T. A. Lipo, "A generalized transformation methodology for polyphase electric machines and networks," in *Proc. 2015 IEEE International Electric Machines & Drives Conference (IEMDC)*, Coeur d'Alene, ID, USA, May 2015, pp. 1–8.
- [24] E. Levi, "A unified approach to main flux saturation modeling in d-q axis models of induction machines," *IEEE Trans. Energy Convers.*, vol. 10, no. 3, pp. 455–461, Sep. 1995.
- [25] S. Chiniforoosh, L. M. Vargas, L. Wang and J. Jatskevich, "Online characterization procedure for induction machines using start-up and loading transients," in *Proc. IEEE Canada Electric Power Conference*, Vancouver, BC, 2008, pp. 1–5.



Navid Amiri (S'11 – M'19) received his B.Sc. and M.Sc. degrees in electrical engineering in the field of power and electrical machines from Isfahan University of Technology, Isfahan, Iran, in 2008 and 2011, and his Ph.D. degree in electrical and computer engineering in 2019 from the University of British Columbia, Vancouver, BC, Canada. He is currently a Postdoctoral Research Fellow in electrical and computer engineering department at the University of British Columbia. His research interests include numerically efficient modeling of electric machines, real-time simulation, electromechanical energy conversion systems, electric machine design, and power electronics.



Seyyedmilad Ebrahimi (S'13–M'19) received the B.Sc. and M.Sc. degrees in electrical engineering in 2010 and 2012 from Sharif University of Technology, Tehran, Iran, and the Ph.D. degree in electrical and computer engineering in 2019 from the University of British Columbia, Vancouver, BC, Canada. He is currently a Postdoctoral Research Fellow in electrical and computer engineering department at the University of British Columbia. His research interests include modeling and analysis of power electronic converters and electrical machines, application of power electronics to power systems, modeling and control of power systems, and simulation of electromagnetic transients.



Juri Jatskevich (M'99–F'17) received the M.S.E.E. and the Ph.D. degrees in electrical engineering from Purdue University, West Lafayette IN, USA, in 1997 and 1999, respectively. Since 2002, he has been at The University of British Columbia, Vancouver, BC, Canada, where he is a Professor in the Department of Electrical and Computer Engineering. His research interests include power electronic systems, electrical machines and drives, and modeling and simulation of electromagnetic transients. Dr. Jatskevich chaired the IEEE CAS Power Systems and Power Electronic Circuits Technical Committee (2009–2010), and had served as an Associate Editor for the IEEE TRANSACTIONS ON POWER ELECTRONICS (2008–2013), the Editor-in-Chief of the IEEE TRANSACTIONS ON ENERGY CONVERSION (2013–2019), and the Editor-in-Chief At-Large for the IEEE PES journals (2019/2020). He was the General Chair for the 2015 IEEE Control and Modeling for Power Electronics (COMPEL) conference. He is also chairing the IEEE Task Force on Dynamic Average Modeling, under Working Group on Modeling and Analysis of System Transients Using Digital Programs.

A VBR Induction Machine Model Implementation for SimPowerSystem Toolbox in Matlab-Simulink

Liwei Wang, *Student Member, IEEE*, Juri Jatskevich, *Senior Member, IEEE*, and Sina Chini Foroosh

Abstract—This paper presents a Voltage-Behind-Reactance (VBR) induction machine model for the Matlab-Simulink toolbox SimPowerSystem, which is widely used for simulations of power and power electronics systems. In the proposed VBR model, the three-phase stator branches are represented by decoupled and constant R - L branches behind voltage sources, which provide a direct interface of the model with the external network-circuit. The rotor subsystem is expressed in the qd coordinates resulting in a high numerical efficiency. The VBR model has full-order and is otherwise algebraically equivalent to the classical qd model. We present case studies of induction machine interfaced with resistive and inductive networks. The studies show that the proposed model significantly improves the machine-network interface and enhances the numerical accuracy and efficiency as compared with the built-in SimPowerSystem qd machine model for the discretized solution of electrical circuit.

Index Terms—Induction machine, qd model, numerical simulation, SimPowerSystem, voltage-behind-reactance, vbr.

I. INTRODUCTION

DYNAMIC simulation of electric machines under various operating conditions is very valuable for power system analysts. The modeling techniques of the electrical machines and the associated software tools are the keys fulfilling those needs. As a widely-used modeling and simulation language, Matlab-Simulink [1] provides a powerful computational engine with many integrated mathematical functions and numerical integration solvers. Based upon the Matlab-Simulink platform, numerous toolboxes and software packages [2]–[5] have been developed specifically for power- and power-electronic systems simulations. SimPowerSystem [2], as one of these applications, integrates a versatile library of electrical machines, power electronics and many other power system components with the advanced analysis and design tools readily available in Matlab-Simulink. The

simulation methodology behind SimPowerSystem is based on state variable approach wherein the global state-space equations of the electrical and control systems are formulated and solved by Simulink.

The induction machine models implemented in SimPowerSystem are based on the well-recognized qd model in arbitrary reference frame [6]. Due to their nonlinear nature, the machine models are represented in the form of voltage-controlled current sources that are outside of the SimPowerSystem's state-space representation of the linear electrical network [7]. Such current-source-based machine-network interface is effective as long as the machines are connected with the non-inductive network. To interface the machine model with an external inductive network, the artificial numerical snubber circuits are often introduced at the machine terminals to avoid the connection of current sources in series with the inductive branches [8]. This indirect interface between the machine model and the network is further complicated by the transformations between the qd and abc coordinates and may deteriorate the overall simulation accuracy and efficiency.

This paper presents a Voltage-Behind-Reactance (VBR) induction machine model implemented in SimPowerSystem in Matlab-Simulink. In the proposed model, the machine's stator is represented as three-phase controlled voltage sources behind constant and decoupled R - L branches that are part of the external circuit and thus form a direct interface. To improve the simulation efficiency, the rotor subsystem is expressed in the arbitrary qd reference frame as a proper state variable model that is outside of the main electrical circuits. To demonstrate the advanced features of this modeling approach, we present numerous case studies of an induction machine interfaced with resistive and inductive networks. When the external network is inductive, the interface with the traditional qd induction machine model in SimPowerSystem (and other state-variable-based toolboxes) may be achieved using auxiliary snubber circuits. However, the VBR model does not require such artificial snubbers and significantly improves the numerical accuracy and efficiency of the simulation as compared with the built-in SimPowerSystem models for the discretized solution of electrical circuit.

Manuscript received Nov. 30, 2007. This work was supported in part by the Natural Sciences and Engineering Research Council (NSERC) of Canada under the Discovery Grant, and in part by the Power Engineering Grant-In-Aid from British Columbia Transmission Corporation and BC Hydro.

L. Wang, J. Jatskevich, and S. Chini Foroosh are with the Department of Electrical and Computer Engineering, University of British Columbia, Vancouver, BC, V6T 1Z4, Canada (e-mails: liweiw@ece.ubc.ca; jurij@ece.ubc.ca; sinach@ece.ubc.ca).

II. MACHINE MODELS

Without loss of generality, this paper assumes a 3-phase symmetrical induction machine with sinusoidally distributed stator and rotor windings [6]. For convenience of analysis, all rotor quantities are referred to the stator side by appropriate turns-ratio. Motor convention is used in the paper which assumes positive machine currents flowing into the machine stator and rotor windings. For the purpose of discussion, two types of models are reviewed in this section, namely, the qd model and the VBR model. For both models, the mechanical dynamics is assumed to be defined by a single rigid body system described by the following equations:

$$p\theta_r = \omega_r \quad (1)$$

$$p\omega_r = \frac{P}{2J}(T_e - T_m) \quad (2)$$

Here, the operator $p = d/dt$; θ_r and ω_r are the rotor position and angular electrical speed, respectively; and T_m and T_e are the combined mechanical torque and the developed electromagnetic torque, respectively.

A. The qd Model

Here, to simplify the analysis of machine dynamics, the induction machine abc variables are transformed into the qd arbitrary reference frame. The voltage equations of the qd model are given as follows:

$$v_{qs} = r_s i_{qs} + \omega \lambda_{ds} + p \lambda_{qs} \quad (3)$$

$$v_{ds} = r_s i_{ds} - \omega \lambda_{qs} + p \lambda_{ds} \quad (4)$$

$$v_{qr} = r_r i_{qr} + (\omega - \omega_r) \lambda_{dr} + p \lambda_{qr} \quad (5)$$

$$v_{dr} = r_r i_{dr} - (\omega - \omega_r) \lambda_{qr} + p \lambda_{dr} \quad (6)$$

For the purpose of this paper, the zero sequence variables are not included as in many applications induction machines have symmetrical rotor circuit and floating stator neutral. The flux linkage equations of the stator and rotor windings are expressed as

$$\lambda_{qs} = L_{ls} i_{qs} + \lambda_{mq} \quad (7)$$

$$\lambda_{ds} = L_{ls} i_{ds} + \lambda_{md} \quad (8)$$

$$\lambda_{qr} = L_{lr} i_{qr} + \lambda_{mq} \quad (9)$$

$$\lambda_{dr} = L_{lr} i_{dr} + \lambda_{md} \quad (10)$$

where the magnetizing fluxes along the q and d axes are

$$\lambda_{mq} = L_m (i_{qs} + i_{qr}) \quad (11)$$

$$\lambda_{md} = L_m (i_{ds} + i_{dr}) \quad (12)$$

The developed electromagnetic torque in transformed qd variables is

$$T_e = \frac{3P}{4} (\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}). \quad (13)$$

B. The Voltage-Behind-Reactance Model

The qd machine model is extensively used in power- and power-electronic systems simulations. However, the presence

of the transformations between the machine's qd variables and the external network abc variables makes such models less convenient.

The VBR model formulation represents the machine stator subsystem in direct abc coordinates and physical variables while expressing the rotor subsystem in transformed qd reference frame. Such simple and elegant formulation has been first proposed in the pioneering work on the synchronous machines [9], and then extended to the induction machines [10]. Several VBR models have been recently proposed in [10] which provide a direct machine-network interface. The basic model structure is depicted in Fig. 1. The model equations are briefly summarized in this subsection for consistency.

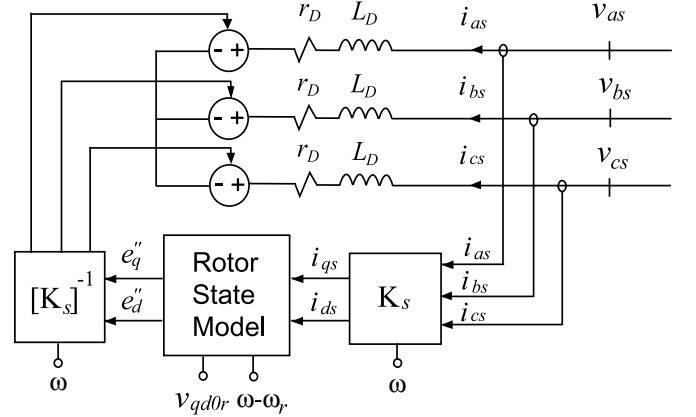


Fig. 1. The voltage-behind-reactance model structure.

The stator voltage equation of the VBR model is expressed as

$$\mathbf{v}_{abcs} = \text{diag}(r_D, r_D, r_D) \mathbf{i}_{abcs} + \text{diag}(L_D, L_D, L_D) p \mathbf{i}_{abcs} + \mathbf{e}''_{abcs} \quad (14)$$

where the constant and diagonal resistance and inductance matrices use the following elements

$$r_D = r_s + \frac{L_m^2}{L_{lr}^2} r_r \quad (15)$$

and

$$L_D = L_{ls} + L_m'' \quad (16)$$

with

$$L_m'' = \left(\frac{1}{L_m} + \frac{1}{L_{lr}} \right)^{-1} \quad (17)$$

The stator subtransient voltage sources in (14) are given as:

$$\mathbf{e}''_{abcs} = [\mathbf{K}_s]^{-1} [e''_q \ e''_d \ 0]^T \quad (18)$$

where

$$e''_q = \omega_r \frac{L_m''}{L_{lr}} \lambda_{dr} + \frac{L_m'' r_r}{L_{lr}^2} \left(\frac{L_m''}{L_{lr}} - 1 \right) \lambda_{qr} + \frac{L_m''}{L_{lr}} v_{qr} \quad (19)$$

$$e''_d = -\omega_r \frac{L_m''}{L_{lr}} \lambda_{qr} + \frac{L_m'' r_r}{L_{lr}^2} \left(\frac{L_m''}{L_{lr}} - 1 \right) \lambda_{dr} + \frac{L_m''}{L_{lr}} v_{dr} \quad (20)$$

The qd to abc coordinates transformation matrix is represented as

$$\mathbf{K}_s = \frac{2}{3} \begin{bmatrix} \cos \theta & \cos\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) \\ \sin \theta & \sin\left(\theta - \frac{2\pi}{3}\right) & \sin\left(\theta + \frac{2\pi}{3}\right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad (21)$$

The speed and angle of the arbitrary reference frame are related as

$$\omega = \frac{d\theta}{dt} \quad (22)$$

For VBR model, the rotor state model of Fig. 1 is formed by the rotor flux linkages state equation with the stator currents as inputs according to (23) and (24).

$$p\lambda_{qr} = -\frac{r_r}{L_{lr}}(\lambda_{qr} - \lambda_{mq}) - (\omega - \omega_r)\lambda_{dr} + v_{qr} \quad (23)$$

$$p\lambda_{dr} = -\frac{r_r}{L_{lr}}(\lambda_{dr} - \lambda_{md}) + (\omega - \omega_r)\lambda_{qr} + v_{dr} \quad (24)$$

where the mutual flux linkages are expressed as

$$\lambda_{mq} = L_m'' \left(i_{qs} + \frac{\lambda_{qr}}{L_{lr}} \right) \quad (25)$$

$$\lambda_{md} = L_m'' \left(i_{ds} + \frac{\lambda_{dr}}{L_{lr}} \right) \quad (26)$$

As the stator fluxes λ_{qs} and λ_{ds} are not calculated directly in the VBR model, the developed electromagnetic torque is calculated in terms of the magnetizing fluxes λ_{mq} and λ_{md} as

$$T_e = \frac{3P}{4} (\lambda_{md} i_{qs} - \lambda_{mq} i_{ds}) \quad (27)$$

III. VBR MODEL IMPLEMENTATION IN SIMPOWERSYSTEM

The selection of machine models depends on the modeling languages and simulation environment. For state-variable-based packages such as plain Simulink without the circuit interface, a direct implementation of the qd model may be simple and effective modeling approach [3]. However, for the circuit-based tools such as SimPowerSystem, which is a widely-used toolbox for simulations of power and power-electronic systems in Simulink environment, the qd model is not solved directly with the state model of the circuit. Instead, the qd machine models are interfaced with the external network as three-phase voltage-controlled current sources [7]–[8]. As a consequence of this model-to-circuit interface, the inductive network branches cannot be connected directly to the machine terminals. To enable the connection of inductive elements with machine models, this toolbox requires connection of resistive or capacitive loads (or artificial snubbers) at the interface terminals. Such snubbers are needed to make the respective state models of the machine and external circuit input-output compatible.

The VBR model formulation offers numerous structural and numerical advantages for implementing and interfacing the electrical machine models in different simulation platforms. Moreover, due to the structural symmetry of

induction machine, the resulting stator branches can be made decoupled among the phases as well as constant – independent of rotor position (see (14)–(15)). This feature makes the implementation of the proposed VBR model in SimPowerSystem very straightforward as shown in Fig. 2. In particular, the stator branches are realized using basic circuit components in SimPowerSystem – three-phase reactance block with decoupled R - L elements and three voltage-controlled voltage sources. The dynamic equations of the rotor and mechanical subsystems are implemented using standard Simulink blocks. The subtransient voltages $\mathbf{e}_{abs}^*$ and the stator currents $\mathbf{i}_{abs}$ are the interface variables between the rotor subsystem and the machine's stator combined with the rest of the circuit, as shown in Fig. 2.

Within the SimPowerSystem, there are two types of numerical solvers that may be applied to the underlying state equation of the circuit, namely the continuous solver or the discretized solver. The continuous solver simply uses the variable-step or fixed-step solver of Simulink. In this case the electrical circuit and the Simulink blocks are solved together. Alternatively, the discretized solver in SimPowerSystem can be used for its built-in components. Under this condition, the built-in qd machine models are solved using the Forward Euler method [2]. The state equation of electrical circuit is discretized using Tustin (Trapezoid) integration rule while the control blocks outside SimPowerSystem are solved by the Simulink solvers. Therefore, in the VBR model, higher-order integration solver may be selected for the rotor and mechanical subsystems compared to the first-order Euler solver used by SimPowerSystem for qd models. This may also improve the model accuracy.

IV. CASE STUDIES

In order to verify the induction machine VBR model proposed for SimPowerSystem, a start-up transient study has been considered here. The machine parameters are given in Appendix. Without loss of generality, the synchronous reference frame has been used in all machine models and the mechanical load torque is assumed to be zero. For comparison purposes, both the built-in qd model and the VBR model have been considered. Note that since the qd model in SimPowerSystem is represented as voltage-controlled current source, such machine model cannot be directly interfaced with the external inductive network. Therefore, two case studies are conducted depending on whether the external network is resistive or inductive.

For the purpose of benchmark comparisons, a reference qd model has been also implemented separately using standard basic Simulink blocks. The reference solution has been obtained using this model solved with 4th-order Runge-Kutta method using a very small fixed step of $1\mu s$. Also, since there is no circuit interface for the reference qd model, the impedance of external source is combined with the machine's stator resistance and leakage inductance, respectively.

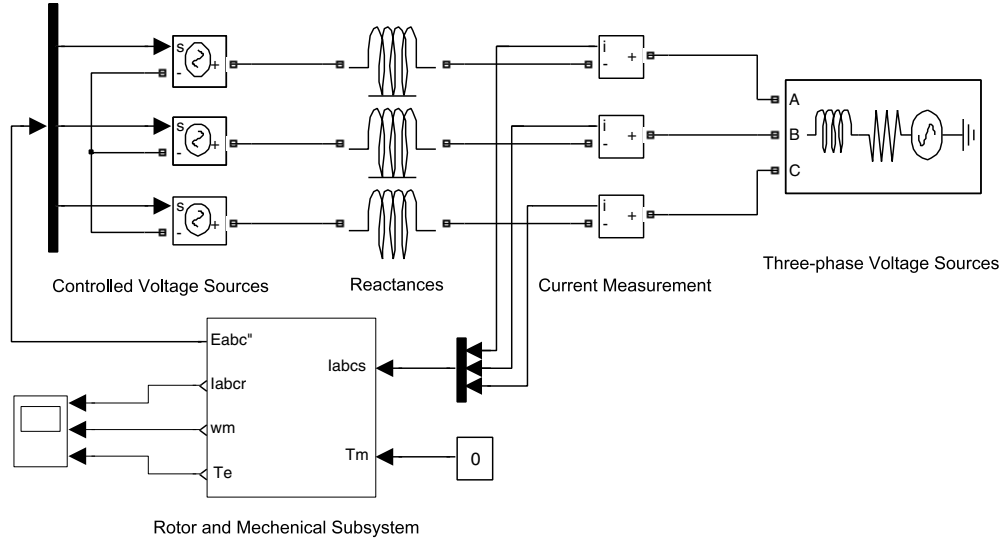


Fig. 2. Implementation of the VBR model for SimPowerSystem in Matlab-Simulink.

A. External Resistive Network

In this study, it is assumed that the induction machine is connected to a resistive network which is represented by the Thevenin equivalent source as shown in Fig. 3. The discretized SimPowerSystem solver with a fixed-step of $10^{-4} s$ is used for the solution of the main circuits of the built-in *qd* model and the VBR model. As the built-in *qd* model is actually discretized by the Forward Euler method to avoid the algebraic loop with SimPowerSystem, the overall system may not achieve the 2nd-order accuracy expected from the Tustin method. However, the VBR model can use high-order ODE solver for the rotor and mechanical subsystems since they are represented as standard Simulink blocks. To be consistent with the 2nd-order Tustin method, a 2nd-order Heun's (explicit Trapezoidal) method has been used in Simulink.

The transient responses predicted by the built-in SimPowerSystem *qd* model and the VBR model obtained using the same time step of $10^{-4} s$ are overlapped in Figs. 4 – 6 and compared with the reference solution. It is seen in Figs. 4 – 6 that the simulation results obtained by the VBR model match the reference solution very well while the built-in *qd* model produces significant errors.

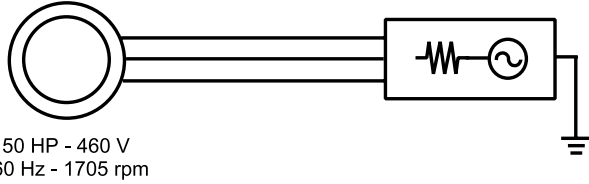


Fig. 3. Induction machine fed from resistive network with voltage source.

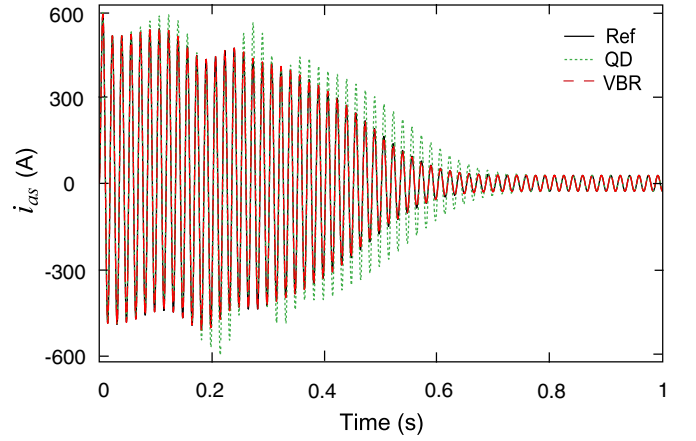


Fig. 4. The stator currents i_{as} during the startup transient.

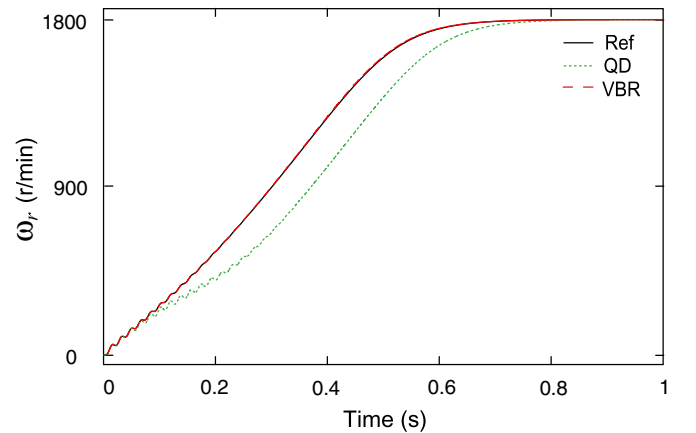


Fig. 5. The rotor speeds ω_r during the startup transient.

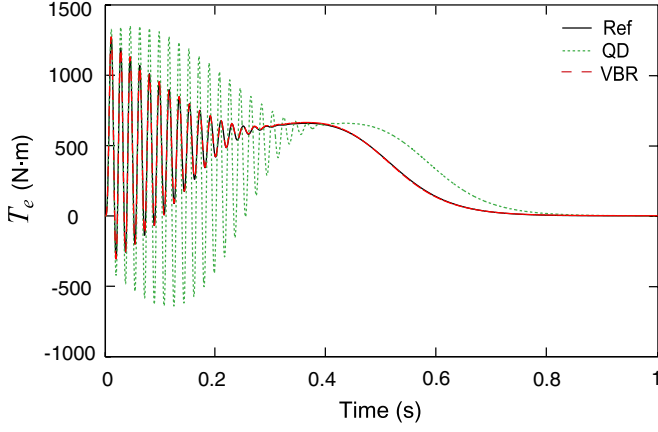


Fig. 6. The electromagnetic torques T_e during the startup transient

In order to further quantify the numerical accuracy of various models, the simulation study has been repeated several times using different time steps from 10^{-6} s to 10^{-4} s. Without loss of generality, the stator current i_{as} is used to demonstrate the simulation accuracy. In particular, a relative error $\varepsilon(h)$ defined as [11]

$$\varepsilon(h) = \frac{\|\tilde{i}_{as} - i_{as}\|_2}{\|\tilde{i}_{as}\|_2} \times 100\% \quad (28)$$

has been calculated for different results. Here, h represents the time step, $\tilde{i}_{as}$ and i_{as} denote the reference and numerical solution trajectories, respectively. The behavior of the relative error is shown in Fig. 7. As it can be clearly observed in Fig. 7, the VBR model greatly improves the numerical accuracy compared with the built-in qd model, and therefore permits using much larger time-steps.

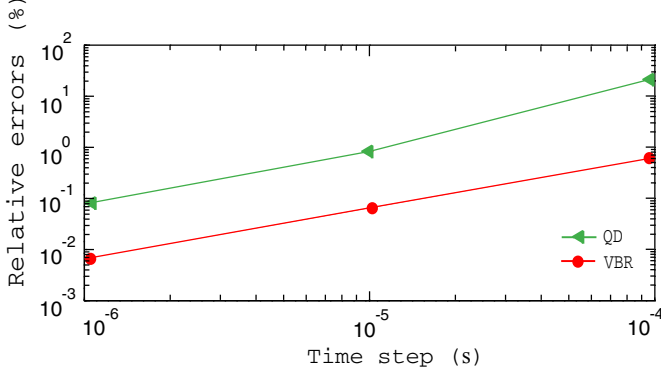


Fig. 7. The relative error in stator current i_{as} for different integration time-steps.

B. External Inductive Network

In the second case study, the induction machine is fed by an external inductive network, represented by the Thevenin equivalent circuits shown in Fig. 8. Because the built-in qd model is interfaced with the external abc phase coordinates as voltage-controlled current source, the direct connection of the machine-network is not possible. To enable this connection, the artificial snubber circuits are typically chosen with very high impedance so that the errors introduced

are negligible. However, for the VBR model, no snubber circuits are required as the stator branches are directly available for the external network.

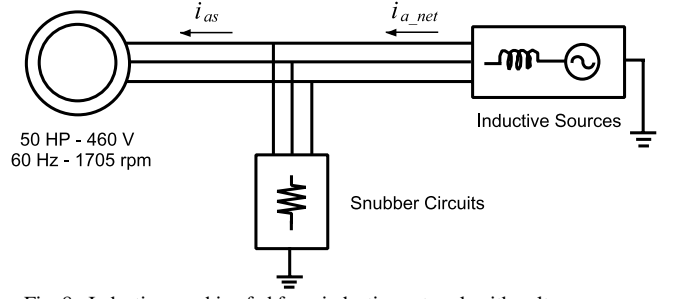


Fig. 8. Induction machine fed from inductive network with voltage source.

In the following study, the three-phase shunt resistors of $10^3 \Omega$ have been used with the built-in qd model. The discretized solver of SimPowerSystem and fixed-step Heun solver of Simulink have been applied for both models. By using different values for the integration time-step size, it is found that the largest time step used by the built-in qd and VBR models were on the order of 10^{-6} s and 10^{-4} s, respectively. The transient responses predicted by these two models using their respective largest time step and the reference solution are shown in Fig. 9. This study also shows that both the built-in qd and VBR models converge to the reference solutions. However, the VBR model may use a much larger time step (by two orders of magnitude) and is therefore more efficient compared to the built-in qd model.

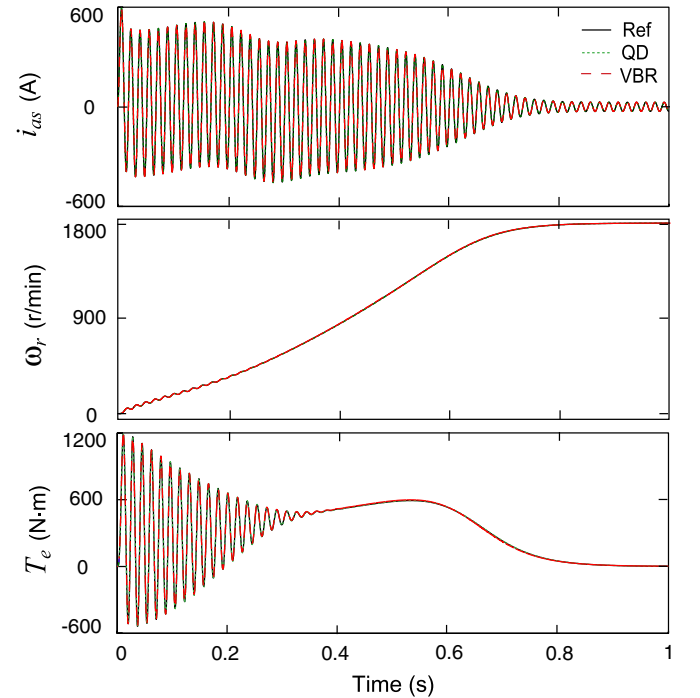


Fig. 9. Startup transient of induction machine interfaced with the inductive network. Artificial snubber resistors are used for the built-in SimPowerSystem qd model.

In general, the addition of snubbers significantly increases the stiffness of the overall system. Consequently, the largest possible time-step has to be decreased to ensure the numerical stability [12]. In order to further illustrate this point, different values of the shunt resistance from 10 to $10^3 \Omega$ are used. Due to the limited space, only the network current i_{a_net} is shown in Fig. 10 corresponding to the time interval close to a steady state. The absolute error in the magnitude of the steady state current shown in Fig. 10 is very large for the snubbers of 10 and 100Ω . As Fig. 10 shows, increasing the snubber resistance leads to better accuracy, with the curve corresponding to $10^3 \Omega$ visibly approaching the reference solution. However, at the same time largest time step that can be used by the built-in *qd* model has also decreased to 10^{-6} .

The relative numerical errors of the network current i_{a_net} (the entire trajectory) are calculated using (28) and plotted in Fig. 11. The results depicted in Fig. 11 are consistent with that shown in Fig. 10.

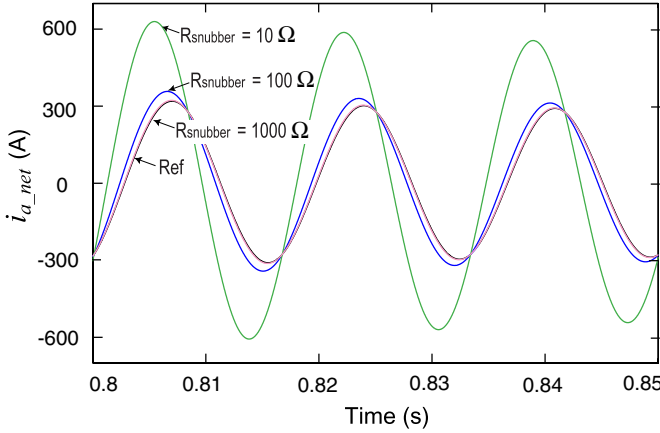


Fig. 10. The steady-state network currents i_{a_net} for the different snubbers.

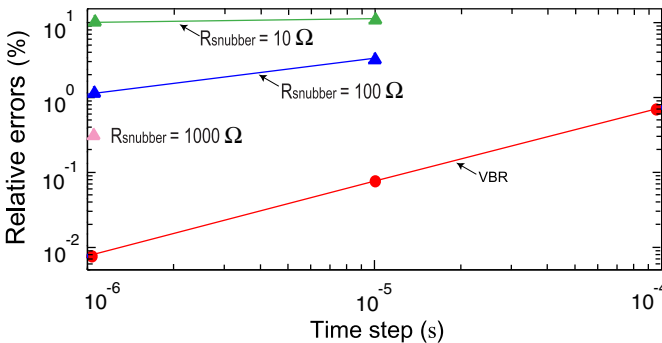


Fig. 11. The error propagations for the stator currents i_{a_net} .

V. CONCLUSION

A voltage-behind-reactance induction machine model has been proposed for direct use with the SimPowerSystem in Matlab-Simulink. The VBR model structure is very simple and elegant which makes its implementation very convenient. Case studies of induction machine startup transients demonstrate that the VBR model significantly improves the

simulation accuracy and efficiency compared to the built-in *qd* machine model for the discretized electric model solution in SimPowerSystem.

VI. APPENDIX

Induction machine parameters [6]: 50-hp, 460V, 1705 rpm, 4 poles, 60-Hz, $r_s = 0.087 \Omega$, $X_{ls} = 0.302 \Omega$, $X_m = 13.08 \Omega$, $r_r = 0.228 \Omega$, $X_{lr} = 0.302 \Omega$, $J = 1.662 \text{ kg} \cdot \text{m}^2$, $r_{th} = 0.1 \Omega$, $X_{th} = 0.1885 \Omega$, $V_{th,LL} = 460 \text{ V}$.

REFERENCES

- [1] *Simulink Dynamic System Simulation Software – Users Manual*, MathWorks, Inc., Natick, Massachusetts, 2005.
- [2] *SimPowerSystems 4 – Reference Manual*, The MathWorks, Inc., Natick, Massachusetts, 2007. Available: www.mathworks.com.
- [3] D. Logue and P. T. Krein, "Simulation of electrical machinery and power electronics interfacing using MATLAB/SIMULINK," *The 7th Workshop on Computers in Power Electronics*, 2000. *COMPEL 2000*, pp. 34–39, July 2000.
- [4] *Piecewise Linear Electrical Circuit Simulation (PLECS) User Manual*, Version. 1.5, Plexim GmbH, 2006. Available at: www.plexim.com.
- [5] *Automated State Model Generator (ASMG), Reference Manual*, Version 2, PC Krause and Associates Inc., 2002. Available at: www.pcka.com.
- [6] P. C. Krause, O. Wasynczuk, and S. D. Sudhoff, *Analysis of Electric Machinery and Drive Systems*, 2nd Edition, IEEE Press, Piscataway, NJ, 2002.
- [7] L.-A. Dessaint, K. Al-Haddad, H. Le-Huy, G. Sybille, and P. Brunelle, "A power system simulation tool based on Simulink," *IEEE Trans. on Industrial Electronics*, vol. 46, no. 6, pp. 1252–1254, Dec. 1999.
- [8] R. Champagne, L.-A. Dessaint, H. Fortin-Blanchette and G. Sybille, "Analysis and validation of a real-time AC drive simulator," *IEEE Trans. on Power Electronics*, vol. 19, no. 2, pp. 336–345, Mar. 2004.
- [9] S. D. Pekarek, O. Wasynczuk, and H. J. Hegner, "An efficient and accurate model for the simulation and analysis of synchronous machine/converter systems," *IEEE Trans. Energy Conversion*, vol. 13, no. 1, pp. 42–48, Mar. 1998.
- [10] L. Wang, J. Jatskevich, and S. Pekarek, "Modeling of induction machines using a voltage-behind-reactance formulation," *To appear in IEEE Trans. on Energy Conversion* (Paper: TEC-00124-2007), In Press.
- [11] W. Gautchi, *Numerical Analysis: An Introduction*, Birkhauser, Boston, Massachusetts, 1997.
- [12] Uri Ascher and Linda R. Petzold, *Computer Methods for Ordinary Differential Equations and Differential-Algebraic Equations*. SIAM, Philadelphia, 1998.

Liwei Wang (S'04) received his M.S. degree in electrical engineering from Tianjin University, China, in 2004. He is currently pursuing a Ph.D. degree in Electrical and Computer Engineering at the University of British Columbia, Vancouver, Canada. His research interests include electrical machines, power and power electronic systems simulation.

Juri Jatskevich (S'97, M'99, SM'07) received the M.S.E.E. and the Ph.D. degrees from Purdue University, West Lafayette IN, USA, in 1997 and 1999, respectively. He stayed at Purdue as a Post-Doctoral Research Associate and Research Scientist, as well as consulted for the P C Krause and Associates, Inc. until 2002. Since 2002, he has been a faculty member at the University of British Columbia, Vancouver, Canada, where he is now an Associate Professor of Electrical and Computer Engineering. His research interests include electrical machines, power electronic systems, average-value modeling and simulation.

Sina Chini Foroosh received his M.S. degree in electrical engineering from Sharif University of Technology, Tehran, Iran, in 2007. He is currently pursuing a Ph.D. degree in Electrical and Computer Engineering at the University of British Columbia, Vancouver, Canada. His research interests include electrical machines and power electronics.

Constant-Parameter Voltage-Behind-Reactance Induction Machine Model Including Main Flux Saturation

Francis Therrien, *Student Member, IEEE*, Mehrdad Chapariha, *Student Member, IEEE*,
and Juri Jatskevich, *Senior Member, IEEE*

Abstract—Interfacing of electrical machine models with ac power networks has a significant impact on numerical accuracy and efficiency in state-variable-based transient simulation programs. This paper continues the recent work in this area by proposing a new explicit constant-parameter voltage-behind-reactance (VBR) induction machine model that includes main flux saturation and allows a direct interface to any external network. The proposed model uses numerical approximations to achieve a decoupled interfacing circuit with constant RL branches that is easy to use in many simulation programs. Computer studies demonstrate that the proposed model provides high numerical accuracy for machines with a diverse range of parameters, even at fairly large integration step sizes.

Index Terms—Induction machine, interfacing circuit, magnetic saturation, state-variable-based program, transient simulation, voltage-behind-reactance (VBR) model.

I. INTRODUCTION

INDUCTION machines normally operate under lightly-to-moderately saturated conditions. As machine saturation is a complex phenomenon, refined analysis techniques such as finite element and/or coupled magnetic circuit methods may be used during the design stage when very precise results are required and the computational cost and time are of a lesser priority. For power system transient studies, lower order general-purpose lumped-parameter models are typically preferred due to their relative simplicity and high numerical efficiency. In such models, magnetic nonlinearities are often represented by saturating the fundamental component of the main magnetizing flux [1]–[10]. In addition to increasing the overall modeling accuracy, adequate representation of main flux saturation is crucial for many indirect rotor field-oriented control schemes [5], [11]. Saturation of leakage inductances can also be used to further increase model fidelity [7], [9], [12], [13], particularly for uncontrolled motor start-ups. However, unlike the main flux saturation characteristic which can be easily obtained from a no-load test, complicated procedures are required to determine the leakage

saturation characteristics [12], [14]. Consequently, many simulation programs assume constant leakage inductances (e.g., see [1] and [2]). To develop a straightforward-to-use model, this paper considers only saturation of the main flux, which offers sufficient accuracy for many practical cases and system-level transient studies [5].

In transient simulation programs [1]–[4], induction machines are typically modeled in a rotating qd -reference frame using transformed variables [5], [15], [16]. The classical qd models offer many numerical advantages including the absence of rotor-position-dependent inductances, magnetically decoupled q - and d -axis circuits, and constant qd -variables in steady state [5].

This paper considers state-variable-based transient simulation programs (e.g., see [1] and [2]), wherein the built-in qd models are typically interfaced with external circuits using controlled current sources [16]. Therefore, when a machine is in series with an inductive or ideal switching component (which is very common in practical systems), a proper state model cannot be generated due to incompatible input–output interfacing [16]. To achieve a compatible interface, fictitious snubber circuits are typically used, which introduce additional error and lead to increased numerical stiffness and longer CPU times [16].

The so-called coupled-circuit phase-domain (CC-PD) models [16], [17] are represented in physical abc -variables (phase coordinates), therefore allowing direct interfacing with any external circuit. However, the CC-PD models remain computationally expensive due to their fully coupled rotor-position-dependent inductance matrices [18], [19]. From a numerical perspective, the voltage-behind-reactance (VBR) models [18]–[25] appear to be a better alternative. In these models, the stator interfacing circuit is in abc -phase coordinates, allowing a direct interface with external networks, while the rotor subsystem is represented in qd -coordinates, yielding better scaled eigenvalues and higher numerical efficiency [18].

It is emphasized that under the same set of assumptions (e.g., representation of saturation or not), the qd , CC-PD, and VBR models are algebraically equivalent. In other words, these models yield identical analytical solutions (when converted to a common reference frame). However, they have different numerical properties (e.g., interfacing circuits and eigenstructures), which can have a considerable impact on numerical efficiency and accuracy when these models are interfaced with power networks inside simulation programs.

Due to dynamic saliency, the interfacing circuit of the original magnetically linear VBR synchronous machine model contains

Manuscript received October 8, 2013; revised April 11, 2014; accepted July 3, 2014. Date of publication August 11, 2014; date of current version February 16, 2015. This work was supported by the Natural Science and Engineering Research Council (NSERC) of Canada under the Discovery Grant. Paper no. TEC-00596-2013.

The authors are with the Department of Electrical and Computer Engineering, University of British Columbia, Vancouver, BC V6T 1Z4, Canada (e-mail: francist@ece.ubc.ca; mehrdadc@ece.ubc.ca; jurij@ece.ubc.ca).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TEC.2014.2342258

variable inductances [18]. While being more efficient than CC-PD models [18], it is nonetheless computationally expensive due to the need for refactorization (or updating) of the state matrices (equations) [20]. Several techniques have since been proposed to accurately neglect dynamic saliency and achieve a constant-parameter VBR synchronous machine model: adding a fictitious high-frequency damper winding [20], applying singular perturbation theory [21], and transferring the time-varying inductances of the interfacing circuit into its subtransient voltage source [22]. In particular, the numerical efficiency of the latter approach can be significantly improved by using continuous- or discrete-time filters to form a proper and explicit state-space model [22]. It is stressed that all of the above techniques/models [20]–[22] are based on the assumption of magnetic linearity. Incorporation of main flux saturation into these models would cause their interfacing circuit inductances to become time-dependent.

Unlike the synchronous machine model [18], the magnetically linear VBR induction machine model has a constant-parameter interfacing circuit since its rotor is isotropic [19]. However, the saturable VBR induction machine models [1], [23] (which are based on the synchronous machine model presented in [24]) have time-varying interfacing circuits. Consequently, these models also suffer from poor numerical efficiency.

The main objective of this paper is to propose a numerically accurate and efficient constant-parameter VBR induction machine model including main flux saturation. This paper is a continuation of the prior work in this area [18]–[25]. The contributions of this paper and the properties of the new model are summarized as follows.

- 1) A constant-parameter VBR induction machine model including main flux saturation is derived. To the best of our knowledge, it is the first time that a saturable ac machine model with a constant-parameter interfacing circuit is proposed.
- 2) The new model uses the unified decoupled *RL* branch stator interfacing circuit presented in [25].
- 3) To obtain an efficient and explicit formulation, the proposed model expands the numerical approximation techniques originally proposed in [22] to take into account saturation in both magnetic axes (including cross-saturation).
- 4) The numerical error introduced by the proposed approximation is analyzed and related to the integration step size, the machine parameters, and the saturation level. Several machines with diverse range of parameters are considered to investigate the applicability of the proposed approach.
- 5) The presented computer studies demonstrate that the proposed model possesses high numerical efficiency and accuracy for machines with diverse parameters even at fairly large integration step sizes.
- 6) It is envisioned that the proposed model will find its application in many commercial-grade transient simulation programs and packages, wherein such models can be used as built-in components readily available to thousands of engineers and users.

II. QD MODEL

To set the stage for the derivation of the new model, it is useful to recall the equations of the classical general-purpose lumped-parameter symmetrical induction machine model in *qd*-coordinates [5]. Herein, the motor sign convention and the arbitrary reference frame are used. All parameters are referred to the stator side. To incorporate the double-cage or deep-bar effects (if desired) [26], the induction machine is assumed to have N equivalent rotor windings with no end-ring common resistance. All windings on the same magnetic axis are assumed to have identical coupling. If necessary, standard double-cage models with $L_{lr1} = 0$ and nonidentical coupling [10] can be converted to equivalent models with identical coupling by following the procedure set forth in [27]. The corresponding voltage equations are [5]

$$v_{qs} = r_s i_{qs} + \omega \lambda_{ds} + p \lambda_{qs} \quad (1)$$

$$v_{ds} = r_s i_{ds} - \omega \lambda_{qs} + p \lambda_{ds} \quad (2)$$

$$v_{0s} = r_s i_{0s} + p \lambda_{0s} \quad (3)$$

$$v_{qrj} = r_{rj} i_{qrj} + (\omega - \omega_r) \lambda_{drj} + p \lambda_{qrj}, \quad j = 1, \dots, N \quad (4)$$

$$v_{drj} = r_{rj} i_{drj} - (\omega - \omega_r) \lambda_{qrj} + p \lambda_{drj}, \quad j = 1, \dots, N \quad (5)$$

$$v_{0rj} = r_{rj} i_{0rj} + p \lambda_{0rj}, \quad j = 1, \dots, N. \quad (6)$$

The flux linkage equations are

$$\lambda_{qs} = L_{ls} i_{qs} + \lambda_{mq} \quad (7)$$

$$\lambda_{ds} = L_{ls} i_{ds} + \lambda_{md} \quad (8)$$

$$\lambda_{0s} = L_{ls} i_{0s} \quad (9)$$

$$\lambda_{qrj} = L_{lrj} i_{qrj} + \lambda_{mq}, \quad j = 1, \dots, N \quad (10)$$

$$\lambda_{drj} = L_{lrj} i_{drj} + \lambda_{md}, \quad j = 1, \dots, N \quad (11)$$

$$\lambda_{0rj} = L_{lrj} i_{0rj}, \quad j = 1, \dots, N \quad (12)$$

where the magnetizing fluxes are given by

$$\lambda_{mq} = L_m(\lambda_m) i_{mq} = L_m(\lambda_m) \left(i_{qs} + \sum_{j=1}^N i_{qrj} \right) \quad (13)$$

$$\lambda_{md} = L_m(\lambda_m) i_{md} = L_m(\lambda_m) \left(i_{ds} + \sum_{j=1}^N i_{drj} \right). \quad (14)$$

Here, $L_m(\lambda_m)$ denotes the saturation-dependent steady-state magnetizing inductance, which can be found from the machine no-load saturation characteristic [6]

$$\lambda_m = F^{-1}(i_m). \quad (15)$$

Since the machine is round and symmetrical, the main magnetizing flux λ_m and current i_m are defined as

$$\lambda_m = \sqrt{\lambda_{mq}^2 + \lambda_{md}^2} \quad \text{and} \quad i_m = \sqrt{i_{mq}^2 + i_{md}^2}. \quad (16)$$

Finally, the developed electromagnetic torque and mechanical equation of single rigid-body motion are given by

$$T_e = \frac{3P}{4}(\lambda_{md}i_{qs} - \lambda_{mq}i_{ds}) \quad (17)$$

$$T_e - T_m = J \left(\frac{2}{P} \right) p\omega_r \quad (18)$$

where P is the number of poles and J denotes the combined rotor-load inertia.

III. VARIABLE-PARAMETER VBR MODEL

The derivation of the saturable variable-parameter VBR induction machine model (VP-VBR) has been presented in [23]. The main steps of this derivation are briefly explained here.

For magnetically linear VBR models, the state variables of the equivalent rotor subsystem are typically chosen to be the rotor flux linkages [18], [19]. When modeling main flux saturation, it is convenient to also consider the magnetizing fluxes λ_{mq} and λ_{md} as state variables [28]. Consequently, one pair of q - and d -axis rotor flux linkages is removed from the set of state variables. Without loss of generality, in this paper, λ_{qr1} and λ_{dr1} are excluded from the state vector and instead are considered auxiliary algebraic variables.

Defining $\Gamma_m(\lambda_m) \equiv 1/L_m(\lambda_m)$, differentiating (13) and (14) with respect to time, and using the chain and quotient rules yield

$$p \begin{bmatrix} i_{mq} \\ i_{md} \end{bmatrix} = \Gamma_{mi}(\lambda_{mqd}) p \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} \quad (19)$$

where

$$\Gamma_{mi}(\lambda_{mqd}) = \begin{bmatrix} \left(\frac{di_m}{d\lambda_m} - \Gamma_m(\lambda_m) \right) \frac{\lambda_{mq}^2}{\lambda_m^2} + \Gamma_m(\lambda_m) & \left(\frac{di_m}{d\lambda_m} - \Gamma_m(\lambda_m) \right) \frac{\lambda_{mq}\lambda_{md}}{\lambda_m^2} \\ \left(\frac{di_m}{d\lambda_m} - \Gamma_m(\lambda_m) \right) \frac{\lambda_{mq}\lambda_{md}}{\lambda_m^2} & \left(\frac{di_m}{d\lambda_m} - \Gamma_m(\lambda_m) \right) \frac{\lambda_{md}^2}{\lambda_m^2} + \Gamma_m(\lambda_m) \end{bmatrix} \quad (20)$$

Based on (13) and (14), the left-hand side of (19) is explicitly defined as

$$p \begin{bmatrix} i_{mq} \\ i_{md} \end{bmatrix} = p \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \sum_{j=1}^N p \begin{bmatrix} i_{qrj} \\ i_{drj} \end{bmatrix}. \quad (21)$$

The rotor currents in (21) must be written as a function of state and input variables. As a first step, the rotor currents can be expressed in terms of rotor flux linkages and magnetizing fluxes by algebraically manipulating (10) and (11), giving

$$\begin{bmatrix} i_{qrj} \\ i_{drj} \end{bmatrix} = \frac{1}{L_{lrj}} \left(\begin{bmatrix} \lambda_{qrj} \\ \lambda_{drj} \end{bmatrix} - \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} \right), \quad j = 1, \dots, N. \quad (22)$$

Substituting (21) and (22) into (19) consecutively and solving for $p\lambda_{mqd}$ yields the following state equations:

$$p \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} = \mathbf{L}_{mi}''(\lambda_{mqd}) \left(p \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \sum_{j=1}^N \frac{p}{L_{lrj}} \begin{bmatrix} \lambda_{qrj} \\ \lambda_{drj} \end{bmatrix} \right) \quad (23)$$

where

$$\mathbf{L}_{mi}''(\lambda_{mqd}) \equiv \begin{bmatrix} L_{mqq}'' & L_{mqd}'' \\ L_{mqd}'' & L_{mdd}'' \end{bmatrix} = \left(\Gamma_{mi}(\lambda_{mqd}) + \sum_{j=1}^N \frac{1}{L_{lrj}} \mathbf{I}_2 \right)^{-1} \quad (24)$$

and $\mathbf{I}_n$ is the n -by- n identity matrix.

To formulate a standard state-space model, (23) must also be rewritten as a function of state and input variables. First, inserting (7), (8), and (23) into (1) and (2) and solving for $p\mathbf{i}_{qds}$ gives

$$p \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} = \begin{bmatrix} L_{ls} + L_{mqq}'' & L_{mqd}'' \\ L_{mqd}'' & L_{ls} + L_{mdd}'' \end{bmatrix}^{-1} \times \left(\begin{bmatrix} v_{qs} \\ v_{ds} \end{bmatrix} - \begin{bmatrix} r_s & \omega L_{ls} \\ -\omega L_{ls} & r_s \end{bmatrix} \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} - \omega \begin{bmatrix} \lambda_{md} \\ -\lambda_{mq} \end{bmatrix} - \begin{bmatrix} e_q \\ e_d \end{bmatrix} \right) \quad (25)$$

where

$$\begin{bmatrix} e_q \\ e_d \end{bmatrix} = \begin{bmatrix} L_{mqq}'' & L_{mqd}'' \\ L_{mqd}'' & L_{mdd}'' \end{bmatrix} \left(\sum_{j=1}^N \frac{p}{L_{lrj}} \begin{bmatrix} \lambda_{qrj} \\ \lambda_{drj} \end{bmatrix} \right). \quad (26)$$

Next, solving (4) and (5) for $p\lambda_{qdrj}$ yields

$$p \begin{bmatrix} \lambda_{qrj} \\ \lambda_{drj} \end{bmatrix} = \begin{bmatrix} v_{qrj} \\ v_{drj} \end{bmatrix} - r_{rj} \begin{bmatrix} i_{qrj} \\ i_{drj} \end{bmatrix} - (\omega - \omega_r) \begin{bmatrix} \lambda_{drj} \\ -\lambda_{qrj} \end{bmatrix}, \quad j = 1, \dots, N \quad (27)$$

where $\mathbf{i}_{qdrj}$ is defined using (22). Due to the choice of state variables, (27) is only integrated for $j = 2, \dots, N$. Finally, in order to obtain an explicit model, λ_{qdr1} must be written in terms of state and input variables. Substituting (22) into (13) and (14) and solving for λ_{qdr1} yields

$$\begin{bmatrix} \lambda_{qr1} \\ \lambda_{dr1} \end{bmatrix} = L_{lr1} \left(\left(\Gamma_m(\lambda_m) + \sum_{j=2}^N \frac{1}{L_{lrj}} \right) \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} - \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} - \sum_{j=2}^N \frac{1}{L_{lrj}} \begin{bmatrix} \lambda_{qrj} \\ \lambda_{drj} \end{bmatrix} \right) + \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix}. \quad (28)$$

The next step is to define the equation for the stator interfacing circuit. Inserting (7), (8), and (23) into (1) and (2) yields

$$\begin{bmatrix} v_{qs} \\ v_{ds} \end{bmatrix} = r_s \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \omega \begin{bmatrix} -L_{mqd}'' & L_{ls} + L_{mqq}'' \\ -L_{ls} - L_{mdd}'' & L_{mqd}'' \end{bmatrix} \times \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \begin{bmatrix} L_{ls} + L_{mqq}'' & L_{mqd}'' \\ L_{mqd}'' & L_{ls} + L_{mdd}'' \end{bmatrix} p \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \begin{bmatrix} e_q'' \\ e_d'' \end{bmatrix} \quad (29)$$

where

$$\begin{bmatrix} e_q'' \\ e_d'' \end{bmatrix} = \omega \begin{bmatrix} L_{mqd}'' & -L_{mqq}'' \\ L_{mdd}'' & -L_{mqd}'' \end{bmatrix} \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \omega \begin{bmatrix} \lambda_{md} \\ -\lambda_{mq} \end{bmatrix} + \begin{bmatrix} e_q \\ e_d \end{bmatrix}. \quad (30)$$

Adding zero-sequence [(3) and (9)] to (29) and applying the inverse Park's transformation [5] yield the machine-network interfacing equation

$$\mathbf{v}_{abcs} = r_s \mathbf{i}_{abcs} + (L_{ls} \mathbf{I}_3 + \mathbf{L}_s''(\theta_r)) \mathbf{p} \mathbf{i}_{abcs} + \mathbf{e}_{abcs}'' \quad (31)$$

where $\mathbf{f}_{abcs} = [f_{as} \ f_{bs} \ f_{cs}]^T$ (f may represent v or i), $\mathbf{e}_{abcs}'' = \mathbf{K}_s^{-1} [e_q'' \ e_d'' \ 0]^T$, and $\mathbf{K}_s$ is Park's transformation matrix [5]. The subtransient inductance matrix $\mathbf{L}_s''(\theta_r)$ is defined as

$$\begin{aligned} \mathbf{L}_s''(\theta_r) &= L_a \begin{bmatrix} 1 & -0.5 & -0.5 \\ -0.5 & 1 & -0.5 \\ -0.5 & -0.5 & 1 \end{bmatrix} \\ &+ \begin{bmatrix} L(\theta_r) & L(\theta_r + 120^\circ) & L(\theta_r - 120^\circ) \\ L(\theta_r + 120^\circ) & L(\theta_r - 120^\circ) & L(\theta_r) \\ L(\theta_r - 120^\circ) & L(\theta_r) & L(\theta_r + 120^\circ) \end{bmatrix} \end{aligned} \quad (32)$$

where the function $L(\varphi)$ is

$$L(\varphi) = L_b \cos(2\varphi) + L_c \sin(2\varphi) \quad (33)$$

and $\varphi = \{\theta_r, \theta_r - 120^\circ, \theta_r + 120^\circ\}$. The equivalent inductances L_a , L_b , and L_c are defined as

$$L_a = \frac{L_{mqq}'' + L_{mdd}''}{3}, \quad L_b = \frac{L_{mqq}'' - L_{mdd}''}{3}, \quad L_c = \frac{2}{3} L_{mqd}''. \quad (34)$$

The resulting interfacing circuit (31) has a constant diagonal resistance matrix; however, its inductance matrix is full and dependent on the rotor angle θ_r . From (32)–(34), it can be observed that $\mathbf{L}_s''(\theta_r)$ will become constant if the following three conditions are satisfied:

$$L_{mqq}'' + L_{mdd}'' = \kappa = \text{const.} \quad \forall t \quad (35)$$

$$L_{mqq}'' - L_{mdd}'' = 0 \quad \forall t \quad (36)$$

$$L_{mqd}'' = 0 \quad \forall t \quad (37)$$

where κ is an arbitrary time-independent scalar. Conditions (35)–(37) are observed in the magnetically linear symmetrical induction machine model [19]. However, when saturation is considered, it can be seen from (20) and (24) that the three conditions are not always met.

It is important to keep in mind that the saturable VP-VBR model derived in this section is algebraically equivalent to the general-purpose lumped-parameter qd model defined in Section II, i.e., it is possible to convert one model into the other using algebraic manipulations. However, unlike the qd model, the VP-VBR model can be interfaced directly to arbitrary external networks.

IV. CONSTANT-PARAMETER VBR MODEL

To obtain the desired constant-parameter (CP-VBR) model, it is necessary to satisfy conditions (35)–(37) at all times. This is achieved if $\mathbf{L}_{mi}''(\lambda_{mqd})$ becomes a constant scalar matrix, i.e., a diagonal matrix with identical nonzeros.

A. Derivation of Constant-Parameter Interfacing Circuit

The first step consists of substituting (21) into (19) and separating $\Gamma_{mi}(\lambda_{mqd})$ into constant and time-dependent terms

$$\begin{aligned} p \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \sum_{j=1}^N p \begin{bmatrix} i_{qrj} \\ i_{drj} \end{bmatrix} &= \Gamma_{mu} p \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} \\ &+ \Delta \Gamma_{mis}(\lambda_{mqd}) p \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} \end{aligned} \quad (38)$$

where $\Gamma_{mu} \equiv 1/L_{mu}$ is the unsaturated magnetizing inverse inductance, and

$$\begin{aligned} \Delta \Gamma_{mis}(\lambda_{mqd}) &= \\ &\begin{bmatrix} \left(\frac{di_m}{d\lambda_m} - \Gamma_m(\lambda_m) \right) \frac{\lambda_{mq}^2}{\lambda_m^2} + \Delta \Gamma_m(\lambda_m) \\ \left(\frac{di_m}{d\lambda_m} - \Gamma_m(\lambda_m) \right) \frac{\lambda_{mq} \lambda_{md}}{\lambda_m^2} \\ \left(\frac{di_m}{d\lambda_m} - \Gamma_m(\lambda_m) \right) \frac{\lambda_{mq} \lambda_{md}}{\lambda_m^2} \\ \left(\frac{di_m}{d\lambda_m} - \Gamma_m(\lambda_m) \right) \frac{\lambda_{md}^2}{\lambda_m^2} + \Delta \Gamma_m(\lambda_m) \end{bmatrix} \end{aligned} \quad (39)$$

where $\Delta \Gamma_m(\lambda_m) = \Gamma_m(\lambda_m) - \Gamma_{mu}$. Substituting (22) into (38) and solving for the $p\lambda_{mqd}$ vectors multiplied by time-independent elements yields

$$\begin{aligned} p \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} &= L_{mu}'' \left(p \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \sum_{j=1}^N \frac{p}{L_{lrj}} \begin{bmatrix} \lambda_{qrj} \\ \lambda_{drj} \end{bmatrix} \right. \\ &\quad \left. - \Delta \Gamma_{mis}(\lambda_{mqd}) p \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} \right) \end{aligned} \quad (40)$$

where

$$L_{mu}'' = \left(\Gamma_{mu} + \frac{1}{L_{\Sigma lr}} \right)^{-1} \quad \text{and} \quad \frac{1}{L_{\Sigma lr}} = \sum_{j=1}^N \frac{1}{L_{lrj}}. \quad (41)$$

Substituting (23) into the right-hand side of (40) yields

$$\begin{aligned} p \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} &= L_{mu}'' \left(p \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \sum_{j=1}^N \frac{p}{L_{lrj}} \begin{bmatrix} \lambda_{qrj} \\ \lambda_{drj} \end{bmatrix} \right. \\ &\quad \left. - \Delta \Gamma_{mis}(\lambda_{mqd}) \mathbf{L}_{mi}''(\lambda_{mqd}) \left(p \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} \right. \right. \\ &\quad \left. \left. + \sum_{j=1}^N \frac{p}{L_{lrj}} \begin{bmatrix} \lambda_{qrj} \\ \lambda_{drj} \end{bmatrix} \right) \right). \end{aligned} \quad (42)$$

Inserting (7), (8), and (42) into (1) and (2) then gives

$$\begin{bmatrix} v_{qs} \\ v_{ds} \end{bmatrix} = r_s \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \omega(L_{ls} + L''_{mu}) \begin{bmatrix} i_{ds} \\ -i_{qs} \end{bmatrix} + (L_{ls} + L''_{mu})p \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \begin{bmatrix} e''_q \\ e''_d \end{bmatrix} \quad (43)$$

where

$$\begin{bmatrix} e''_q \\ e''_d \end{bmatrix} = \omega \begin{bmatrix} \lambda_{md} \\ -\lambda_{mq} \end{bmatrix} + L''_{mu} \left(\omega \begin{bmatrix} -i_{ds} \\ i_{qs} \end{bmatrix} + \sum_{j=1}^N \frac{p}{L_{lrj}} \begin{bmatrix} \lambda_{qrj} \\ \lambda_{drj} \end{bmatrix} - \Delta \Gamma_{mis}(\lambda_{mqd}) \mathbf{L}''_{mi}(\lambda_{mqd}) \left(p \begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} + \sum_{j=1}^N \frac{p}{L_{lrj}} \begin{bmatrix} \lambda_{qrj} \\ \lambda_{drj} \end{bmatrix} \right) \right). \quad (44)$$

Combining (43) with (3) and (9) and applying the inverse Park's transformation results in the following constant-parameter machine-network interfacing equation:

$$\mathbf{v}_{abcs} = r_s \mathbf{i}_{abcs} + (L_{ls} \mathbf{I}_3 + \mathbf{L}''_{cs}) p \mathbf{i}_{abcs} + \mathbf{e}''_{abcs} \quad (45)$$

where the new constant subtransient inductance matrix is

$$\mathbf{L}''_{cs} = \frac{2}{3} L''_{mu} \begin{bmatrix} 1 & -0.5 & -0.5 \\ -0.5 & 1 & -0.5 \\ -0.5 & -0.5 & 1 \end{bmatrix}. \quad (46)$$

The constant-parameter VBR model (45) has been achieved without modifying the rotor subsystem state equations (23) and (27).

The final step consists of modifying (45) to achieve an interfacing circuit comprised only of decoupled RL branch elements [25]. Assuming a wye-connected (grounded or floating neutral) machine, the terminal stator voltage $\mathbf{v}_{abcs}$ is decomposed into two vectors by following the procedure set forth in [25]

$$\mathbf{v}_{abcs} = \mathbf{v}_{abcn} + \mathbf{v}_{ng} \quad (47)$$

where

$$\mathbf{v}_{abcn} = r_s \mathbf{i}_{abcs} + L_D p \mathbf{i}_{abcs} + \mathbf{e}''_{abcs} \quad (48)$$

and $\mathbf{v}_{ng} = [v_{ng} \ v_{ng} \ v_{ng}]^T$. The voltage v_{ng} is defined as

$$v_{ng} = L_0 p i_{ng} \quad (49)$$

where $i_{ng} = i_{as} + i_{bs} + i_{cs}$. Finally, the newly introduced inductances L_D and L_0 are related to L_{ls} and L''_{mu} by

$$L_D = L_{ls} + L''_{mu} \quad \text{and} \quad L_0 = -\frac{L''_{mu}}{3}. \quad (50)$$

Equations (47)–(50) define the decoupled RL branch interfacing circuit [25, Fig. 1]. Up to this point, the CP-VBR model is also algebraically equivalent to the qd and VP-VBR models.

B. Numerical Approximation

The procedure presented in Section IV-A is analogous to moving the time-varying elements of the algebraically exact inductance matrix $\mathbf{L}''_s(\theta_r)$ defined in (32) into the subtransient

voltage source $\mathbf{e}''_{abcs}$. As a consequence, the term $\mathbf{e}''_{qd}$ is a function of the current derivative $p \mathbf{i}_{qds}$.

One method to calculate $p \mathbf{i}_{qds}$ in (44) is to rewrite it in terms of state and input variables using (25), as is done in (23). However, since $\mathbf{e}''_{qd}$ is an output of the equivalent rotor subsystem, this approach results in an algebraic feedthrough between $\mathbf{e}''_{qd}$ and $\mathbf{v}_{qds}$. In the case where $\mathbf{v}_{qds}$ is an output of the external circuit subsystem (e.g., if the machine is connected to an inductive branch), $\mathbf{v}_{qds}$ will be algebraically related to $\mathbf{e}''_{qd}$. Consequently, the state model will have an algebraic loop containing $\mathbf{e}''_{qd}$ and $\mathbf{v}_{qds}$ [22]. Models with algebraic loops generally require special iterative solvers which are not available in some simulation programs (e.g., see [1]), reduce simulation robustness [29], and increase the overall computational time [6], [30].

An alternative approach is to approximate $p \mathbf{i}_{qds}$ in (44) using numerical techniques. Specifically, $p \mathbf{i}_{qds}$ is herein approximated using backward differentiation formulas (BDFs) [22], [31]. A similar approach was originally presented in [22] to achieve an explicit constant-parameter magnetically linear VBR synchronous machine model. Therein, due to dynamic saliency, only $p i_{qs}$ needed to be approximated; in addition, $p i_{qs}$ was multiplied by a constant value [22]. This is in contrast to the proposed model, wherein due to saturation (and cross-saturation), $p \mathbf{i}_{qds}$ is multiplied by a time-dependent coupled matrix [see (44)].

Assuming a variable-step solver for generality, the first-order backward differentiation formula (BDF1) is

$$p \tilde{\mathbf{i}}_{qds}^{(k)} = \frac{\mathbf{i}_{qds}^{(k)} - \mathbf{i}_{qds}^{(k-1)}}{\Delta t^{(k)}} \quad (51)$$

where the superscript (k) indicates the value at the k th time step, and $\Delta t^{(k)} = t^{(k)} - t^{(k-1)}$. Here, the tilde symbol ($\sim$) denotes an approximation as opposed to the exact value. If a more accurate approximation is desired, the second-order BDF (BDF2) can also be used

$$\begin{aligned} p \tilde{\mathbf{i}}_{qds}^{(k)} = & \left(\mathbf{i}_{qds}^{(k)} - \mathbf{i}_{qds}^{(k-1)} \right) \left(\frac{1}{\Delta t^{(k)}} + \frac{1}{\Delta t^{(k)} + \Delta t^{(k-1)}} \right) \\ & - \frac{\Delta t^{(k)}}{\Delta t^{(k-1)}} \left(\frac{\mathbf{i}_{qds}^{(k-1)} - \mathbf{i}_{qds}^{(k-2)}}{\Delta t^{(k)} + \Delta t^{(k-1)}} \right). \end{aligned} \quad (52)$$

Replacing $p \mathbf{i}_{qds}$ in (44) by (51) or (52), an explicit constant-parameter VBR induction machine model with main flux saturation is obtained. A diagram depicting the implementation of this new model is shown in Fig. 1, wherein the interfacing circuit defined by (47)–(50) is contained within the shaded area. In general, other approximations (e.g., see [22]) could also be used instead of (51) or (52).

V. ERROR ANALYSIS

If the solver uses a high-order integration rule and the integration step size is sufficiently small, it can be assumed that the numerical error in the proposed model will mostly come from the approximation of $L''_{mu} \Delta \Gamma_{mis}(\lambda_{mqd}) \mathbf{L}''_{mi}(\lambda_{mqd}) p \mathbf{i}_{qds}$ in (44) using BDFs [(51) or (52)]. This overall error is thus determined



The values of $L_{mu}/L_{\Sigma lr}$ and χ^{-1} for several machines with a diverse range of parameters and power ratings are summarized in Table I. As this table shows, $L_{mu}/L_{\Sigma lr}$ and χ^{-1} are fairly correlated and generally increase with machine size. These

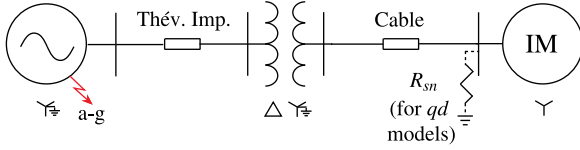


Fig. 2. Example network used for the voltage sag transient study.

coefficients are also typically larger for machines modeled with two rotor windings instead of one. Finally, based on (58)–(60) and the magnitude of these coefficients in Table I, it can be concluded that $\zeta(\lambda_m)$ should remain relatively small even for machines with less favorable parameters under heavy saturation. This is a direct result and benefit of the VBR formulation which uses the subtransient inductance, wherein the magnetizing inductance $L_m(\lambda_m)$ appears in parallel with the typically much smaller equivalent rotor leakage inductance $L_{\Sigma lr}$.

VI. COMPUTER STUDIES

The proposed CP-VBR model (see Section IV) and the algebraically exact VP-VBR model (see Section III) have been implemented in MATLAB/Simulink [29] using the PLECS toolbox [1]. Two versions of the proposed model are considered. In the first one, the current derivative $\dot{\mathbf{i}}_{qds}$ in (44) is computed using (51) (i.e., BDF1); in the second one, $\dot{\mathbf{i}}_{qds}$ is computed using (52) (i.e., BDF2). The resulting formulations are referred to as CP-VBR-BDF1 and CP-VBR-BDF2, respectively. To provide a meaningful comparison, the classical qd model (with winding flux linkages as the state variables) [5] is also implemented, wherein saturation is represented using the explicit flux correction method [6]. All machine models are simulated in the synchronous reference frame. For consistency, all simulations are executed on a PC with a 2.83-GHz Intel CPU running Windows XP.

In the considered case study, a low voltage (LV) wye-connected induction motor is connected to a medium voltage distribution (MV) network (represented by a three-phase Thévenin equivalent) through a cable and a transformer. The single-line diagram of the network is shown in Fig. 2. The motor is initially assumed to operate in steady state with applied base mechanical torque T_b . The source voltage V_s is set to 1.1 p.u. as to emulate a strongly saturated operation and thus increase the numerical error as explained in Section V-B. While this voltage is high for grid-connected induction motors, it is not uncommon for machine-converter systems to operate at even higher voltages [33]. At $t = 0.04$ s, a single-line-to-ground fault is assumed to occur upstream in the network, resulting in an unbalanced voltage sag in the test system. The voltage sag is modeled by decreasing the voltage on phase a of the equivalent Thévenin source to 0.5 p.u. The fault is cleared 100 ms later, and V_s is restored back to 1.1 p.u. This study has been chosen because of the highly unbalanced currents and severe transients occurring as a result of the voltage sag. Consequently, according to Section V-A, the approximation error will be maximized under such conditions, thus validating the model in its less accurate operating region.

The network of Fig. 2 has been implemented using PLECS' built-in circuit components. Since the motor is in series with an inductive cable, only the VBR models can be interfaced directly to the external network among the test case models [16]. The qd model is represented using controlled current sources and is interfaced using fictitious resistive snubbers R_{sn} [16]. To demonstrate the effect of snubbers, two values (large and small) of R_{sn} have been considered. The corresponding models are referred to as $qd1$ (which is numerically stiff) and $qd2$ (which is less stiff), respectively.

A reference solution is obtained by solving the saturable VP-VBR model using Simulink's general-purpose solver ode45 with the following stringent settings: maximum and minimum step sizes of 10^{-5} and 10^{-8} s, respectively, and absolute and relative error tolerances of 10^{-5} . Under these conditions, the numerical error is very small and the simulation results are assumed to be the reference solution. To achieve a fair comparison, due to the numerical stiffness of the $qd1$ model, all subject models are simulated using the stiffly stable solver ode15s [29]. Typical settings are used: maximum and minimum step sizes of 10^{-3} and 10^{-7} s, respectively, and absolute and relative error tolerances of 10^{-4} .

In this paper, the 2-norm (cumulative) relative error [34] of the solution trajectories is adopted to quantify the models' numerical accuracy. For example, the 2-norm relative error of i_{as} is given by

$$\varepsilon(i_{as}) = \frac{\|\mathbf{i}_{as}^{\text{ref}} - \mathbf{i}_{as}\|_2}{\|\mathbf{i}_{as}^{\text{ref}}\|_2} \times 100\% \quad (61)$$

where $\mathbf{i}_{as}^{\text{ref}}$ and $\mathbf{i}_{as}$ are the reference and calculated solution trajectories of i_{as} , respectively, for the duration of the considered transient study.

To verify the proposed approach, a few motors with different parameters are considered.

A. Induction Machine #1 (IM1)

The subject transient study is first executed using a 10-hp machine (IM1) with two rotor windings [10]. The machine parameters and saturation curve are summarized in Appendix A. For this machine, the ratio of unsaturated magnetizing inductance over equivalent rotor leakage inductance $L_{mu}/L_{\Sigma lr}$ (214) and the coefficient χ^{-1} (999) are very large (see Table I). Therefore, as shown in Section V-B, the proposed CP-VBR models should yield excellent numerical accuracy. Snubber resistances R_{sn} of 250 Ω and 5 Ω are used in the $qd1$ and $qd2$ models, respectively.

The simulated stator current i_{as} , rotor electrical speed ω_r , electromagnetic torque T_e , and main flux λ_m are shown in Fig. 3, wherein the unbalanced conditions created by the single-phase voltage sag are apparent. Only phase a of the stator current is presented since it has the largest peaks during the disturbance. To better visualize the performance of the models, a magnified view of the first peak of i_{as} is also reproduced in Fig. 4. As observed in Figs. 3 and 4, all models except $qd2$ accurately predict the machine behavior in both steady-state and transient

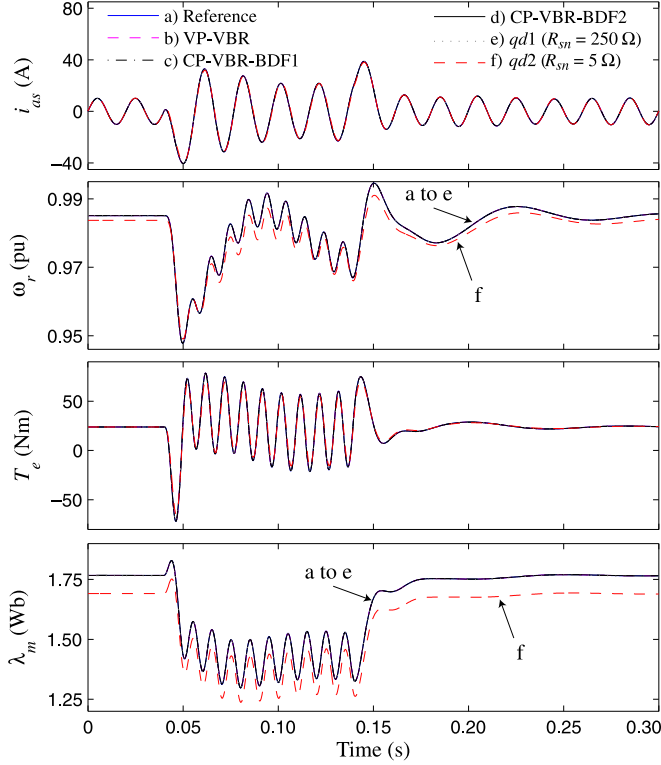


Fig. 3. Transients in stator current i_{as} , rotor electrical speed ω_r , electromagnetic torque T_e , and main flux λ_m during a single-phase voltage sag (IM1).

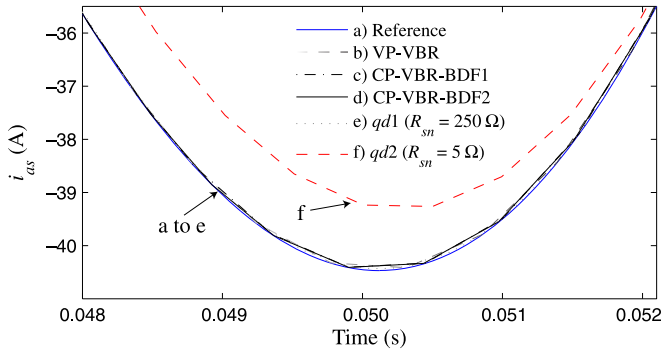


Fig. 4. Magnified view of the first peak of stator current i_{as} during a single-phase voltage sag (IM1).

conditions. The inaccuracy of $qd2$ is caused by its fairly small snubber.

In Section V-B, the level of saturation was found to affect the solution accuracy. Specifically, the more saturated the machine is, the larger the numerical error should be. To verify that IM1 is well saturated, the dynamic values of λ_m (see Fig. 3) are depicted on the steady-state saturation curve in Fig. 5. The air-gap line is also shown in Fig. 5 to better visualize the level of saturation at different operating points. Analyzing Figs. 3 and 5, it is seen that IM1 is well saturated in steady state ($\lambda_m = 1.77$ Wb), and that it saturates even further ($\lambda_m = 1.83$ Wb) shortly after the fault. The machine remains saturated during the rest of the sag.

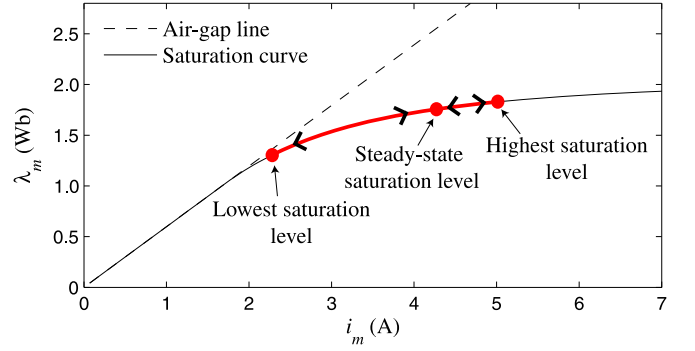


Fig. 5. Saturation curve of IM1 and its operating range (denoted by the thick line) during a single-phase voltage sag.

TABLE II
CALCULATED 2-NORM RELATIVE ERRORS OF i_{as} , ω_r , T_e , AND λ_m
FOR THE SINGLE-PHASE VOLTAGE SAG TRANSIENT (IM1)

Model	VP-VBR	CP-VBR-BDF1	CP-VBR-BDF2	$qd1$ ($R_{sn} = 250\Omega$)	$qd2$ ($R_{sn} = 5\Omega$)
Var.					
i_{as} (%)	0.035	0.056	0.052	0.153	5.578
ω_r (%)	0.001	0.002	0.002	0.004	0.208
T_e (%)	0.033	0.060	0.055	0.205	7.895
λ_m (%)	0.004	0.010	0.009	0.090	4.445

To quantify the accuracy achieved by the various models, the 2-norm (cumulative) relative errors of i_{as} , ω_r , T_e , and λ_m with respect to their reference trajectories are summarized in Table II. As this table shows, all three VBR models yield errors below 0.1% for all variables, and no significant difference is observed between CP-VBR-BDF1 and CP-VBR-BDF2. The $qd1$ model is also highly accurate, as its 2-norm errors are all smaller than 0.21%. However, as observed in Figs. 3 and 4 and summarized in Table II, the $qd2$ model is much less accurate with errors of 5.578% for i_{as} and 7.895% for T_e .

B. Induction Machine #2 (IM2)

The same voltage sag transient study is repeated here using a 5-hp machine (IM2) [9]. The machine parameters are summarized in Appendix B. For this machine, the very low values of $L_{mu}/L_{\Sigma l_r}$ (14.3) and χ^{-1} (31.6) indicate that the proposed approximation method should be less accurate. In this study, R_{sn} is set to 300 Ω and 3 Ω for the $qd1$ and $qd2$ models, respectively.

The predicted stator current i_{as} , rotor electrical speed ω_r , electromagnetic torque T_e , and main flux λ_m are presented in Fig. 6. A magnified view of the first peak of i_{as} after the clearing of the upstream fault is also shown in Fig. 7. Despite the unfavorable machine parameters, all models except $qd2$ again predict electrical and mechanical variables with sufficient numerical accuracy.

The dynamic values of λ_m during the study (see Fig. 7) are depicted on the steady-state saturation curve in Fig. 8 along with the air-gap line. The machine IM2 is also well saturated in steady state ($\lambda_m = 0.473$ Wb). Higher values of flux are reached at the

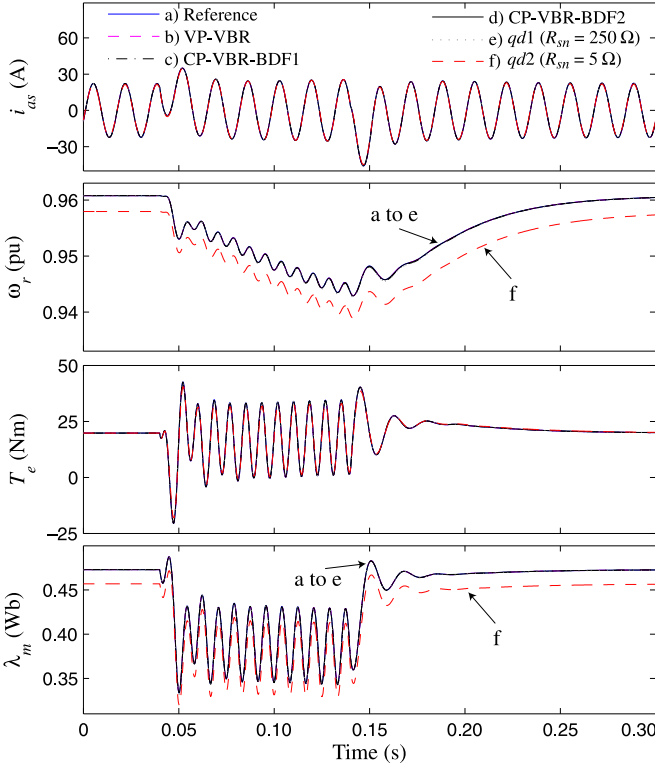


Fig. 6. Transients in stator current i_{as} , rotor electrical speed ω_r , electromagnetic torque T_e , and main flux λ_m during a single-phase voltage sag (IM2).

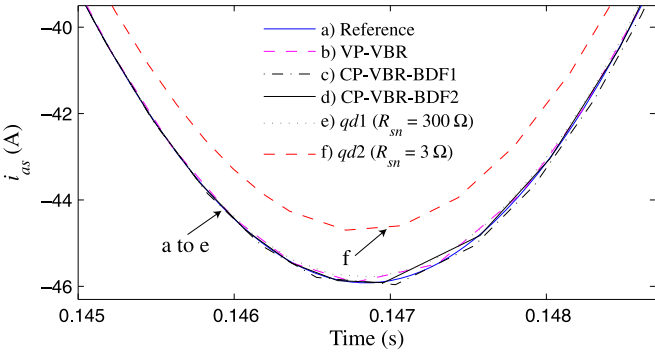


Fig. 7. Magnified view of the first peak of stator current i_{as} following the clearing of the single-phase fault (IM2).

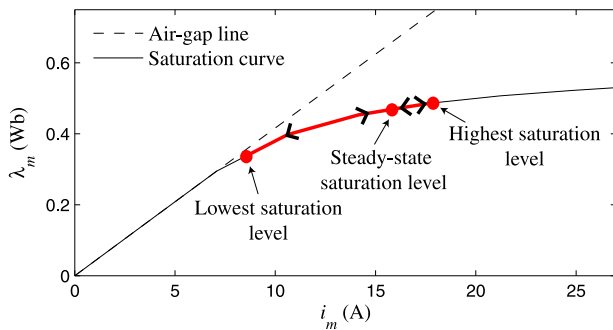


Fig. 8. Saturation curve of IM2 and its operating range (denoted by the thick line) during a single-phase voltage sag.

TABLE III
CALCULATED 2-NORM RELATIVE ERRORS OF i_{as} , ω_r , T_e , AND λ_m
FOR THE SINGLE-PHASE VOLTAGE SAG (IM2)

Model	VP-VBR	CP-VBR-BDF1	CP-VBR-BDF2	qd1 ($R_{sn} = 300\Omega$)	qd2 ($R_{sn} = 3\Omega$)
Var.					
i_{as} (%)	0.069	0.754	0.094	0.473	3.64
ω_r (%)	0.002	0.009	0.002	0.017	0.371
T_e (%)	0.074	0.273	0.086	0.256	4.406
λ_m (%)	0.013	0.068	0.014	0.086	3.694

beginning and the end of the voltage sag ($\lambda_m = 0.488$ and 0.484 Wb, respectively). The motor also remains saturated throughout the rest of the sag.

The 2-norm relative errors of i_{as} , ω_r , T_e , and λ_m with respect to the reference trajectories are summarized in Table III. All models except $qd2$ provide accurate results with errors smaller than 1%. However, there is now a noticeable difference between the two CP-VBR formulations. Specifically, CP-VBR-BDF2 yields more accurate results than CP-VBR-BDF1 (e.g., 0.094% error in i_{as} compared to 0.754%) due to its second-order approximation of pi_{qds} . Nevertheless, the numerical accuracy attained by the CP-VBR-BDF1 model remains satisfactory for most practical cases. This study demonstrates that highly accurate results can be achieved with the proposed constant-parameter model even for well saturated machines with unfavorable parameters.

C. Model Accuracy versus Integration Step Size

It is also useful to validate the accuracy of the proposed models as a function of a constant step size Δt . For this purpose, the same voltage sag transient study is repeated here using the proposed CP-VBR-BDF1 and CP-VBR-BDF2 models. To focus only on the machine models, the whole ac network (Thévenin equivalent, transformer, and cable) is replaced by an infinite bus. A third machine (IM3), whose value of $L_{mu}/L_{\Sigma lr}$ is in the middle range (43.3), is also considered. The parameters of IM3 are summarized in Appendix C. To enforce a fixed step size using PLECS [1], all simulations were executed using Simulink's ode45 solver with relaxed error tolerances.

The resulting 2-norm relative errors of i_{as} using CP-VBR-BDF1 and CP-VBR-BDF2 are plotted in Fig. 9 for the three machines. As it can be observed, the errors increase with the step size, which was predicted in Section V-A. The errors also converge towards zero for small step sizes, which demonstrates the consistency [31] of the proposed method. At the same time, the errors grow faster for machines with lower ratios $L_{mu}/L_{\Sigma lr}$. This emphasizes the correlation between model accuracy and machine parameters noted in Section V-B and observed in Sections VI-A and VI-B. It can also be seen in Fig. 9 that as the error increases, CP-VBR-BDF2 is clearly more accurate than CP-VBR-BDF1. For example, for the machine IM2 and $\Delta t = 0.5$ ms, the CP-VBR-BDF2 model yields an error twice as small as the

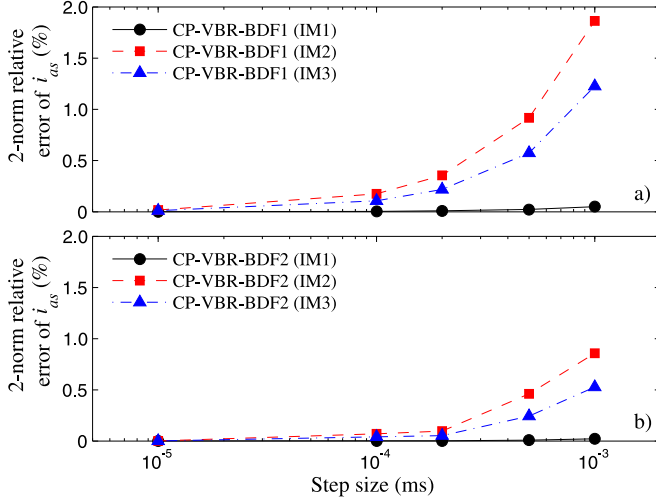


Fig. 9. Cumulative 2-norm relative error of stator current i_{as} for constant step sizes using: (a) CP-VBR-BDF1 and (b) CP-VBR-BDF2.

TABLE IV
NUMERICAL EFFICIENCY OF THE CONSIDERED MODELS
FOR THE SINGLE-PHASE VOLTAGE SAG STUDY

		VP-VBR	CP-VBR-BDF1	CP-VBR-BDF2	$qd1$	$qd2$
IM1	No. time steps	603	567	578	2528	741
	CPU time (ms)	836	179	184	422	131
	CPU time per time step (μ s)	1386	316	318	167	177
IM2	No. time steps	777	816	793	3602	1308
	CPU time (ms)	1079	184	197	433	158
	CPU time per time step (μ s)	1389	225	248	120	121

CP-VBR-BDF1 model (0.46% and 0.92%, respectively). Another important point is that all errors remain relatively small (below 2%) even at a fairly large integration step size of one millisecond with machine IM2 having unfavorable parameters (i.e., low $L_{mu}/L_{\Sigma lr}$). In particular, in this study, the error never exceeds 1% for the CP-VBR-BDF2 model.

D. Computational Performance

To evaluate the models' computational performance, the resulting number of time steps, CPU time, and average CPU time per time step from the studies of Sections VI-A and VI-B are summarized in Table IV. For a given machine, all three VBR models take similar numbers of time steps. However, the CPU time of VP-VBR is much higher than that of CP-VBR. For example, for IM1, the VP-VBR model takes more than 4 times longer than CP-VBR-BDF1 or CP-VBR-BDF2 (836 ms versus 179 ms or 184 ms, respectively). This is explained by the presence of variable inductances in VP-VBR. Table IV also illustrates that using BDF2 instead of BDF1 to approximate $p i_{qds}$ adds little

overhead. Additionally, the computational cost of the qd model per time step is much less than that of CP-VBR (approximately 172 μ s versus 317 μ s for the small system with IM1), which is due in part to its efficient implementation of saturation [6]. However, to obtain an accurate solution, the large snubber required to interface the qd model considerably increases the system's numerical stiffness, which in turn results in the use of smaller step sizes and increased computational burden. The proposed CP-VBR model (in particular the CP-VBR-BDF2 implementation) therefore offers the best combination of numerical accuracy and efficiency.

VII. DISCUSSION

A. Classical General-Purpose qd Model

Equations (1)–(18) summarize the traditional/classical general-purpose lumped-parameter symmetrical induction machine model in qd -coordinates. This well-established model (or its variations) has (have) been extensively used for several decades to evaluate the dynamics of systems containing induction machines [5], [15], [35]. It has been experimentally validated in numerous publications for various machines and transient phenomena [8], [10], [35], [36]. To this day, the general-purpose lumped-parameter symmetrical induction machine model remains the default model in many commercial-grade electromagnetic transient simulation programs such as PLECS [1], SimPowerSystems [2], EMTP-RV [3], and PSCAD/EMTDC [4], which are widely used by engineers and researchers in industry and academia.

At the same time, there are many higher-fidelity and/or higher-order models capable of reproducing the dynamics of physical induction machines more accurately [7], [9], [13], [33], [37]–[40]. In some cases, the general-purpose lumped-parameter model is modified as to take into account specific phenomena, e.g., main flux saturation harmonics [33], [37], [38] and leakage inductance saturation [7], [9], [13]. Other models are based on the finite element [39] or magnetic equivalent circuit [40] approaches, which typically allow for a finer representation of the physical structure of machines. Such higher-fidelity and/or higher-order models generally require more parameters and corresponding sophisticated procedures for their determination, and therefore are typically used only when the appropriate high fidelity is necessary. Such models also increase the computational cost of the solution and may not be justified for system-level studies.

B. Impact of Main Flux Saturation

Including main flux saturation in induction machine models is very common. The unsaturated models tend to underestimate steady-state stator currents, especially under low load conditions. To demonstrate this point, the voltage sag study of Section VI-B is conducted again with a low load ($T_m = 0.2$ p.u.) using saturated and unsaturated models, and sufficiently small integration step sizes as to ensure negligible numerical error. The resulting stator current i_{as} is presented in Fig. 10, wherein it is seen that in steady state the unsaturated model predicts a

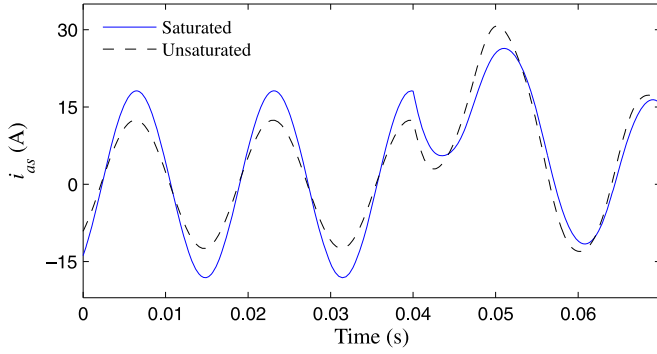


Fig. 10. Transients in stator current i_{as} for the study of Section VI-A under low-load condition as predicted by unsaturated and saturated induction machine models.

current roughly 30% smaller than the saturated model. There is also a noticeable difference in terms of phase and magnitude between the transient currents predicted by both models after the voltage sag. Another simple and effective mean to observe the effect of main flux saturation on machine variables is to apply a voltage step which brings the machine into the saturated region. One such study is presented in [41, Figs. 6–9], in which significant differences between the saturated and unsaturated stator currents and main fluxes are observed. The study in [41] also underlines the much more limited effect of saturation on the rotor speed and electromagnetic torque of grid-connected machines.

More specifically, representation of main flux saturation is essential when simulating self-excited induction generators [8], [10], [36]. Its impact on the study of reclosing transients of induction machines with power factor correction shunt capacitors has also been recognized for a long time [35]. Additionally, some motor drive schemes take main flux saturation into account [5], [11]. For example, it is shown in [5, Section 14.7] that an erroneous estimation of L_m (e.g., arising from a change in the level of saturation) yields degraded transient performance with indirect rotor field-oriented control approaches. Machine models with main flux saturation are thus required when simulating/testing these schemes. For these reasons, the built-in induction machine models found in the state-of-the-art electromagnetic transient programs [1]–[4] all have provisions to incorporate saturation of the main flux's fundamental component.

The main disadvantage of representing main flux saturation is the additional computational cost incurred. To quantify this overhead, the study of Section VI-B is repeated using the CP-VBR-BDF2 model and the standard unsaturated VBR model [19]. It is seen that for this scenario the CPU time per time step increases by roughly 60% when representing saturation. While the added computational burden is not negligible, it is believed to be a fair compromise to improve model accuracy.

C. Benefits and Potential Use of the Proposed Model

The proposed CP-VBR model requires the same typical machine parameters as the traditional lumped-parameter qd models currently available in commercial programs. Consequently,

users do not need to provide new input data. Furthermore, unlike the qd model, the CP-VBR model does not require interfacing snubbers, whose parameters would have to be selected as a compromise between numerical accuracy and efficiency. The constant-parameter decoupled interfacing circuit of the new saturable model is not only beneficial from a computational cost point of view (as evidenced by Table IV), it is also practical since the components library of some simulation programs (such as SimPowerSystems [2]) do not include time-dependent inductances. Moreover, as demonstrated in Section VI, the gains in numerical efficiency of the proposed model come at the expense of very little numerical error.

It is envisioned that the proposed saturable model could replace or complement the existing qd and VP-VBR models currently available in commonly used state-variable-based simulation programs such as SimPowerSystems and PLECS, with essentially no structural changes to the programs. The resulting gains in simulation speed can have a significant impact for many industrial and research users, in particular when simulating large networks and/or running batch simulations, e.g., Monte Carlo analyses and parametric sweeps.

VIII. CONCLUSION

A new explicit constant-parameter VBR induction machine model (CP-VBR) incorporating main flux saturation has been presented. The model is obtained by transferring the saturation-dependent (time-varying) part of the equivalent stator inductance into the subtransient voltage source and approximating its corresponding derivative using backward differentiation formulas (BDFs). The numerical error introduced by this approximation is analyzed. It is shown that the error depends on machine parameters and saturation level, and decreases very rapidly with the integration step size. Simulation studies of machines with a diverse range of parameters are presented. It is shown that even for machines with unfavorable parameters (low $L_{mu}/L_{\Sigma lr}$ ratio), the introduced cumulative error remains very small (under 1 to 2%) for step sizes as large as 1 ms. The proposed CP-VBR model is more efficient than the state-of-the-art models since it does not increase the system's stiffness (as does the snubber-interfaced classical qd models) and does not require to change/update the state matrices (as does the algebraically exact variable-parameter VBR model). Overall, the new model offers an excellent combination of numerical accuracy and efficiency, while using a constant-parameter interfacing circuit which is easy to use in most state-variable-based transient simulation programs. It is envisioned that the proposed model will find its application in many commercial-grade transient simulation programs used by thousands of users.

APPENDIX A

INDUCTION MACHINE IM1 [10]

- 1) *Machine parameters (converted using [27]):* 7.5 kW, 50 Hz, 2 poles, wye-connected, $V_{lls} = 658$ V, $r_s = 1.969$ Ω , $L_{ls} = 10.2$ mH, $L_m = 598$ mH, $r_{r1} = 1.093$ Ω , $L_{lr1} = 8.98$ mH, $r_{r2} = 5.727$ Ω , $L_{lr2} = 4.07$ mH, and $J = 0.04$ J $\cdot$ s².

2) Saturation Characteristic (peak values):

λ_m (Wb)	1.237	1.579	1.674	1.754	1.812	1.851	1.881	1.903	1.924
i_m (A)	2.121	3.182	3.677	4.243	4.808	5.303	5.798	6.222	6.718

- 3) Transformer: $S_B = 50$ kVA, $V_H = 12.47$ kV, $V_L = 658$ V, $Z_0 = Z_1 = 1.5 + 2j\%$.
 4) Cable: $Z_0 = 0.15 + j0.15 \Omega$, $Z_1 = 0.05 + j0.05 \Omega$.
 5) Thévenin equivalent: $V_{nom} = 12.47$ kV, $Z_0 = 0.75 + j1.95 \Omega$, $Z_1 = 0.3 + 0.6j \Omega$.

APPENDIX B

INDUCTION MACHINE IM2 [9]

- 1) Machine parameters: 5 hp, 60 Hz, 4 poles, wye-connected, $V_{lls} = 230$ V, $r_s = 0.4122 \Omega$, $X_{ls} = 1.1 \Omega$, $X_m = 15.7 \Omega$, $r_r = 0.4976 \Omega$, $X_{lr} = 1.1 \Omega$, and $J = 0.11 \text{ J} \cdot \text{s}^2$.
 2) Saturation characteristic (peak values):

λ_m (Wb)	0.147	0.295	0.398	0.454	0.486	0.522	0.535	0.543	0.553
i_m (A)	3.536	7.071	10.61	14.41	17.68	24.75	28.28	31.82	35.82

- 3) Transformer: $S_B = 25$ kVA, $V_H = 12.47$ kV, $V_L = 230$ V, and $Z_0 = Z_1 = 1.5 + j2\%$.
 4) Cable: $Z_0 = 0.15 + j0.15 \Omega$, $Z_1 = 0.05 + j0.05 \Omega$.
 5) Thévenin equivalent: $V_{nom} = 12.47$ kV, $Z_0 = 0.75 + j1.95 \Omega$, $Z_1 = 0.3 + j0.6 \Omega$.

APPENDIX C

INDUCTION MACHINE IM3 [5]

- 1) Machine parameters: 50 hp, 60 Hz, 4 poles, wye-connected, $V_{lls} = 460$ V, $r_s = 0.087 \Omega$, $X_{ls} = 0.302 \Omega$, $X_m = 13.08 \Omega$, $r_r = 0.228 \Omega$, $X_{lr} = 0.302 \Omega$, and $J = 1.662 \text{ J} \cdot \text{s}^2$.
 2) Saturation characteristic (peak values):

λ_m (Wb)	0.38	0.48	0.60	0.68	0.75	0.79	0.82	0.86	0.92
i_m (A)	11.20	14.33	18.33	21.31	24.42	26.74	29.07	33.21	40.78
λ_m (Wb)	0.96	1.01	1.06	1.10	/				
i_m (A)	46.21	53.21	60.33	66.10	/				

REFERENCES

- [1] Plexim GmbH. (2013). *Piece-wise linear electrical circuit simulation (PLECS) user manual version 3.4*, Plexim GmbH, Zurich, Switzerland [Online]. Available: www.plexim.com
- [2] *SimPowerSystems R2009a—User's Guide*, Hydro-Québec and The MathWorks, Inc., Natick, MA, USA, 2009.
- [3] CEA Technologies, Inc. (2007). *Electromagnetic Transient Program, EMTF-RV* [Online]. Available: www.emtp.com
- [4] Manitoba HVDC Research Centre and RTDS Technologies, Inc. (2009). *PSCAD/EMTDC V4.2 On-Line Help* [Online]. Available: www.pscad.com
- [5] P. C. Krause, O. Wasynczuk, and S. D. Sudhoff, *Analysis of Electric Machinery and Drive Systems*, 2nd ed. Piscataway, NJ, USA: IEEE Press, 2002.
- [6] F. Therrien, L. Wang, J. Jatskevich, and O. Wasynczuk, "Efficient explicit representation of AC machines main flux saturation in state-variable-based transient simulation packages," *IEEE Trans. Energy Convers.*, vol. 28, no. 2, pp. 380–393, Jun. 2013.
- [7] S. D. Sudhoff, D. C. Aliprantis, B. T. Kuhn, and P. L. Chapman, "An induction machine model for predicting inverter-machine interaction," *IEEE Trans. Energy Convers.*, vol. 17, no. 2, pp. 203–210, Jun. 2002.
- [8] E. Levi, "A unified approach to main flux saturation modeling in d-q axis models of induction machines," *IEEE Trans. Energy Convers.*, vol. 10, no. 3, pp. 455–461, Sep. 1995.
- [9] T. A. Lipo and A. Consoli, "Modeling and simulation of induction motors with saturable leakage reactances," *IEEE Trans. Ind. Appl.*, vol. IA-20, no. 1, pp. 180–189, Jan./Feb. 1984.
- [10] E. Levi, "Main flux saturation modelling in double-cage and deep-bar induction machines," *IEEE Trans. Energy Convers.*, vol. 11, no. 2, pp. 305–311, Jun. 1996.
- [11] E. Levi and M. Wang, "Online identification of the mutual inductance for vector controlled induction motor drives," *IEEE Trans. Energy Convers.*, vol. 18, no. 2, pp. 299–305, Jun. 2003.
- [12] L. Monjo, F. Córcoles, and J. Pedra, "Saturation effects on torque- and current-slip curves of squirrel-cage induction motors," *IEEE Trans. Energy Convers.*, vol. 28, no. 1, pp. 243–254, Mar. 2013.
- [13] G. J. Rogers and D. Shirmohammadi, "Induction machine modelling for electromagnetic transient program," *IEEE Trans. Energy Convers.*, vol. 2, no. 4, pp. 622–628, Dec. 1987.
- [14] S. D. Sudhoff, D. C. Aliprantis, B. T. Kuhn, and P. L. Chapman, "Experimental characterization procedure for use with an advanced induction machine model," *IEEE Trans. Energy Convers.*, vol. 18, no. 1, pp. 48–56, Mar. 2003.
- [15] P. C. Krause and C. H. Thomas, "Simulation of symmetrical induction machinery," *IEEE Trans. Power App. Syst.*, vol. PAS-84, no. 11, pp. 1038–1053, Nov. 1965.
- [16] L. Wang, J. Jatskevich, V. Dinavahi, H. W. Dommel, J. A. Martinez, K. Strunz, M. Rioual, G. W. Chang, and R. Iravani, "Methods of interfacing rotating machine models in transient simulation programs," *IEEE Trans. Power Del.*, vol. 25, no. 2, pp. 891–903, Apr. 2010.
- [17] O. Wasynczuk and S. D. Sudhoff, "Automated state model generation algorithm for power circuits and systems," *IEEE Trans. Power Syst.*, vol. 11, no. 4, pp. 1951–1956, Nov. 1996.
- [18] S. D. Pekarek, O. Wasynczuk, and H. J. Hegner, "An efficient and accurate model for the simulation and analysis of synchronous machine/converter systems," *IEEE Trans. Energy Convers.*, vol. 13, no. 1, pp. 42–48, Mar. 1998.
- [19] L. Wang, J. Jatskevich, and S. D. Pekarek, "Modeling of induction machines using a voltage-behind-reactance formulation," *IEEE Trans. Energy Convers.*, vol. 23, no. 2, pp. 382–392, Jun. 2008.
- [20] S. D. Pekarek and E. A. Walters, "An accurate method of neglecting dynamic saliency of synchronous machines in power electronic based systems," *IEEE Trans. Energy Convers.*, vol. 14, no. 4, pp. 1177–1183, Dec. 1999.
- [21] S. D. Pekarek, M. T. Lemanski, and E. A. Walters, "On the use of singular perturbations to neglect the dynamic saliency of synchronous machines," *IEEE Trans. Energy Convers.*, vol. 17, no. 3, pp. 385–391, Sep. 2002.
- [22] M. Chapariha, F. Therrien, J. Jatskevich, and H. W. Dommel, "Explicit formulations for constant-parameter voltage-behind-reactance interfacing of synchronous machine models," *IEEE Trans. Energy Convers.*, vol. 28, no. 4, pp. 1053–1063, Dec. 2013.
- [23] F. Therrien, M. Chapariha, and J. Jatskevich, "Generalized state-space saturable induction machine model using a voltage-behind-reactance formulation," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, Vancouver, BC, Canada, Jul. 21–25, 2013, pp. 1–5.
- [24] D. C. Aliprantis, O. Wasynczuk, and C. D. Rodríguez Valdez, "A voltage-behind-reactance synchronous machine model with saturation and arbitrary rotor network representation," *IEEE Trans. Energy Convers.*, vol. 23, no. 2, pp. 499–508, Jun. 2008.
- [25] M. Chapariha, L. Wang, J. Jatskevich, H. W. Dommel, and S. D. Pekarek, "Constant-parameter RL-branch equivalent circuit for interfacing AC machine models in state-variable-based simulation packages," *IEEE Trans. Energy Convers.*, vol. 27, no. 3, pp. 634–645, Sep. 2012.
- [26] S. J. Chapman, *Electric Machinery Fundamentals*, 4th ed. New York, NY, USA: McGraw-Hill, 2005.
- [27] J. Pedra, I. Candela, and L. Sainz, "Modelling of squirrel-cage induction motors for electromagnetic transient programs," *IET Elect. Power Appl.*, vol. 3, no. 2, pp. 111–122, Mar. 2009.
- [28] S. D. Pekarek, E. A. Walters, and B. T. Kuhn, "An efficient and accurate method of representing magnetic saturation in physical-variable models of synchronous machines," *IEEE Trans. Energy Convers.*, vol. 14, no. 1, pp. 72–79, Mar. 1999.
- [29] *Simulink 7—User's Guide*, The MathWorks Inc., Natick, MA, USA, 2009.

- [30] M. Chapariha, F. Therrien, J. Jatskevich, and H. W. Dommel, "Implementation of induction machine VBR model with optional zero-sequence in SimPowerSystems, ASMG, and PLECS toolboxes," in *Proc. Int. Conf. Power Syst. Transients*, Vancouver, BC, Canada, Jul. 18–20, 2013, pp. 1–5.
- [31] U. M. Ascher and L. R. Petzold, *Computer Methods for Ordinary Differential Equations and Differential-Algebraic Equations*. Philadelphia, PA, USA: SIAM, 1998.
- [32] E. Levi, "Impact of cross-saturation on accuracy of saturated induction machine models," *IEEE Trans. Energy Convers.*, vol. 12, no. 3, pp. 211–216, Sep. 1997.
- [33] X. Tu, L.-A. Dessaint, R. Champagne, and K. Al-Haddad, "Transient modeling of squirrel-cage induction machine considering air-gap flux saturation harmonics," *IEEE Trans. Ind. Electron.*, vol. 55, no. 7, pp. 2798–2809, Jul. 2008.
- [34] U. M. Ascher and C. Greif, *A First Course in Numerical Methods*. Philadelphia, PA, USA: SIAM, 2011.
- [35] F. P. De Mello and G. W. Walsh, "Reclosing transients in induction motors with terminal capacitors," *AIEE Trans. Power App. Syst.*, vol. 80, pp. 1206–1213, Feb. 1961.
- [36] K.-E. Hallenius, P. Vas, and J. E. Brown, "The analysis of saturated self-excited asynchronous generator," *IEEE Trans. Energy Convers.*, vol. 6, no. 2, pp. 336–345, Jun. 1991.
- [37] J. C. Moreira and T. A. Lipo, "Modeling of saturated ac machines including air-gap flux harmonic components," *IEEE Trans. Ind. Appl.*, vol. 28, no. 2, pp. 343–349, Mar./Apr. 1992.
- [38] D. Bispo, L. M. Neto, J. T. de Resende, and D. A. de Andrade, "A new strategy for induction machine modeling taking into account the magnetic saturation," *IEEE Trans. Ind. Appl.*, vol. 37, no. 6, pp. 1710–1719, Nov./Dec. 2001.
- [39] S. J. Salon, *Finite Element Analysis of Electrical Machines*. Norwell, MA, USA: Kluwer, 1995.
- [40] S. D. Sudhoff, B. T. Kuhn, K. A. Corzine, and B. T. Branecky, "Magnetic equivalent circuit modeling of induction motors," *IEEE Trans. Energy Convers.*, vol. 22, no. 2, pp. 259–270, Jun. 2007.
- [41] L. Wang and J. Jatskevich, "Including magnetic saturation in voltage-behind-reactance induction machine model for EMTP-type solution," *IEEE Trans. Power Syst.*, vol. 25, no. 2, pp. 975–987, May 2010.



Francis Therrien (S'12) received the B.Eng. degree in electrical engineering from the Université de Sherbrooke, Sherbrooke, QC, Canada, in 2010. He is currently working toward the Ph.D. degree in electrical and computer engineering at the University of British Columbia, Vancouver, BC, Canada.

He was a Power Systems Researcher for CYME International T&D until 2011. His current research interests include modeling of electrical distribution systems, electrical machines, and simulation of electromagnetic transients.



Mehrdad Chapariha (S'08) received the B.Sc. and M.Sc. degrees in electrical engineering from the Isfahan University of Technology, Isfahan, Iran, in 2006 and 2009, respectively, and the Ph.D. degree in electrical engineering from the University of British Columbia, Vancouver, BC, Canada, in 2013.

He is currently a Postdoctoral Fellow and a Sessional Lecturer at the University of British Columbia. His current research interests include modeling electrical machinery, simulation of power systems transients, modeling and design of electrical drive and power electronic systems, and applications of power electronics in power systems.



Juri Jatskevich (M'99–SM'07) received the M.S.E.E. and the Ph.D. degrees in electrical engineering from Purdue University, West Lafayette, IN, USA, in 1997 and 1999, respectively.

Since 2002, he has been a Faculty Member with the University of British Columbia, Vancouver, BC, Canada, where he is currently a Professor of Electrical and Computer Engineering. His current research interests include power electronic systems, electrical machines and drives, and modeling and simulation of electromagnetic transients.

Dr. Jatskevich chaired the IEEE CAS Power Systems and Power Electronic Circuits Technical Committee in 2009–2010. He was as an Associate Editor for the IEEE TRANSACTIONS ON POWER ELECTRONICS from 2008 to 2013, and is currently the Editor-in-Chief of the IEEE TRANSACTIONS ON ENERGY CONVERSION and the Editor of the IEEE POWER ENGINEERING LETTERS. He is the Chair of the IEEE Task Force on Dynamic Average Modeling, under the Working Group on Modeling and Analysis of System Transients Using Digital Programs.

High-Frequency Modeling of the Long-Cable-Fed Induction Motor Drive System Using TLM Approach for Predicting Overvoltage Transients

Liwei Wang, *Student Member, IEEE*, Carl Ngai-Man Ho, *Member, IEEE*, Francisco Canales, *Member, IEEE*, and Juri Jatskevich, *Senior Member, IEEE*

Abstract—Induction motor drive systems fed with cables are widely used in many industrial applications. Accurate prediction of motor terminal overvoltage, caused by impedance mismatch between the long cable and the motor, plays an important role for motor dielectric insulation and optimal design of dv/dt filters. In this paper, a novel modeling methodology for the investigation of long-cable-fed induction motor drive overvoltage is proposed. An improved high-frequency motor equivalent circuit model is developed to represent the motor high-frequency behavior for the time- and frequency-domain analyses. The motor equivalent circuit parameters for the differential mode (DM) and common mode (CM) are extracted based on the measurements. A high-frequency cable model based on improved high-order multiple- π sections is proposed. The cable model parameters are identified from the DM impedances in open circuit (OC) and short circuit (SC). To obtain a computationally efficient solution that could potentially be integrated with the motor drive controller, the system equations are discretized and solved using transmission-line modeling (TLM) approach. The proposed methodology is verified on an experimental 2.2-kW ABB motor drive benchmark system. The motor overvoltage transients predicted by the proposed model is in excellent agreement with the experimental results and represents a significant improvement compared with the conventional models.

Index Terms—High-frequency modeling, impedance measurement, induction motor drives, long cable, overvoltage transients, transmission-line modeling (TLM).

I. INTRODUCTION

THE ADJUSTABLE speed drive systems are undergoing significant changes with the successful application of novel power semiconductor devices. The rise and fall switching times of newer devices have been decreased in order to reduce switching losses and increase the overall efficiency. However, very fast switching times coupled with the long cables can cause voltage reflection at the motor terminals due to cable-motor surge impedance mismatch [1], [2]. In some cases, the motor terminal overvoltage can reach two times (or even higher) of the

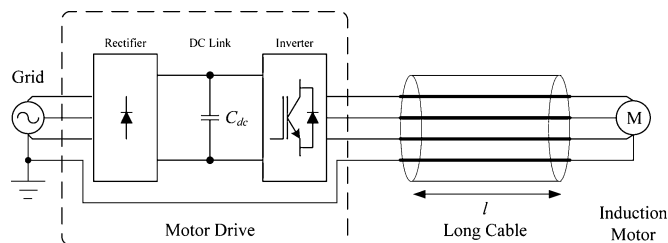


Fig. 1. Typical motor drive system with connecting cable.

dc-bus voltage [3], [4], which may seriously harm the motors' dielectric insulation leading to subsequent failures [5], [6]. Therefore, accurate modeling of the cable-fed motor drive systems in the high-frequency range is crucial for predicting the motor terminal overvoltage and designing mitigation filters [7]–[10].

A typical system composed of a drive feeding an induction machine through a cable is shown in Fig. 1. In order to facilitate the high-frequency cable and motor modeling and characterization, two modes of operations, common mode (CM) and differential mode (DM) [11], are usually considered. The CM is formed as a two-port network between the parallel-connected three phases and ground wire. The DM is regarded as a two-port network between two parallel-connected phases and the third phase of the cable motor system.

Numerous high-frequency motor models have been proposed in the literature for overvoltage and electromagnetic interference (EMI) analysis [12]–[18]. A pioneering basic phase-belt winding model was proposed in [12] to represent motor single-phase coils. Therein, a high-frequency model was established based on series connections of four basic phase-belt windings. Subsequent research [13]–[18] on high-frequency motor models has focused on constructing more “easy-to-use” motor equivalent circuits and straightforward parameter identification procedures. One [13], [15], [17] or two [14], [16] phase-belt winding motor models are proposed, which greatly simplifies the high-frequency motor model. In [17], an efficient high-frequency motor model is proposed for the studies of both DM and CM behaviors in time- and frequency-domain. Therein, a simple and easy-to-follow model parameterization procedure is included based on DM and CM impedance measurements. In [18], a universal induction machine model with low-to-high frequency-response characteristics is proposed. This model is quite sophisticated as the DM, CM, and bearing models are all integrated into one three-phase motor equivalent circuit.

Manuscript received October 6, 2009; revised February 12, 2010; accepted March 12, 2010. Date of current version September 17, 2010. Recommended for publication by Associate Editor B. Ferreira.

L. Wang and J. Jatskevich are with the Department of Electrical and Computer Engineering, University of British Columbia, Vancouver, BC V6T1Z4, Canada.

C. N.-M. Ho and F. Canales are with the ABB Corporate Research Center, Baden-Daettwil 5405, Switzerland (e-mail: carl.ho@ch.abb.com).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TPEL.2010.2047027

Modeling of long cables for analysis of motor voltage reflection phenomena has been studied as well [7]–[9], [19]–[23]. It has been recognized in [15] and [20]–[23] that a realistic and accurate cable model is of great importance for the accurate prediction of motor terminal overvoltage. Various cable models were proposed including conventional multiple π -section model with second-order per section [7], [8], lossless [9], [19], lossy [20]–[22], and frequency-dependent [15], [23] transmission-line models (TLMs). In [21], a lossy but distortionless cable model is used, which was shown to improve the accuracy of the predicted overvoltage compared with the lossless cable model. In [15], a multiple- π cable model was proposed, where the cable parallel branch is represented as a high-order capacitive circuit to include the frequency-dependent dielectric losses. Oriti and Julian [23] represented the entire cable–motor system in frequency domain to model the frequency-dependent effects of the cable and motor. However, to the best of our knowledge, a time-domain frequency-dependent cable model, which includes both the skin and proximity effects and the dielectric losses for the high-frequency range, have not been developed for the motor overvoltage studies. This paper proposes such a model and makes the overall contributions that can be summarized in the following.

- 1) An improved motor model, based on [17], is proposed that accurately captures the high-frequency DM and CM impedance characteristics from hundreds of hertz to tens of megahertz. We also provide detailed and easy-to-reproduce parameter identification and characterization procedure for the improved equivalent circuit model.
- 2) A more accurate frequency-dependent cable model is developed using a higher-order multiple π -sections, which includes the skin and proximity effects as well as dielectric losses. The model parameters are identified based on the measured DM impedance characteristics in a wide frequency range from 100 Hz to 10 MHz.
- 3) For efficient calculation of the time-domain transient solution, the continuous-time cable–motor system is discretized using the TLM approach [24]–[27]. Therefore, the formulation and integration of differential equations is avoided, and the entire cable–motor system (with arbitrary number of cable sections) could be implemented in a stand-alone program and hardware for control algorithms and/or firmware applications.
- 4) The proposed methodology is verified experimentally as well as compared against two conventional models [7], [8]. It is shown that the proposed model represents an appreciable improvement in predicting the motor overvoltage transients.

II. HIGH-FREQUENCY INDUCTION MOTOR MODELING

Accurate modeling of induction motors in high-frequency range plays an important role in investigating the motor drive overvoltage and EMI problems [12]–[18]. In many cases, motor equivalent circuit and the associated parameters are established from the motor impedance characteristics and/or geometry. The high-frequency motor model used in this paper is based on per-

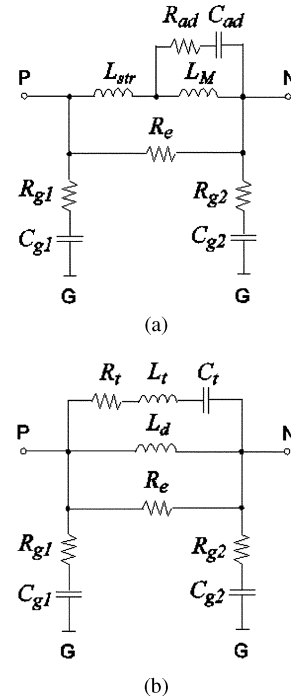


Fig. 2. High frequency per-phase motor equivalent circuit (a) model defined in [17] and (b) proposed model.

phase equivalent circuit proposed in [17]. The circuit diagram of the motor per-phase high-frequency model is illustrated in Fig. 2(a). This circuit provides relatively simple model structure. However, the parameters R_{ad} and C_{ad} in Fig. 2(a) are very difficult to express using analytical formulas [17]. Instead, trials and adjustments of R_{ad} and C_{ad} are usually required with the help of frequency-domain simulations to achieve satisfactory results [17]. In this paper, an improved high-frequency model is proposed as shown in Fig. 2(b). The proposed circuit uses similar model structure and parameterization procedure as in [17]. However, a series R_t , L_t , and C_t branch is introduced [see Fig. 2(b)] to replace the R_{ad} and C_{ad} branch [see Fig. 2(a)] without sacrificing model accuracy. Since this new series branch has its own resonance, analytical calculation of parameters R_t , L_t , and C_t can be carried out in a more straightforward way from the measured impedances, which represents an advantage over identification of the R_{ad} and C_{ad} branch in Fig. 2(a). The detailed derivations for the branch parameters are not included here due to limited space, whereas interested reader can find more background information in [13] and [15].

The equivalent circuit depicted in Fig. 2(b) and its parameters also have physical meaning and significance. In particular, R_{g1} and C_{g1} represent the parasitic resistance and capacitance between the stator winding and the motor frame; R_{g2} and C_{g2} represent the parasitic resistance and capacitance between the stator neutral and motor frame; L_d represents the stator winding leakage inductance; R_e represents the high-frequency iron loss of the stator winding; L_t and C_t are introduced to capture the second resonance in the motor impedance characteristic, which may be caused by the skin effect and interturn capacitance of the stator windings.

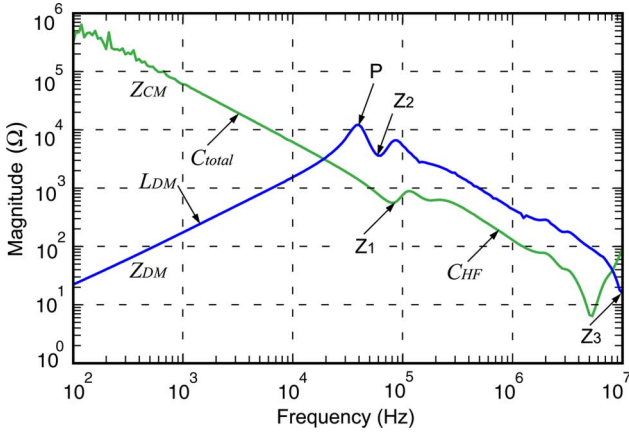


Fig. 3. Measured DM and CM impedances of the induction motor under study.

The overall motor high-frequency three-phase equivalent circuit model can be obtained assuming a wye connection of three single-phase circuits of Fig. 2(b). Such assumption is widely used in high-frequency machine modeling [13]–[18], and it has an advantage of simpler model structure and easier identification of the associated parameters. In particular, the parameters of the per-phase equivalent circuit in Fig. 2(b) are identified through the DM and CM impedance characteristics measured in frequency domain. However, it is also noted that the proposed three-phase high-frequency motor model is valid independent of the actual physical connection of the stator windings (i.e., delta- or wye-connection). In other words, whether the actual stator winding is delta- or wye-connected, the procedure presented in this paper will give an equivalent wye-connected circuit-model that will adequately represent the considered overvoltage phenomena.

The motor under investigation is the four-pole, delta-connected, 2.2-kW, ABB M2AA100LA. The detailed DM and CM measurements setup and test procedures have been well documented in the literature, e.g., [17]. Similar measurement procedures have been used here. In particular, the motor impedances have been measured when the motor is stationary and disconnected from the ac source (drive) [13]–[18]. As it has been shown in [18] and [28], the rotor speed or the stator current has no significant influence on the motor impedance characteristics in mid-to-high frequency range. The underlying reason is that the flux penetrating into the rotor magnetic circuit at high frequency is very small [28], [29]. For the measurements presented here, an HP/Agilent 4294 A impedance analyzer [30] is used to evaluate the impedances in the range from 100 Hz to 10 MHz. The measured DM and CM impedances Z_{DM} and Z_{CM} are shown in Fig. 3, respectively.

To calculate the parameters of the high-frequency motor model, several characteristic points are chosen from the measured DM and CM impedances of Fig. 3. First, the low- and high-frequency CM capacitances (slopes) C_{total} and C_{HF} are extracted from the measured CM impedance Z_{CM} , respectively. Second, the resonance frequency and impedance magnitude of the first zero point (Z_1) from Z_{CM} are identified. Third, the low-frequency DM inductance L_{DM} is evaluated from the measured

TABLE I
HIGH-FREQUENCY MOTOR PARAMETERS

Parameter	Value	Parameter	Value
R_{g1}	10.57 Ω	C_{g1}	462 pF
R_{g2}	42.94 Ω	C_{g2}	422 pF
R_t	2.9 k Ω	C_t	147 pF
R_e	8.11 k Ω	L_t	43.2 mH
L_{zu}	411 nH	L_d	17.2 mH

DM impedance Z_{DM} . Finally, the resonance frequencies and impedance magnitudes of the pole (P), the second (Z_2), and the third (Z_3) zero points are measured from Z_{DM} . Given these characteristic measurements, the parameters of the high-frequency motor model in Fig. 2(b) are readily calculated according to the equations summarized in Appendix A. For example, the parameters R_t , L_t , and C_t are derived from the second resonant point (Z_2) in the DM impedance Z_{DM} characteristic shown in Fig. 3. The corresponding formulas are given in Appendix A (A8)–(A10) for consistency. The resulting high-frequency motor parameters for the considered induction motor are summarized in Table I.

To verify the effectiveness of the proposed equivalent circuits and the procedure for calculating parameters, the measured DM and CM impedances are compared to the corresponding impedances calculated from the high-frequency motor equivalent circuits. The high-frequency DM motor equivalent circuit is constructed here by connecting two of the phases in parallel and the remaining third phase. Based on the per-phase equivalent circuit in Fig. 2(b), the combined DM motor equivalent circuit is shown in Fig. 4(a). The high-frequency CM motor equivalent circuit is built by connecting all the three phases in parallel. Based on per-phase equivalent circuits of Fig. 2(b), the resulting CM motor equivalent circuit from phase to ground is shown in Fig. 4(b). The measured and fitted results using the constructed equivalent circuits are superimposed in Fig. 5 for better clarity. As can be observed in Fig. 5, the fitted DM and CM frequency characteristics have good agreement with the measured responses. These results verify the effectiveness of the proposed three-phase wye-connected equivalent circuit model for representing the high-frequency characteristics of the motor windings (although the stator winding is delta-connected).

It is noted that the motor line-to-line overvoltage ringing transient is mostly associated with DM, since this mode defines the physical path along which the insulated gate bipolar transistor (IGBT)-generated voltage pulse travels through the cable and the stator winding of the motor. Since the line-to-line overvoltage predominantly impacts the insulation of windings and cables, this type of overvoltage is primarily important for the purpose of this paper. Therefore, the DM equivalent circuit in Fig. 4(a) is used here for motor drive overvoltage studies.

III. HIGH-FREQUENCY CABLE MODELING

Accurate modeling of power cable is also essential for predicting motor terminal overvoltage. Since the overvoltage transient

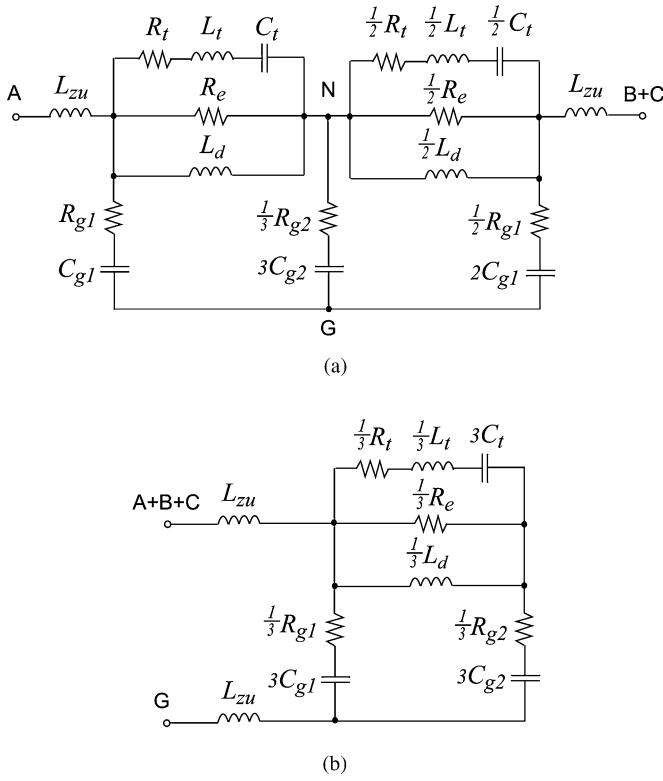


Fig. 4. High-frequency equivalent circuit motor model: (a) for DM assuming phases B and C are connected in parallel and (b) for CM between the three connected phases and ground.

at the motor end may contain oscillations in wide range of frequencies from several hundreds of hertz to several megahertz depending on the cable characteristic and length, the cable model should also include the frequency-dependent phenomena such as skin and proximity effects, dielectric losses. The available methods for modeling cables at this level and extracting parameters include finite-element analysis [22] or direct impedance measurement [15], [31] in frequency domain using precision impedance analyzers. In this paper, the same impedance analyzer was also used to characterize the cable.

The cable used here is an unshielded, PVC-insulated, four-core cable with the conductor area of 1.5 mm^2 . The short circuit (SC) and open circuit (OC) tests in DM [22], [31] have been performed upon which the cable parameters were calculated. The measured shunt capacitance and conductance (OC test), and the series resistance and inductance (SC test) for 1-m sample cable section are also shown in Fig. 6. As can be observed in Fig. 6, the series resistance and inductance, as well as the parallel conductance and capacitance, all appear strongly frequency-dependent, especially in the range of high frequencies. However, this phenomenon is often ignored in most of the published literature on the motor drive overvoltage studies. When such assumption is made, the resulting per-section equivalent circuit is just a conventional second-order per-section cable model shown in Fig. 7(a) [7], [8].

The advantage of equivalent circuit of Fig. 7(a) is of course its simplicity. The parameters for this circuit are typically identified at the resonant frequency of motor terminal overvoltage.

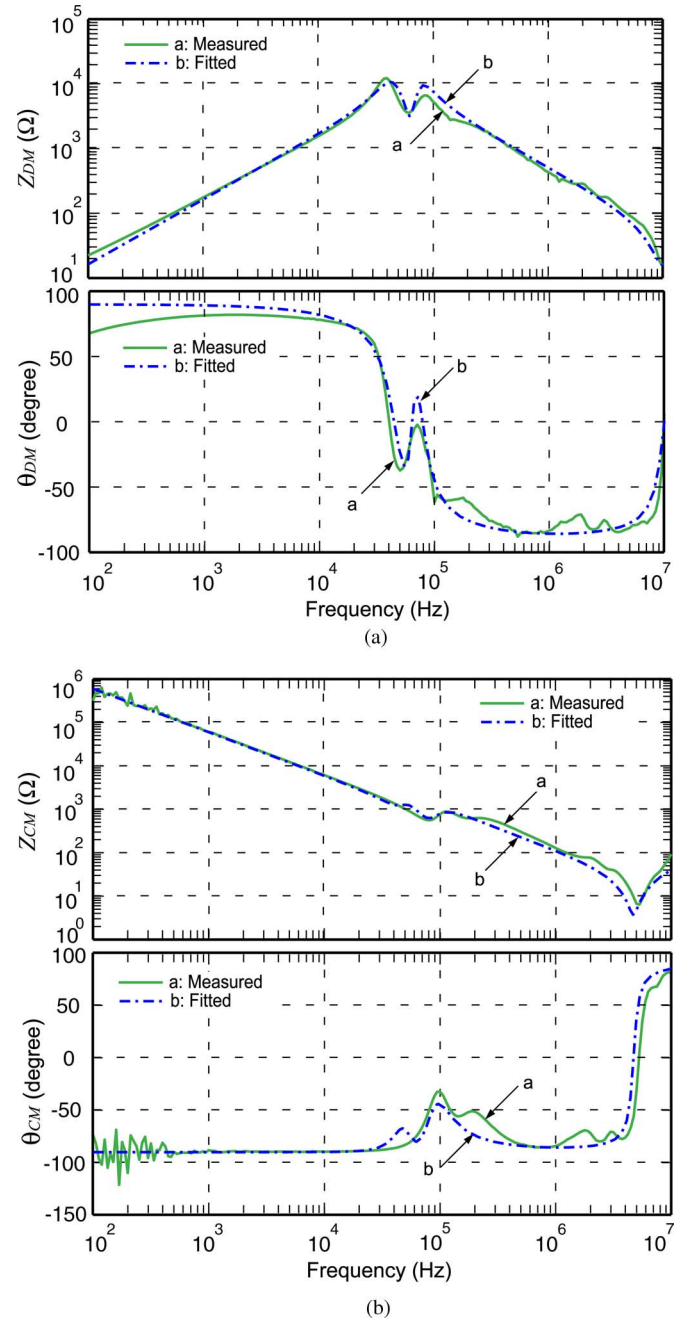


Fig. 5. Measured and fitted motor impedances for (a) DM and (b) CM.

However, the resonant frequency itself will also depend on the cable length and may be difficult to predict with good accuracy. An improved cable model was proposed in [15], which can represent the dielectric losses using a higher order parallel branch as shown in Fig. 7(b). Further improvements are proposed in this paper in order to include the skin and proximity effects. Such effects are not taken into account with the use of first-order series branch. Therefore, a high-order model is proposed in Fig. 7(c). This section model combines the parallel branches as proposed in [15] and utilizes higher order series branches to represent skin and proximity effects. Overall, one additional inductor and

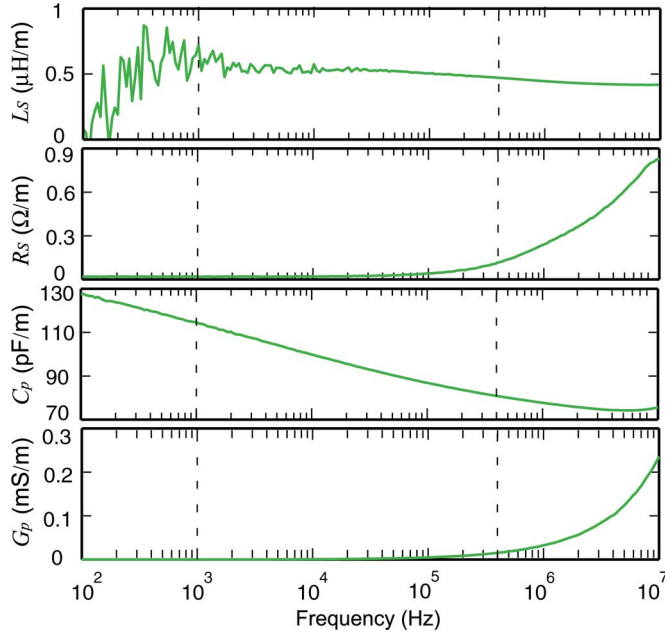


Fig. 6. Frequency-dependent parameters of the cable as measured in DM.

capacitor are utilized in the per-section model compared with the classic second-order constant parameter cable model.

To identify the parameters of the circuit of Fig. 7(c), the circuit response is considered in the low- and high-frequency ranges, respectively. In particular, in the low-frequency range, the SC and OC circuits would appear as depicted in Fig. 8(a) and (c), respectively. Similarly, for high-frequency range, the SC and OC circuits will appear as in Fig. 8(b) and (d), respectively. The separation into low- and high-frequency ranges facilitates the parameter identification procedure for the proposed section circuit of Fig. 7(c) from the measured impedances. To illustrate this point, the measured SC and OC impedances Z_{SC} and Z_{OC} of the 1-m cable section in DM are depicted in Fig. 9. As shown in Fig. 9, the measured SC impedance Z_{SC} can be approximated over low- and high-frequency ranges by the first-order inductance transfer functions with the asymptotes of 20-dB/decade slope. The low- and high-frequency SC inductances L_{SC-LF} and L_{SC-HF} are directly extracted from the magnitude of Z_{SC} . As it can be observed in Fig. 9, the inductance value of L_{SC-HF} is smaller than L_{SC-LF} . This difference is attributed due to the skin and proximity effects, which again verifies the frequency-dependent nature of the measured cable impedance, which is represented by the series branch in the equivalent circuit. A similar conclusion applies to the measured OC cable impedance Z_{OC} as shown in Fig. 9.

Based on SC results Z_{SC} , the corresponding inductances can be determined from the 20-dB/decade slope in the low- and high-frequency regions, L_{SC-LF} and L_{SC-HF} , using the following relationships:

$$L_{s1} + L_{s2} = L_{SC-LF} \quad (1)$$

$$L_{s1} = L_{SC-HF}. \quad (2)$$

Similarly, based on the results of Fig. 9 in OC Z_{OC} , the corresponding capacitances can be determined from the –20-dB/decade slope in the low- and high-frequency regions as

$$C_{p1} + C_{p2} = C_{OC-LF} \quad (3)$$

$$C_{p1} = C_{OC-HF} \quad (4)$$

where C_{OC-LF} and C_{OC-HF} are the low- and high-frequency open-circuit cable capacitance shown in Fig. 9, respectively.

The aforementioned procedure readily establishes parameters of the equivalent circuit section of Fig. 7(c). More detailed formulas for cable parameter calculation are given in Appendix B and the resulting parameters are summarized in Table II. To show the effectiveness of this procedure, the measured and the fitted SC and OC impedances are superimposed in Fig. 10. As can be observed in Fig. 10, the results produced by the proposed equivalent circuit of Fig. 7(c) are in good agreement with the measured characteristics for the actual cable segment. It is observed in Fig. 10(b) that the absolute fitting error of the phase angle θ_{OC} is within $\pm 5^\circ$ starting from 400 Hz to MHz range (which is within the targeted frequency range of the equivalent circuit for motor overvoltage studies).

IV. TLM FORMULATION OF THE LONG-CABLE-FED MOTOR DRIVE SYSTEM

In this paper, the TLM approach is used to formulate the difference equations of the overall long-cable-fed motor drive system for time-domain simulation as a stand-alone program. This approach also provides a convenient means to consider arbitrary number of cable sections for accurate representation of cables of different lengths. Depending on component characteristics, stub lines with different termination conditions are utilized to represent various circuit components. For linear resistors, the algebraic equation given by Ohm's law is used in TLM representations. The inductor TLM model is usually represented as a Thevenin equivalent circuit as shown in Fig. 11(a). For an inductor L , a short-circuited stub line is used, given the surge impedance of $Z_L = 2L/\Delta t$, where Δt is the simulation time step and also round-trip traveling time along the stub line. The inductor branch voltage equation is then formulated as [24]–[27]

$$v_L(t) = Z_L i_L(t) + 2v_L^i(t) \quad (5)$$

where $v_L^i(t)$ is the incident voltage pulse at the present time step t . For the calculation purpose, $v_L^i(t)$ is required to be represented by the branch voltage and incident voltage pulse at the previous time step, e.g., $v_L(t - \Delta t)$ and $v_L^i(t - \Delta t)$. For the short-circuited stub, the reflected voltage pulse at the previous time step $v_L^r(t - \Delta t)$ is inverted when it travels back from the line end after the round-trip traveling time Δt . Therefore, the incident voltage pulse at the present time step $v_L^i(t)$ is represented as

$$v_L^i(t) = -v_L^r(t - \Delta t) = v_L^i(t - \Delta t) - v_L(t - \Delta t). \quad (6)$$

Similarly, a capacitive branch is modeled as an open-circuited stub line depicted in Fig. 11(b) with surge impedance of $Z_C = \Delta t/2C$. The capacitor is also represented as a Thevenin

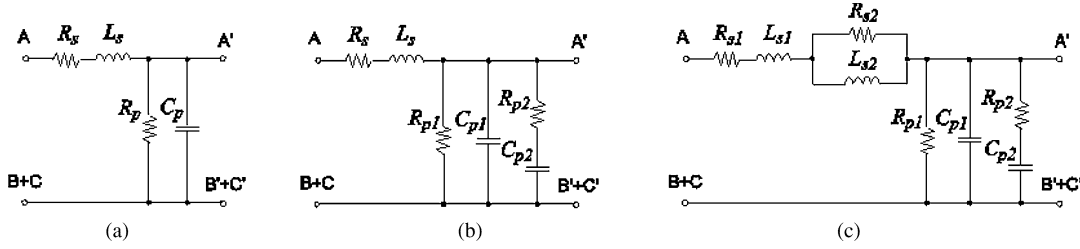


Fig. 7. Per-section DM cable models. (a) Conventional. (b) improved model [15]. (c) Proposed model.

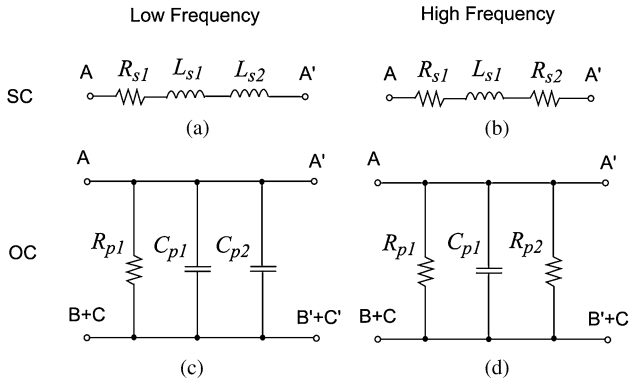


Fig. 8. SC and OC cable model representations as viewed in low- and high frequencies.

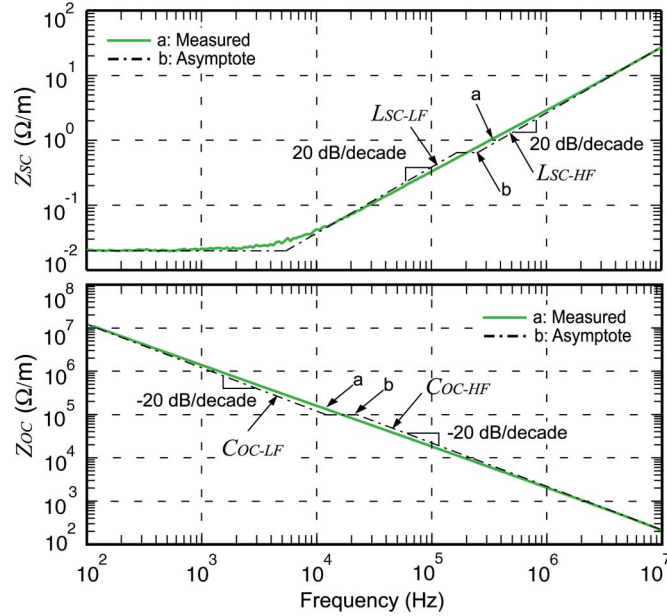
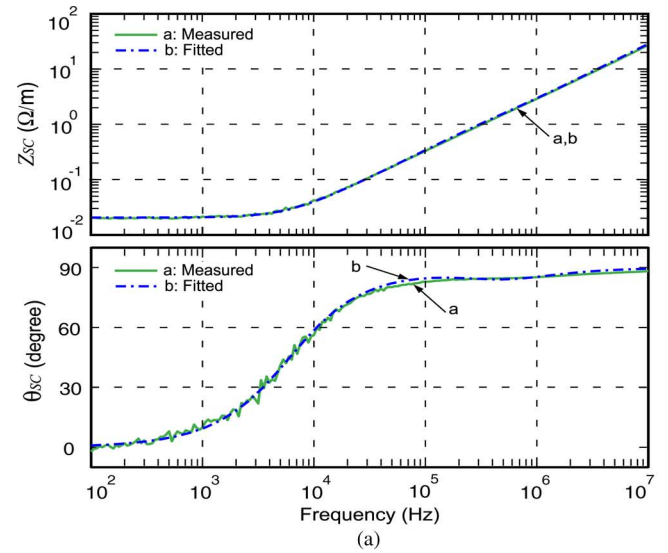


Fig. 9. Measured SC and OC impedances of the 1-meter cable section in DM.

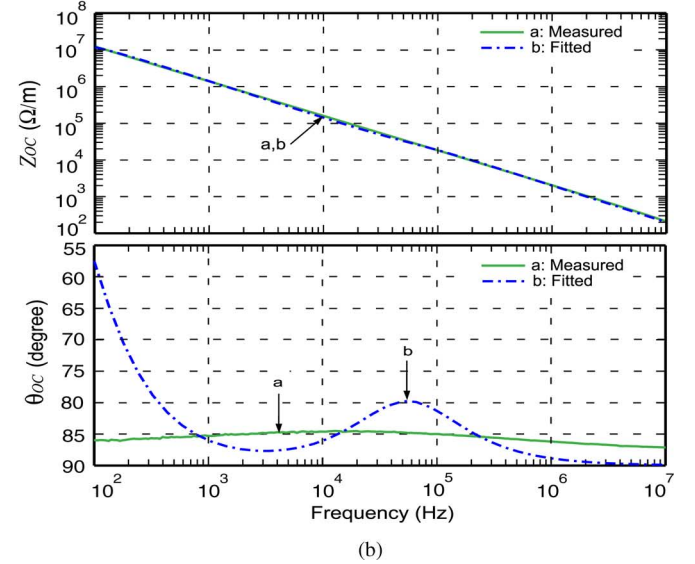


Fig. 10. Measured and fitted cable impedances. (a) SC. (b) OC.

TABLE II
PROPOSED CABLE PER-METER PARAMETERS

Parameter	Value	Parameter	Value
R_{s1}	21 mΩ	L_{s1}	451.8 nH
R_{s2}	271.2 mΩ	L_{s2}	89.32 nH
R_{p1}	22 MΩ	C_{p1}	79.6 pF
R_{p2}	100 kΩ	C_{p2}	34.11 pF

equivalent circuit shown in Fig. 11(b). The branch voltage equations is given as

$$v_C(t) = Z_C i_C(t) + 2v_C^i(t). \quad (7)$$

As the capacitive stub is open circuited, the reflected voltage pulse at the previous time step $v_C^r(t - \Delta t)$ travels back without inversion and becomes the incident voltage pulse at the present

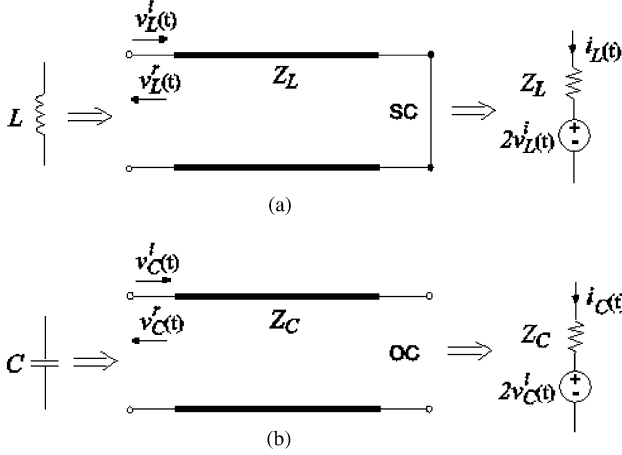


Fig. 11. TLM models of (a) inductance and (b) capacitance.

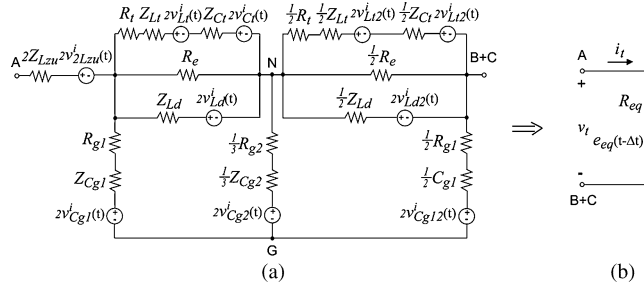


Fig. 12. TLM representation of the DM motor model. (a) Original circuit. (b) Reduced Thevenin equivalent.

time step after the round-trip traveling time Δt as

$$v_C^i(t) = v_C^r(t - \Delta t) = v_C(t - \Delta t) - v_C^i(t - \Delta t). \quad (8)$$

In order to represent the DM motor and per-section cable models using TLM technique, the inductances and capacitances in Figs. 4(a) and 7(c) are replaced by their corresponding impedances and incident voltage sources. The resultant TLM representations of the DM motor and per-section cable model are formulated in Figs. 12(a) and 13(a), respectively. For the purpose of convenience, the Thevenin equivalent circuits of Figs. 12(a) and 13(a) are calculated and are given as Figs. 12(b) and 13(b), respectively. Therein, $e_{eq}(t - \Delta t)$, $e_s(t - \Delta t)$, and $e_p(t - \Delta t)$ are dependent on the previous time step as the incident voltages in Figs. 12(a) and 13(a) are calculated by (6) and (8), respectively. Due to space limitation, the detailed expressions for the Thevenin impedances and voltage sources in Figs. 12(b) and 13(b) are not included (these are easily calculated using network reduction techniques).

Assuming an arbitrary number of cable sections, the entire DM long-cable motor drive system in discrete time domain is represented in Fig. 14. At each time step, the loop current equations are then used to solve the system of Fig. 14 as

$$\mathbf{Z}\mathbf{I} = \mathbf{V} \quad (9)$$

where

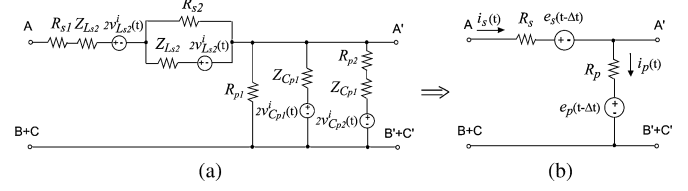


Fig. 13. TLM representation of the per-section DM cable model. (a) Original circuit. (b) Thevenin equivalent.

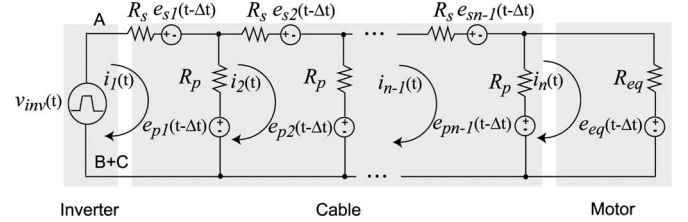


Fig. 14. Final formulation of DM long-cable motor drive models in discrete time domain.

$$\mathbf{Z} = \begin{bmatrix} R_s + R_p & -R_p & & & \\ -R_p & R_s + 2R_p & -R_p & & \\ & -R_p & \ddots & -R_p & \\ & & -R_p & R_s + 2R_p & -R_p \\ & & & -R_p & R_p + R_{eq} \end{bmatrix} \quad (10)$$

$$\mathbf{V} = [v_{inv}(t) - e_{s1}(t - \Delta t) - e_{p1}(t - \Delta t) \\ e_{p1}(t - \Delta t) - e_{s2}(t - \Delta t) - e_{p2}(t - \Delta t) \cdots \\ e_{p,n-2}(t - \Delta t) - e_{s,n-1}(t - \Delta t) - e_{p,n-1}(t - \Delta t) \\ e_{p,n-1}(t - \Delta t) - e_{eq}(t - \Delta t)]^T \quad (11)$$

$$\mathbf{I} = [i_1(t) \ i_2(t) \ \cdots \ i_{n-1}(t) \ i_n(t)]^T. \quad (12)$$

Here, $\mathbf{Z}$ is constant system resistance matrix; $\mathbf{V}$ is the loop voltage vector; $\mathbf{I}$ is the unknown loop current vector. It is noted that $\mathbf{V}$ is known for the present time step as the Thevenin equivalent voltage sources in (11) are known from the previous time step and the inverter voltage $v_{inv}(t)$ is given.

Since the matrix $\mathbf{Z}$ is tridiagonal, sparse matrix technique may be applied to solve the system equations very efficiently. At each time step, the loop currents are solved first from (9). The branch currents and voltages are then calculated according to Figs. 12(a) and 13(a). The incident voltages are finally updated using (6) or (8), depending on the inductive or capacitive branches, for the next time step.

V. EXPERIMENTAL VERIFICATION

To validate the methodology described in this paper, the experimental tests with long-cable-fed motor drive system have been carried out and are presented in this section. The drive system is ABB 2.2-kW ACS50-type with switching frequency of 16 kHz. The rise and fall times of IGBT switches are approximately 190 μ s. The power cable is an unshielded, PVC-insulated, four-core cable with the conductor area of

TABLE III
CONVENTIONAL CABLE PER-METER PARAMETERS
(a) AT 1 kHz. (b) AT 0.4 MHz

(a)			
Parameter	Value	Parameter	Value
R_s	21 m Ω	L_s	580 nH
R_p	16.9 M Ω	C_p	114.6 pF
(b)			
Parameter	Value	Parameter	Value
R_s	120 m Ω	L_s	530 nH
R_p	66.2 k Ω	C_p	80.9 pF

1.5 mm². The same four-pole, 2.2-kW, ABB (model number: M2AA100LA) induction machine was used in all tests. The measured impedances of the cable and the motor have been illustrated in Sections II and III.

In order to demonstrate the importance of including frequency-dependent effects in cable modeling, two conventional models and the proposed model have been considered. All considered models use multi- π sections equivalent circuits to approximate the distributed parameter effects, wherein the length of each cable section is assumed to be 0.1 m. The conventional models use second-order per-section equivalent circuit shown in Fig. 7(a) with parameters R_s , L_s , R_p , and C_p measured at low- and high-frequency (1 kHz, Conventional model 1) and high-frequency (0.4 MHz, Conventional model 2), respectively. For consistency, the corresponding parameters are summarized in Table III. The high-frequency measurement at 0.4 MHz is chosen because the overvoltage oscillations for the considered cable were in the same frequency range. The proposed fourth-order per-section cable model has also been used. The model parameters were identified according to the experimental procedure described in Section III and results are summarized in Table II. For investigation of motor overvoltage transients, the cable lengths of 20, 25, 50, 60, 70, 100, and 120 m have been studied.

To demonstrate the overvoltage phenomena on the considered motor drive system with a 100-m cable, the measured line-to-line voltages at the inverter side and the motor terminal are shown in Fig. 15. As can be seen in this figure, the rectangular pulses generated by the inverter switching produce oscillatory transient and significant overvoltage at the motor terminal. A fragment of Fig. 15 has been magnified and superimposed with the simulated responses produced by the proposed and conventional models as depicted in Fig. 16(a) and (b), respectively. As can be seen in Fig. 16(a), the proposed model predicts more accurately both the waveform and the damping than either of the conventional models in Fig. 16(b). The models are further compared to the experimental results for the 70- and 120-m cables, as shown in Figs. 17 and 18, respectively. It is observed in Figs. 16(b), 17(b), and 18(b) that the conventional models 1 and 2 either over- or under estimate the overvoltage peaks. Also, since the attenuation and distortion effect due to frequency de-

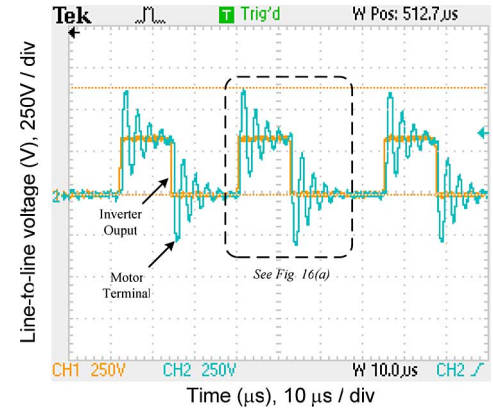


Fig. 15. Measured voltage waveforms at the inverter and motor terminals for 100-m cable.

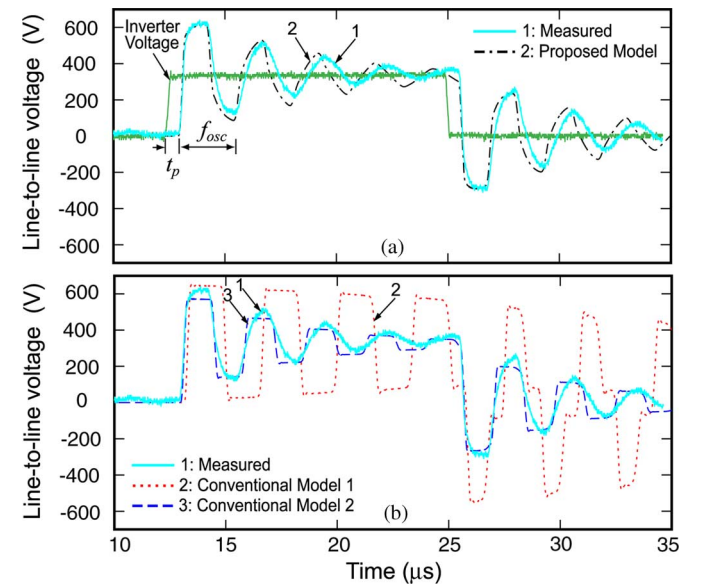


Fig. 16. Measured voltage waveforms at the motor terminals for 100-m cable. (a) Proposed model. (b) Conventional models 1 and 2.

pendence are not represented, the shape of the overvoltages predicted by models 1 and 2 are also more square-like. The overall damping time of the overvoltage transient predicted by the model 1 is also much longer compared with model 2. This also clearly shows that using the second-order section model with either low- or high-frequency parameters (see Table III) is not adequate to cover the spectrum of the overvoltage phenomena. At the same time, the proposed model demonstrates good results in predicting the overvoltage peaks, shape, as well as damping as can be clearly seen in Figs. 16(a)–18(a).

To further evaluate the considered models in predicting the overvoltage transient phenomena, the cable delay time t_p , oscillation frequency f_{osc} , and peak overvoltage have been compared with the experimental results for different cable lengths. The results of these comparisons are summarized in Figs. 19–21. The delay time t_p is defined with respect to the initial pulse produced by the inverter as shown in Fig. 16(a). The oscillation frequency f_{osc} is defined for the voltage on the motor terminal also shown

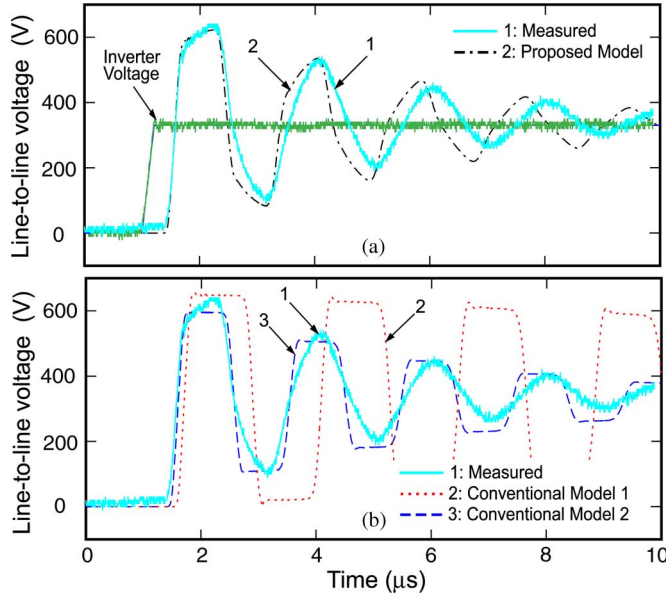


Fig. 17. Measured voltage waveforms at the motor terminals for 70-m cable. (a) Proposed model. (b) Conventional models 1 and 2.

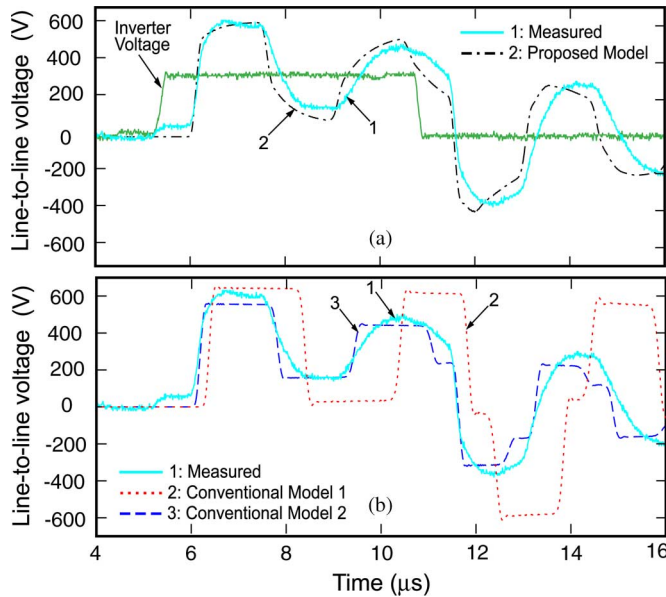


Fig. 18. Measured voltage waveforms at the motor terminals for 120-m cable. (a) Proposed model (b) Conventional models 1 and 2.

in Fig. 16(a). Both t_p and f_{osc} can also be expressed as [20]

$$t_p = l\sqrt{L_s C_p} \quad (13)$$

$$f_{osc} = \frac{1}{4l\sqrt{L_s C_p}} \quad (14)$$

where l is cable length, L_s and C_p are equivalent per-meter inductance and capacitance for a specific traveling wave frequency.

As seen in Fig. 19, the delay times are almost linear, which is consistent with (13). The conventional model 1 has a higher slope than the measured delay times because it relies on pa-

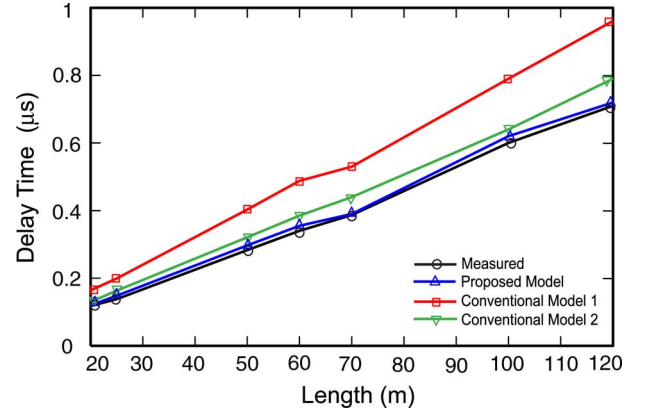


Fig. 19. Measured and predicted delay times for different cable lengths.

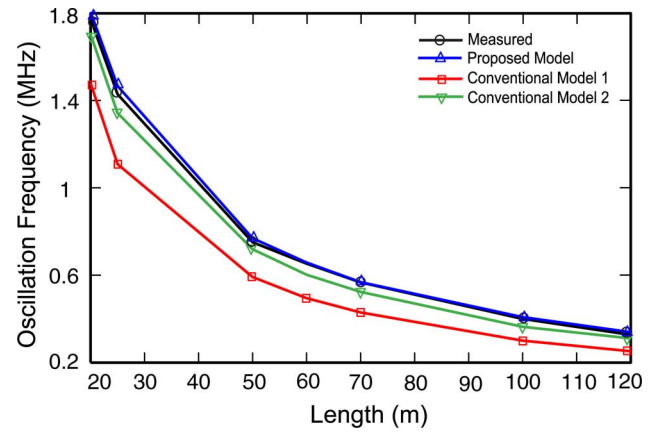


Fig. 20. Measured and predicted oscillation frequencies for different cable lengths.

rameters extracted at low frequency. The conventional model 2 gives more accurate delay times, since its parameters were extracted at high frequency that is closer to the ringing frequency. However, the proposed model predicts delay times with an even greater accuracy. Figs. 20 and 21 show the oscillation frequencies and the peak of the line-to-line overvoltage as measured from the drive-cable-motor system and predicted by various models. A similar consistent conclusion applies here. The lower order conventional models 1 and 2 are less accurate and the proposed model predicts the overvoltage phenomena in terms of both the oscillation frequencies and the peak of the line-to-line overvoltage with greater accuracy.

VI. DISCUSSION

A. Overvoltage Range

The typical motor terminal overvoltages demonstrated experimentally and predicted by the developed model reached nearly two times of the dc-bus voltage. This basic phenomenon can be easily explained by transmission-line theory assuming propagation of a single pulse from cable to motor and mismatch of the terminating impedances. However, under some conditions, the motor terminal overvoltage can reach even higher levels. For example, as has been reported in [3], [4], [20], and [21],

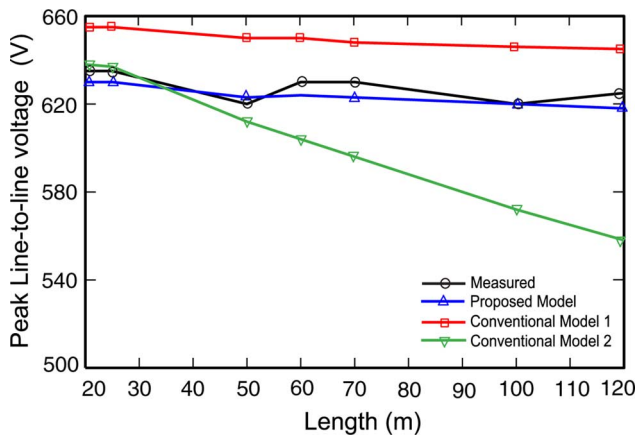


Fig. 21. Measured and predicted peak of line-to-line overvoltage for different cable lengths.

the motor terminal overvoltage can reach three to four times the dc-bus voltage. Such very high level (higher than two times) of the motor terminal overvoltage may occur when, in addition to a long cable, the inverter operation produces double pulsing [3] or successive pulsing [4], pulsing polarity reversal [3] or bipolar pulsing [4], etc. Under such conditions, the preceding overvoltage transient has not fully decayed before the successive voltage pulses are injected by the inverter. The overlapped pulses traveling in the cable could result in higher terminal overvoltages. The contributing factors to this phenomenon include (but not limited to) drive modulation techniques, carrier frequency selected, cable natural frequency of oscillation, and cable high-frequency damping losses [3]. As has been pointed out in [3], special considerations and/or adjustments of the switching strategy should be used to avoid such conditions. The voltage stress may seriously degrade motor dielectric capability [32] and even causes motor winding insulation failures.

B. Definition of Cable DM

Depending on the numbers of cable internal conductors, the DM may be defined as a two-port or multiport network between/among two or more phase conductors [22]. As the focus of the paper is the line-to-line overvoltage transients, the DM assumed in this paper is defined by connecting two cable phases (conductors) in parallel and the remaining third phase becomes the returning path. It is also noted that the aforementioned two-port DM cable model is asymmetrical. Similar definition of the DM has been used in [4], [15], and [19]–[21], and it is chosen here, since it corresponds to the physical path along which the IGBT-generated voltage pulse travels in the considered hardware system.

The advantage of using the two-port DM network is that a single-phase TLM can be utilized [4], [15], [19]–[21] to represent the DM cable characteristic, which further simplifies the three-phase cable modeling. Since the cable characteristics are directly measured in DM as a two-port network, the parameters of the single-phase TLM are directly derived from the measured DM characteristics. Therefore, there is no need to measure the three-phase coupling parameters and reducing the three-phase

model to a single-phase DM model. Although the asymmetrical DM cable model gives the current in return path, once the model is available, it is possible to calculate the currents in the two parallel phases (conductors). However, it is not required for the purpose of studies considered herein.

C. CM Overvoltage

The CM overvoltage can also appear at motor terminals and create CM current through the stator-winding-to-frame and cable-to-ground parasitic capacitances. The CM current, as a high-frequency noise, could have harmful effects including the EMI for the surrounding equipments [33], and can also cause oil film dielectric breakdown inside the motor bearings and subsequently reduction of the bearing's lifetime [34]–[37]. Interested reader can find an equivalent circuit for representing the bearing-shaft path ([36], Fig. 2), which in general could be integrated with the CM equivalent circuit described in this paper. However, properly addressing this subject would require a dedicated publication and is beyond the scope of the paper.

D. Factors Influencing the Accuracy of Cable Parameter Measurement

It is noted that the accuracy of cable impedance measurement is potentially influenced by many factors including the length of the sample cable, measurement layout (i.e., cable layout with curves, straight lines, nearby conductors, grounding surfaces, proximity to magnetic materials, leads/fixture/adaptor of the impedance analyzer, etc.). Therefore, a very careful measurement setup is designed such that the effect of these influencing factors is minimized. Also, the preferred sample cable length should not be too short [15], [20]–[22] in order to be within the acceptable range of accuracy for a typical impedance analyzer. If a shorter cable length is used, the measured parameters could be too small, and therefore, less accurate. A 1-meter cable section was used for the measurements using the HP 4294 A analyzer with the type 16047E fixture/adaptor [30], [38]. To minimize the influence of external parasitics, this fixture can directly clamp the wires of the cable segment under test without using additional leads.

VII. CONCLUSION

This paper has proposed a high-frequency modeling methodology for the long-cable-fed motor drive system to predict the overvoltage transients. An improved high-frequency motor model was proposed that accurately represents the high-frequency DM and CM impedance characteristics from hundreds of hertz to tens of megahertz. A frequency-dependent cable model is developed that includes the skin- and proximity effects as well as dielectric losses. The model parameters are easily identified from the measured DM impedance characteristics in a wide frequency range from 100 Hz to 10 MHz. The TLM technique was applied to the modeling of long-cable-fed motor drive system in discrete time domain, which avoided the otherwise laborious formulation and integration of the overall system ordinary differential equations. The proposed

methodology is verified experimentally as well as compared against two conventional modeling techniques. It is shown that the proposed model represents an appreciable improvement in predicting the long-cable-fed motor drive overvoltage transients.

ACKNOWLEDGMENT

The authors would like to thank R. S. Y. Hui (City University of Hong Kong) for giving valuable comments in TLM modeling.

APPENDIX A

The motor high-frequency model parameters are calculated from the measured DM and CM characteristics as follows [17]:

$$C_{g1} = \frac{1}{3} C_{HF} \quad (A1)$$

$$C_{g2} = \frac{1}{3} (C_{total} - C_{HF}) \quad (A2)$$

$$L_{CM} = (12\pi^2 C_{g2} f_{Z1}^2)^{-1} \quad (A3)$$

$$R_e \approx \frac{2}{3} |Z_P| \quad (A4)$$

$$R_{g1} = \frac{2}{3} |Z_{Z3}| \quad (A5)$$

$$L_d = L_{CM} + \frac{4}{9} L_{DM} \quad (A6)$$

$$R_{g2} \approx \frac{1}{3} |Z_{Z1}| \quad (A7)$$

$$C_t \approx \frac{1}{6} (C_{g1} + C_{g2}) \quad (A8)$$

$$L_t = \frac{1}{C_t} \left(\frac{1}{2\pi f_{Z2}^2} \right)^2 \quad (A9)$$

$$R_t = |Z_{Z2}| \cos(\theta_{Z2}) \quad (A10)$$

$$L_{zu} = 3 (16\pi^2 C_{g1} f_{Z3}^2)^{-1} \quad (A11)$$

APPENDIX B

The cable high-frequency model parameters are calculated from the measured SC and OC characteristics as follows:

$$L_{s1} = L_{SC-HF} \quad (B1)$$

$$L_{s2} = L_{SC-LF} - L_{s1} \quad (B2)$$

$$R_{s1} = |Z_{SC-LF}| \cos(\theta_{SC-LF}) \quad (B3)$$

$$R_{s2} = |Z_{SC-HF}| \cos(\theta_{SC-HF}) - R_{s1} \quad (B4)$$

$$C_{p1} = C_{OC-HF} \quad (B5)$$

$$C_{p2} = C_{OC-LF} - C_{p1} \quad (B6)$$

$$R_{p1} = |Z_{OC-LF}| [\cos(\theta_{OC-LF})]^{-1} \quad (B7)$$

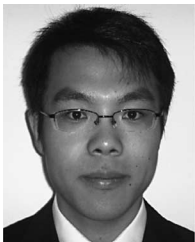
$$R_{p1/p2} = |Z_{OC-HF}| [\cos(\theta_{OC-HF})]^{-1} \quad (B8)$$

$$R_{p2} = [(R_{p1/p2})^{-1} - (R_{p1})^{-1}]^{-1} \quad (B9)$$

REFERENCES

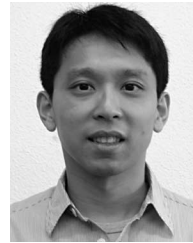
- [1] E. Persson, "Transient effects in application of PWM inverters to induction motors," *IEEE Trans. Ind. Appl.*, vol. 28, no. 5, pp. 1095–1101, Sep./Oct. 1992.
- [2] L. A. Saunders, G. L. Skibinski, S. T. Evon, and D. L. Kempkes, "Riding the reflected wave – IGBT drive technology demands new motor and cable considerations," in *Proc. IEEE Petroleum Chem. Ind. Conf. 1996*, Philadelphia, PA, Sep. 23–25, pp. 75–84.
- [3] R. Kerkman, D. Legagte, and G. Skibinski, "Interaction of drive modulation and cable parameters on AC motor transients," *IEEE Trans. Ind. Appl.*, vol. 33, no. 3, pp. 722–731, May/Jun. 1997.
- [4] S. Amarir and K. Al-Haddad, "Mathematical analysis and experimental validation of transient over-voltage higher than 2 per unit along industrial ASDM long cables," in *Proc. IEEE PESC 2008*, Montreal, QC, Canada, pp. 1846–1851.
- [5] A. H. Bonnett, "Analysis of the impact of pulse-width modulated inverter voltage waveforms on AC induction motors," *IEEE Trans. Ind. Appl.*, vol. 32, no. 2, pp. 386–392, Mar./Apr. 1996.
- [6] ABB Technical Guide No. 102: Effects of AC Drives on Motor Insulation – Knocking Down the Standing Wave, ABB Industrial Systems, Inc., New Berlin, WI, USA, 1997.
- [7] N. Aoki, K. Satoh, and A. Nabae, "Damping circuit to suppress motor terminal overvoltage and ringing in PWM inverter-fed AC motor drive systems with long cable leads," *IEEE Trans. Ind. Appl.*, vol. 35, no. 5, pp. 1014–1020, Sep./Oct. 1999.
- [8] A. Jouanne, D. A. Rendusara, P. N. Enjeti, and J. W. Gray, "Filtering techniques to minimize the effect of long motor leads on PWM inverter-fed AC motor drive systems," *IEEE Trans. Ind. Appl.*, vol. 32, no. 4, pp. 919–926, Jul./Aug. 1996.
- [9] S. Lee and K. Nam, "Overvoltage suppression filter design methods based on voltage reflection theory," *IEEE Trans. Power Electron.*, vol. 19, no. 2, pp. 264–271, Mar. 2004.
- [10] A. F. Moreira, P. M. Santos, T. A. Lipo, and G. Venkataramanan, "Filter networks for long cable drives and their influence on motor voltage distribution and common-mode currents," *IEEE Trans. Ind. Electron.*, vol. 52, no. 2, pp. 515–522, Apr. 2005.
- [11] J. D. Irvin, *The Industrial Electronics Handbook*. Boca Raton, FL/Piscataway, NJ: CRC Press/IEEE Press, 1997.
- [12] E. Zhong and T. A. Lipo, "Improvements in EMC performance of inverter-fed motor drives," *IEEE Trans. Ind. Appl.*, vol. 31, no. 6, pp. 1247–1256, Nov./Dec. 1995.
- [13] A. Boglietti and E. Carpaneto, "Induction motor high frequency model," in *Proc. Conf. Rec. IEEE IAS Annu. Meeting*, Phoenix, AZ, Oct. 1999, pp. 1551–1558.
- [14] G. Grandi, D. Casadei, and A. Massarini, "High frequency lumped-parameter model for AC motor windings," in *Proc. EPE 1997*, Trondheim, Norway, vol. 2, pp. 578–583.
- [15] A. F. Moreira, T. A. Lipo, G. Venkataramanan, and S. Bernet, "High-frequency modeling for cable and induction motor overvoltage studies in long cable drives," *IEEE Trans. Ind. Appl.*, vol. 38, no. 5, pp. 1297–1306, Sep./Oct. 2002.
- [16] S. Weber, E. Hoene, S. Guttowski, W. John, and H. Reichl, "Modeling induction machines for EMC-analysis," in *Proc. 35th IEEE Annu. Power Electron. Spec. Conf.*, Aachen, Germany, 2004, pp. 94–98.
- [17] M. Schinkel, S. Weber, S. Guttowski, W. John, and H. Reichl, "Efficient HF modeling and model parameterization of induction machines for time and frequency domain simulations," in *Proc. IEEE APEC*, Dallas, TX, 2006, pp. 1181–1186.
- [18] B. Mirafzal, G. Skibinski, R. Tallam, D. Schlegel, and R. Lukaszewski, "Universal induction motor model with low-to-high frequency-response characteristics," *IEEE Trans. Ind. Appl.*, vol. 43, no. 5, pp. 1233–1246, Sep./Oct. 2007.
- [19] S. Amarir and K. Al-Haddad, "A modeling technique to analyze the impact of inverter supply voltage and cable length on industrial motor-drives," *IEEE Trans. Power Electron.*, vol. 23, no. 2, pp. 753–762, Mar. 2008.
- [20] G. Skibinski, D. Leggate, and R. Kerkman, "Cable characteristics and their influence on motor overvoltages," in *Proc. 12th Annu. Appl. Power Electron. Conf. Expo.*, vol. 1, pp. 114–121.
- [21] G. Skibinski, R. Kerkman, D. Leggate, J. Pankau, and D. Schlegel, "Reflected wave modeling techniques for PWM AC motor drives," in *Proc. IEEE APEC 1998*, Anaheim, CA, Feb. 15–19, vol. 2, pp. 1021–1029.
- [22] G. Skibinski, R. Tallam, R. Reese, B. Buchholz, and R. Lukaszewski, "Common-mode and differential-mode analysis of 3 phase cables for PWM AC drives," in *Proc. Conf. Rec. IEEE IAS Annu. Meeting*, Tampa, FL, Oct. 6, 2006, pp. 880–888.

- [23] G. Oriti and A. L. Julian, "Application of the transmission line theory to the frequency domain analysis of the motor voltage stress caused by PWM inverters," in *Proc. IEEE IAS Annu. Meeting 2004*, Seattle, WA, Oct. 1996–2002.
- [24] S. Y. R. Hui and C. Christopoulos, "A discrete approach to the modeling of power electronic switching networks," *IEEE Trans. Power Electron.*, vol. 5, no. 4, pp. 398–403, Oct. 1990.
- [25] S. Y. R. Hui and C. Christopoulos, "Computer simulation of a converter-fed DC drive using the transmission-line modeling technique," *IEEE Trans. Power Electron.*, vol. 6, no. 4, pp. 636–644, Oct. 1991.
- [26] C. Christopoulos and S. Y. R. Hui, "The application of transmission-line modeling to the simulation of an induction motor drive," *IEEE Trans. Energy Convers.*, vol. 11, no. 2, pp. 287–297, Jun. 1996.
- [27] C. Christopoulos, *The Transmission-Line Modeling Method: TLM*. Oxford, U.K.: Oxford Univ. Press, 1995.
- [28] B. Mirafzal, G. Skibinski, and R. Tallam, "Determination of parameters in the universal induction motor model," *IEEE Trans. Ind. Appl.*, vol. 45, no. 1, pp. 142–151, Jan./Feb. 2009.
- [29] G. Suresh, A. Toliyat, A. Rendusara, and N. Enjeti, "Predicting the transient effects of PWM voltage waveform on the stator windings of random wound induction motors," *IEEE Trans. Power Electron.*, vol. 14, no. 1, pp. 23–30, Jan. 1999.
- [30] HewlettPackard/Agilent Model HP4294 A Precision Impedance Analyzer 40 Hz–110 MHz. (2009). Agilent Technologies Inc., Santa Clara, CA, USA. [Online]. Available: www.agilent.com
- [31] Y. Weens, N. Idir, R. Bausiere, and J. J. Franchaud, "Modeling and simulation of unshielded and shielded energy cables in frequency and time domains," *IEEE Trans. Magn.*, vol. 42, no. 7, pp. 1876–1882, Jul. 2006.
- [32] G. Skibinski, J. Erdman, and J. Pankau, "Assessing AC motor dielectric withstand capability to reflected voltage stress using corona testers," in *Proc. IEEE IAS Electric Mach. Conf.*, San Diego, CA, 1996, pp. 694–702.
- [33] G. Skibinski, R. J. Kerkman, and D. Schlegel, "EMI emissions of modern PWM ac drives," *IEEE Ind. Appl. Mag.*, vol. 5, no. 6, pp. 47–81, Nov./Dec. 1999.
- [34] J. M. Erdman, R. J. Kerkman, D. W. Schlegel, and G. L. Skibinski, "Effect of PWM inverters on AC motor bearing currents and shaft voltages," *IEEE Trans. Ind. Appl.*, vol. 32, no. 2, pp. 250–259, Mar./Apr. 1996.
- [35] D. F. Busse, J. M. Erdman, R. J. Kerkman, D. W. Schlegel, and G. L. Skibinski, "The effects of PWM voltage source inverters on the mechanical performance of rolling bearings," *IEEE Trans. Ind. Appl.*, vol. 33, no. 2, pp. 567–576, Mar./Apr. 1997.
- [36] D. F. Busse, J. M. Erdman, R. J. Kerkman, D. W. Schlegel, and G. L. Skibinski, "System electrical parameters and their effects on bearing currents," *IEEE Trans. Ind. Appl.*, vol. 33, no. 2, pp. 577–584, Mar./Apr. 1997.
- [37] D. F. Busse, J. M. Erdman, R. J. Kerkman, D. W. Schlegel, and G. L. Skibinski, "An evaluation of the electrostatic shielded induction motor: A solution for rotor shaft voltage buildup and bearing current," *IEEE Trans. Ind. Appl.*, vol. 33, no. 6, pp. 1563–1570, Nov./Dec. 1997.
- [38] Agilent LCR Meters Impedance Analyzers and Test Fixtures Selection Guide. (2009). Agilent Technologies Inc., Santa Clara, CA, USA. [Online]. Available: www.agilent.com



Liwei Wang (S'04) received the M.S. degree in electrical engineering from Tianjin University, Tianjin, China, and the Ph.D. degree in electrical and computer engineering from the University of British Columbia, Vancouver, BC in 2004 and 2010, respectively.

He is currently a Postdoctoral Research Fellow in the Department of Electrical and Computer Engineering, University of British Columbia. From February 2009 to July 2009, he was an Internship Researcher at the ABB Corporate Research Center, Baden-Dättwil, Switzerland. His research interests include power system analysis and operation, electrical machine and drives, power electronic system and controls, renewable energy sources, and distributed generation.



Carl Ngai-Man Ho (M'07) received the B.Eng. and M.Eng. degrees in 2002 and the Ph.D. degree in 2007 in electronic engineering from the City University of Hong Kong, Kowloon, Hong Kong.

He was involved in research on the development of dynamic voltage regulation and restoration technology during his Ph.D. degree. From 2002 to 2003, he was a Research Assistant at the City University of Hong Kong. From 2003 to 2005, he was an Engineer at e.Energy Technology Ltd., Hong Kong. In May 2007, he joined ABB Corporate Research Center, Baden-Dättwil, Switzerland, as a Scientist. His research interests include power electronics, power quality, modeling and control of power converters, and characterization of wide bandgap power semiconductor devices and applications of those. He holds two U.K. and one U.S. patents in the area of lighting applications.

Dr. Ho was a Reviewer for the IEEE TRANSACTIONS ON POWER ELECTRONICS, the IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS, the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS, and several conferences.



Francisco Canales (M'95) received the B.S. degree in mechanical and electrical engineering from the Universidad Veracruzana, Veracruz, Mexico, in 1989, the M.Sc. degree in electronic engineering from the Centro Nacional de Investigación y Desarrollo Tecnológico (CENIDET), Cuernavaca, Mexico, in 1994, and the Ph.D. degree in electrical engineering from the Virginia Polytechnic Institute and State University (Virginia Tech), Blacksburg, in 2003.

He was a Senior Research Assistant at the Center for Power Electronics Systems (CPES), Virginia Tech, where he was involved in core research and several industry-sponsored projects during his Ph.D. degree. He was an Associate Professor in the Department of Electronic Engineering, CENIDET. Currently he is a Principal Scientist at ABB Corporate Research Center, Baden-Dättwil, Switzerland, where he is working on high-density traction converters and industrial applications.



Juri Jatskevich (M'99–SM'07) received the M.S.E.E. and the Ph.D. degrees in electrical engineering from Purdue University, West Lafayette, IN, in 1997 and 1999, respectively.

Since 2002, he has been a Faculty Member at the University of British Columbia, Vancouver, where he is currently an Associate Professor of electrical and computer engineering. His research interests include power electronic systems, electrical machines and drives, average-value modeling and simulation.

Dr. Jatskevich is currently a Chair of IEEE CAS Power Systems & Power Electronic Circuits Technical Committee, an Editor of the IEEE TRANSACTIONS ON ENERGY CONVERSION, the IEEE POWER ENGINEERING LETTERS, and an Associate Editor of the IEEE TRANSACTIONS ON POWER ELECTRONICS. He is also the Chair of the IEEE Task Force on Dynamic Average Modeling, under Working Group on Modeling and Analysis of System Transients Using Digital Programs.

Wide-Band Constant-Parameter Voltage-Behind-Reactance Model of Squirrel-Cage Induction Machines

Sheraz Baig, *Student Member, IEEE*, Seyyedmilad Ebrahimi, *Member, IEEE*, Juri Jatskevich, *Fellow, IEEE*, Liwei Wang, *Senior Member, IEEE*, and Aria Fani, *Student Member, IEEE*

Abstract—Variable frequency drives are widely utilized in many commercial and industrial applications. Therein, typically, a three-phase squirrel-cage induction machine (IM) is fed from an inverter through a cable. To design and tune such systems, various models of IMs have been proposed in the literature to study the motor-converter interactions and the low-to-high frequency phenomena. This paper presents a wide-band model for the IMs that is valid in the frequency range from DC to ten MHz. The new model is obtained by incorporating the low-mid frequency voltage-behind-reactance model with the universal high-frequency stator and bearing circuit models. The performance of the proposed model is demonstrated using computer simulations and experimental results obtained from a 7.5hp induction motor connected to a drive system through a cable. The new model is demonstrated to accurately predict the common mode (CM) and differential mode (DM) impedances, reflected transient over-voltages, CM currents, and bearing voltages, all representing an advantage over the conventional/existing IM models.

Index Terms—Bearing circuit, common-mode (CM) impedance, differential-mode (DM) impedance, high-frequency (HF) modeling, induction machine, low-frequency (LF) modeling, voltage-behind-reactance (VBR) model.

I. INTRODUCTION

VARIABLE frequency drives (VFDs) are found in many commercial and industrial applications, including offshore platforms, oil and gas drilling,

mining, automotive, ship propulsion, aerospace, railway traction, hybrid/fully electric vehicle systems [1]–[4], etc. Power-electronic converters based on wide-band-gap (WBG) semiconductor devices are used in such drive systems to enhance the system performance. The application of WBG switches in VFDs can also pose challenges, as such devices may produce steep-fronted pulse voltages, cause electromagnetic interference (EMI) issues and wave propagation effects [5]–[7] due to their high dv/dt . The wave-propagation effects can also occur in long-cable-fed machine-drive systems due to a mismatch between the cable characteristic impedance and the motor surge impedance [5], causing the high-frequency PWM voltage pulses to reflect back and forth between the inverter and the machine. This phenomenon, in turn, results in reflected over-voltage ringing across the machine terminals [1], [2], [4]–[6], [8], [9]. Due to fast switching, the reflected over-voltages may be observed in drives even with short cables [10].

It is also important to keep in mind that the EMI and wave propagation effects are significantly determined by the switching speed (fast rise/fall voltage edge times, high dv/dt) of IGBTs or other switching devices that create steep-fronted pulse voltages (and not so much only on the inverter PWM switching frequency). During the fast rise and fall switching transitions, the parasitic capacitances of the cable and motor are charged and discharged, contributing to high-frequency oscillations and traveling wave phenomena [1].

Typically, the switching frequency in cable-fed VFD systems used in high power and remote/offshore oil, gas drilling, and mining applications (i.e., where VFD and motor are located at distant locations and interconnected through power cables) is not very high and may range from around a kilohertz up to 20 kHz. However, due to the significant parasitic capacitances of cable and the machine and wave propagation effects, the inverter switching can cause high-frequency reflected over-voltage and current transients. Moreover, with the use of newer WBG semiconductor devices, i.e. SiC with higher dv/dt , the EMI and wave-propagation-related effects may become pronounced even in short-cable-fed VFD systems [10]. Therefore, the frequency bandwidth/spectra of voltage and current signals at the motor terminals may range from $\sim$ DC up to 10MHz.

The EMI and wave-propagation phenomena can cause

Manuscript received September 13, 2023; revised April 19, 2024, and May 26, 2024; accepted June 13, 2024. This work was supported by the Natural Science and Engineering Research Council (NSERC) of Canada under the Collaborative Research and Development Grant. Paper no. TEC-01052-2023. (Corresponding author: Juri Jatskevich).

S. Baig was with the Electrical and Computer Engineering Department, University of British Columbia, Vancouver, BC V6T 1Z4, Canada. He is now with Powertech Labs Inc., Surrey, BC V3W 7R7, Canada (e-mail: sheraz.baig@powertechlabs.com).

S. M. Ebrahimi, and J. Jatskevich are with the Electrical and Computer Engineering Department, University of British Columbia, Vancouver, BC V6T 1Z4, Canada (e-mail: ebrahimi@ece.ubc.ca; juri@ece.ubc.ca).

L. Wang, and A. Fani are with the School of Engineering, University of British Columbia Okanagan, Kelowna, BC V1V 1V7, Canada (e-mail: liwei.wang@ubc.ca; aria.fani@ubc.ca).

Color versions of one or more of the figures in this paper are available online at <https://ieeexplore.ieee.org>.

unfavorable effects, leading to the deterioration of machine bearings [6], [7] due to the induced nonzero CM voltages and discharging bearing currents [1], [5], [11], micro-arcing [12], failure of dielectric insulation [1], [5]–[7] leading to inter-turn faults in stator windings and accelerated aging [5], [6], leakage CM current EMI issues [1], [6], [7], [11], high voltage stress [11], etc. Therefore, developing wide-band models for IMs is crucial to facilitate the design, analysis, and control of cable-fed VFD systems. Such models may be utilized to test different inverters and their switching algorithms, to design dv/dt and di/dt mitigation filters, predict and reduce the over-voltage and current transients, mitigate EMI issues to comply with device electromagnetic compatibility (EMC) requirements, predict and reduce stresses on motor winding insulation and bearing failure.

At lower frequencies, induction machines are generally inductive [13] and can be modeled using the so-called lumped-parameter models such as the phase domain (PD) [14], [15], classical $qd0$ [16], and voltage-behind-reactance (VBR) models [17]. The PD model becomes computationally expensive [18] due to its rotor position-dependent and time-varying inductances. Similarly, the classical $qd0$ model may suffer from numerical error and simulation inefficiency due to its indirect interface (requires additional snubbers) with the inductive and switching networks [18]. Moreover, it may also underestimate the current switching ripples in inverter-machine interaction studies [19]. As an alternative to the PD and $qd0$ models, the decoupled constant-parameter VBR (DCPVBR) model [17] enables direct interface with arbitrary inductive or switching circuits without numerical snubber circuits. The DCPVBR model has advantageous numerical properties and is suitable for system-level studies [17].

At higher frequencies, the behavior of IMs becomes capacitive due to the stray capacitances between the stator winding and motor frame [6], [8], [12], [13]. Several high-frequency models [3], [5]–[9], [11], [20]–[26] were developed experimentally from the corresponding machine's DM and CM impedances in the frequency domain. Those are referred to as high-frequency (HF) behavioral equivalent circuit models, which are typically used to capture the machine's high-frequency behavior only. Such models [3], [20], [21], [24], [26] possess high-order and complex structures and require complicated parameterization.

Numerous research efforts were made to propose simpler and easy-to-parametrize HF models, namely the single- [8], [23], [25] or multiple- [22], [24] section phase-belt winding models, which are capable of representing the machine impedance characteristics over extended frequency bandwidth. In [5], a universal HF model was proposed based on a single-section phase-belt winding model [25] as an extension of the model in [9]. Specifically, the universal HF model [5] possesses a relatively simple and unified structure that can represent arbitrary stator windings (i.e., Y-, Δ -, series- and parallel-) and is accurate in the mid-high frequency range (i.e.,

50kHz to 10MHz) and also has a straightforward characterization.

In IMs, bearing voltages are induced through high-frequency stray capacitances, causing discharging bearing currents under the PWM operation of the VFDs [11], [27]. To model and analyze this phenomenon, various HF equivalent circuit models [6], [11], [27]–[33] have been proposed in the literature. Traditionally, CM equivalent circuit models may be utilized [6] since these effects are the consequences of the induced CM voltages and the flow of CM currents [27], [33]. In [31], a lumped-parameter capacitive equivalent circuit model was proposed to predict bearing voltages through a capacitive voltage divider circuit consisting of several stray capacitances.

The DM and CM models proposed in the literature address only the HF EMI phenomena [6], [7], and use parameters that may not have a direct correlation to the geometry and physical behavior [6]. At the same time, it is very desirable to integrate the low- and high-frequency IM models to capture the machine's behavior over a wide range of frequencies in the VFD systems. Additionally, incorporating the effects of the motor slip variations in the high-frequency models would be advantageous to observe the low-frequency electromagnetic transients [14]–[17]. For this purpose, in [34], a HF distributed circuit model was connected in parallel to the classical $qd0$ model through a high pass filter. However, a high pass filter could introduce inaccuracies in the model results. In [3], a three-phase IM model was proposed, considering the inductive and capacitive coupling effects between the stator phases and ground. This resulted in a high-order model structure and complicated parametrization procedure. In [6], a three-phase IM model was proposed, which combines the IEEE recommended T-circuit model [35] with HF branches to capture its low-high frequency response characteristics. However, since the T-equivalent circuit is a steady-state model, it does not capture the low-frequency dynamic behavior [5]. Some full-order IM models [19], [36] were proposed to predict the dynamic behavior by incorporating the distributed rotor circuit represented as a high-order transfer function in addition to the machine flux saturation for better accuracy. However, the research focus of those models was on the low-mid frequency range (i.e., below 50kHz). Also, the increased complexity of using a high-order transfer function and several parameters result in a complicated parametrization procedure [37].

To formulate a wide-band IM model for efficient simulation of VFD systems in the widely-used electromagnetic transient (EMT) simulation programs, this paper incorporates the DCPVBR model [17], the universal HF model [5], and the bearing circuit model [31] to represent a squirrel-cage IM with Y-stator winding configuration. The salient features of the proposed model with respect to the existing state-of-the-art models are summarized in Table I. The essential properties of the proposed model and the main

TABLE I SUMMARY OF FEATURES OF THE PROPOSED INDUCTION MACHINE MODEL WITH RESPECT TO THE EXISTING STATE-OF-THE-ART MODELS

Models		PD [14], [15]	qd0 [16]	DCPVBR [17]	[3], [20], [21], [24], [26]	[19]	[5]	[6]	Proposed WBDCPVBR Model
Structure		complex	simple	simple	complex	complex	simple	complex	simple
Interfacing circuit		variable & direct	constant & indirect	constant & direct	constant & direct	constant & indirect	constant & direct	constant & direct	constant & direct
Parameterization method		easy	easy	easy	complicated	complicated	easy	complicated	easy
Valid frequency range (Hz)		~DC-5k	~DC-5k	~DC-5k	~5k~10M	~1k~50k	50k-10M	10-10M	~DC-10M
Low-frequency behavior	Transient	Yes	Yes	Yes	No	Yes	No	No	Yes
	Steady-state	Yes	Yes	Yes	No	Yes	No	Yes	Yes
	Motor slip variation	Yes	Yes	Yes	No	Yes	No	Yes	Yes
High-frequency behavior	Transient	No	No	No	Yes	Yes	Yes	Yes	Yes
	Steady-state	No	No	No	Yes	Yes	Yes	Yes	Yes
	DM impedance	low-freq. range	low-freq. range	low-freq. range	mid-high freq. range	–	mid-high freq. range	low-high freq. range	low-high freq. range
	CM impedance	No	No	No	Yes	–	Yes	Yes	Yes
	CM current	No	No	No	–	–	No	No	Yes
	Reflected transient over-voltages	No	No	No	Yes	–	Yes	Yes	Yes
	Bearing voltages	No	No	No	No	No	No	Yes	Yes

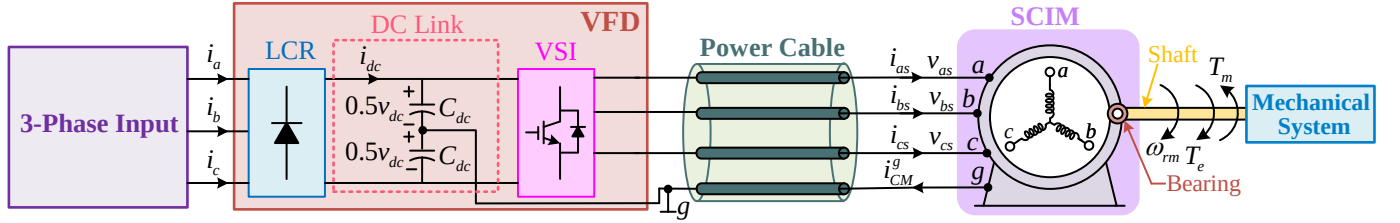


Fig. 1. A typical three-phase VFD system consisting of a line-commutated rectifier (LCR) and a voltage-source-inverter (VSI), feeding a squirrel-cage induction motor (SCIM) through a power cable.

contributions of this paper are summarized as follows:

- 1) A wide-band decoupled constant-parameter VBR (WBDCPVBR) model for three-phase magnetically linear squirrel-cage induction machines with Y-connected stator windings is developed, which extends the research work in [5], [17], [31].
- 2) The proposed model captures the machine's DM and CM impedances accurately from ~DC to ~10MHz.
- 3) The proposed WBDCPVBR IM model has a direct and efficient machine-network interface in the EMT programs due to its constant-parameter and decoupled stator interfacing using simple *RLC* branches.
- 4) The new model enables the implementation of inverter control algorithms using time-domain simulations and simultaneous analysis of the common- and differential-mode response in predicting the bearing voltages, CM and DM currents/voltages, current ripples, motor loading effects and reflected transient over-voltages across the motor terminals.
- 5) The proposed WBDCPVBR model is validated using an experimental 7.5hp induction motor setup connected to an ABB drive [38]. Compared to the existing HF machine

models, the new model improves the prediction of transient responses in long- and short-cable-fed motor-drive systems.

To the best of the authors' knowledge, such a comprehensive IM model does not exist in the literature. It is envisioned that the desirable features of the proposed model can make it a versatile asset for state-variable-based and nodal-analysis-based electromagnetic transient simulation programs for modeling the behavior of induction machines over a very wide range of frequency spectrum [39].

II. MODELLING OF INDUCTION MACHINE

In this paper, a typical VFD system is considered, as shown in Fig. 1. Therein, a line-commutated rectifier (LCR) is used to produce a DC voltage from the three-phase power supply, which is then converted into variable frequency/amplitude three-phase AC voltages using a three-phase, two-level voltage-source inverter (VSI) to feed a squirrel-cage induction machine (SCIM) through a cable. The voltages across the upper and lower capacitors in the DC link are specified by $\pm 0.5v_{dc}$. The IM is assumed to have Y-

connected stator winding. The magnetic saturation is not considered since VFDs generally prevent over-saturation and may operate in a constant flux region (i.e., equivalent to Volts/Hz control). The mechanical system acting as the load is coupled to the shaft of the IM. Additionally, the shaft of the IM is linked with the stator frame through bearings and the induced voltage across the bearings is specified by v_B . The stator frame is connected to the middle point of the DC link through a cable lead, and the current that flows out of the stator frame is specified as CM ground current (i_{CM}^g). The proposed WBDPVBR IM model comprises three parts, i.e., the low-frequency sub-circuit and the mechanical subsystem, the high-frequency stator sub-circuit, and the bearing circuit, which are explained subsequently. To simplify the notations, all variables are referred to the stator side.

A. Low-Frequency DCPVBR Model [17]

At lower frequencies (i.e., DC to ~10kHz), the IM behavior is predominantly inductive, and the energy conversion is mainly realized through the air-gap flux, which is defined by the machine's stator and rotor currents. Therefore, the transient and steady-state behavior of the induction machine may be well represented by the lumped-parameter low-frequency models. Specifically, for the purpose of this paper, the decoupled constant-parameter voltage-behind-reactance (DCPVBR) [17] is selected to represent the Y-connected induction machine in the low-frequency range. This model possesses a constant-parameter direct interfacing with an arbitrary external network in state-variable-based EMT simulation programs. For completeness, a summary of this model is briefly presented.

In the DCPVBR model [17], the stator voltage equation, considering the zero-sequence, has the following form

$$\mathbf{v}_{abc} = r_D \mathbf{i}_{abc} + L_D p \mathbf{i}_{abc} + \mathbf{e}_{abc}'' + \underbrace{3r_0 i_{0s} + 3L_0 p i_{0s}}_{r_0 i_{ng} + L_0 p i_{ng}}, \quad (1)$$

Here, i_{ng} denotes the neutral current, and i_{0s} denotes the stator zero-sequence current. The neutral current equals three times the zero-sequence current, i.e. $i_{ng} = 3 \times i_{0s}$. In (1), $\mathbf{v}_{abc} = [v_{as} \ v_{bs} \ v_{cs}]^T$ represents the stator terminal voltages, $\mathbf{i}_{abc} = [i_{as} \ i_{bs} \ i_{cs}]^T$ represents the stator currents, and $\mathbf{e}_{abc}'' = [e_{as}'' \ e_{bs}'' \ e_{cs}'']^T$ represents the so-called sub-transient back-emf voltages. The lower-case letter p denotes the Heaviside operator $p = d/dt$. The constant and decoupled RL branch resistances and inductances (r_D and L_D), in addition to the zero-sequence resistances and inductances (r_0 and L_0), are defined as [17]

$$\left. \begin{aligned} r_D &= r_s + L_m''^2 L_{lr}^{-2} r_r, & L_D &= L_{ls} + L_m'', \\ r_0 &= -\frac{1}{3} L_m''^2 L_{lr}^{-2} r_r, & L_0 &= -\frac{1}{3} L_m''. \end{aligned} \right\} \quad (2)$$

Here, r_s and L_{ls} denote the original stator phase winding resistance and leakage inductance, respectively; r_r and L_{lr} denote the rotor winding resistance and leakage inductance, respectively. Also, L_m'' is the sub-transient magnetizing inductance defined as [17]

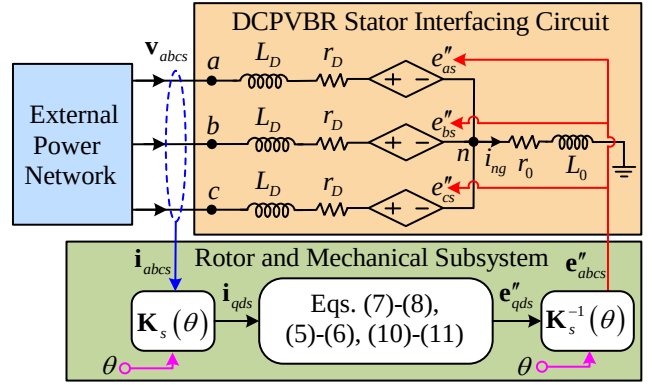


Fig. 2. Implementation of the DCPVBR model of a three-phase Y-connected IM with decoupled and constant RL branches for stator interfacing circuit [17].

$$L_m'' = (L_m^{-1} + L_{lr}^{-1})^{-1}, \quad (3)$$

where L_m is the magnetizing inductance of the machine. In (1), $\mathbf{e}_{abc}''$ represents the transformed sub-transient back-emf voltages in the abc coordinates and is obtained as

$$\mathbf{e}_{abc}'' = \begin{bmatrix} e_{as}'' & e_{bs}'' & e_{cs}'' \end{bmatrix}^T = \mathbf{K}_s^{-1}(\theta) \begin{bmatrix} e_{qs}'' & e_{ds}'' & e_{0s}'' \end{bmatrix}^T, \quad (4)$$

where $\mathbf{K}_s^{-1}(\theta)$ is the inverse Park's transformation matrix [16], and θ is the angle of the arbitrary reference frame. The q - and d - axes sub-transient back-emf voltages are [17]

$$e_{qs}'' = \omega_r L_m'' L_{lr}^{-1} \lambda_{dr} + r_r L_m'' L_{lr}^{-2} (L_m'' L_{lr}^{-1} - 1) \lambda_{qr}, \quad (5)$$

$$e_{ds}'' = -\omega_r L_m'' L_{lr}^{-1} \lambda_{qr} + r_r L_m'' L_{lr}^{-2} (L_m'' L_{lr}^{-1} - 1) \lambda_{dr}. \quad (6)$$

The state-space equations of the rotor subsystem flux linkages in the qd coordinates can be expressed as

$$p \lambda_{qr} = r_r L_{lr}^{-1} (L_m'' L_{lr}^{-1} - 1) \lambda_{qr} - (\omega - \omega_r) \lambda_{dr} + r_r L_{lr}^{-1} L_m'' i_{qs}, \quad (7)$$

$$p \lambda_{dr} = r_r L_{lr}^{-1} (L_m'' L_{lr}^{-1} - 1) \lambda_{dr} + (\omega - \omega_r) \lambda_{qr} + r_r L_{lr}^{-1} L_m'' i_{ds}. \quad (8)$$

Here, ω is the arbitrary reference frame's electrical speed, and ω_r is the rotor's electrical speed, respectively. Also, i_{qs} and i_{ds} are the stator currents in qd coordinates, which are obtained as

$$\mathbf{i}_{qds} = \begin{bmatrix} i_{qs} & i_{ds} & i_{0s} \end{bmatrix}^T = \mathbf{K}_s(\theta) \begin{bmatrix} i_{as} & i_{bs} & i_{cs} \end{bmatrix}^T. \quad (9)$$

Here, $\mathbf{K}_s(\theta)$ is Park's transformation matrix [16].

The electromagnetic torque and the mechanical dynamics are expressed as

$$T_e = \left(\frac{3P}{4} \right) \left(\frac{L_m''}{L_{lr}} \right) (\lambda_{dr} i_{qs} - \lambda_{qr} i_{ds}), \quad (10)$$

$$p \omega_{rm} = \frac{1}{J} (T_e - T_m), \quad \omega_r = \frac{P}{2} \omega_{rm}, \quad (11)$$

where the upper-case letter P denotes the number of poles, J is the moment of inertia, and T_e and T_m are the electromagnetic and the mechanical torque of the machine, respectively. The implementation of the DCPVBR model of a Y-connected IM is demonstrated in Fig. 2.

B. Universal High-Frequency Model [5]

The universal high-frequency model [5] was derived based on [9] using a single-section phase-belt winding equivalent circuit and is depicted in Fig. 3 as a per-phase equivalent

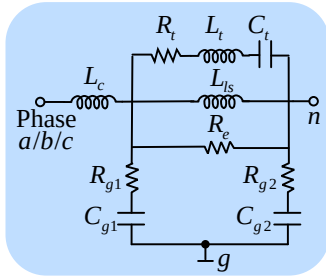


Fig. 3. Per-phase equivalent circuit of the universal high-frequency induction machine model [5] for arbitrary stator windings connections.

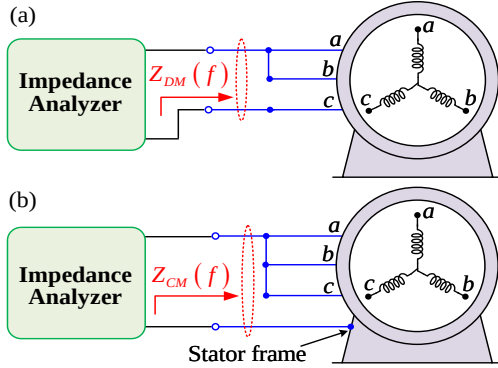


Fig. 4. Setup for measuring the impedance characteristics of a three-phase squirrel-cage IM: (a) DM impedance (Z_{DM}), and (b) CM impedance (Z_{CM}).

circuit. As can be seen, it possesses a simple and unified structure for representing arbitrary stator winding configurations such as Y-, Δ -, series- and parallel- for induction machines. Additionally, the associated parameters have physical definitions and importance [5], [9]. Specifically, C_{g1} and C_{g2} represent the phase-ground and neutral-ground equivalent stray capacitances, respectively; R_{g1} and R_{g2} represent the copper skin-effect and proximity-effect resistances; L_{ls} represents the stator leakage inductance; R_e accounts for power losses due to high-frequency eddy-currents; and the series branch elements R_t , L_t , and C_t capture the second resonance in the measured DM impedance as a result of the stator windings' inter-turn capacitive effects. Notably, at mid-high frequencies, the voltage distribution throughout different coils of the stator windings becomes non-uniform [40], which causes the entire surge to fall in the first few turns of the coils [6], [40]. In the universal model [5], the effects of the combined stator leakage inductance of the first few slot turns and the HF inductance of the internal feed conductors are incorporated using the inductance L_c , as shown in Fig. 3. Nevertheless, this model does not consider the phase-phase capacitive couplings due to the relatively weak coupling which can be neglected [26].

The universal HF IM model has a straightforward parameterization procedure, details of which can be found in [5], although the parameters are influenced by different machine stator winding configurations [5]. Generally, the DM and CM impedances are considered to parametrize the high-frequency IM model and deemed effective in analyzing its behavior in a wide frequency range. It was also noted that the

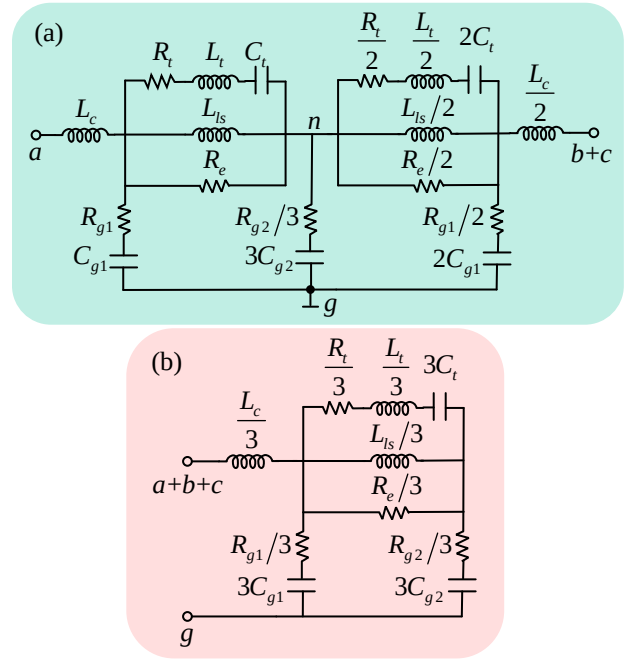


Fig. 5. The universal high-frequency equivalent circuits of the induction machine: (a) DM equivalent circuit, and (b) CM equivalent circuit.

DM impedance strongly influences the prediction of the reflected overvoltage ringings [5], whereas bearing or shaft voltages/currents are strongly influenced by the CM impedance [6]. Therefore, accurate representation of both the DM and CM impedances is critical for an accurate prediction of the HF phenomena associated with the VFDs.

Typically, the DM and CM impedances are measured when the machine is disconnected from the AC supply and is at a standstill [5]–[7], [9], [24] because the rotor speed and stator currents have no significant impact in the mid-high frequency range [1], [5]–[7], [9]. The DM impedance is obtained by shorting two phases of the stator windings and measuring the impedance between them and the third phase, as illustrated in Fig. 4(a). An impedance analyzer or a frequency response analyzer can be used to measure the impedances. The CM impedance is obtained by connecting all three phases of the stator windings together and measuring the impedance between them and the motor frame, as illustrated in Fig. 4(b). These measured DM and CM impedances depend on different machine stator winding configurations and the machine geometry [5], [9], [11].

According to the arrangements in Fig. 4, the universal HF CM and DM equivalent circuits can be realized based on the per-phase equivalent circuit in Fig. 3, as demonstrated in Fig. 5. This model can accurately depict the DM impedances in the mid-high frequency range (i.e. 50kHz–10MHz) [5]. However, some discrepancies in the DM impedances at the low-frequency range were observed in this model. Moreover, it also does not predict the bearing voltages, ripples in the currents, and torque waveforms due to the changes in motor slip and loading effects. At the same time, this type of phase belt-winding model offers more sensitivity to fast voltage

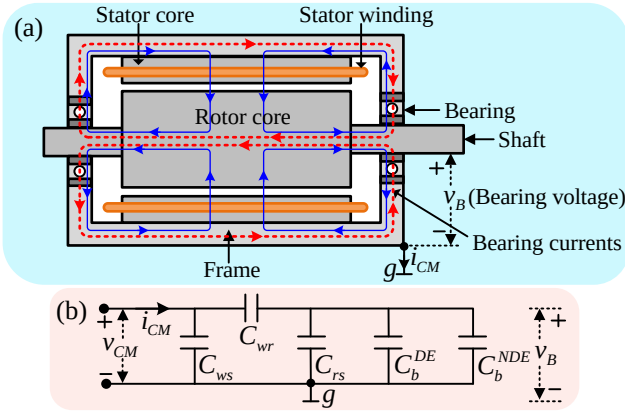


Fig. 6. High-frequency bearing phenomenon modeling: (a) cross-section of an induction machine depicting induced bearing voltage and currents, and (b) equivalent high-frequency bearing circuit model [30], [31].

changes, and CM current calculation using this model is closer to the CM current measurements [11]. Therefore, it is useful to integrate this universal HF model with the LF full-order models.

C. High-Frequency Bearing Circuit Model [30], [31]

A cross-section of an induction machine depicting the inverter-induced bearing voltage and currents is shown in Fig. 6(a). Based on that, it was noted the induced CM voltage influences the bearing voltage/currents [31], [33]. As a result, a lumped-parameter equivalent bearing circuit model was proposed in [30], [31] in the form of a capacitive voltage divider circuit consisting of several HF stray capacitances, as illustrated in Fig. 6(b). Specifically, the bearing voltage mirrors the CM voltage waveforms based on the adopted modulation schemes through the bearing voltage ratio (BVR) as

$$v_B = \underbrace{\left(\frac{C_{wr}}{C_{wr} + C_{rs} + C_b^{DE} + C_b^{NDE}} \right)}_{\text{BVR}} \times v_{CM}. \quad (12)$$

The stray capacitances in (12) and the bearing circuit model of Fig. 6(b) have physical significance. Specifically, C_{ws} represents the winding-to-stator frame stray capacitance, C_{wr} represents the winding-to-rotor stray capacitance, C_{rs} represents the rotor-to-stator frame stray capacitance, C_b^{DE} and C_b^{NDE} represent the bearing stray capacitances at the drive and non-drive ends [30], [31]. Various methods can be employed to extract these parameters, e.g., analytically based on design [31], experimentally [27], or through FEA methods [29], [30].

D. Proposed Wide-Band DCPVBR Model

To address the limitations of the prior/existing models, a wide-band constant-parameter voltage-behind-reactance (WBDCPVBR) model is proposed, which integrates the advantageous properties of the low-frequency DCPVBR model [17], universal high-frequency model [5], as well as the bearing circuit model [31]. The term “wide-band” refers to the capability of the model to represent the machine’s DM and CM behavior over a wide range of frequencies ($\sim$ DC to

10MHz). The equivalent circuit and implementation of the proposed WBDCPVBR model are shown in Fig. 7.

As seen in Fig. 7, the proposed WBDCPVBR model merges the stator interfacing circuits of the low-frequency DCPVBR model in Fig. 2 and the universal high-frequency model in Fig. 3. Specifically, most of the HF model is conveniently added on top of the LF model, except that the stator winding leakage inductance L_{ls} of the HF model is omitted because its effect is already taken into account in the interfacing inductance L_D of the LF model. The rotor and mechanical subsystems are based on the DCPVBR model, which computes the sub-transient back EMFs, based on (5)–(8), as inputs to the controlled voltage sources in the stator interfacing circuit. It is worth noting that, at the mid-high frequencies, the magnetic flux penetration into the rotor of the squirrel-cage IM is minimal [6], [7], [9]. Therefore, the low-frequency parameters can be used for the rotor subsystem and remain unchanged in the mid-high frequency range [6], [7].

The proposed WBDCPVBR model also includes the lumped-parameter bearing equivalent circuit of Fig. 6 to predict the effect of high frequency bearing voltage and CM currents phenomena induced through stray capacitances. This is done by representing the branch consisting of a parallel combination of the stator HF R_{g2} and C_{g2} elements as the equivalent bearing circuit of Fig. 6, shown in the red-highlighted boxes in Fig. 7.

In agreement with the LF-DCPVBR and universal HF models, the parameters associated with the proposed WBDCPVBR model possess the same physical significance, whose parametrization procedure is also very simple and straightforward. Specifically, the low-frequency parameters (r_s , r_r , L_{ls} , L_{lr} , L_m) of the proposed WBDCPVBR can be obtained analytically from motor nameplate data or experimentally from standard DC resistance, blocked-rotor, and no-load tests performed on the machine. The IEEE standard test procedure to experimentally obtain these low-frequency parameters is explained in [35]. Meanwhile, the high-frequency parameters (C_{g1} , C_{g2} , R_{g1} , R_{g2} , R_e , R_t , L_t , C_t , L_c) are calculated by utilizing the exact same HF parametrization procedure developed in [5] and [9] for the universal HF model. Generally, the HF parametrization method, as developed in [5] and [9], calculates the HF parameters by fitting the measured DM and CM impedances based on analytical formulas. For this purpose, the DM and CM machine impedances must be measured using an impedance or a frequency response analyzer. The procedure is based on identifying four characteristic points and three slopes on the DM and CM impedance plots and fitting the corresponding capacitances and inductances using simple analytical equations. Due to limited space, the detailed HF parametrization procedure is not presented here. However, interested readers are referred to [5] for a detailed procedure to calculate the HF stator interfacing circuit parameters of the proposed WBDCPVBR model. Moreover, the bearing circuit

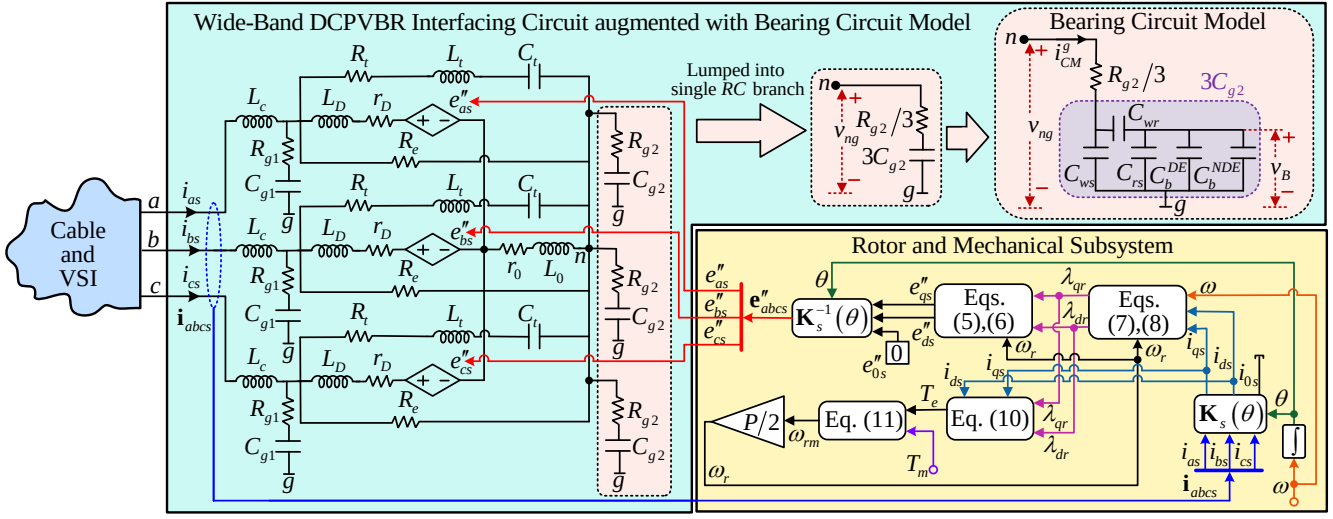


Fig. 7. Implementation of the proposed WBDCPVBR model of induction machines augmented with bearing circuit model.

parameters (C_{ws} , C_{wr} , C_{sr} , $C_{b,DE}$, $C_{b,NDE}$) of the proposed WBDCPVBR model can be obtained analytically based on design [31], experimentally [27], or through FEA methods [29], [30]. However, these parameters can also be obtained by fitting the $3C_{g2}$, BVR values, and typical ratios of a few parasitic capacitances (C_{sr} , $C_{b,DE}$, $C_{b,NDE}$) in the bearing circuit, which was the method considered in this paper.

Additionally, with the inclusion of the zero-sequence branch in the proposed WBDCPVBR model, the zero-sequence currents are included when needed. Overall, the proposed WBDCPVBR model possesses a convenient structure and interfacing circuit using a constant-parameter, decoupled RLC branches, and controlled voltage sources, enabling its efficient and direct interface with arbitrary external networks when implemented in the EMT simulation programs.

III. MODEL VALIDATION

To verify the effectiveness of the proposed WBDCPVBR IM model, the VFD system shown in Fig. 1 is considered, where a three-phase 4-pole, dual-voltage (230V/460V), nine-lead, 7.5hp squirrel cage IM (Model: Baldor EM3710T) is connected to an ABB drive (ACS150-03U-09A8-2) through power cables. The experimental laboratory setup is shown in Fig. 8. Table II in Appendix A summarizes the system data, including the low-, high-frequency, and bearing circuit parameters for the subject series-Y and parallel-Y IM stator winding configurations.

For computer studies and comparisons with the experimental measurements, the case-study system has also been implemented in MATLAB/Simulink using the proposed WBDCPVBR model and its conventional counterpart models, i.e., low-frequency DCPVBR [17] and universal HF [5] models.

A. Experimental Verification of DM and CM Impedances

The DM and CM impedance response characteristics of the

proposed WBDCPVBR model are verified here against the existing IM models and the experimental measurements. It is considered that the rotor is at a standstill, and the stator terminals are disconnected from the three-phase AC power supply for impedance extraction.

In this paper, for benchmarking purposes, the measured DM and CM impedances of the subject IM series-Y winding connection from 100Hz to 10MHz have been obtained using a Venable frequency response analyzer (FRA) according to the experimental setup shown in Fig. 4. The CM and DM impedance characteristics of the universal HF model and proposed WBDCPVBR model were calculated through frequency sweep method applied to the simulated circuits, instead of deriving the analytical equations. In general, deriving the CM and DM impedances is possible, but it would be cumbersome and less practical due to the high order of the resulting circuits (series/parallel combinations in DM and CM).

Instead, the frequency sweep is also a precise and straightforward method because it avoids the laborious work of converting the per-phase equivalent circuits of the universal HF model and the proposed WBDCPVBR model into their

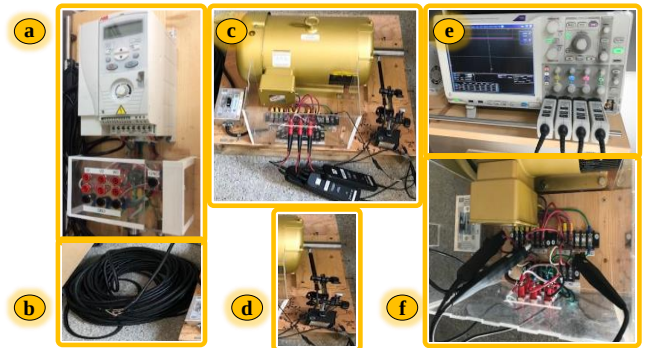


Fig. 8. Experimental setup of the VFD system: (a) ABB ACS150 drive with safety block, (b) power cable, (c) three-phase 7.5hp Baldor EM3710T IM with arbitrary stator winding connections and terminal voltage measurement setup, (d) bearing voltage measurement setup; (e) oscilloscope for displaying and recording output waveforms; (f) phase and CM current measurement setup.

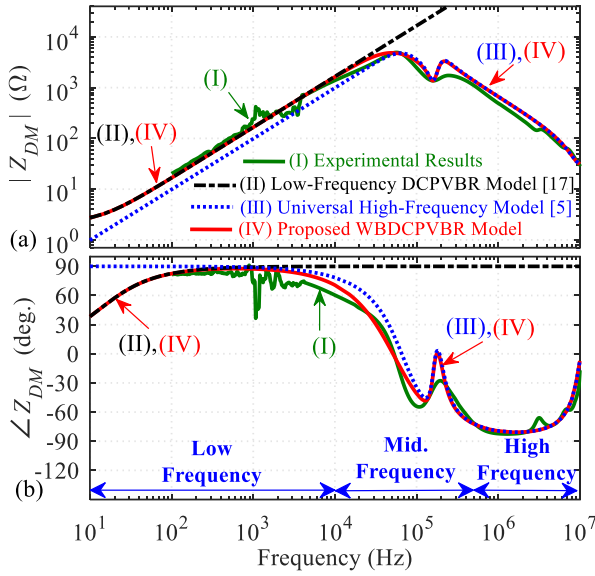


Fig. 9. Superimposed measured and simulated DM impedance characteristics of the proposed WBDCPVBR model for the series-Y winding connection: (a) magnitude of the DM impedance, and (b) phase of the DM impedance.

equivalent CM circuits and then deriving the impedances analytically. Also, considering that the motor is stationary, the rotor speed in the proposed WBDCPVBR model should be set to zero when calculating these DM and CM impedances for consistency with experimental measurements.

The resulting DM and CM impedance characteristics (i.e. magnitude and phase spectrums) are shown in Figs. 9 and 10, respectively. In the same figure, the simulation results of the low-frequency DCPVBR, universal HF, and the proposed WBDCPVBR IM models are superimposed with the experimental results. These characteristics are divided into three regions: low-frequency range (10Hz-10kHz), mid-frequency range (10kHz-500kHz), and high-frequency range (500kHz-10MHz).

As observed in Fig. 9 (a) and (b), the DM impedance shows inductive behavior in the low-mid frequency range (i.e. 10Hz-50kHz) with its corresponding phase starting from around 38° (for LF-DCPVBR and proposed WBDCPVBR models at 10Hz) and 81° (for measurement results at 100Hz). However, the DM impedance starts to demonstrate a capacitive behavior in the high-frequency range (i.e. 500kHz-10MHz) with its corresponding phase angle ending around -7.4° (for universal HF and proposed WBDCPVBR models at 10MHz) and 18° (for measurement results at 10MHz). Meanwhile, in the mid-frequency range, DM impedances alternate between inductive and capacitive behavior. While, as observed in Fig. 10 (a) and (b), the CM impedances overall behave capacitively in a broad frequency range with few exceptions at mid frequencies (due to the resonance phenomenon) as reflected by the corresponding negative phase angles starting from around -90° (for universal HF and proposed WBDCPVBR models at 10Hz) and -75° (for measurement results at 100Hz).

It is also observed in Fig. 9 that the LF-DCPVBR model

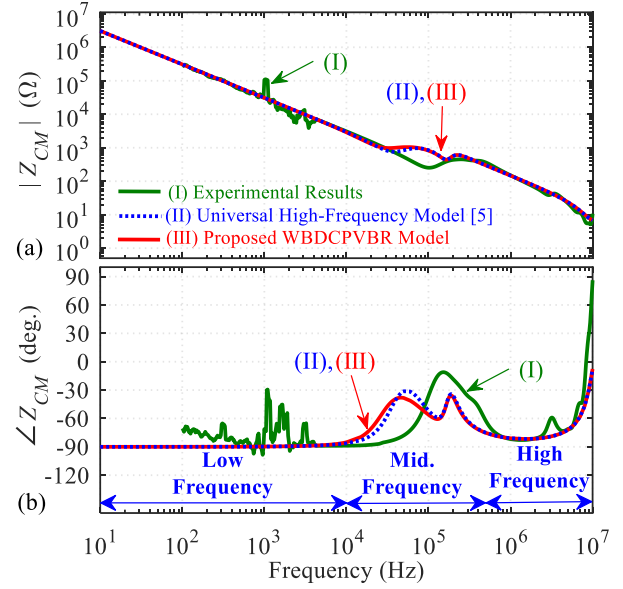


Fig. 10. Superimposed measured and simulated CM impedance characteristics of the proposed WBDCPVBR model for series-Y winding connection: (a) magnitude of the CM impedance, and (b) phase of the CM impedance.

accurately predicts the DM impedances only in the low-frequency range and does not predict the DM impedance in the mid-high frequency range. In addition, the LF-DCPVBR model also does not predict the CM impedances at all in the entire spectrum due to the lack of the stator frame connection point and open-circuit in the LF-DCPVBR model for extracting CM impedance based on the configuration in Fig. 4(b). On the other hand, the universal HF model predicts the CM impedances reasonably well in the wide frequency range. However, the universal HF model accurately captures the DM impedances only at mid-high frequencies and gives a noticeable error in the low-frequency range. Meanwhile, the proposed WBDCPVBR model offers higher accuracy compared to the other models and can accurately capture the machine's measured DM and CM impedances across almost the entire frequency spectrum from $\sim$ DC to 10MHz. Moreover, the proposed WBDCPVBR model exhibits consistent behavior and accurately matches the experimental impedance measurements while having an excellent agreement between the LF-DCPVBR model results (i.e., in the low-frequency range) and universal HF model results (i.e., in the mid-high frequency range).

It is also noted that there is a slight difference in the DM and CM impedances between the experimental and simulation results at some frequencies in the mid-frequency range. However, the accuracy of simulations can be further improved by utilizing advanced parameter optimization methods to estimate the model parameters with higher accuracy.

Overall, the frequency domain impedance analysis results verify the effectiveness of the proposed WBDCPVBR model in accurately predicting the wide-band impedance characteristics of IMs from $\sim$ DC to 10MHz, consistent with the experimental measurements, which represents a significant improvement over the conventional/existing state-of-the-art

IM models.

B. High-Frequency DM and CM Analysis in VFD System

A test case study has been carried out on a short cable-fed induction machine drive system to validate the high-frequency DM and CM responses in the time domain. It is assumed that the machine operates under steady state and no-load condition (approximately corresponding to 40Hz stator input frequency). The measurements are taken from the experimental setup in Fig. 8 when the IM is configured in parallel-Y stator windings and is driven by the ABB VFD (Model: ACS150-03U-09A8-2) through a short cable of 1m in length. The VFD has a two-level inverter topology, and its switching frequency is 4kHz. The CM ground current is measured using a current probe as the current that flows out of the motor frame to the floating ground point of the VFD, as shown in Fig. 1. The bearing voltage is measured with carbon brushes through the AEGIS's bearing voltage measurements test setup mounted on the machine shaft as the potential difference between the shaft at the drive-end and the machine stator frame, as shown in Fig. 8.

For adequate/fair comparison, the phase-to-ground voltages $\mathbf{v}_{abcg}$ of the three-phase VFD output are measured from the experiments and then fed into the MATLAB/Simulink simulations using controlled voltage sources for the proposed model verification. The waveform comparison for the system variables, i.e. phase a voltage and current, neutral-ground voltage, bearing voltage, CM and DM currents as obtained from the experimental measurements and the proposed WBDCPVBR model are presented in Figs. 11–13, respectively.

As it is observed in Figs. 11–13, the proposed WBDCPVBR model yields a reasonably accurate prediction of the corresponding system variables, and the simulation results are in good agreement with the experimental measurements. Since short cables are used (and the ABB VFD does not have WBG semiconductor devices), the over-voltage reflections are not pronounced in Figs. 11–13. Moreover, it is also seen that the measured phase-ground voltages are asymmetric, i.e., slightly fluctuate, which is due to the floating ground point (i.e., middle point between the DC link capacitors C_1 & C_2) and varying DC link capacitor voltage between $\pm 177\text{V}$ in the ABB VFD system.

The observed CM current in Figs. 11(e) and 12(c) have spikes and are capacitive in nature, which is due to the switching as well as the capacitive coupling among the phases and the ground terminals in the IM. Also, the observed ripples in the phase current waveform predicted by the proposed model are reasonably close to the experimental measurements, as seen in Fig. 11(b). It is also observed from Figs. 11–13 that the bearing voltage mirrors the CM voltage v_{CM} and the neutral-to-ground voltage v_{ng} through a bearing-voltage-ratio (BVR) such that $v_B = \text{BVR} \times v_{CM}$, and it can be readily predicted by the proposed WBDCPVBR model augmented with a capacitive voltage divider bearing circuit. The BVR of

the subject motor was calculated to be $\sim 7.14\%$ using measured v_B and v_{CM} waveforms. It is noted that the CM and neutral-to-

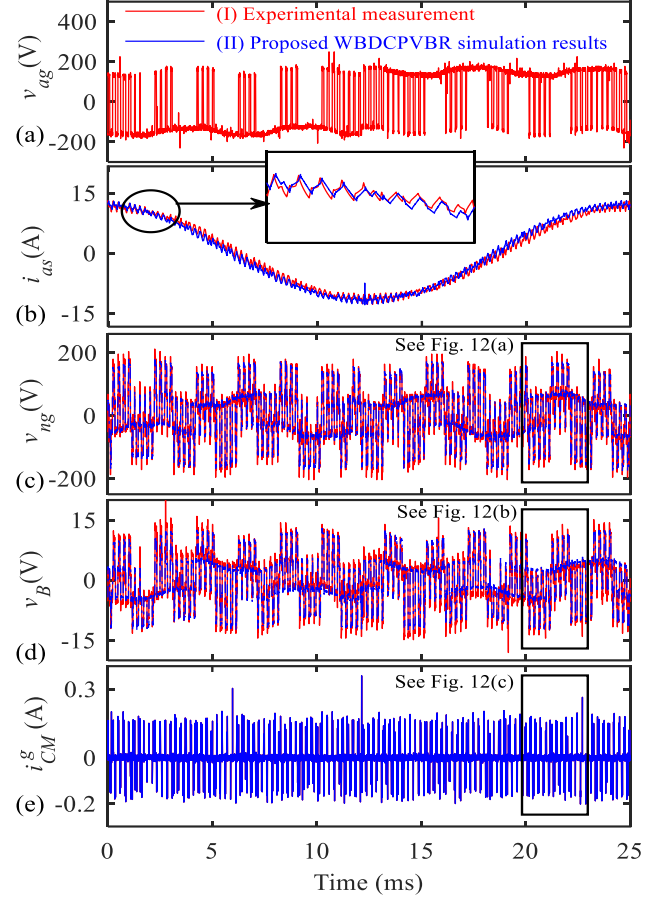


Fig. 11. Response of several system variables as obtained by the experimental measurements and the proposed model under no-load condition for a 1m power cable: (a) v_{ag} , (b) i_{as} , (c) v_{ng} , (d) v_B , and (e) i_{CM}^g .

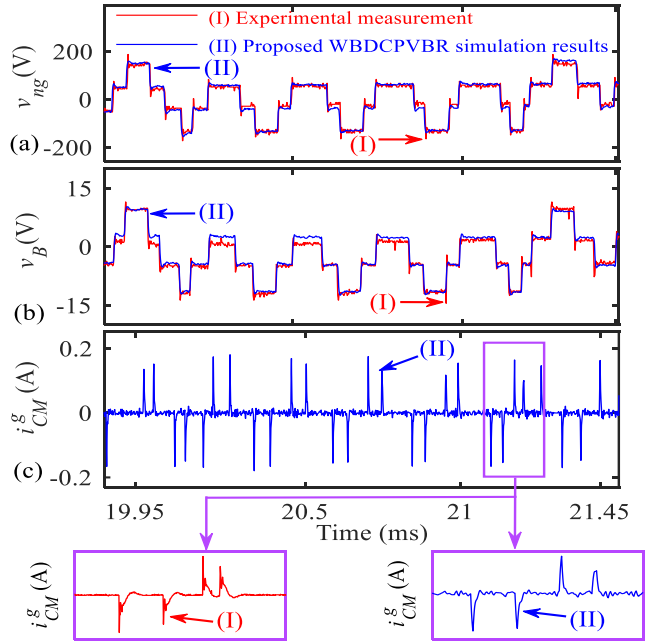


Fig. 12. Magnified view of the corresponding variables from Fig. 11 as obtained by the experimental measurements and the proposed model under no-load condition for a 1m power cable: (a) v_{ng} , (b) v_B , and (c) i_{CM}^g .

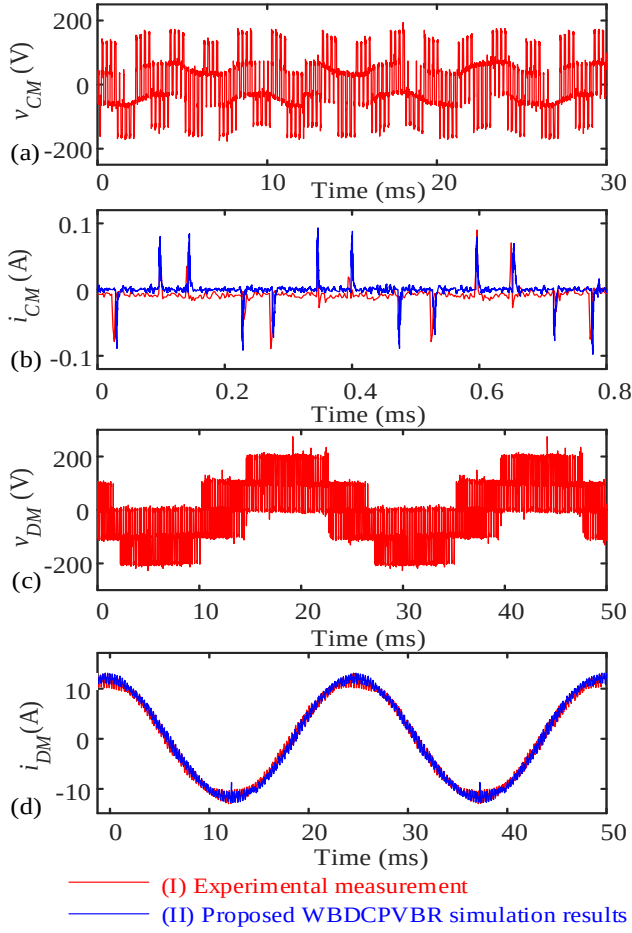


Fig. 13. CM and DM response of parallel-Y IM configuration as obtained by the experimental measurements and the proposed WBDCPVBR model under no-load condition for a 1m power cable: (a) v_{CM} , (b) i_{CM} , (c) v_{DM} , and (d) i_{DM} .

ground voltages are equal in a balanced three-phase system but are slightly different in an unbalanced system.

The accuracy of the proposed model can be further verified from the magnified plot comparisons of selected variables, i.e., neutral-to-ground voltage, bearing voltage, and CM ground current, as depicted in Fig. 12. Therein, it is observed that the results of the proposed model match closely the experimental measurements despite small errors which are mainly introduced due to fitting of high-frequency DM and CM impedances.

Furthermore, Figs. 13(a)–(d) demonstrate the agreement between the measured and simulated waveforms of the CM and DM currents of the IM-drive system, which are calculated from the typical DM and CM formulas listed in Appendix-B section (B1), (B2). It is also worth noting that the fundamental 40Hz component of the DM voltage waveform, as shown in Fig. 13(c) for the parallel-Y IM winding connection, is calculated to be 72.9 V_{rms}. Meanwhile, simulations with the proposed model resulted in DM current's (as shown in Fig. 13(d)) fundamental component of 7.4 A_{rms}, corresponding to a DM impedance of 9.8Ω at 40Hz under steady-state operation. However, it was also noted that at the standstill condition, the fundamental DM current component obtained by the proposed

model becomes 42.8A_{rms}, thus resulting in a DM impedance of 1.7Ω at 40Hz.

This observation is consistent with the DM impedance response characteristics of the proposed WBDCPVBR and LFVBR model at 40Hz standstill condition as depicted in Fig. 9(a), wherein $Z_{DM,series-Y}$ equals to $4 \times Z_{DM,parallel-Y}$ in the low-frequency range. Therefore, the case study verifies the proposed WBDCPVBR model for short-cable fed IM drive against the experimental results.

C. Long-Cable Fed Induction Machine Drive System

To demonstrate the reflected transient overvoltage phenomenon caused by the wave propagation effects in the long cable-fed IM drive, a series-Y configuration of the same motor fed through a 50m cable is considered in this study. The measured waveforms of the phase-to-ground voltages and the bearing voltage are presented in Fig. 14. It is seen in Fig. 14 that the overvoltage ringing with peaks roughly equal to 2.1 times the DC link voltage, and are caused by the wave propagation effects between the motor and VFD.

In the simulation of this case study, a three-phase distributed parameter line model of length 50m in the Simscape Electrical toolbox is inserted between the motor drive and the proposed WBDCPVBR model. The line model parameters are shown in Table III of Appendix A. The response of the three phase-to-ground voltages and the bearing voltage, as obtained by the proposed IM model, are depicted in Fig. 15. As seen, the overvoltage reflections across the IM terminals are consistent with the experimental results in Fig. 14. It may also be expected that the proposed WBDCPVBR model can yield more accurate results when a more precise high-frequency cable model is used instead (which is outside of the scope for this paper).

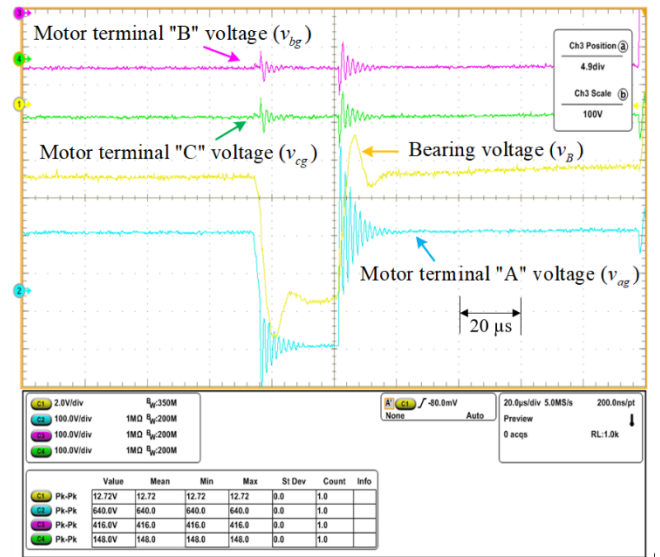


Fig. 14. Measured reflected overvoltage waveforms at the motor terminals and bearing/shaft for a 50m power cable and 7.5hp series-Y motor configuration.

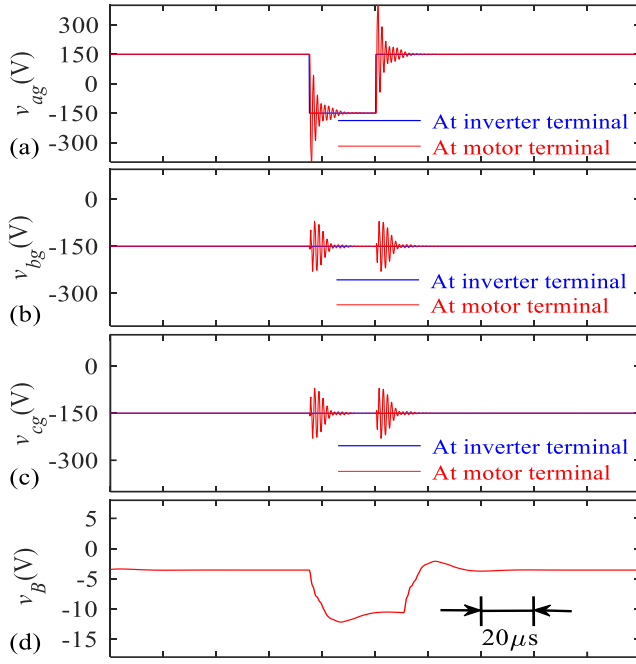


Fig. 15. Reflected overvoltage waveforms at the motor terminals and bearing/shaft for a 50m power cable and 7.5hp series-Y motor configuration as obtained by the proposed model: (a) v_{ag} , (b) v_{bg} , (c) v_{cg} , and (d) v_B .

IV. CONCLUSION

Wide-frequency modeling of IMs is essential in studying machine-converter interactions. This paper developed a wide-band, decoupled constant-parameter VBR (WBDCPVBR) model for three-phase squirrel-cage induction machines with Y-connected stator windings. The new model is achieved by incorporating the low-frequency DCPVBR, the universal high-frequency, and the bearing circuit models. The proposed model was demonstrated to predict various phenomena associated with the VFD systems, such as reflected wave transient over-voltages, DM and CM impedances, and bearing/shaft voltages. The proposed model also offers advantages in terms of its machine-network interface, simple structure, straightforward parameterization procedure, and accuracy in predicting the transient and steady-state phenomena from $\sim$ DC to $\sim$ 10 MHz. Due to these features, it is envisioned that the proposed model may become an advantageous asset for studying VFD systems using offline and real-time EMT simulation programs.

The proposed model may also be extended to represent the Δ -connected stator windings. The basic methodology will include the conversion of the high-frequency circuits from Y to an equivalent Δ , as developed in [5, see Section IV-A], combined with a different version of the low-frequency voltage-behind-reactance model in [17, see Section IV] that is also suitable for Δ connection. However, many details on this extension cannot be included in this paper due to limited space, and this future work will be considered in a follow-up publication.

APPENDIX A

TABLE II. BALDOR INDUCTION MOTOR PARAMETERS FOR SERIES-Y AND PARALLEL-Y STATOR WINDING

PARAMETERS	SERIES-Y	PARALLEL-Y
Rated Power / Voltage / Poles	7.5hp / 460V / 4	7.5hp / 230V / 4
Rated Current / Speed	9.5A / 1770rpm	19A / 1770rpm
Inertia / Frequency	0.0394kg.m ² / 60Hz	0.0394kg.m ² / 60Hz
LOW-FREQUENCY PARAMETERS		
r_s, r_r	1.0283 Ω , 0.4422 Ω	0.2571 Ω , 0.1106 Ω
L_{ls}, L_{lr}, L_m (unsaturated)	9.1mH, 9.1mH, 143.8mH	2.3mH, 2.3mH, 36mH
HIGH-FREQUENCY PARAMETERS [5]		
C_{q1}, C_{q2}	356.24pF, 1.3624nF	712.49pF, 1.3624nF
R_{q1}, R_{q2}	20.04 Ω , 757.2 Ω	10.02 Ω , 378.6 Ω
L_{ls}, R_e	10.4mH, 3.2893k Ω	2.6mH, 822.3 Ω
R_t, L_t	1368 Ω , 5.6mH	684 Ω , 2.8mH
C_t, L_c	171.86pF, 66 μ H	171.86pF, 33 μ H
BEARING CIRCUIT PARAMETERS		
C_{ws}, C_{wr}	2.0435nF, 2.2007nF	2.0435nF, 2.2007nF
$C_{sr}, C_{b,DE}, C_{b,NDE}$	28.202nF, 204pF, 204pF	28.202nF, 204pF, 204pF

TABLE III. DISTRIBUTED PARAMETER CABLE MODEL

PARAMETERS	VALUE
Length	50m
r_1, r_0	10.184 $\mu\Omega$ /m, 0.3091 $\times 10^{-3}$ Ω /m
l_1, l_0	1.4939 μ H/m, 3.3011 μ H/m
c_1, c_0	20.384 pF/m, 6.2008 pF/m

APPENDIX B

Calculations of CM and DM Voltages and Currents of Three-Phase IM-Drive System [26]

The equations to calculate the CM and DM voltages and currents of a three-phase variable frequency drive system can be expressed as follows:

$$v_{CM} = \frac{1}{3}(v_{ag} + v_{bg} + v_{cg}), \quad i_{CM} = \frac{1}{3}(i_{as} + i_{bs} + i_{cs}), \quad (B1)$$

$$v_{DM} = \frac{1}{3}(2v_{ag} - v_{bg} - v_{cg}), \quad i_{DM} = \frac{1}{3}(2i_{as} - i_{bs} - i_{cs}). \quad (B2)$$

REFERENCES

- [1] P. Xie, G. Vakil, and C. Gerada, "Electric drive systems with long feeder cables," *IET Electr. Power Appl.*, vol. 14, no. 1, pp. 16–30, Jan. 2020.
- [2] B. Narayanasamy, *et al.*, "Reflected wave phenomenon in SiC motor drives: Consequences boundaries and mitigation," *IEEE Trans. Power Electron.*, vol. 35, no. 10, pp. 10629–10642, Oct. 2020.
- [3] J. Sun and L. Xing, "Parameterization of three-phase electric machine models for EMI simulation," *IEEE Trans. Power Electron.*, vol. 29, no. 1, pp. 36–41, Jan. 2014.
- [4] C. Purcarea, P. Mutschler, O. Magdun, A. Rocks, and A. Binder, "Time domain simulation models for inverter-cable-motor system in electrical drives," in *Proc. 13th Eur. Conf. Power Electron. Appl.*, pp. 1–10, 2009.
- [5] M. S. Toulabi, L. Wang, L. Bieber, S. Filizadeh, and J. Jatskevich, "A universal high-frequency induction machine model and characterization

- method for arbitrary stator winding connections,” *IEEE Trans. Energy Convers.*, vol. 34, no. 3, pp. 1164–1177, Sep. 2019.
- [6] B. Mirafzal, *et al.*, “Universal induction motor model with low-to-high frequency-response characteristics,” *IEEE Trans. Ind. Appl.*, vol. 43, no. 5, pp. 1233–1246, Sep./Oct. 2007.
 - [7] B. Mirafzal, G. L. Skibinski, and R. M. Tallam, “Determination of parameters in the universal induction motor model,” *IEEE Trans. Ind. Appl.*, vol. 45, no. 1, pp. 142–151, Jan./Feb. 2009.
 - [8] A. Moreira, *et al.*, “High frequency modeling for cable and induction motor overvoltage studies in long cable drives,” in *Proc. IEEE Ind. Appl. Conf. 36th IAS Annu. Meeting*, vol. 3, pp. 1787–1794, 2001.
 - [9] L. Wang, C. Ngai-Man Ho, F. Canales, and J. Jatskevich, “High-frequency modeling of the long-cable-fed induction motor drive system using TLM approach for predicting overvoltage transients,” *IEEE Trans. Power Electron.*, vol. 25, no. 10, pp. 2653–2664, Oct. 2010.
 - [10] M. J. Scott, *et al.*, “Reflected wave phenomenon in motor drive systems using wide bandgap devices,” in *Proc. IEEE Workshop Wide Bandgap Power Devices Appl.*, pp. 164–168, 2014.
 - [11] O. Magdun and A. Binder, “High-frequency induction machine modeling for common mode current and bearing voltage calculation,” *IEEE Trans. Ind. Appl.*, vol. 50, no. 3, pp. 1780–1790, May/Jun. 2014.
 - [12] T. Plazenet, *et al.*, “A comprehensive study on shaft voltages and bearing currents in rotating machines,” *IEEE Trans. Ind. Appl.*, vol. 54, no. 4, pp. 3749–3759, Jul./Aug. 2018.
 - [13] F. A. Kharanaq, A. Emadi, and B. Bilgin, “Modeling of conducted emissions for EMI analysis of power converters: State-of-the-art review,” *IEEE Access*, vol. 8, pp. 189313–189325, 2020.
 - [14] H. C. Stanley, “An analysis of the induction machine,” *Trans. Amer. Inst. Elect. Eng.*, vol. 57, no. 12, pp. 751–755, Dec. 1938.
 - [15] J. R. Martí and T. O. Myers, “Phase-domain induction motor model for power system simulators,” in *Proc. IEEE Commun. Power Comput. Conf.*, vol. 2, pp. 276–282, May 1995.
 - [16] P. C. Krause, *et al.*, *Analysis of electric machinery and drive systems*, 3rd Edition, IEEE Press, Piscataway, NJ, 2013.
 - [17] L. Wang, J. Jatskevich, and S. D. Pekarek, “Modeling of induction machines using a voltage-behind-reactance formulation,” *IEEE Trans. Energy Convers.*, vol. 23, no. 2, pp. 382–392, Jun. 2008.
 - [18] L. Wang, *et al.*, “Methods of interfacing rotating machine models in transient simulation programs,” *IEEE Trans. Power Del.*, vol. 25, no. 2, pp. 891–903, Apr. 2010.
 - [19] S. D. Sudhoff, D. C. Aliprantis, B. T. Kuhn, and P. L. Chapman, “An induction machine model for predicting inverter-machine interaction,” *IEEE Trans. Energy Convers.*, vol. 17, no. 2, pp. 203–210, Jun. 2002.
 - [20] I. Stevanovic, B. Wunsch, and S. Skibin, “Behavioral high-frequency modeling of electrical motors,” in *Proc. Appl. Power Electron. Expo. Conf.*, pp. 2547–2550, Mar. 2013.
 - [21] E. Zhong and T. A. Lipo, “Improvements in EMC performance of inverter-fed motor drives,” *IEEE Trans. Ind. Appl.*, vol. 31, no. 6, pp. 1247–1256, Nov. 1995.
 - [22] G. Grandi, D. Casadei, and A. Massarini, “High frequency lumped-parameter model for AC motor windings,” in *Proc. Eur. Power Electron. Appl. Conf.*, vol. 2, pp. 578–583, 1997.
 - [23] A. Boglietti and E. Carpaneto, “Induction motor high frequency model,” in *Proc. IEEE Ind. Appl. Conf. 34th IAS Annu. Meeting*, vol. 3, pp. 1551–1558, 1999.
 - [24] S.-P. Weber, E. Hoene, S. Guttowski, W. John, and H. Reichl, “Modeling induction machines for EMC-analysis,” in *Proc. IEEE 35th Annu. Power Electron. Spec. Conf.*, vol. 1, pp. 94–98, 2004.
 - [25] M. Schinkel, S. Weber, S. Guttowski, W. John, and H. Reichl, “Efficient HF modeling and model parameterization of induction machines for time and frequency domain simulations,” in *Proc. 21st Annu. IEEE Appl. Power Electron. Expo. (APEC) Conf.*, pp. 1181–1186, Mar. 2006.
 - [26] G. Grandi, *et al.*, “Analysis of common- and differential-mode HF current components in PWM inverter-fed ac motors,” in *Proc. IEEE Power Electron. Spec. Conf.*, pp. 1146–1151, 1998.
 - [27] D. Busse, *et al.*, “Bearing currents and their relationship to PWM drives,” *IEEE Trans. Power Electron.*, vol. 12, no. 2, pp. 243–252, Mar. 1997.
 - [28] M. Asefi and J. Nazarzadeh, “Survey on high-frequency models of PWM electric drives for shaft voltage and bearing current analysis,” *IET Elect. Syst. Transp.*, vol. 7, no. 3, pp. 179–189, 2017.
 - [29] P. Han, *et al.*, “On the modeling of bearing voltage and current in PWM converter-fed electric machines using electromagnetic finite element analysis,” in *Proc. 2021 IEEE Energy Convers. Cong. Expo. (ECCE)*, pp. 4606–4610, 2021.
 - [30] P. Han, *et al.*, “Modeling of bearing voltage in electric machines based on electromagnetic FEA and measured bearing capacitance,” *IEEE Trans. Indus. Appl.*, vol. 57, no. 5, pp. 4765–4775, Jun. 2021.
 - [31] A. Muetze and A. Binder, “Calculation of motor capacitances for prediction of the voltage across the bearings in machines of inverter-based drive systems,” *IEEE Trans. Ind. Appl.*, vol. 43, no. 3, pp. 665–672, May/Jun. 2007.
 - [32] J. M. Erdman, *et al.*, “Effect of PWM inverters on AC motor bearing currents and shaft voltages,” *IEEE Trans. Ind. Appl.*, vol. 32, no. 2, pp. 250–259, Mar./Apr. 1996.
 - [33] S. Quabeck, V. Grau, and R. W. De Doncker, “Modeling and mitigation of bearing currents in electrical traction drives,” in *Proc. IEEE 23rd Int. Elect. Mach. Syst. Conf.*, pp. 1101–1106, 2020.
 - [34] L. Ran, S. Gokani, J. Clare, K. J. Bradley, and C. Christopoulos, “Conducted electromagnetic emissions in induction motor drive systems. Part I. Time domain analysis and identification of dominant modes,” *IEEE Trans. Power Electron.*, vol. 13, no. 4, pp. 757–767, Jul. 1998.
 - [35] IEEE, 112-2017, “IEEE Standard Test Procedure for Polyphase Induction Motors and Generator”, pp. 1–115, Feb. 2018.
 - [36] Y. Xu and D. C. Aliprantis, “Wound-rotor induction machine model with saturation and high-frequency effects,” in *Proc. Int. Elect. Mach. (ICEM) Conf.*, pp. 1838–1844, 2014.
 - [37] S. D. Sudhoff, *et al.*, “Experimental characterization procedure for use with an advanced induction machine model,” *IEEE Trans. Energy Convers.*, vol. 18, no. 1, pp. 48–56, Mar. 2003.
 - [38] ABB ACS150 Micro Drive, 2020. [Online]. Available: <https://new.abb.com/drives/low-voltage-ac/micro/acs150>
 - [39] E. Mostajeran, N. Amiri, S. Ebrahimi, and J. Jatskevich, “Electrical machines in electromagnetic transient simulations: Focusing on efficient and accurate models,” *IEEE Electrification Magazine*, vol. 11, no. 4, pp. 38–53, Dec. 2023.
 - [40] A. Narang, B. K. Gupta, E. P. Dick, and D. Sharma, “Measurement and analysis of surge distribution in motor stator windings,” *IEEE Trans. Energy Convers.*, vol. 4, no. 1, pp. 126–134, Mar. 1989.



Sheraz Baig (S’22) received the B.Sc. degree in electrical engineering from the Capital University of Science and Technology, Islamabad, in 2019, and the M.A.Sc. degree in electrical and computer engineering from The University of British Columbia (UBC), Vancouver, BC, Canada, in 2023. He is currently with Engineering Applications Group, Powertech Labs Inc., Surrey, BC, Canada. His research interests include modeling and analysis of electrical machines and drives, and power system operation and simulation.



Seyyedmilad Ebrahimi (S’13–M’19) received the B.Sc. and M.Sc. degrees in electrical engineering from the Sharif University of Technology, Tehran, Iran, in 2010 and 2012, respectively, and the Ph.D. degree in electrical and computer engineering from The University of British Columbia (UBC), Vancouver, BC, Canada, in 2019. He is currently a Postdoctoral Teaching and Research Fellow with the Department of Electrical and Computer Engineering, UBC. His research interests include modeling and analysis of power electronic converters and electrical machines, application of power electronics to power systems, modeling and control of power systems, and simulation of electro-magnetic transients.



Juri Jatskevich (M'99–F'17) received the M.S.E.E. and Ph.D. degrees in electrical engineering from Purdue University, West Lafayette IN, USA, in 1997 and 1999, respectively. Since 2002, he has been with The University of British Columbia, Vancouver, BC, Canada, where he is currently a Professor

with the Department of Electrical and Computer Engineering. His research interests include power electronic systems, electrical machines and drives, modeling and simulation of electromagnetic transients. Dr. Jatskevich served as an Associate Editor for IEEE TRANSACTIONS ON POWER ELECTRONICS during 2008–2013, the Editor-in-Chief of IEEE TRANSACTIONS ON ENERGY CONVERSION during 2013–2019, and the Editor-in-Chief At-Large of the IEEE PES journals during 2019–2020. He chaired the IEEE Task Force on Dynamic Average Modeling. He was the recipient of the 2022 IEEE PES Cyril Veinott Electromechanical Energy Conversion Award and the 2023 Nagamori Award.



Liwei Wang (S'04–M'10) received the Ph.D. degree in electrical and computer engineering from the University of British Columbia, Vancouver, BC, Canada, in 2010. In August 2010, he joined the ABB Corporate Research Center, Västerås, Sweden, as a Scientist and then as a Senior Scientist. Since July 2014, he has

been with the School of Engineering, the University of British Columbia, Kelowna, BC, Canada, where he is presently an Associate Professor. His research interests include power system modeling and simulation; electrical machines and drives; utility power electronics applications and distributed generation.



Aria Fani received the Master of Applied Science degree in electrical engineering from the University of British Columbia, Kelowna, BC, Canada, in 2022. He also received the Bachelor of Technology degree in electronics and Diploma of Technology associate degree in electronics from British Columbia

Institute of Technology, Burnaby, BC, Canada in 2015 and 1994 respectively. He has more than 25 years of experience working in telecommunications and power industries in Canada. Since April 2016, he has worked for the School of Engineering, the University of British Columbia, Kelowna, BC, Canada, where he has supervised the research and teaching laboratories. His research interests include power system modeling and simulation, electrical machines and drives, utility power electronics applications, and distributed generation.